Signals and Systems

Fifth Revised Edition

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Foreword

Without question one of the most important aspect in this text is the manner in which the content lends itself to typical syllabus content.

The order of the chapters is continuous, in sequential presentation with important concepts and methods of analysis. The number of worked example is rich, simpler in presentation and good in conceptual presentation.

On discussion with numerous students in graduate level the global requirement is a need of a book that gives a more systems approach, developed to support students who need to adapt themselves in applications of a real time engineering world.

There is no question that a primary goal of any publication is to give a book free of errors, a possibility. Solution in the book are confirming to the highest level of accuracy obtainable.

Specific area of the text are an extension of a sequential introduction of systems approach to signal analysis with an increased emphasis on the frequency response.

In precise the text is categorised into components of system with continuous signal analysis and discrete signal analysis. In any event the text "builds" from, the early chapter.

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1

Introduction to Signals

1.1 A Signal

A signal is defined as a physical quantity that varies with time, space or any other independent variable. The signal may depend on one or more independent variables. If a signal depends on only one variable, then it is known as one dimensional signal. For example, *ac* power supply signal, speech signal, Electrocardiagram and the variation of room temperature are one dimensional signals. Throughout this text we concentrate on this type of signals. If a signal depends on two independent variables then the signal is known as two dimensional signal. Examples are pictures, X-ray images and sonograms. Multidimensional signals depends on many variables.

Definition: A signal is defined as a physical quantity that varies with time, space or any other independent variables.

1.2 Signal Modeling

The representation of a signal by mathematical expression is known as signal modeling. If a signal can be represented by mathematical equation then the signal is said to be deterministic signal. Otherwise the signal is known as random signal.

1.3 Continuous-time, Discrete-time and Digital Signals

1.3.1 Continuous-time signal

Continuous-time signals are defined for all values of time t and is represented by $x(t)$. A continuous-time signal is also called an analog signal. Most of the signals encountered in practice are continuous-time signals. Fig. 1.1 shows different types of continuous-time signals. The temperature recorded over an interval of time is shown in Fig. 1.1a. It is defined at every time instant. Other example of continuous-time signals are the waveform of AC power supply and ECG waveform.

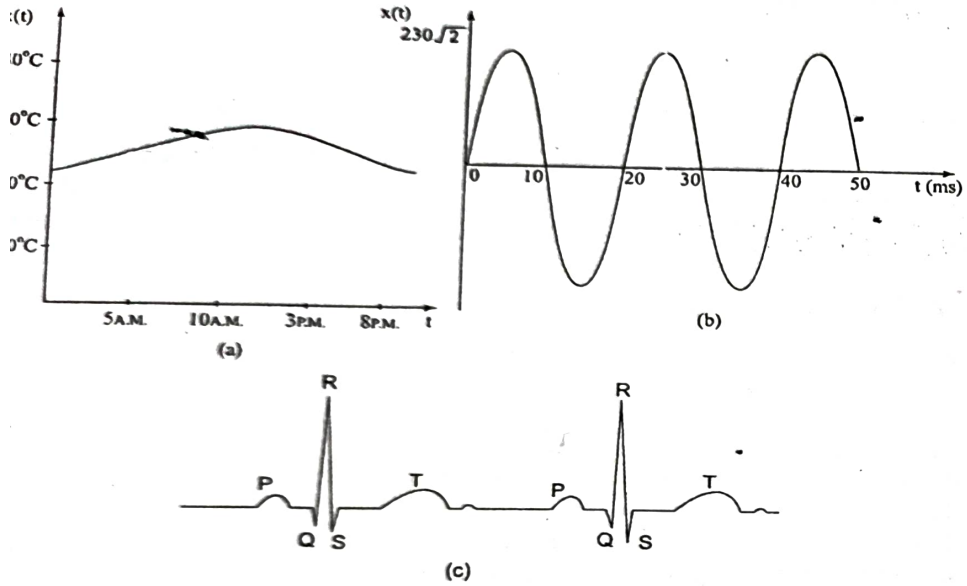

Fig. 1.1 Examples for continuous-time signals.

3.2 Discrete-time signal

e discrete-time signals are defined at discrete-instant of time and is represented $x(n)$, where n is index. Some signals are discrete in nature and some signals y be discrete representation of continuous-time signal.

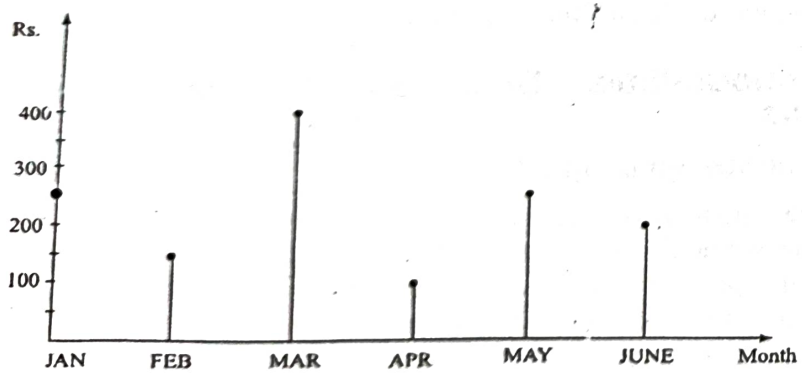

Fig. 1.2 A Discrete-time signal.

For example, the amount deposited every month in a savings account is discrete. The signal is shown in Fig. 1.2.

The other type of discrete-time signal is obtained by sampling a continuous-time signal at regular intervals. That is, the amplitude of the continuous-time signal is defined at particular intervals. For example, Fig. 1.1a shows the temperature in a room which is continuous-in-time. If the temperature is recorded at an interval of one hour then we obtain a discrete-time signal as shown in Fig. 1.3.

The discrete-time signal can be denoted by

$$x(nT) = x(t)\big|_{t=nT} \tag{1.1}$$

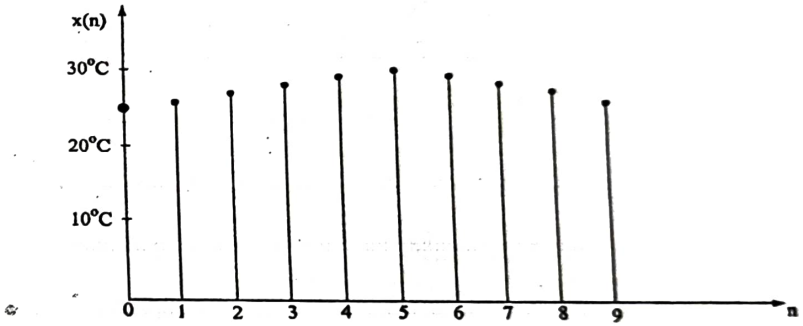

Fig. 1.3 Sampled sequence of signal shown in Fig. 1.1a.

where T is called the sampling period and n is an integer ranging from $-\infty$ to ∞ called the time index. We call $x(nT)$ as the sampled signal or sampled sequence of $x(t)$. The instants at which the signal appear are called sampling instants. For convenience we write $x(nT) = x(n)$, $n = 0, \pm 1, \pm 2$. Thus a discrete-time signal is represented by the sequence numbers $\ldots x(-2), x(-1), x(0), x(1), x(2) \ldots$

If we use an analog thermometer to measure the temperature then the temperature may be any value in a continuous range. But if we use a digital thermometer that has only two digits, the temperature recorded for the same time is different. For example, if the temperature reading at a point using analog thermometer is 37.2°C, then using digital thermometer with two digits shows the temperature as 37°C. In this case, the temperature (magnitude) value is said to be quantized. Here the values of temperature assumes a finite set of numbers (30, 31, 32 ..., 38, 39 etc.) and are called quantization levels. The difference between two adjacent quantization levels is called the quantization step. A digital themometer with two digits, that display only integers has quantization step equal to 1. If the temperature is displayed upto the first decimal point, then the quantization step is 0.1.

A signal that is discretized in time and quantized in amplitude is known as digital signal. In processing and transmission, digital signals are usually expressed in binary form, a string of zeros and ones.

Definition:

Continuous-time signals: The signals that are defined for every instant of time are known as continuous-time signals. They are denoted by $x(t)$.

Discrete-time Signals: The signals that are defined at discrete-instant of time are known as discrete-time signals. The discrete-time signals are continuous in amplitude and discrete-in-time. They are denoted by $x(n)$.

Digital signals: The signals that are discrete-in-time and quantized in amplitude are digital signals.

Solved Problem 1.1 Sketch the continuous-time signal $x(t) = 2\sin \pi t$ for an interval $0 \le t \le 2$. Sample the continuous-time signal with a sampling period $T = 0.2$ sec and sketch the discrete-time signal.

Solution:

Given

$$x(t) = 2\sin \pi t$$

To sketch the signal, we have to find the values of $x(t)$ for different values of t, that is

$$\begin{aligned}
x(0) &= 2\sin\pi(0) = 0\\
x(0.25) &= 2\sin\pi(0.25) = 2\left(\frac{1}{\sqrt{2}}\right) = \sqrt{2}\\
x(0.5) &= 2\sin\pi(0.5) = 2\\
x(0.75) &= 2\sin\pi(0.75) = \sqrt{2}\\
x(1) &= 2\sin\pi = 0\\
x(1.25) &= 2\sin\pi(1.25) = -\sqrt{2}\\
x(1.5) &= 2\sin\pi(1.5) = -2\\
x(1.75) &= 2\sin\pi(1.75) = -\sqrt{2}\\
x(2) &= 2\sin 2\pi = 0
\end{aligned}$$

Take t on x-axis and $x(t)$ on y-axis. Mark the above points and join all the points by a smooth curve to get a continuous-time signal as shown in Fig. 1.4a.

Given sampling period $T = 0.2$.

$$\begin{aligned}
x(n) = x(nT) &= x(t)\big|_{t=nT}\\
&= x(t)\big|_{t=0.2n}\\
&= x(0.2n)\\
\Rightarrow \quad x(n) &= 2\sin\pi(0.2n) = 2\sin(0.2\pi n)
\end{aligned}$$

$x(0) = 0;$
$x(1) = 2\sin(0.2\pi) = 1.175;$
$x(2) = 2\sin(0.4\pi) = 1.902;$
$x(3) = 2\sin(0.6\pi) = 1.902;$
$x(4) = 2\sin(0.8\pi) = 1.175;$
$x(5) = 2\sin\pi = 0$
$x(6) = 2\sin(1.2\pi) = -1.175$
$x(7) = 2\sin(1.4\pi) = -1.902$
$x(8) = 2\sin(1.6\pi) = -1.902$
$x(9) = 2\sin(1.8\pi) = -1.175$
$x(10) = 2\sin(2\pi) = 0$

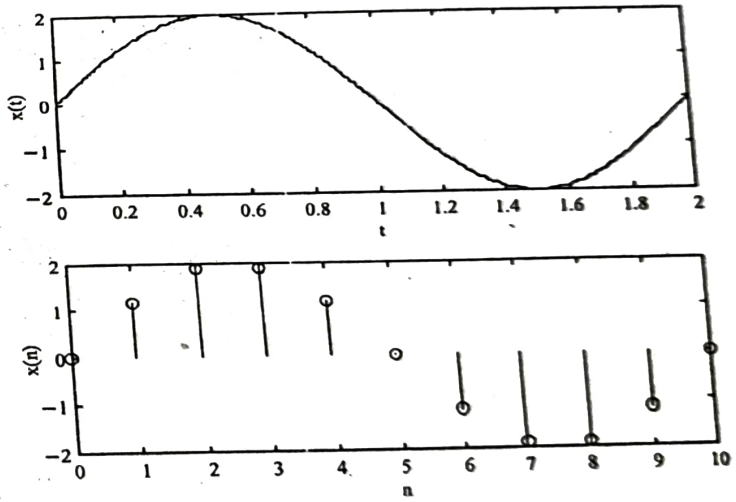

Fig. 1.4 (a) Signal $2\sin \pi t$ (b) Sampled signal of $2\sin \pi t$.

The sequence $x(n)$ can be written as

$$x(n) = \{0,\ 1.175,\ 1.902,\ 1.902,\ 1.175,\ 0,\ -1.175,\ -1.902,\ -1.902,\ -1.175,\ 0\}$$

Take the index n on x-axis and $x(n)$ on y-axis and draw the sequence as shown in Fig. 1.4b.

Solved Problem 1.2 Sketch the signal $x(t) = e^{-t}$ for an interval $0 \le t \le 2$. Sample the signal with a sampling period $T = 0.2$ second and sketch the discrete-time signal.

Solution:

Given

$x(t) = e^{-t}$; $x(0) = 1$; $x(0.5) = 0.606$; $x(1) = 0.36768$; $x(1.5) = 0.2237$; $x(2) = 0.1353$

To get the signal $x(t)$, mark the above points and draw a smooth curve joining all points as shown in Fig. 1.5a.

$$\begin{aligned} x(nT) &= x(t)\big|_{t=nT} \\ &= x(t)\big|_{t=0.2n} \\ &= x(0.2n) = e^{-0.2n} \end{aligned}$$

$x(0) = 1$
$x(1) = e^{-0.2} = 0.818$
$x(2) = e^{-0.4} = 0.67$
$x(3) = e^{-0.6} = 0.5488$
$x(4) = e^{-0.8} = 0.449$
$x(5) = e^{-1} = 0.3678$
$x(6) = e^{-1.2} = 0.3012$
$x(7) = e^{-1.4} = 0.2466$
$x(8) = e^{-1.6} = 0.2022$
$x(9) = e^{-1.8} = 0.1653$
$x(10) = e^{-2} = 0.1353$

The sequence values of $x(n)$ can be represented as

$x(n) = \{1, 0.818, 0.67, 0.5488, 0.449, 0.3678, 0.3012, 0.2466, 0.2022, 0.1653, 0.1353\}$

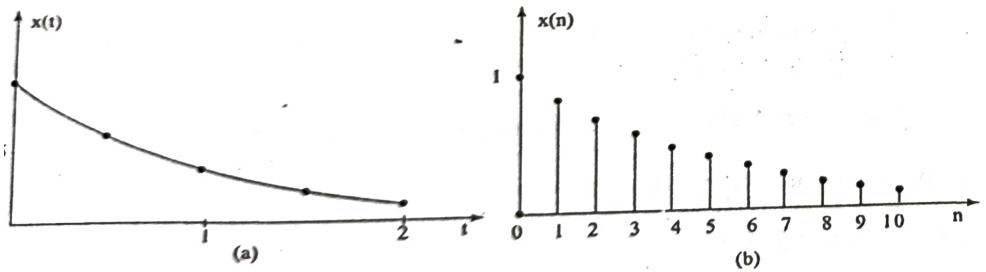

Fig. 1.5 (a) Signal e^{-t} (b) Sampled sequence.

Review Questions

1. Define signal.
2. Define (i) Continuous-time signal (ii) Discrete-time signal (iii) Digital signal

Practice Problem 1.1

Sketch the following signals

i) $x(t) = \sin\left(\frac{\pi}{4}t + \frac{\pi}{4}\right)$

ii) $x(t) = e^{2t}$ for $-2 \le t \le 2$

iii) $x(n) = 0.1\cos\left(\frac{\pi}{2}n + \frac{\pi}{3}\right)$

iv) $x(n) = (0.8)^n$ for $0 \le n \le 10$
$= 0$ otherwise

1.4 Elementary Continuous Time Signals

In section (1.2) we discussed that representing a signal using mathematical expression is known as signal modeling. In this section we are going to study about some of the signals, that are used as input for studying the system's behaviour. These signals include, step, impulse, ramp, sinusoidal and exponential functions. The above signals can be used as building blocks for the modeling of more complex signals.

1.4.1 Unit step function

The step function is an important signal used for analysis of many systems. For example when we turn the ignition key of a car, we actually introducing a step voltage (a step function) to the starting motor. Likewise when we apply brake to an automobile, we are applying a constant force (a step function). If a step function has unity magnitude then it is called unit step function.

The unit step function is defined as

$$\begin{aligned} u(t) &= 1 \quad \text{for} \quad t \ge 0 \\ &= 0 \quad \text{for} \quad t < 0 \end{aligned} \tag{1.2}$$

and is shown in Fig. 1.6a.

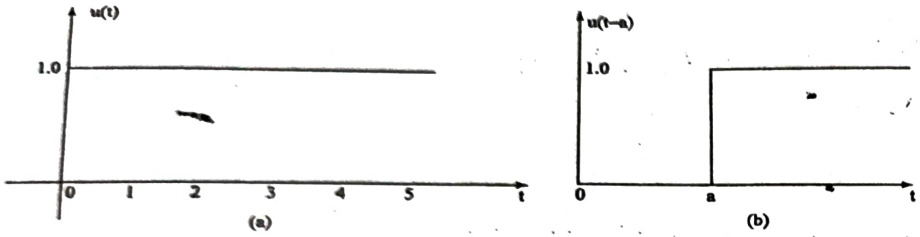

Fig. 1.6 (a) Unit step signal (b) Delayed unit step signal.

From Eq. (1.2) we can find that when the argument t in $u(t)$ is less than zero, then the unit step function is zero, and when the argument t in $u(t)$ is greater than or equal to zero then the unit step function is unity.

Similarly for the shifted unit step function $u(t-a)$ is zero if the argument $(t-a)$ is <0 or $t<a$ and is one if the argument $(t-a)>0$ or $t>a$.

That is

$$u(t-a)=1 \quad \text{for} \quad t \geq a$$
$$=0 \quad \text{for} \quad t<a. \tag{1.3}$$

1.4.2 Unit ramp function

The unit ramp function is defined as

$$r(t)=t \quad \text{for} \quad t \geq 0$$
$$=0 \quad \text{for} \quad t<0 \tag{1.4}$$

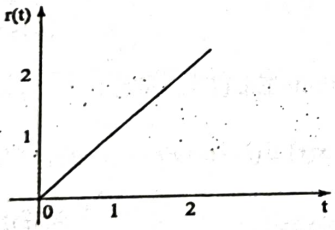

Fig. 1.7 Unit ramp signal

or

$$r(t)=tu(t) \tag{1.5}$$

The ramp function can be obtained by applying unit step function to an integrator.

$$r(t)=\int u(t)dt=\int dt=t \quad \text{(in the interval } t>0)$$

In otherwords, the unit step function can be obtained by differentiating the unit ramp function.

Thus

$$u(t)=\frac{d}{dt}r(t) \tag{1.6}$$

1.4.3 Unit parabolic function

The unit parabolic function is given by

$$p(t)=\frac{t^2}{2} \quad \text{for} \quad t \geq 0$$
$$=0 \quad \text{for} \quad t<0 \tag{1.7}$$

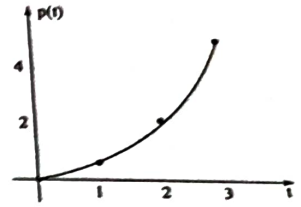

Fig. 1.8 Unit parabolic function

or

$$p(t)=\frac{t^2}{2}u(t) \tag{1.8}$$

The unit parabolic function can be obtained by integrating the ramp function.

$$p(t)=\int r(t)dt=\int t\,dt=\frac{t^2}{2} \qquad \text{(for} \quad t \geq 0\text{)}$$

In otherwords, the ramp function is derivative of the parabolic function.

Thus

$$r(t)=\frac{dp(t)}{dt} \tag{1.9}$$

1.4.4 Impulse function

The impulse function occupies an important place in signal analysis. It is defined as

$$\int_{-\infty}^{\infty}\delta(t)dt=1$$

and

$$\delta(t)=0 \quad \text{for} \quad t \neq 0 \tag{1.10}$$

That is the impulse function has zero amplitude everywhere except at $t=0$. At $t=0$, the amplitude is infinity such that the area under the curve is equal to one. The unit impulse function can be obtained by using a limiting process on the rectangular pulse function shown in Fig. 1.9. Note that the area under the rectangular pulse is equal to unity.

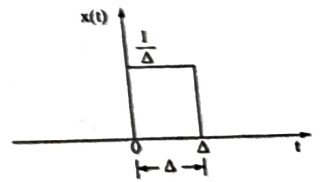

Fig. 1.9 Rectangular pulse function

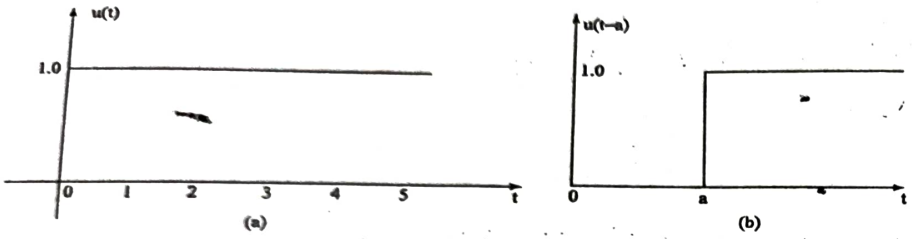

Fig. 1.6 (a) Unit step signal (b) Delayed unit step signal.

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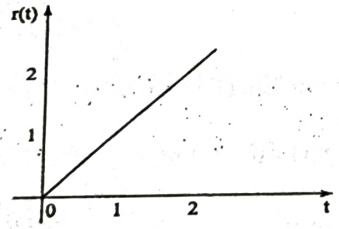

Fig. 1.7 Unit ramp signal

or

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In otherwords, the unit step function can be obtained by differentiating the unit ramp function.

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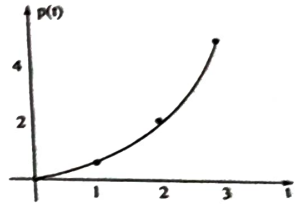

Fig. 1.8 Unit parabolic function

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In otherwords, the ramp function is derivative of the parabolic function.

Thus

$$r(t) = \frac{dp(t)}{dt} \tag{1.9}$$

1.4.4 Impulse function

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$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

and

$$\delta(t) = 0 \quad \text{for} \quad t \neq 0 \tag{1.10}$$

That is the impulse function has zero amplitude everywhere except at $t = 0$. At $t = 0$, the amplitude is infinity such that the area under the curve is equal to one. The unit impulse function can be obtained by using a limiting process on the rectangular pulse function shown in Fig. 1.9. Note that the area under the rectangular pulse is equal to unity.

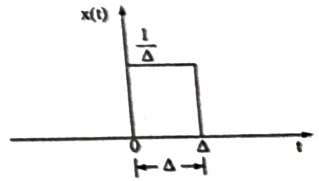


Fig. 1.9 Rectangular pulse function

The rectangular pulse function of Fig. 1.9 can be expressed mathematically as

$$x(t) = \frac{1}{\Delta}[u(t) - u(t-\Delta)] \qquad (1.11)$$

The impulse function can be obtained from Fig. 1.9 when the width $\Delta \to 0$. That is

$$\delta(t) = \lim_{\Delta \to 0} x(t)$$

$$= \lim_{\Delta \to 0} \frac{1}{\Delta}[u(t) - u(t-\Delta)] \qquad (1.12)$$

The delayed unit impulse function is defined as

$$\int_{-\infty}^{\infty} \delta(t-a) = 1 \quad \text{for} \quad t = a \qquad (1.13)$$

and

$$\delta(t-a) = 0 \quad \text{for} \quad t \neq a$$

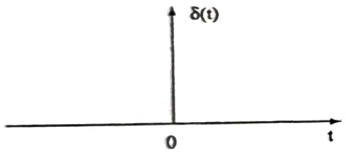
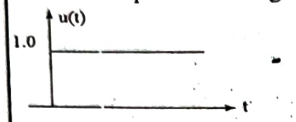
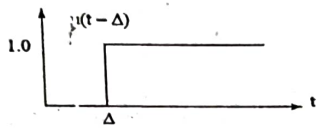


Fig. 1.10 Unit impulse function

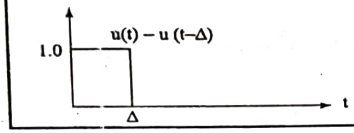
The unit step function is given by



The unit step function shifted by 'Δ' is



Subtracting u(t − Δ) from u(t) we get a rectangular function



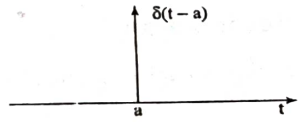


Fig. 1.11 Delayed unit impulse function

Properties of unit impulse

1.

$$\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = x(0) \qquad (1.14)$$

Let us consider the product of $x(t)$ and $\delta(t)$ which is

$$x(t)\,\delta(t).$$

Let the signal $x(t)$ is continuous at $t = 0$; the value of $x(t)$ at $t = 0$ is $x(0)$. The impulse exists only at $t = 0$.

Therefore we can write

$$x(t)\delta(t) = x(0)\delta(t) \qquad (1.15)$$

Substituting Eq. (1.15) in Eq. (1.14) we obtain

$$\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = \int_{-\infty}^{\infty} x(0)\,\delta(t)\,dt$$

$$= x(0)\int_{-\infty}^{\infty} \delta(t)\,dt$$

$$= x(0)$$

provided $x(t)$ is continuous at $t = 0$.

$$\boxed{\int_{-\infty}^{\infty} x(t)\,\delta(t)\,dt = x(0)}$$

2.

$$x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0) \qquad (1.16)$$

Let the signal $x(t)$ is continuous at $t = t_0$ and the value of $x(t)$ at $t = t_0$ is $x(t_0)$. The impulse exists only at $t = t_0$. Therefore

$$\boxed{x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)}$$

3.

$$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0) \qquad (1.17)$$

Substitute Eq. (1.16) in Eq. (1.17) we get

$$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = \int_{-\infty}^{\infty} x(t_0)\,\delta(t-t_0)\,dt$$

$$= x(t_0)\int_{-\infty}^{\infty} \delta(t-t_0)dt$$

Let $t - t_0 = \lambda$
$\therefore dt = d\lambda$

$$= x(t_0)\int_{-\infty}^{\infty} \delta(\lambda)d\lambda$$

$$= x(t_0)$$

$$\boxed{\int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt = x(t_0)}$$

1.4.5 Rectangular pulse function

The rectangular pulse function is defined as

$$\Pi(t) = 1 \quad \text{for} \quad |t| \le \frac{1}{2}$$
$$= 0 \quad \text{otherwise} \qquad (1.24)$$

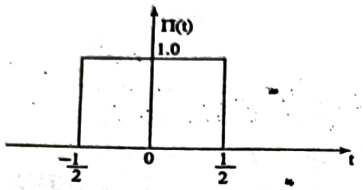


Fig. 1.12 Rectangular pulse function.

1.4.6 Triangular pulse function

The unit triangular pulse function is defined as

$$\Delta_a(t) = \begin{cases} 1 - \dfrac{|t|}{a} & |t| \le a \\ 0 & |t| > a \end{cases} \qquad (1.25)$$

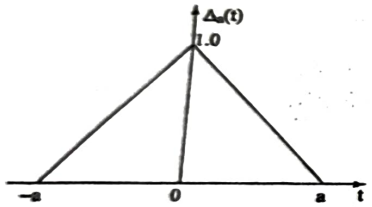


Fig. 1.13 Triangular pulse function.

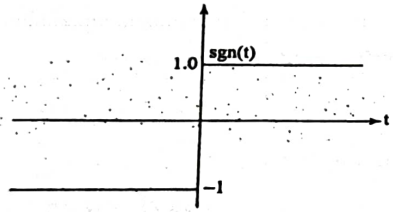


Fig. 1.14 Signum function.

1.4.7 Signum function

The unit signum function is defined by

$$\text{sgn}(t) = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases} \qquad (1.26)$$

This function can be expressed in terms of unit step function as

$$\text{sgn}(t) = -1 + 2u(t) \qquad (1.27)$$

1.4.8 Sinc function

The sinc function defined by the expression

$$\text{sinc}(t) = \frac{\sin t}{t} \quad -\infty < t < \infty \qquad (1.28)$$

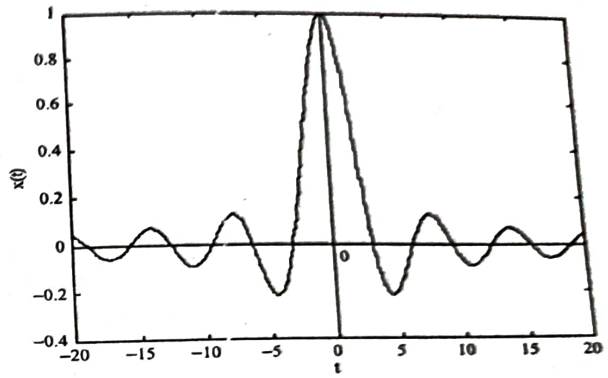


Fig. 1.15 Sinc function.

The sinc function oscillates with period 2π and decays with increasing t. Its value is zero at $n\pi, n = \pm 1, \pm 2, \ldots$. It is an even function of t.

1.4.9 Gaussian function

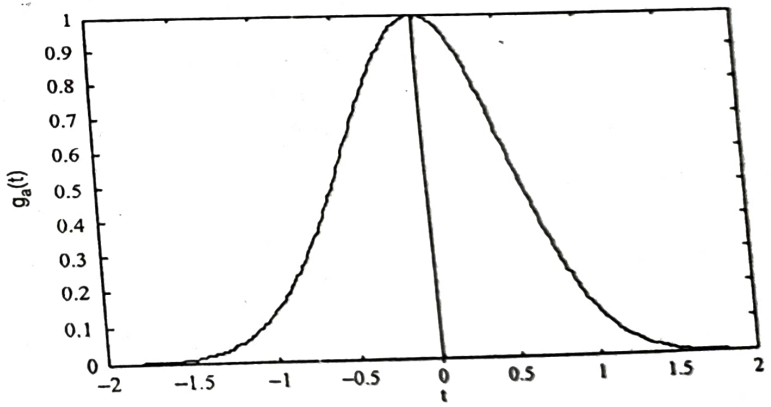


Fig. 1.16 Gaussian function.

The Gaussian function is defined by the expression

$$g_a(t) = e^{-at^2} \quad -\infty < t < \infty \qquad (1$$

This function is extremely used in probability theory.

1.4.10 Sinusoidal signal

A continuous-time sinusoidal signal is given by

$$x(t) = A\sin(\Omega t + \theta) \tag{1.30}$$

where A is the amplitude, Ω is the frequency in radians per second and θ is the phase angle in radians. Fig. 1.17 shows the waveform of a sinusoidal signal with $A = 1, \theta = \frac{-\pi}{3}$ with a time period of $\frac{2\pi}{\Omega}$.

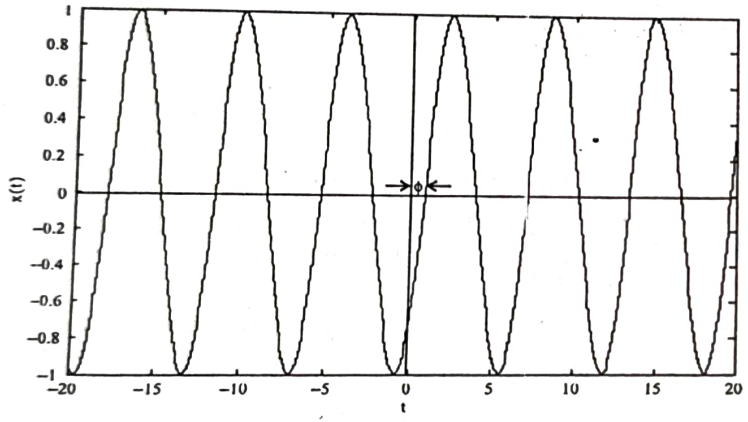


Fig. 1.17 A Sinusoidal signal.

1.4.11 Real exponential signals

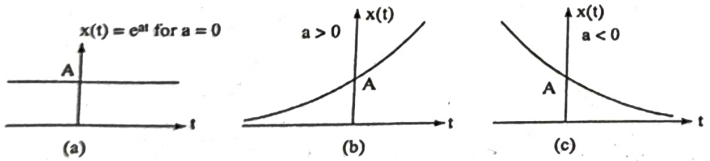


Fig. 1.18 (a) A dc signal (b) exponentially growing signal (c) exponentially decaying signal.

A real exponential signal is defined as

$$x(t) = Ae^{at} \tag{1.31}$$

where both A and a are real. Depending on the value of 'a' we get different signals. If a is positive the signal $x(t)$ is a growing exponential. One such type of signal is illustrated in Fig. 1.18b. If a is negative, the signal $x(t)$ is a decaying exponential. For $a = 0$, $x(t)$ is constant.

1.4.12 Complex exponential signal

The most general form of complex exponential is given by

$$x(t) = e^{st} \tag{1.32}$$

where 's' is a complex variable defined as

$$s = \sigma + j\Omega \tag{1.33}$$

Therefore

$$\begin{aligned} x(t) = e^{st} &= e^{(\sigma + j\Omega)t} \\ &= e^{\sigma t}e^{j\Omega t} \end{aligned} \tag{1.34}$$

Using Euler's identity we can expand

$$e^{j\Omega t} = \cos\Omega t + j\sin\Omega t \tag{1.35}$$

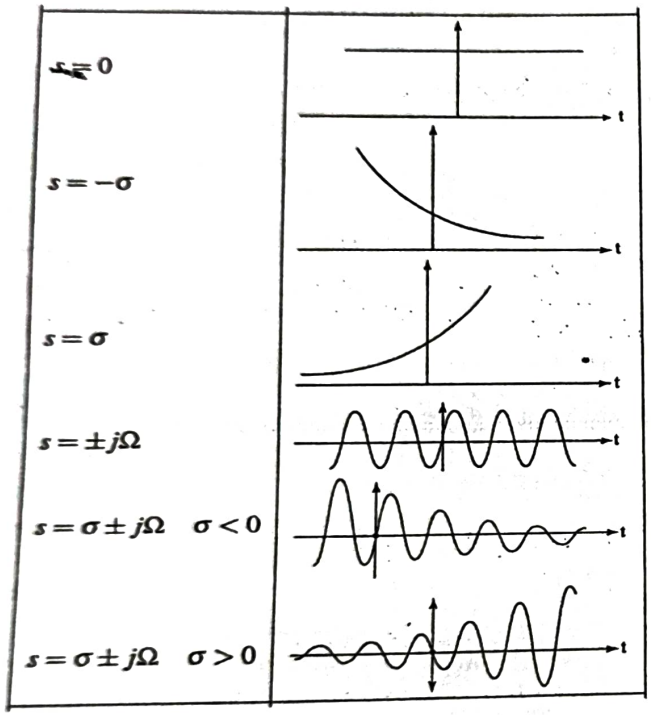
Substituting Eq. (1.35) in Eq. (1.34) yields

$$x(t) = e^{\sigma t}[\cos\Omega t + j\sin\Omega t] \tag{1.36}$$

Depending on the values of σ and Ω, we get different signals.

1. If $\sigma = 0$ and $\Omega = 0; x(t) = 1$; that is the signal $x(t)$ is a pure DC signal.
2. If $\Omega = 0$, then $s = \sigma$. $x(t) = e^{\sigma t}$, which decays exponentially for $\sigma < 0$ and grows exponentially for $\sigma > 0$.
3. If $\sigma = 0$ then $s = \pm j\Omega$ gives $x(t) = e^{j\Omega t}$ a sinusoidal signal with $\phi = 0$.
4. If $\sigma < 0$ with finite Ω, we get a exponentially decaying sinusoidal signal.
5. If $\sigma > 0$ with finite Ω, we get a exponentially growing sinusoidal signal.

Table 1.1



Solved Problem 1.3 Evaluate the following integrals.

(i) $\int_{-\infty}^{\infty} e^{-\alpha t^2}\delta(t-10)dt$ (ii) $\int_{0}^{\infty} t^2\delta(t-3)dt$

Solution:

(i) Given

$$\int_{-\infty}^{\infty} e^{-\alpha t^2}\delta(t-10)dt$$

From the property

$$\int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt = x(t_0)$$

We have

$$\int_{-\infty}^{\infty} e^{-\alpha t^2}\delta(t-10)dt = e^{-\alpha t^2}\Big|_{t=10} = e^{-100\alpha}$$

$\delta(t-10) = 1$ for $t = 10$
$= 0$ otherwise

(ii) Given

$$\int_{0}^{\infty} t^2\delta(t-3)dt = t^2\Big|_{t=3} = 9$$

$\delta(t-3) = 1$ for $t = 3$
$= 0$ otherwise

Solved Problem 1.4 Evaluate the following integrals.

i) $\int_{0}^{5} \delta(t)\sin 2\pi t dt$ ii) $\int_{-\infty}^{\infty} \delta(t+3)e^{-t}dt$

iii) $\int_{-\infty}^{\infty} (t-3)^2\delta(t-3)dt$ iv) $\int_{-\infty}^{\infty} [\delta(t)\cos t + \delta(t-1)\sin t]dt$

v) $\int_{-\infty}^{\infty} \delta(t)e^{-j\Omega t}dt$

Solution:

(i)

$$\int_{0}^{5} \delta(t)\sin 2\pi t dt$$

$\delta(t) = 1$ for $t = 0$
$= 0$ for $t \neq 0$

$$= \sin 2\pi t\Big|_{t=0} = 0$$

(ii)

$$\int_{-\infty}^{\infty} \delta(t+3)e^{-t}dt$$

$\delta(t+3) = 1$ for $t = -3$
$= 0$ for $t \neq -3$

$$= e^{-t}\Big|_{t=-3} = e^3$$

(iii)

$$\int_{-\infty}^{\infty} (t-3)^2\delta(t-3)dt$$

$\delta(t-3) = 1$ for $t = 3$
$= 0$ for $t \neq 3$

$$= (t-3)^2\Big|_{t=3} = 0$$

(iv)

$$\int_{-\infty}^{\infty}[\delta(t)\cos t+\delta(t-1)\sin t]dt$$

$$=\int_{-\infty}^{\infty}\delta(t)\cos t dt+\int_{-\infty}^{\infty}\delta(t-1)\sin t dt$$

$$=1+\sin 1$$

(v)

$$\int_{-\infty}^{\infty}\delta(t)e^{-j\Omega t}dt$$

$$=1$$

Practice Problem 1.2

Evaluate the following integrals

(i) $\int_{-\infty}^{\infty}\delta(\tau)x(t-\tau)d\tau$ (ii) $\int_{-\infty}^{\infty}\delta(t-2)\sin\pi t dt$

(iii) $\int_{-\infty}^{\infty}\delta(t)\cos\pi t dt$ (iv) $\int_{-\infty}^{\infty}x(2-t)\delta(3-t)dt$

(v) $\int_{-\infty}^{\infty}(t^3+4)\delta(1-t)dt$ (vi) $\int_{-\infty}^{\infty}e^{-t}\delta(2t-2)dt$

Ans: (i) $x(t)$ (ii) 0 (iii) 1 (iv) $x(-1)$ (v) 5 (vi) e^{-1}

Practice Problem 1.3

Show that

(i) $\int_{-\infty}^{\infty}\delta(at-b)dt=\frac{1}{|a|}$ (ii) $\int_{-\infty}^{\infty}x(t)\delta(t-t_0)dt=x(t_0)$

(iii) $\int_{-\infty}^{\infty}x(\tau)\delta(a\tau-b)d\tau=\frac{1}{|a|}x\left(\frac{b}{a}\right)$

Solved Problem 1.5 Prove that

(i) $\delta(at)=\frac{1}{|a|}\delta(t)$ (ii) $\int_{-\infty}^{\infty}x(t)\delta(at)dt=\frac{1}{|a|}x(0)$

(iii) $\int_{-\infty}^{\infty}\dot{\delta}(t)x(t)dt=-\dot{x}(0)$

(i) $\delta(at)=\frac{1}{|a|}\delta(t)$

See fourth property of impulse function

(ii) $\int_{-\infty}^{\infty}x(t)\delta(at)dt=\frac{1}{|a|}x(0)$

See fourth property of impulse function

(iii) $\int_{-\infty}^{\infty}\dot{\delta}(t)x(t)dt=-\dot{x}(0)$ $\left[\dot{\delta}(t)=\frac{d}{dt}\delta(t)\right]$

Let $u=x(t)$ $dv=\dot{\delta}(t)$

$$\Rightarrow v=\delta(t) \qquad \boxed{\int udv=uv-\int vdu}$$

$$du=\dot{x}(t)$$

$$\int_{-\infty}^{\infty}\dot{\delta}(t)x(t)dt=x(t)\delta(t)\Big|_{-\infty}^{\infty}-\int_{-\infty}^{\infty}\dot{x}(t)\delta(t)dt$$

If $x(t)$ and $\dot{x}(t)$ are continuous at $t=0$ and $x(t)\to 0$ at $t\pm\infty$ then

$$\int_{-\infty}^{\infty}\dot{\delta}(t)x(t)dt=-\int_{-\infty}^{\infty}\dot{x}(t)\delta(t)dt$$

$$=-\dot{x}(0) \qquad (1.37)$$

Review Questions

1. Define and plot the following signals. **AU, April 2004**
   (i) Ramp (ii) Step (iii) Pulse (iv) Impulse (v) Exponential
2. Explain the properties of impulse function.
3. Explain in detail about the following functions.
   (a) Triangular pulse function
   (b) Signum function
   (c) Sinc function
   (d) Gaussian function
4. Sketch the sinusoidal signal $x(t) = A \sin(\Omega t + \phi)$.

1.5 Representation of Discrete-time Signals

There are different types of representations for discrete-time signals. They are

1. Graphical representation
2. Functional representation
3. Tabular representation
4. Sequence representation

1.5.1 Graphical representation

Let us consider a signal $x(n)$ with values $x(-1) = 1; x(0) = 2; x(1) = 2; x(2) = 0.5$ and $x(3) = 1.5$. This discrete-time signal can be represented graphically as shown in Fig. 1.19.

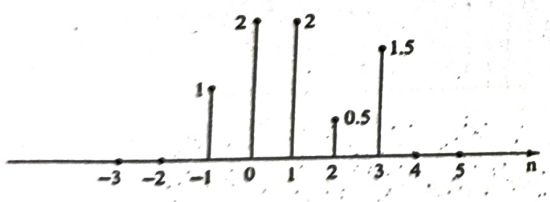


Fig. 1.19 Graphical representation of a discrete-time signal.

1.5.2 Functional representation

The discrete-time signal shown in Fig. 1.19 can be represented using functional representation as below.

$$x(n) = \begin{cases} 1 & \text{for } n = -1 \\ 2 & \text{for } n = 0, 1 \\ 0.5 & \text{for } n = 2 \\ 1.5 & \text{for } n = 3 \\ 0 & \text{otherwise} \end{cases}$$

1.5.3 Tabular representation

The discrete-time signal can also be represented as

n	-1	0	1	2	3
$x(n)$	1	2	2	0.5	1.5

1.5.4 Sequence representation

A finite duration sequence with time origin ($n = 0$) indicated by the symbol $\uparrow$ is represented as

$$x(n) = \{1, \underset{\uparrow}{2}, 2, 0.5, 1.5\}$$

A finite duration sequence can be represented as

$$x(n) = \{\ldots 0.2, \underset{\uparrow}{1}, -1, 3, 2 \ldots\}$$

A finite duration sequence that satisfies the condition $x(n) = 0$ for $n < 0$ can be represented as

$$x(n) = \{2, 4, 6, 8, -3\}$$

1.6 Elementary Discrete-Time Signals

1.6.1 Unit step sequence

The unit step sequence is defined as

$$\begin{aligned} u(n) &= 1 \quad \text{for} \quad n \geq 0 \\ &= 0 \quad \text{for} \quad n < 0 \end{aligned} \tag{1.38}$$

The graphical representation of $u(n)$ is shown in Fig. 1.20.

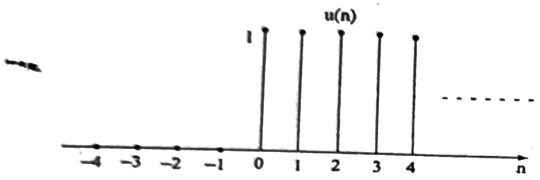


Fig. 1.20 Unit step sequence.

1.6.2 Unit ramp sequence

The unit ramp sequence is defined as

$$\begin{aligned} r(n) &= n \quad \text{for} \quad n \geq 0 \\ &= 0 \quad \text{for} \quad n < 0 \end{aligned} \tag{1.39}$$

The graphical representation of $r(n)$ is shown in Fig. 1.21.

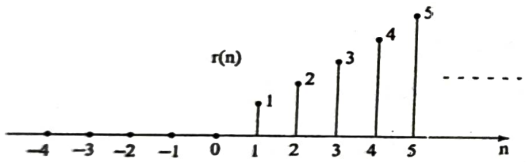


Fig. 1.21 Unit ramp sequence.

1.6.3 Unit-sample sequence (unit impulse sequence)

The unit-sample sequence is defined as

$$\begin{aligned} \delta(n) &= 1 \quad \text{for} \quad n = 0 \\ &= 0 \quad \text{for} \quad n \neq 0 \end{aligned} \tag{1.40}$$

The graphical representation of $\delta(n)$ is shown in Fig. 1.22.

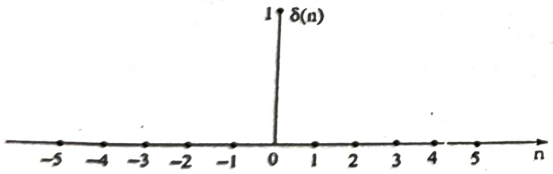


Fig. 1.22 Unit impulse sequence.

The unit impulse function has the following properties

$$\delta(n) = u(n) - u(n-1) \tag{1.41}$$

$$u(n) = \sum_{k=-\infty}^{n} \delta(k) \tag{1.42}$$

$$\sum_{n=-\infty}^{\infty} x(n)\delta(n-n_0) = x(n_0) \tag{1.43}$$

1.6.4 Exponential sequence

The exponential signal is a sequence of the form

$$x(n) = a^n \text{ for all } n \tag{1.44}$$

Fig. 1.23 illustrates different types of discrete-time exponential signals. When the value of $a > 1$ the sequence grows exponentially and when the value is $0 < a < 1$, the sequence decay exponentially. Note also that when $a < 0$, the discrete-time exponential signal takes alternating signs.

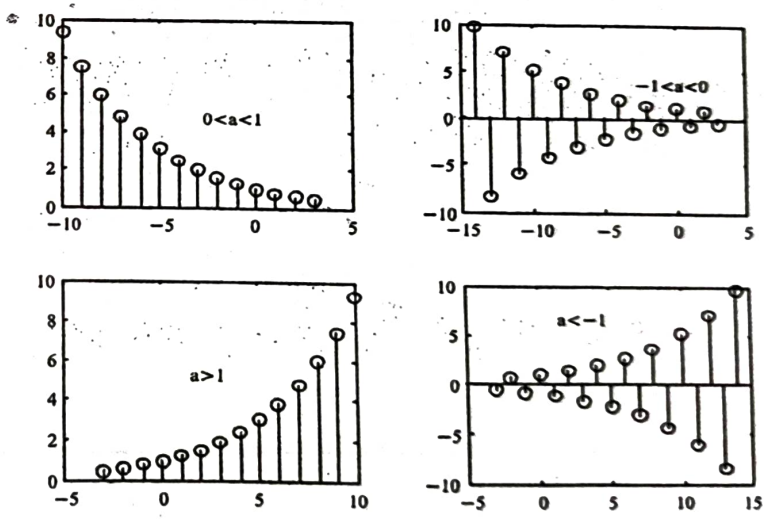


Fig. 1.23 Exponential sequences.

1.6.5 Sinusoidal signal

The discrete time sinusoidal signal is given by

$$x(n) = A\cos(\omega_0 n + \phi) \tag{1.45}$$

where ω_0 is the frequency (in radians per sample) and ϕ is the phase (in radians).

Using Euler's identity, we can write

$$A\cos(\omega_0 n + \phi) = \frac{A}{2}e^{j\phi}e^{j\omega_0 n} + \frac{A}{2}e^{-j\phi}e^{-j\omega_0 n} \tag{1.46}$$

Since $|e^{j\omega_0 n}|^2 = 1$, the energy of the signal is infinite and the average power of the signal is 1.

An example of discrete-time sinusoidal signal is shown in Fig. 1.24a.

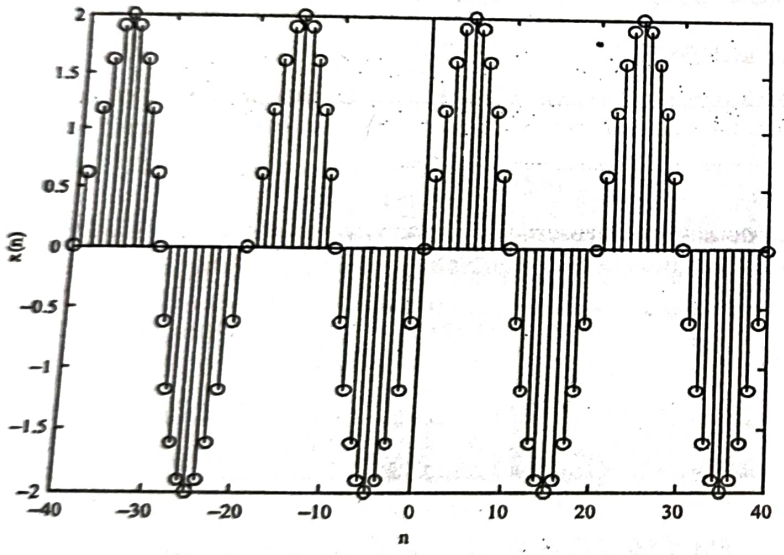


Fig. 1.24 (a) A sinusoidal sequence.

1.6.6 Complex exponential signal

The discrete-time complex exponential signal is given by

$$x(n) = a^n e^{j(\omega_0 n + \phi)}$$

$$= a^n\cos(\omega_0 + \phi) + ja^n\sin(\omega_0 n + \phi) \tag{1.47}$$

For $|a| = 1$, the real and imaginary parts of complex exponential sequence are sinusoidal. For $|a| < 1$, the amplitude of the sinusoidal sequence decays exponentially and for $|a| > 1$, the amplitude of the sinusoidal sequences increases exponentially. Examples of these signals are given in Fig. 1.24b.

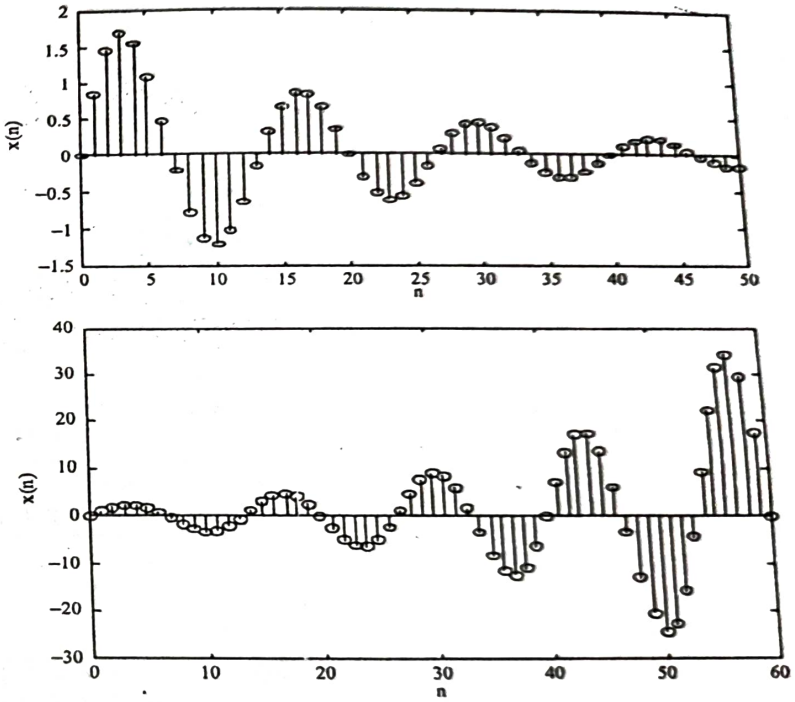


Fig. 1.24(b) Complex exponential sequences

Review Questions

1. Explain different types of representations for discrete-time signals.
2. Define and sketch the following discrete-time signals
   (a) unit step sequence
   (b) unit ramp sequence
   (c) unit-sample sequence
   (d) exponential sequence

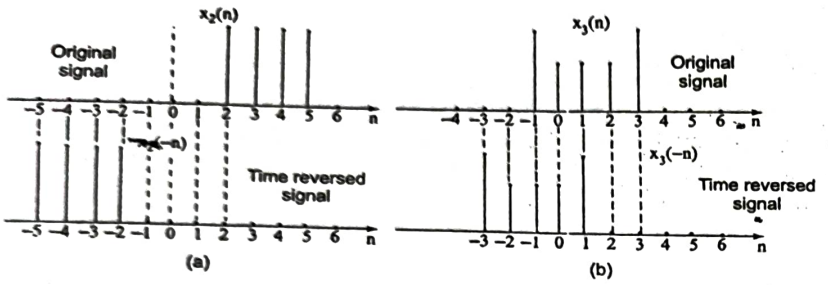


Fig. 1.30 Time reversal operation on DT signals.

Solved Problem 1.7 Sketch the following signals

(i) $u(-t+1)$ (ii) $-2u(t-1)$ (iii) $3r(t-1)$

(iv) $-2r(t)$ (v) $r(-t+2)$ (vi) $\Pi(t+3)$

Solution:

Given $x(t) = u(-t+1)$.

The signal $x(t)$ can be obtained by folding the unit step signal $u(t)$ about $t = 0$ and shifting the signal right by one unit of time.

Step 1. First draw the unit step signal

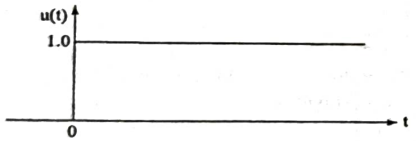


Fig. 1.31 Unit step signal.

Step 2. Fold the signal about $t = 0$ as shown in Fig.1.32.

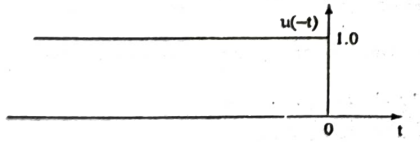


Fig. 1.32 Time reversal of unit step signal.

Step 3. Shift the signal $u(-t)$ right by one unit of time as shown in Fig.1.33.

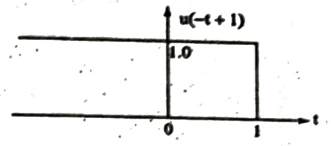


Fig. 1.33

(ii) $x(t) = -2u(t-1)$

Step 1. Draw the unit step signal.

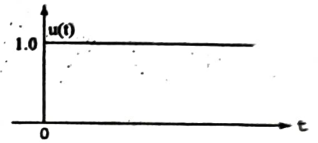


Fig. 1.34

Step 2. Shift the signal $u(t)$ right by one unit of time as shown in Fig.1.35.

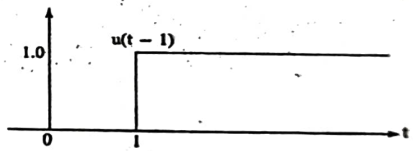


Fig. 1.35

Step 3. Multiply the signal $u(t-1)$ with -2.

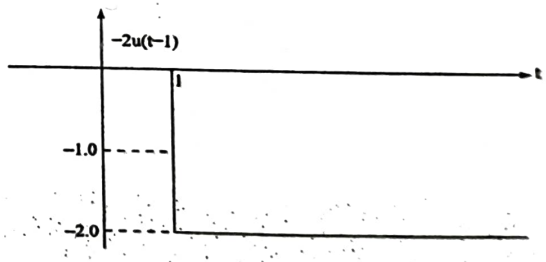


Fig. 1.36

Solved Problem 1.6 Find the following summations

(i) $\sum_{n=-\infty}^{\infty} e^{2n}\delta(n-2)$ (ii) $\sum_{n=-\infty}^{\infty} \delta(n-1)\sin 2n$

(iii) $\sum_{n=-\infty}^{\infty} n^2\delta(n+2)$ (iv) $\sum_{n=0}^{\infty} \delta(n-1)e^{n^2}$

(v) $\sum_{n=0}^{5} \delta(n+1)2^n$

Solution:

(i) Given

$$\sum_{n=-\infty}^{\infty} e^{2n}\,\delta(n-2) \qquad \boxed{\delta(n-2)=1 \text{ for } n=2 \\ =0 \text{ for } n\neq 2}$$

$$= e^{2n}\Big|_{n=2} = e^4$$

(ii)

$$\sum_{n=-\infty}^{\infty} \delta(n-1)\sin 2n = \sin 2n\Big|_{n=1} = \sin 2$$

$$\boxed{\because \delta(n-1) = 1 \text{ for } n=1 \\ =0 \text{ for } n\neq 1}$$

(iii)

$$\sum_{n=-\infty}^{\infty} n^2\delta(n+2) = n^2\Big|_{n=-2} = (-2)^2 = 4$$

$$\boxed{\delta(n+2)=1 \text{ for } n=-2 \\ =0 \text{ for } n\neq -2}$$

(iv)

$$\sum_{n=0}^{\infty} \delta(n-1)e^{n^2} = e^{n^2}\Big|_{n=1} = e$$

(v)

$$\sum_{n=0}^{5} \delta(n+1)2^n = 0 \qquad \boxed{\because \delta(n+1) = 1 \text{ for } n=-1 \\ =0 \text{ for } n\neq -1}$$

Practice Problem 1.4

Find the following summations

(i) $\sum_{n=2}^{\infty} \delta(n-1)\sin 2n$ (ii) $\sum_{n=0}^{\infty} x(n)\delta(n-2)$ (iii) $\sum_{n=-\infty}^{\infty} a^{n-2}\delta(n+3)$

Ans: (i) 0 (ii) $x(2)$ (iii) a^{-5}

1.7 Basic Operations on Signals

The basic set of operations on signals are

1. Time shifting
2. Time reversal
3. Time scaling
4. Amplitude scaling
5. Signal multiplier
6. Signal addition

1.7.1 Time shifting

Let us consider a signal $x(t)$ shown in Fig. 1.25a. The time shifting of $x(t)$ may delay or advance the signal in time. Mathematically this can be represented by

$$y(t) = x(t-T) \tag{1.48}$$

If T is positive the shifting delays the signal as shown in Fig. 1.25(c) and if T is negative the shifting advances the signal, as shown in Fig. 1.25(b)

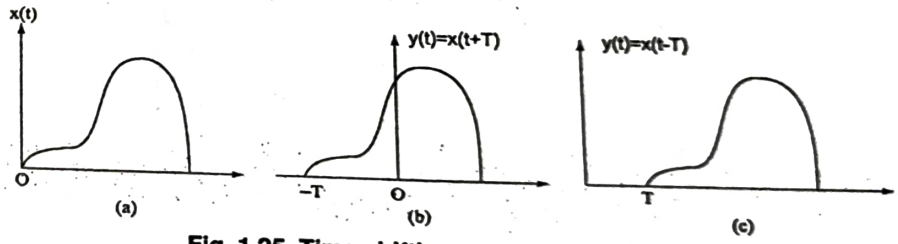


Fig. 1.25 Time shifting operation of CT signal.

Similarly the time shifting operation of a discrete-time signal is represented by

$$y(n) = x(n-k) \tag{1.49}$$

which shows that the signal $y(n)$ can be obtained by shifting $x(n)$ by k units. If k is positive, the shift is to the right (delay) and if k is negative, the shift is to the left (advance). Let us consider a signal $x(n)$ shown in Fig. 1.26a. The signal

$x(n-2)$ is obtained by shifting $x(n)$ right by 2 units of time. The result is shown in Fig. 1.26b. On the other hand, the signal $x(n+3)$ is obtained by shifting $x(n)$ left by three units of time.

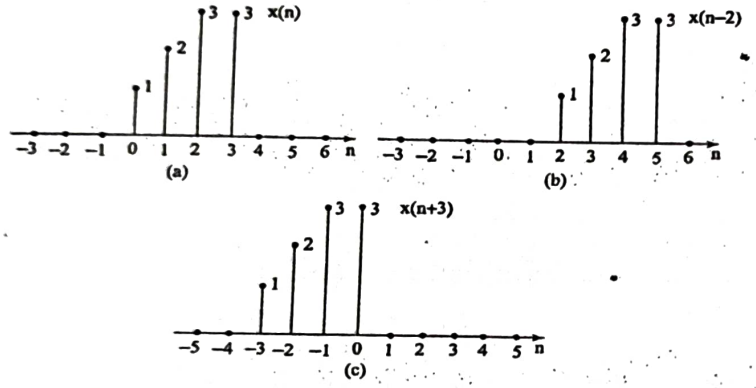


Fig. 1.26 Time shifting operation of a DT signal.

1.7.2 Time reversal

The time reversal of a signal $x_1(t)$ can be obtained by folding the signal about $t = 0$. It is denoted by $x(-t)$. The signal $x_1(t)$ and its reflection $x_1(-t)$ are illustrated in Fig. 1.27a and 1.27b respectively.

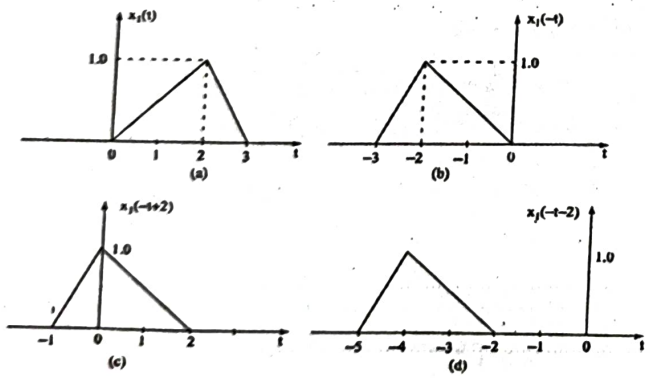


Fig. 1.27 Time reversal operation.

The signal $x_1(-t+2)$ is obtained by delaying $x_1(-t)$ by two units of time and $x_1(-t-2)$ is obtained by advancing $x_1(-t)$ by two units of time. The signals $x_1(-t+2)$ and $x_1(-t-2)$ are shown in Fig. 1.27c and Fig. 1.27d respectively.

The other examples for time reversal operation are shown in Fig. 1.28.

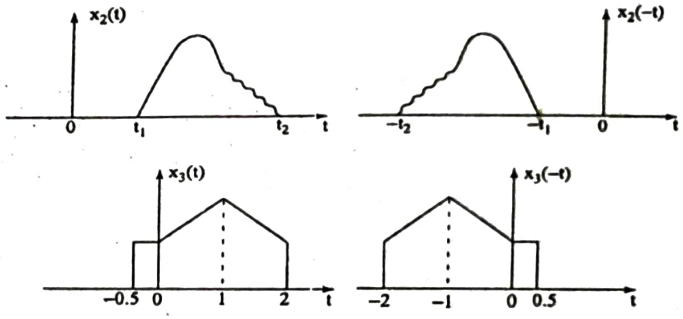


Fig. 1.28 Time reversal operation on CT signals.

The time reversal of a discrete-time signal $x(n)$ can be obtained by folding the sequence $x(n)$ about $n = 0$. For the signal $x_1(n)$ in Fig. 1.29a the time reversal signal $x_1(-n)$ is shown in Fig. 1.29b.

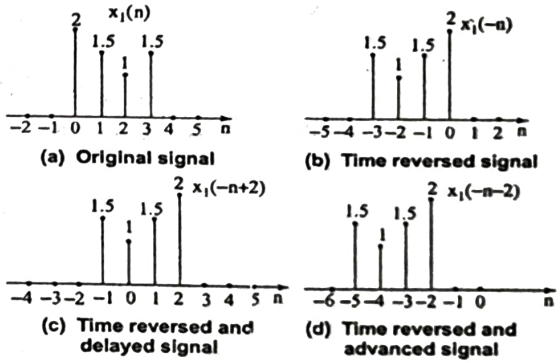


Fig. 1.29 Time reversal operation on DT signals.

The signal $x_1(-n+2)$ is obtained by delaying $x_1(-n)$ by two units of time and $x_1(-n-2)$ is obtained by advancing $x_1(-n)$ by two units of time. The other examples for time reversal of discrete-time signals are illustrated in Figs. 1.30a and 1.30b.

(iii) Given

$$x(t) = 3r(t-1)$$

The signal $r(t)$ is a unit ramp signal with slope equal to 1 and the signal $Ar(t)$ is a ramp signal with slope equal to A. To sketch the signal $x(t)$, first sketch the signal $3r(t)$ as shown in Fig.1.37 and then shift it by one unit of time as shown in Fig.1.38.

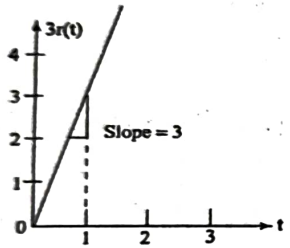


Fig. 1.37

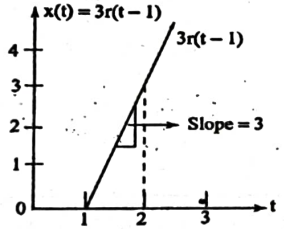


Fig. 1.38

(iv) Given $x(t) = -2r(t)$

The signal $x(t)$ is a ramp signal with slope $= -2$ as shown in Fig.1.39.

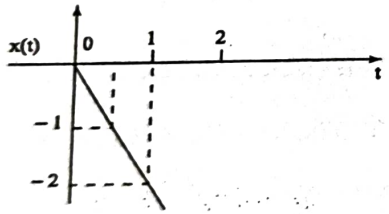


Fig. 1.39

(v) $r(-t+2)$

Step 1. Sketch the unit ramp signal as shown in Fig.1.40.

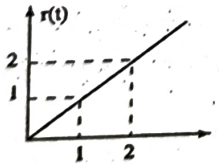


Fig. 1.40

Step 2. Fold the signal about $t = 0$ as shown in Fig.1.41.

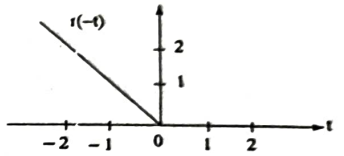


Fig. 1.41

Step 3. Shift the signal $r(-t)$ right by two units of time as shown in Fig.1.42.

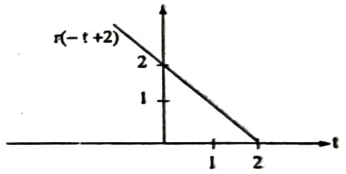


Fig. 1.42

(vi) $x(t) = \Pi(t+3)$

The signal $\Pi(t)$ is shown in Fig. 1.43.

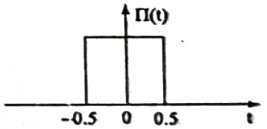


Fig. 1.43

The signal $x(t)$ can be obtained by shifting $\Pi(t)$ left by three units of time as shown in Fig.1.44.

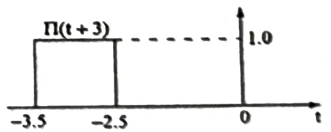


Fig. 1.44

Solved Problem 1.8 Sketch the following signals

(i) $u(n+2)-u(n-3)$ (ii) $u(-n+2)\cdot u(n)$

Solution:

(i) Step 1. Sketch the signal $u(n+2)$, by shifting $u(n)$ left by two samples as shown in Fig.1.45(a).

Step 2. Sketch the signal $u(n-3)$ by shifting $u(n)$ right by three units of time as shown in Fig.1.45(b).

Step 3. Substract $u(n-3)$ from $u(n+2)$. The samples for $n \geq 3$ gets cancel and we get the sequence as shown in Fig. 1.45(c).

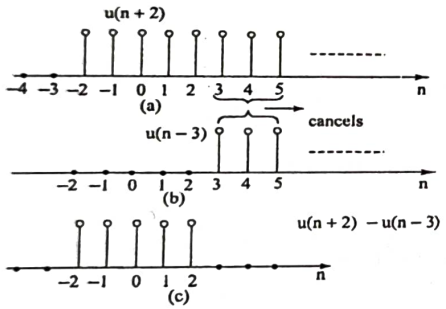


Fig. 1.45

(ii) The signal $u(-n+2)$ can be obtained by folding the sequence $u(n)$ about $n=0$ and then shifting the sequence right by two samples as shown in Fig.1.46.

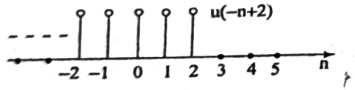


Fig. 1.46

If we multiply the above sequence with $u(n)$, then we get the signal as shown in Fig. 1.47.

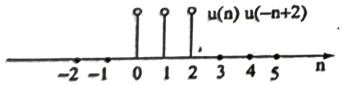


Fig. 1.47

1.7.3 Amplitude scaling

Consider a signal $x(t)$ that is fed to an amplifier with a gain 3; then the output of an amplifier can be represented by

$$y(t) = 3x(t) \tag{1.50}$$

That is the output $y(t)$ is identical in shape to the input signal $x(t)$ but its amplitude is thrice everywhere. This concept is illustrated in Fig. 1.48.

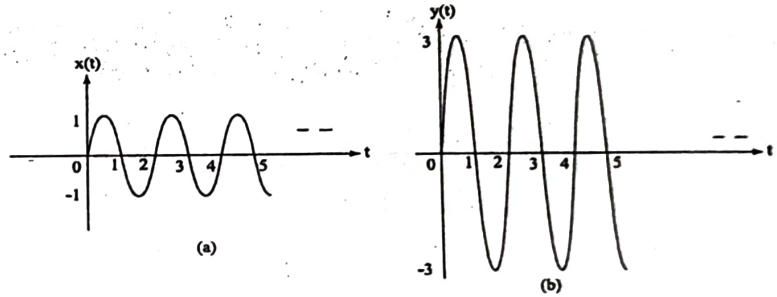


Fig. 1.48 Amplitude scaled in CT signal.

From Fig. 1.48 we find that the amplitude is rescaled hence the name amplitude scaling.

Similarly, the amplitude scaling of a discrete-time signal can be represented by

$$y(n) = ax(n) \tag{1.51}$$

If $x(n)$ is as shown in Fig. 1.49a and $a=2$, then $y(n)$ is as shown in Fig. 1.49b.

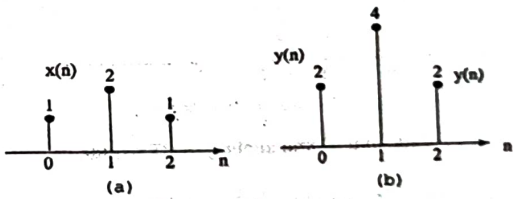


Fig. 1.49 Amplitude scaling in DT signal.

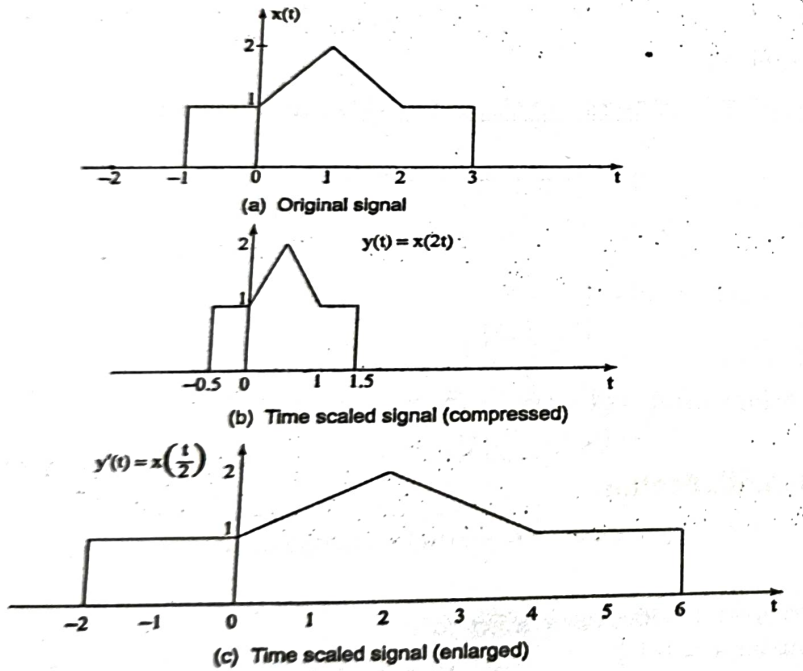
1.7.4 Time scaling

Let us consider a signal $x(t)$ shown in Fig. 1.50a. The time scaling of a signal can be accomplished by replacing t by at in the signal $x(t)$.

If $a = 2$, we get a new signal

$$y(t) = x(2t)$$

Observe that $x(t)$ is switching from 0 to 1 at $t = -1$ and switching from 1 to 0 at $t = 3$. When $y(t) = x(2t)$, $y(t)$ switches from 0 to 1 at $t = -0.5$ and switches from 1 to 0 at $t = 1.5$. In the original signal $x(t)$, magnitude is equal to 1 at $t = 0$ and at $t = 2$. But in the time scaled signal $y(t)$, magnitude is equal to 1 at $t = 0$ and at $t = 1$. This behaviour shows that $x(t)$ is compressed in time by a factor 2.



(a) Original signal

(b) Time scaled signal (compressed)

(c) Time scaled signal (enlarged)

Fig. 1.50 Time scaling in CT signal.

The resulting signal $x(2t)$ is shown in Fig. 1.50b. Let $x(t)$ be an audio signal. If $x(t)$ were recorded on a magnetic tape then $x(2t)$ is the signal played back at twice the recording speed.

Now consider another signal $y(t) = x(at)$ with $a = \frac{1}{2}$. Fig. 1.50c shows $x\left(\frac{t}{2}\right)$ which is expanded in time by a factor 2. It is similar to playback the audio signal at half the recording speed.

In the discrete-time case, we write

$$y(n) = x(an)$$

where $a > 0$.

If $a = 2$

$$y(n) = x(2n)$$

Let $x(n)$ be a sequence shown in Fig. 1.51. Then we can plot time scaled signal $y(n)$ by substituting different values for n.

For $n = -1$

$$y(-1) = x(-2) = 2$$

Similarly

$$y(0) = x(0) = 4$$
$$y(1) = x(2) = 2$$
$$y(2) = x(4) = 0$$

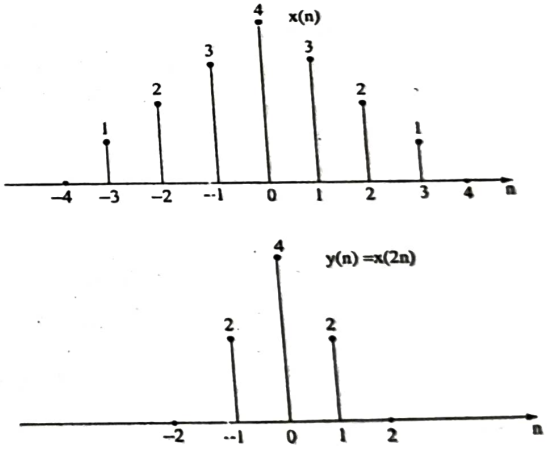


Fig. 1.51 Time scaling in DT signal.

From the above values, we can conclude that, to plot $y(n)$ we have to skip the odd-numbered samples in $x(n)$ and retain even-numbered samples. The resulting sequence is shown in Fig. 1.51. Here we can observe that $y(n)$ is compressed by a factor of 2.

1.7.5 Signal addition

The sum of two continuous-time signals can be obtained by adding their values at every instant. Similarly the subtraction of two continuous-time signals can be obtained by subtracting their values at every instant.

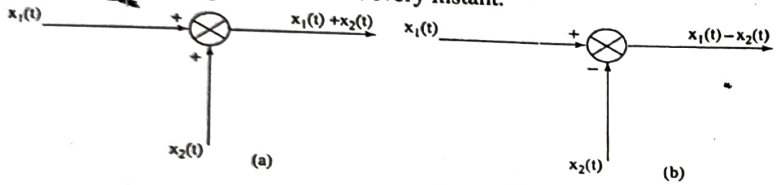


Fig. 1.52 (a) Addition of CT signals (b) Subtraction of CT signals.

Let us consider addition and subtraction of two signals $x_1(t)$ and $x_2(t)$ as shown in Fig. 1.53a and Fig. 1.53b respectively.

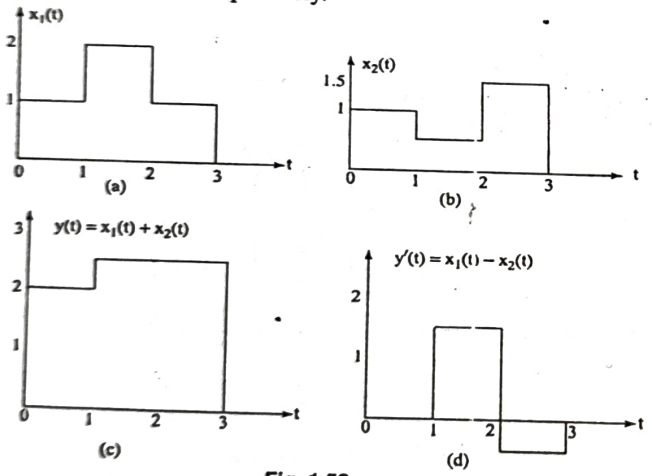


Fig. 1.53

The addition of two signals $x_1(t)$ and $x_2(t)$ can be obtained by considering each time interval separately as follows.

$x_1(t) = 1$ for $0 \le t \le 1$ and $x_2(t) = 1$ for $0 \le t \le 1$
Hence $x_1(t) + x_2(t) = 2$ for $0 \le t \le 1$

$x_1(t) = 2$ for $1 \le t \le 2$ and $x_2(t) = 0.5$ for $1 \le t \le 2$
Hence $x_1(t) + x_2(t) = 2.5$ for $1 \le t \le 2$

$x_1(t) = 1$ for $2 \le t \le 3$ and $x_2(t) = 1.5$ for $2 \le t \le 3$
Hence $x_1(t) + x_2(t) = 2.5$ for $2 \le t \le 3$

The sketch for $y(t) = x_1(t) + x_2(t)$ is shown in Fig. 1.53c.

In the same way, the subtraction of $x_2(t)$ from $x_1(t)$ can be obtained. By inspection we can find

$$y'(t) = x_1(t) - x_2(t) = 0 \text{ for } 0 \le t \le 1$$
$$= 1.5 \text{ for } 1 \le t \le 2$$
$$= -0.5 \text{ for } 2 \le t \le 3$$

For discrete-time signals the sum of two sequences can be obtained by adding the sequence values at the same index n for which the sequences are defined.

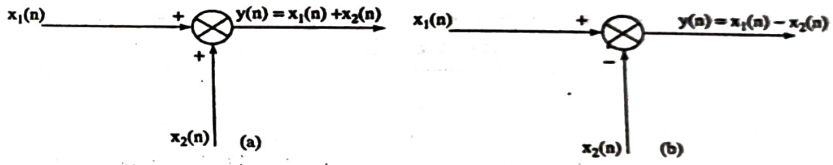


Fig. 1.54 (a) Addition of two discrete-time signals (b) Subtraction of two discrete-time signals.

If

$$x_1(n) = \{1, 3, 2, 1\} \quad \text{and}$$
$$x_2(n) = \{1, -2, 3, 2\}$$

then

$$x_1(n) + x_2(n) = \{1+1, 3-2, 2+3, 1+2\}$$
$$= \{2, 1, 5, 3\}$$

and

$$x_1(n) - x_2(n) = \{1-1, 3-(-2), 2-3, 1-2\}$$
$$= \{0, 5, -1, -1\}$$

1.7.6 Signal multiplication

The multiplication of two signals can be obtained by multiplying their values at every instant.

The multiplication process is illustrated in Fig. 1.56.

In the interval $0 \le t \le 1$

$$x_1(t) = t;\ x_2(t) = 1 \quad \text{Therefore } y(t) = x_1(t)x_2(t) = t$$

In the interval $1 \le t \le 2$

$$x_1(t) = 1 \quad \text{and } x_2(t) = 0.5 \quad \Rightarrow \quad y(t) = 0.5$$

In the interval $2 \le t \le 3$

$$x_1(t) = 0.5 \quad \text{and } x_2(t) = 1.5 \quad \Rightarrow \quad y(t) = 0.75$$

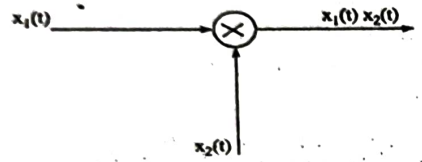


Fig. 1.55 Multiplication of two continuous-time signals.

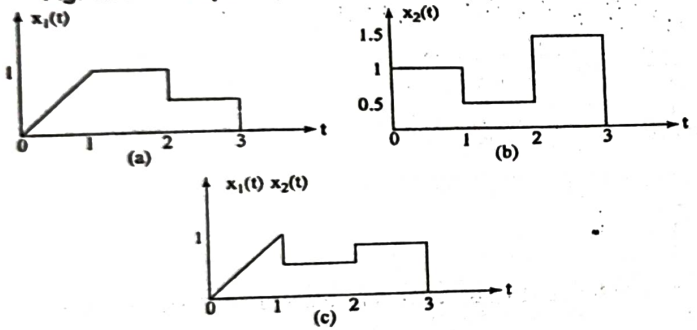


Fig. 1.56

Fig. 1.57 illustrates the multiplication of two sequences to form another sequence $y(n)$.

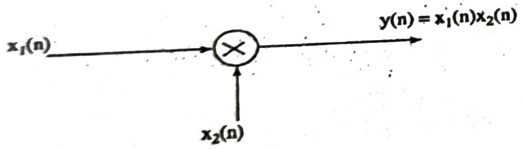


Fig. 1.57

If $x_1(n) = \{1, 2, -2, 3\}$ and $x_2(n) = \{1, 0.5, 0.5, 1\}$ then

$$y(n) = x_1(n) \cdot x_2(n) = \{1.1, 2.(0.5), -2.(0.5), 3.1\}$$
$$= \{1, 1, -1, 3\}$$

Solved Problem 1.9 For the signal $x(t)$ shown in Fig. 1.58(a) find the signals **AU Nov 2009**

(i) $x(t-2)$
(ii) $x(2t+3)$
(iii) $x\left(\frac{3}{2}t\right)$

(iv) $x(-t+1)$

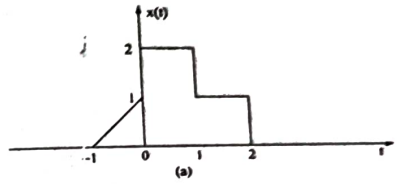


Fig. 1.58(a)

Solution:

(i) The signal $x(t-2)$ can be obtained by delaying the signal by 2 units of time.

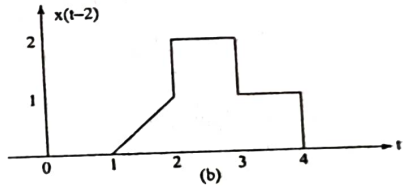


Fig. 1.58(b)

(ii) The signal $x(2t+3)$ can be obtained first shifting the sequence $x(t)$ left by 3 units of time as shown in Fig. 1.58c and then applying time scaling by factor 2. The resultant signal is shown in Fig. 1.58d.

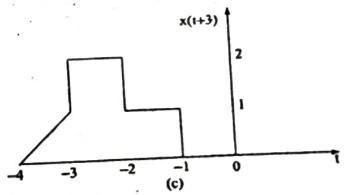


Fig. 1.58(c)

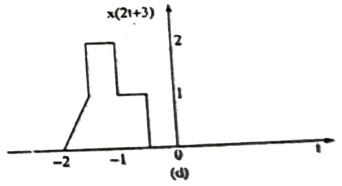


Fig. 1.58(d)

(iii) $x\left(\frac{3}{2}t\right)$
The signal $x\left(\frac{3}{2}t\right)$ can be obtained by compressing the signal $x(t)$ by $\frac{3}{2}$ times.

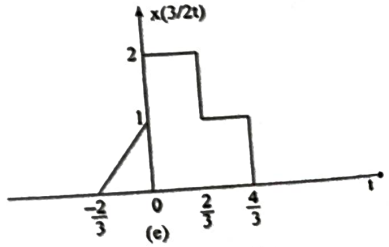


Fig. 1.58(e)

(iv) $x(-t+1)$

This signal can be obtained by folding $x(t)$ about $t = 0$ then shifting right by one unit of time.

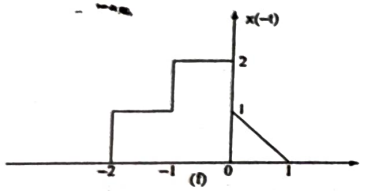


Fig. 1.58(f)

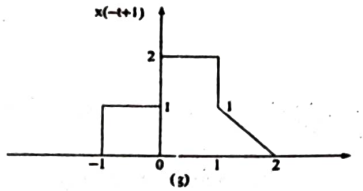


Fig. 1.58(g)

Solved Problem 1.10 Sketch the following signals

(i) $u(t) - u(t-2)$

(ii) $\Pi\left(t - \frac{1}{2}\right)$

(iii) $\Pi\left(\frac{t-1}{2}\right) + \Pi(t-1)$

(iv) $r(t) - 2r(t-1) + r(t-2)$

(v) $r(t)u(2-t)$

(vi) $r(-0.5t+2)$

Solution:

(i) Given the signal $u(t) - u(t-2)$

The signal $u(t)$ is shown in Fig. 1.59a.

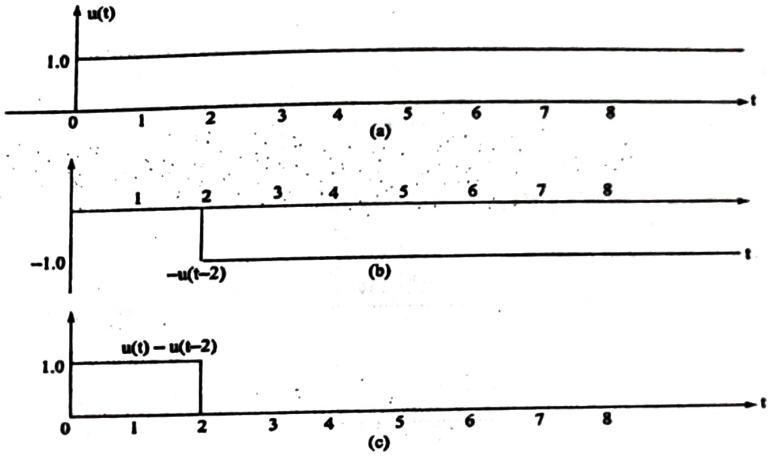


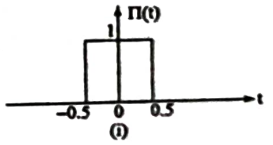
Fig. 1.59

The signal $-u(t-2)$ can be obtained by shifting $u(t)$ to right by 2 units of time and multiplying with -1.

$$u(t) = 1 \quad \text{for} \quad t \geq 0$$
$$\text{and}$$
$$u(t-2) = 1 \quad \text{for} \quad t \geq 2$$

Therefore

$$u(t) - u(t-2) = 1 \quad \text{for} \quad 0 \leq t \leq 2$$
$$= 0 \quad \text{otherwise}$$



(ii) Given $\Pi\left(t - \frac{1}{2}\right)$

We know

$$\Pi(t) = 1 \quad \text{for} \quad |t| \leq \frac{1}{2}$$
$$= 0 \quad \text{otherwise}$$

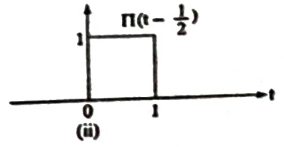


Fig. 1.60(a)

The signal $\Pi(t - \frac{1}{2})$ can be obtained by shifting $\Pi(t)$ right by 0.5 units of time as shown in Fig. 1.60a.

(iii) $\Pi\left(\frac{t-1}{2}\right)+\Pi(t-1)$

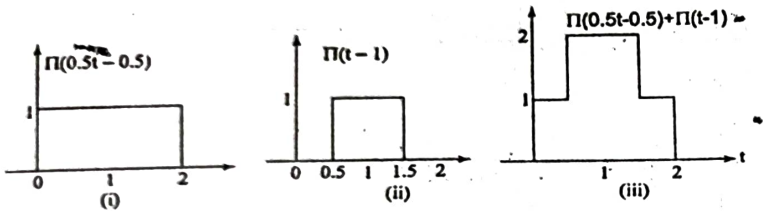


Fig. 1.60b

$\Pi(0.5t-0.5)$ can be obtained first by shifting $\Pi(t)$ by 0.5 units right and then applying times scaling.

$\Pi(t-0.5)$ is shown in Fig. 1.60a. $\Pi(0.5t-0.5)$ can be obtained by applying time scaling to the signal shown. That is by expanding the signal $\Pi(t-\frac{1}{2})$ by two times we can get $\Pi(0.5t-0.5)$. $\Pi(t-1)$ is shown in Fig. 1.60b(ii)

The sketch of $\Pi\left(\frac{t-1}{2}\right)+\Pi(t-1)$ is shown in Fig. 1.60b(iii)

(iv) Given $x(t)=r(t)-2r(t-1)+r(t-2)$

First draw the signals $r(t), -2r(t-1), r(t-2)$ as shown in Fig. 1.61a. $r(t)$ is a signal passing through origin with a slope 1. $-2r(t-1)$ is a signal starting at $t=1$; with a slope -2 and $r(t-2)$ is a signal starting at $t=2$ with a slope 1. When we add all these three signals we can find the following.

In the interval $0 \le t < 1$, the amplitudes of $-2r(t-1)$ and $r(t-2)$ are zero. Therefore the resultant output is $r(t)$ alone. In the interval $1 \le t < 2$, the amplitude of $-r(t-2)$ is zero. Therefore the resultant output is sum of $-2r(t-1)$ and $r(t)$. That is, the slope is changed from 1 to -1. In the interval $2 \le t \le \infty$, the amplitude of resultant output is sum of $-2r(t-1), r(t)$ and $r(t-2)$. It is equal to zero.

$$A = A_1 + A_3 = 1.5 - 1 = 0.5$$
$$B = B_1 + B_2 + B_3 = 2 + 0 - 2 = 0$$
$$C = C_1 + C_2 + C_3 = 2.5 + 0.5 - 3 = 0$$
$$D = D_1 + D_2 + D_3 = 3 + 1 - 4 = 0$$
$$E = E_1 + E_2 + E_3 = 3.5 + 1.5 - 5 = 0$$

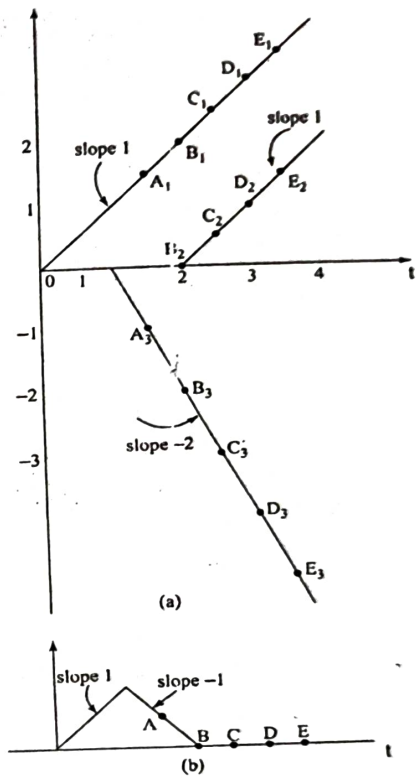


Fig. 1.61

(v) Given $r(t)u(2-t)$

$$u(2-t)=0 \quad \text{for} \quad t>2 \quad \text{and} \quad u(2-t)=1 \quad \text{for} \quad t \le 2$$
$$r(t)=0 \quad \text{for} \quad t<0 \quad \text{and} \quad r(t)=t \quad \text{for} \quad t \ge 0$$

The signals $r(t)$ and $u(2-t)$ are sketched in Fig. 1.62a.

From Fig. 1.62a we can find that

$$r(t)u(2-t)=0 \quad \text{for} \quad t>2 \quad \text{and} \quad t<0$$

Now we can write

$$r(t)u(2-t)=r(t) \quad \text{for} \quad 0 \le t \le 2$$
$$=0 \quad \text{otherwise}$$

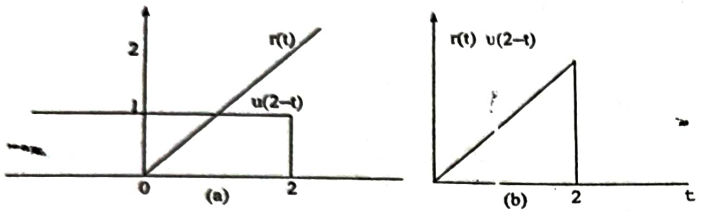


Fig. 1.62

(vi) Given $r(-0.5t+2)$. Let $x(t) = r(-0.5t+2)$

$$r(-0.5t+2) = -0.5t+2 \quad \text{for} \quad -0.5t+2 \geq 0$$
$$= 0 \qquad\qquad \text{otherwise}$$
$$\Rightarrow \quad r(-0.5t+2) = -0.5t+2 \quad \text{for} \quad t \leq 4$$
$$= 0 \qquad\qquad \text{otherwise}$$
$$r(-0.5t+2) = 0 \quad \text{for} \quad t > 4$$
$$= 0.5 \quad \text{for} \quad t = 3$$
$$= 2 \quad \text{for} \quad t = 0$$
$$= 3 \quad \text{for} \quad t = -2$$
$$\text{so on}$$

Now we can plot $r(-0.5t+2)$ as shown in Fig. 1.63.
(or) Find $r(-t)$, shift right by two units of time and apply a time scaling factor of 0.5.

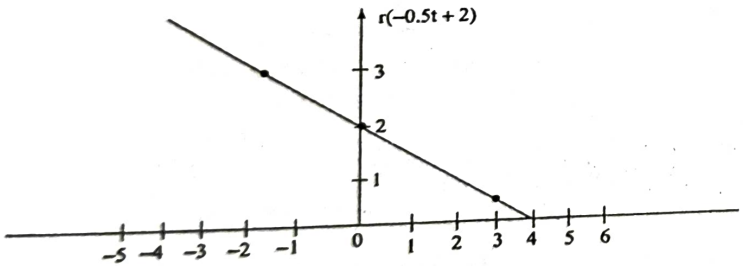


Fig. 1.63

Solved Problem 1.11 Express the following signals as sum of singularity functions

Solution:

(i)

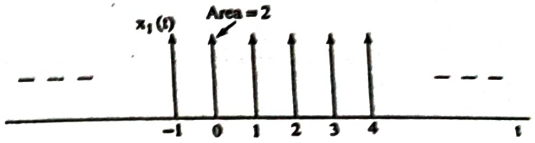


Fig. 1.64

(ii)

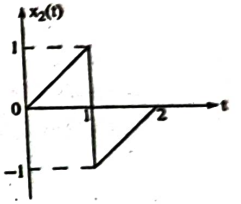


Fig. 1.65

(iii)

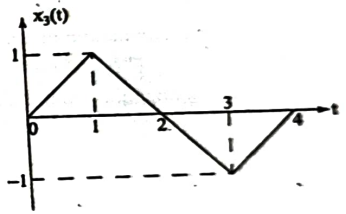


Fig. 1.66

(iv)

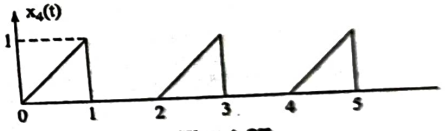


Fig. 1.67

(v)

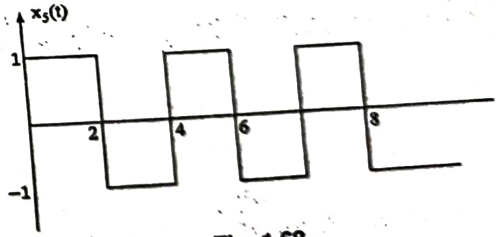


Fig. 1.68

(i) The signal $x_1(t)$ can be expressed as

$$x_1(t) = \ldots 2\delta(t+1)+2\delta(t)+2\delta(t-1)+2\delta(t-2)+\ldots$$

$$= 2\sum_{k=-\infty}^{\infty}\delta(t-k)$$

(ii) Given

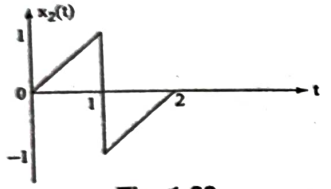


Fig. 1.69

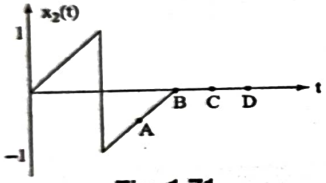


Fig. 1.71

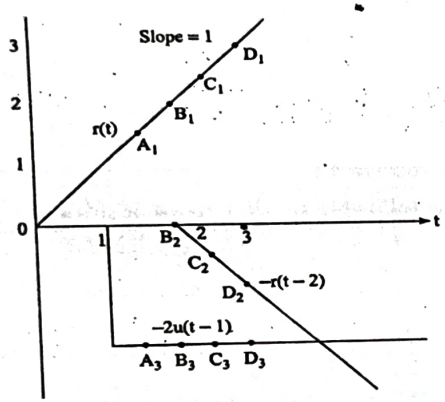


Fig. 1.70

$A = A_1 + A_3 = 1.5 - 2 = -0.5; B = B_1 + B_2 + B_3 = 2 + 0 - 2 = 0$

$C = C_1 + C_2 + C_3 = 2.5 - 0.5 - 2 = 0; D = D_1 + D_2 + D_3 = 3 - 1 - 2 = 0$

From Fig. 1.69 we can observe that the signal in the interval $0 \le t \le 1$ is $r(t)$. At $t = 1$; The signal changes from 1 to -1 and linearly increases with a slope 1). This can be obtained by subtracting $2u(t-1)$ from $r(t)$. For $t > 2$ the signal is zero. This can be achieved by subtracting $r(t-2)$. Therefore the signal is $x_2(t) = r(t) - 2u(t-1) - r(t-2)$

(iii) The signal given is shown in Fig. 1.72.

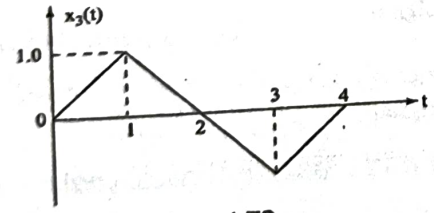


Fig. 1.72

The signal is starting from origin with a slope 1. Therefore the part of signal in the interval $0 \le t \le 1$ is a unit ramp signal $r(t)$. At $t = 1$, the slope of signal is changing from 1 to -1. Therefore add $-2r(t-1)$ to get the signal with a slope of -1. At $t = 3$ the signal slope is changing from -1 to 1. Hence add $2r(t-3)$. At $t = 4$ the signal slope is changing from 1 to zero. So add $-r(t-4)$.

Now the signal can be expressed as

$$x_3(t) = r(t) - 2r(t-1) + 2r(t-3) - r(t-4)$$

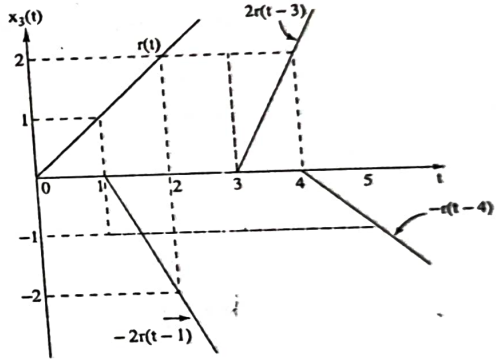


Fig. 1.73

(iv)

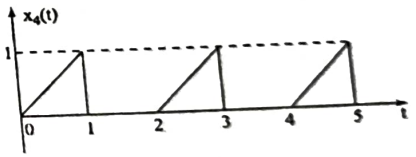


Fig. 1.74

First draw a unit ramp signal. At $t = 1$ the signal value is changing from 1 to zero, and remains the same value until $t = 2$. Therefore first substract $r(t-1)$ from $r(t)$. The resultant signal is shown in Fig. 1.75b which has value 1 for $t > 1$. But in the given signal the value between the interval $1 \le t \le 2$ is zero. So add $-u(t-1)$ to the signal.

The resultant signal due to the combination of three signals is shown in Fig. 1.75(c).

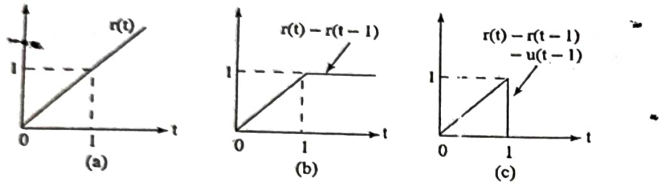


Fig. 1.75

The signal in Fig. 1.74 in the interval $0 \leq t \leq 2$ can be expressed as

$$r(t) - r(t-1) - u(t-1).$$

Repeating the above procedure, we can express $x_4(t)$ as

$$x_4(t) = r(t) - r(t-1) - u(t-1) + r(t-2) - r(t-3) - u(t-3) \ldots$$

(v)

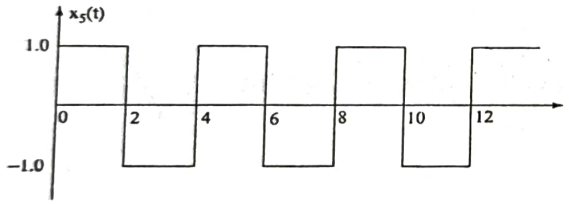


Fig. 1.76

First draw a unit step signal $u(t)$ as shown in Fig. 1.77a. In the given signal the signal amplitude is changing from 1 to -1 at $t = 2$ and maintain same amplitude until $t = 4$. Therefore add $-2u(t-2)$ to $u(t)$ to get a signal as shown in Fig. 1.77b.

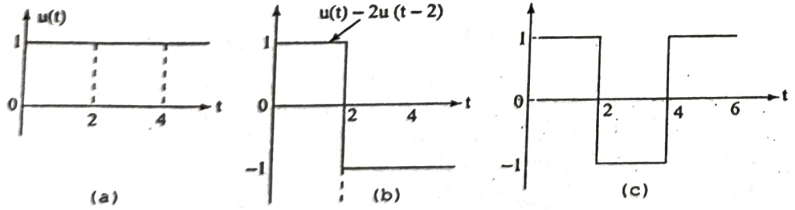


Fig. 1.77

At $t = 4$ the signal amplitude is changing from -1 to 1. Therefore add $2u(t-4)$. If we repeat this process we can express $x_5(t)$ as

$$x_5(t) = u(t) - 2u(t-2) + 2u(t-4) - 2u(t-6) + 2u(t-8) - 2u(t-10) \ldots$$

Practice Problem 1.5

Sketch the following signals

(i) $u\left(\frac{t-1}{4}\right)$ (ii) $\Pi(-2t+2)$ (iii) $r\left(\frac{t+1}{3}\right)$ (iv) $e^{-2t}u(-2+t)$

Practice Problem 1.6

Express the following signal in terms of singularity functions

$$\Pi\left(\frac{t-1}{3}\right) + \Pi\left(\frac{t+2}{3}\right)$$

Ans: $u\left(t+\frac{7}{2}\right) - u\left(t-\frac{5}{2}\right)$

Review Questions

1. Explain in detail about different types of operations on signals.

1.8 Classification of Signals

The signals are classified according to their characteristics. Some of them are

1. Continuous-time and discrete-time signals
2. Deterministic and random signals
3. Periodic and aperiodic signals
4. Even and odd signals
5. Energy and power signals
6. Causal and non-causal signals

1.8.1 Continuous-time and discrete-time signals

These signals are discussed in detail in section (1.3).

1.8.2 Deterministic and random signals

Fig. 1.78

Deterministic signal: A deterministic signal is a signal exhibiting no uncertainity of value at any given instant of time. Its instantaneous value can be accurately predicted by mathematical equation.

Ex: $A\sin\Omega t$

Random signal: A random signal is a signal characterized by uncertainity before its actual occurrence. Some random signals are shown in Fig.1.79.

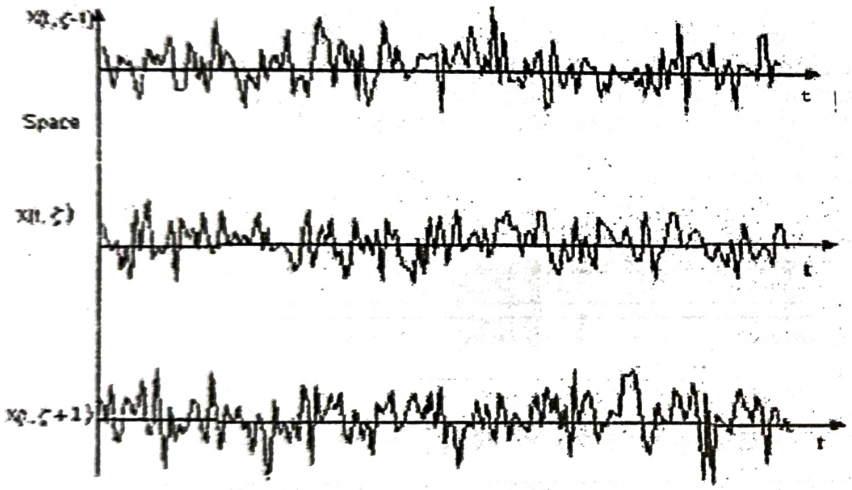


Fig. 1.79

1.8.3 Periodic and aperiodic signals

A continuous-time signal $x(t)$ is said to be periodic if it satisfies the condition

$$x(t+T) = x(t) \quad \text{for all} \quad t \tag{1.52}$$

The smallest value of T that satisfies the above condition is known as fundamental period. A signal is aperiodic if the condition in Eq. (1.52) is not satisfied for atleast one value of t. An example for periodic signal is shown in Fig. 1.80.

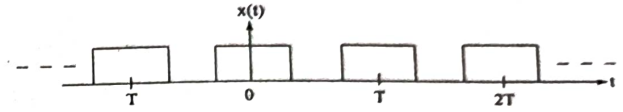


Fig. 1.80 A periodic signal.

A discrete time signal $x(n)$ is said to be periodic if it satisfies the condition

$$x(n+N) = x(n) \quad \text{for all} \quad N \tag{1.53}$$

The smallest value of N for which Eq. (1.53) holds is known as fundamental period. If Eq. (1.53) does not satisfy for atleast one value of n then the discrete-time signal is aperiodic. An example for periodic discrete-time signal is in Fig. 1.81. Here the sequence is repeating after every four samples.

Fig. 1.81 A discrete-time periodic signal.

Therefore the period of the signal $N = 4$. Complex exponential and simusoidal signals are examples for continuous-time periodic signals. Consider a sinusoidal signal $x(t) = A\sin(\Omega_0 t + \theta)$ where A, Ω_0 and θ are amplitude, frequency and phase of the signal respectively.

The unit of Ω_0 is radians per sec and θ is radians. The frequency term is $\Omega_0 = 2\pi f_0$, where f_0 has the units of cycles per second or hertz (Hz). For a periodic signal we have

$$x(t+T) = x(t)$$

$$\text{For} \quad x(t) = A\sin(\Omega_0 t + \theta) \tag{1.54}$$

$$x(t+T) = A\sin[\Omega_0(t+T)+\theta] \tag{1.55}$$

$$= A\sin[\Omega_0 t + \Omega_0 T + \theta) \tag{1.56}$$

The Eq. (1.56) and Eq. (1.54) are equal if

$$\Omega_0 T = 2\pi$$

or

$$T = \frac{2\pi}{\Omega_0} \tag{1.57}$$

The term T is known as fundamental period or time period T and Ω_0 is referred as fundamental frequency. Consider a complex exponential signal

$$x(t) = e^{j\Omega_0 t} \tag{1.58}$$

For a periodic signal $x(t+T) = x(t)$.

Therefore

$$x(t+T) = e^{j\Omega_0(t+T)} = e^{j\Omega_0 t} e^{j\Omega_0 T} \tag{1.59}$$

Eq. (1.58) and Eq. (1.59) are equal if $e^{j\Omega_0 T} = 1$ from which we have

$$\Omega_0 T = 2\pi$$

or

$$T = \frac{2\pi}{\Omega_0} \tag{1.60}$$

Solved Problem 1.12 Find the fundamental period T of the following continuous-time signals.

(i) $x(t) = je^{j5t}$

(ii) $\sin 50\pi t$

(iii) $20\cos\left(10\pi t + \frac{\pi}{6}\right)$

Solution:

(i) Given $x(t) = je^{j5t}$

Comparing this equation with standard form (1.58) we have $\Omega_0 = 5$,

Time period $T = \frac{2\pi}{\Omega_0} = \frac{2\pi}{5} = 0.4\pi$ sec

(ii) Given $x(t) = \sin 50\pi t$; $\Omega_0 = 50\pi$

Time period $T = \frac{2\pi}{50\pi} = \frac{1}{25}$ sec

(iii) Given $x(t) = 20\cos\left(10\pi t + \frac{\pi}{6}\right)$ from which

$$\Omega_0 = 10\pi$$

$$\text{Time period} \quad T = \frac{2\pi}{\Omega_0} = \frac{2\pi}{10\pi} = \frac{1}{5} \text{ sec}$$

Practice Problem 1.7

Find the fundamental period T of the following signals if they are periodic.

(i) $x_1(t) = 4\cos 5\pi t$

(ii) $x_2(t) = \sin 10\pi t u(t)$

(iii) $x_3(t) = e^{-|t|}$

Ans:

(i) Periodic with period $\frac{2}{5}$ sec.

(ii) Not a periodic signal.

(iii) Not a periodic signal.

The sum of two periodic signals $x_1(t)$ and $x_2(t)$ with periods T_1 and T_2 may or may not be periodic depending on the relation between T_1 and T_2. If the sum to be periodic, then the ratio of periods $\frac{T_1}{T_2}$ must be a rational number or ratio of two integers. Otherwise the sum is not periodic. For example, let us take two signals with time periods $T_1 = 5$ and $T_2 = 7$. The ratio of time periods $T_1/T_2 = 5/7$ is a rational number. Therefore the addition of two signals is periodic and the period is given by $T = 7T_1 = 5T_2$. On the other hand if the time periods of two signals are $T_1 = 3$ and $T_2 = \sqrt{7}$, the ratio of time periods $\frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$ is not a rational number. Therefore the sum of two signals is not periodic.

Review Questions

1. Define periodic and aperiodic signals.
2. Explain the condition for sum of two periodic signals to be periodic.

Solved Problem 1.13 Find whether the following signals are periodic or not.

(i) $x(t) = 2\cos(10t+1) - \sin(4t-1)$

(ii) $\cos 60\pi t + \sin 50\pi t$

(iii) $2u(t) + 2\sin 2t$

(iv) $3\cos 4t + 2\sin 2\pi t$

(v) $u(t) - \frac{1}{2}$

(vi) $\sin^2 t$ AU, Nov 2008

Solution:

(i) Given $x_1(t) = 2\cos(10t+1) - \sin(4t-1)$

Time period of $2\cos(10t+1)$ is $T_1 = \frac{2\pi}{10} = \frac{\pi}{5}$ sec

Time period of $\sin(4t-1)$ is $T_2 = \frac{2\pi}{4} = \frac{\pi}{2}$ sec

The ratio of two periods is

$$\frac{T_1}{T_2} = \frac{\pi/5}{\pi/2} = \frac{2}{5}$$

is a rational number. Therefore the sum of two signals are periodic and the period is given by

$$T = 2T_2 = 5T_1$$

$$= \pi \text{ sec}$$

(ii) Given $x_2(t) = \cos 60\pi t + \sin 50\pi t$

$$\text{Period of } \cos 60\pi t \text{ is } T_1 = \frac{2\pi}{60\pi} = \frac{1}{30}$$

$$\text{Period of } \sin 50\pi t \text{ is } T_2 = \frac{2\pi}{50\pi} = \frac{1}{25}$$

$$\text{The ratio } \frac{T_1}{T_2} = \frac{1/30}{1/25} = \frac{5}{6}$$

is a rational number. Hence $x_2(t)$ is a periodic signal

$$\text{the time period} = 6T_1 = 5T_2$$

$$= \frac{1}{5} \text{ sec}$$

(iii) $2u(t) + 2\sin 2t$

The signal is sketched as shown in Fig. 1.82.

Period of the signal $2\sin 2t$ is π sec. The signal $2u(t)$ is aperiodic.

Therefore the signal $2u(t) + 2\sin 2t$ is aperiodic.

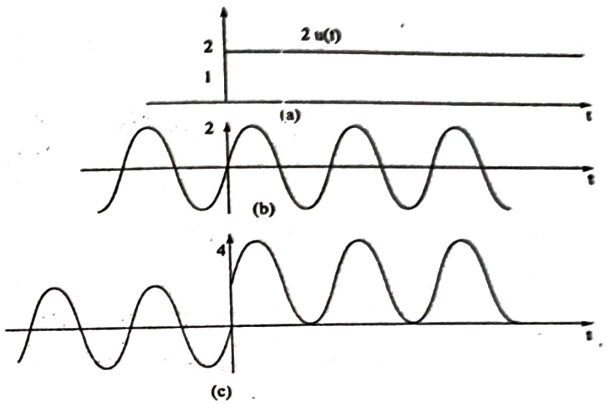


Fig. 1.82

(iv) $3\cos 4t + 2\sin \pi t$

The period of the signal $3\cos 4t$ is $T_1 = \frac{2\pi}{4} = \frac{\pi}{2}$ sec and

the period of the signal $2\sin \pi t$ is $T_2 = \frac{2\pi}{1\pi} = 2$ sec.

The ratio of time periods $\frac{T_1}{T_2} = \frac{\frac{\pi}{2}}{2} = \frac{\pi}{4}$ is not a rational number.

Therefore the signal is not periodic.

(v) Given $u(t) - \frac{1}{2}$

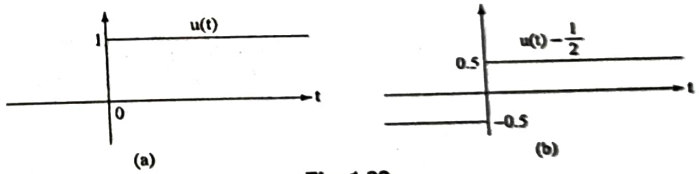


Fig. 1.83

The signal is not periodic.

(vi) The signal $x(t) = \sin^2 t$

$$= \frac{1 - \cos 2t}{2}$$

The period of the signal is $T = \frac{2\pi}{\omega_0} = \frac{2\pi}{2} = \pi$ sec. The signal is periodic.

Practice Problem 1.8

Find whether the following signals are periodic or not?

(i) $3\cos\left(17\pi t+\frac{\pi}{3}\right)+2\sin\left(19\pi t-\frac{\pi}{3}\right)$

Ans: Periodic with period 2 sec

(ii) $u(t)-u(t-10)$ Ans: Aperiodic

(iii) $\cos\left(\frac{1}{3}t\right)+\sin\left(\frac{1}{4}t\right)$ Ans: Periodic with period 24π

(iv) $12e^{j\left(4t+\frac{\pi}{2}\right)}+9e^{j\left(3\pi t+\frac{\pi}{3}\right)}$ Ans: Aperiodic

Condition for Discrete-time sinusoidal signal to be periodic

In case of continous-time signals, all sinusoidal signals are periodic. But in discrete-time case, not all the sinusoidal sequences are periodic. Consider a discrete-time signal given by

$$x(n)=A\sin(\omega_0 n+\theta) \tag{1.61}$$

where A is amplitude, ω_0 and θ are frequency and phase shift respectively.

A discrete-time signal is periodic if and only if

$$x(n+N)=x(n) \quad \text{for all } n \tag{1.62}$$

From Eq. (1.61) we can obtain

$$\begin{aligned} x(n+N) &= A\sin[\omega_0(n+N)+\theta] \\ &= A\sin[\omega_0 n+\omega_0 N+\theta] \end{aligned} \tag{1.63}$$

Eq. (1.61) and Eq. (1.63) are equal if

$$\omega_0 N=2\pi m$$

That is there must be an integer m such that

$$\omega_0=2\pi m \tag{1.64}$$

$$=2\pi\left[\frac{m}{N}\right] \quad \text{or} \tag{1.65}$$

$$N=2\pi\left[\frac{m}{\omega_0}\right]$$

From Eq. (1.65) we find that, for the discrete-time signal to be periodic, the fundamental frequency ω_0 must be a rational multiple of 2π. Otherwise the discrete-time signals is aperiodic. The smallest value of positive integer N, for some integer m, is the fundamental period.

Review Questions

1. What is the condition for discrete-time signals to be periodic?

Solved Problem 1.14 Find whether the following are periodic or not.

(i) $\cos(.1\pi n)$ AU, May 2008

(ii) $e^{j6\pi n}$

(iii) $\sin\left(\frac{6\pi}{7}n+1\right)$

(iv) $e^{j\frac{2\pi}{3}n}+e^{j\frac{3\pi}{4}n}$

(v) $\frac{3}{5}e^{j3\pi\left(n+\frac{1}{2}\right)}$ AU, Nov 2006

(vi) $12\cos(20n)$

(vii) $\cos\left(\frac{1}{4}n\right)$ AU, Nov 2008

Solution:

Given

(i) $x(n)=\cos(0.1\pi n);\ \omega_0=0.1\pi$

$$N=\frac{2\pi}{\omega_0}m=\frac{2\pi}{0.1\pi}m=20m$$

The N takes smallest positive integer value, when $m=1$. Therefore $N=20$ and the sequence is periodic.

(ii) $x(n)=e^{j6\pi n}$

$$\omega_0=6\pi$$

$$N=\frac{2\pi}{\omega_0}m=\frac{2\pi}{6\pi}m=\frac{1}{3}m$$

The smallest value m for which N is an integer is 3.

$$N=\frac{1}{3}(3)=1$$

$N=1$. The given signal is periodic.

(iii) $x(n) = \sin\left(\frac{6\pi}{7}n + 1\right)$

$$\omega_0 = \frac{6\pi}{7}$$

$$N = \left(\frac{2\pi}{\omega_0}\right)m = \left(\frac{2\pi}{\frac{6\pi}{7}}\right)m = \left(\frac{7}{3}\right)m$$

The smallest value of m for which N is an integer is 3.

$N = \left(\frac{7}{3}\right)3 \quad = 7$

$N = 7.$ The given signal is periodic

(iv) $e^{j(\frac{2\pi}{3})n} + e^{j(\frac{3\pi}{4})n}$

The time period of $e^{j(\frac{2\pi}{3})n}$ is

$$N_1 = \left(\frac{2\pi}{2\pi/3}\right)m = 3m$$

for $m = 1$

$N_1 = 3$

The time period of $e^{j\frac{3\pi}{4}n}$ is

$$N_2 = \left(\frac{2\pi}{3\pi/4}\right)m = \frac{8}{3}m$$

for $m = 3$

$N_2 = 8$

The ratio of time periods

$$\frac{N_1}{N_2} = \frac{3}{8}$$

$$\Rightarrow \quad 8N_1 = 3N_2$$

$$N = 8N_1 = 3N_2 = 24$$

$$\boxed{N = 24}$$

(v) $x(n) = \frac{3}{5}\, e^{j\,3\pi\,(n+\frac{1}{2})}$

$$\omega_0 = 3\,\pi$$

$$N = \frac{2\pi}{3\pi}m = \frac{2}{3}m$$

The smallest value of m for which N is an integer is 3

$$\Rightarrow \quad N = \left(\frac{2}{3}\right)3 = 2$$

$$N = 2$$

(vi) $x(n) = 12\cos(20n)$

$$\omega_0 = 20$$

$$N = \frac{2\pi}{\omega_0}m$$

$$= \frac{2\pi}{20}$$

$$= \frac{m\pi}{20}$$

For any value of m, N is not an integer. Therefore, $12\cos(20n)$ is a aperiodic sequence.

(vii) $x(n) = \cos\left(\frac{n}{4}\right)$

The frequency of the discrete-time signal is $\frac{1}{4}$. Since the frequency is not a multiple of π, the signal is aperiodic.

Practice Problem 1.9

Find the fundamental period of the following signals

(i)	$x(t) = 2\sin(3t+1) + 3\sin(4t-1)$	Ans: 2π
(ii)	$x(t) = \sin\left(\frac{7\pi}{3}t\right)$	Ans: 6/7
(iii)	$x(n) = \sin 2\pi n + \sin 6\pi n$	Ans: 1
(iv)	$x(n) = e^{j\frac{7\pi}{3}n}$	Ans: 6

1.8.4 Symmetric (Even) and Anti-symmetric (Odd) signals

Symmetric (Even) signal: A continuous-time signal $x(t)$ is said to be symmetric (even) signal if it satisfies the condition

$$x(-t) = x(t) \quad \text{for all} \quad t \qquad (1.66)$$

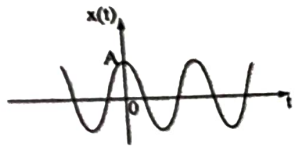


Fig. 1.84

Example of a symmetric signal is $x(t) = A\cos t$ as shown in Fig.1.84

Anti-symmetric (Odd) signal: A continuous-time signal $x(t)$ is said to be anti-symmetric (odd) signal if it satisfies the condition

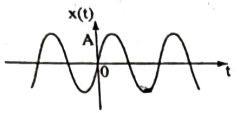


Fig. 1.85

$$x(-t) = -x(t) \quad \text{for all} \quad t \qquad (1.67)$$

Example of an anti-symmetric signal is $A \sin t$ as shown in Fig.1.85

Any signal $x(t)$ can be expressed as sum of even and odd components. That is

$$x(t) = x_e(t) + x_o(t) \qquad (1.68)$$

where $x_e(t)$ is even component of the signal and $x_o(t)$ is odd component of the signal.

Replacing t by $-t$ in Eq. (1.68) yields

$$\begin{aligned} x(-t) &= x_e(-t) + x_o(-t) \\ &= x_e(t) - x_o(t) \end{aligned} \qquad (1.69)$$

Adding Eq. (1.68) and Eq. (1.69) we get

$$2x_e(t) = x(t) + x(-t)$$

Therefore

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] \qquad (1.70)$$

Similarly subtracting Eq. (1.69) from Eq. (1.68) we obtain

$$\begin{aligned} 2x_o(t) &= x(t) - x(-t) \\ \Rightarrow \quad x_o(t) &= \frac{1}{2}[x(t) - x(-t)] \end{aligned} \qquad (1.71)$$

A discrete-time signal $x(n)$ is even if it satisfies the condition

$$x(-n) = x(n) \quad \text{for all} \quad n \qquad (1.72)$$

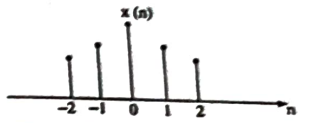


Fig. 1.86 An even sequence.

An example of even signal is shown in Fig. 1.86.
The signal $x(n)$ is said to be an odd signal if

$$x(-n) = -x(n) \quad \text{for all} \quad n \qquad (1.73)$$

An example of odd signal is shown in Fig. 1.87. For an odd signal $x(0) = 0$.
The even and odd parts of a signal $x(n)$ can be obtained using the following relations

$$x_e(n) = \frac{1}{2}[x(n) + x(-n)] \qquad (1.74)$$

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)] \qquad (1.75)$$

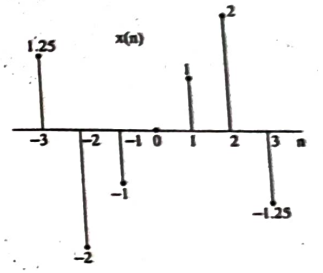


Fig. 1.87 An odd sequence.

Review Question

1. Define symmetric and anti-symmetric signals.

Solved Problem 1.15 Find the even and odd components of the following signals.

(i) $x(t) = \cos t + \sin t + \cos t \sin t$

(ii) $x(n) = \left\{-2, 1, \underset{\uparrow}{2}, -1, 3\right\}$

Solution:

(i) Given

$$x(t) = \cos t + \sin t + \cos t \sin t$$

$$\begin{aligned} x(-t) &= \cos(-t) + \sin(-t) + \cos(-t)\sin(-t) \\ &= \cos t - \sin t - \cos t \sin t \end{aligned}$$

$$x_e(t) = \frac{1}{2}[x(t)+x(-t)]$$
$$= \frac{1}{2}[\cos t + \sin t + \cos t \sin t + \cos t - \sin t - \cos t \sin t] = \cos t$$
$$x_0(t) = \frac{1}{2}[x(t)-x(-t)]$$
$$= \frac{1}{2}[\cos t + \sin t + \cos t \sin t - \cos t - \sin t - \cos t \sin t]$$
$$= \sin t + \cos t \sin t$$

(ii)
$$x(n) = \{-2, 1, \underset{\uparrow}{2}, -1, 3\}$$
$$n = -2, -1, 0, 1, 2$$
$$x_e(n) = 0.5\,[x(n)+x(-n)]$$
$$x_e(0) = 0.5\,[x(0)+x(-0)] = 0.5\,[2+2] = 2$$
$$x_e(1) = \frac{1}{2}[x(1)+x(-1)] = \frac{1}{2}[-1+1] = 0$$
$$x_e(2) = \frac{1}{2}[x(2)+x(-2)] = \frac{1}{2}(-2+3) = 0.5$$
$$\Rightarrow \quad x_e(n) = \{0.5, 0, \underset{\uparrow}{2}, 0, 0.5\}$$
$$x_o(0) = \frac{1}{2}[x(n)-x(-n)] = \frac{1}{2}(2-2) = 0$$
$$x_o(1) = \frac{1}{2}[x(1)-x(-1)] = \frac{1}{2}[-1-1] = -1$$
$$x_o(2) = \frac{1}{2}[x(2)-x(-2)] = \frac{1}{2}[3+2] = 2.5$$
$$x_o(n) = [-2.5, 1, \underset{\uparrow}{0}, -1, 2.5]$$

Practice Problem 1.10

Find even and odd components of the following signals

(i) $x(t) = \sin t + 2\sin t + 2\sin^2 t \cos t$ Ans: $x_e(t) = 2\sin^2 t \cos t$, $x_0(t) = \sin t + 2\sin t$

(ii) $x(n) = \{1, 0, -1, 2, 3\}$

Ans: $x_e(n) = \{1.5, 1, -0.5, \underset{\uparrow}{0}, 1, 0, -0.5, 1, 1.5\}$

$x_0(n) = \{-1.5, -1, 0.5, 0, \underset{\uparrow}{0}, 0, -0.5, 1, 1.5\}$

Review Questions

1. Define periodic signal, odd signal and even signal with an example each. AU, Nov 2008

1.8.5 Energy and power signals

Consider a voltage $v(t)$ across a resistance R producing a current $i(t)$. The instantaneous power dissipated in the resistance R is defined by

$$p(t) = v(t)\, i(t)$$
$$= v(t)\frac{v(t)}{R} = \frac{v^2(t)}{R} \tag{1.76}$$

or

$$p(t) = i^2(t)R \tag{1.77}$$

For a resistance of 1Ω Eq. (1.76) and Eq. (1.77) shows that the instantaneous power $p(t)$ is equal to the square of the amplitude of the signal.

Integrating Eq. (1.77) over the interval $|t| \le T$, we can express the total energy and average power of a signal as

$$E = \lim_{T\to\infty} \int_{-T}^{T} i^2(t)\, dt \text{ joules} \qquad (R = 1\Omega) \tag{1.78}$$

and

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} i^2(t)\, dt \text{ watts} \tag{1.79}$$

For any signal $x(t)$, the total energy is defined as

$$E = \lim_{T\to\infty} \int_{-T}^{T} |x(t)|^2\, dt \text{ joules} \tag{1.80}$$

and the average power is defined as

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2\, dt \text{ watts} \tag{1.81}$$

The square root of P is known as RMS value of the signal $x(t)$.
For a discrete-time signal $x(n)$ the energy E is defined as

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2 \tag{1.82}$$

The average power of a discrete-time signal $x(n)$ is defined as

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2 \tag{1.83}$$

A signal is an energy signal if and only if the total energy of the signal is finite. For an energy signal $P = 0$. Similarly the signal is said to be power signal if the average power of the signal is finite. For a power signal $E = \infty$. The signals that does not satisfy above properties are neither energy nor power signal.

Definition:

A signal $x(t)$ is called an energy signal if the energy satisfies $0 < E < \infty$. For an energy signal $P = 0$.

A signal $x(t)$ is called an power signal if the power satisfies $0 < P < \infty$. For a power signal $E = \infty$.

Review Questions

1. Define energy and power of a signal.
2. When a signal is called
   (i) energy signal (ii) power signal?

Solved Problem 1.16 Determine the power and RMS value of the signal $x(t) = A\cos(\Omega_0 t + \theta)$;

Solution:

Given $x(t) = A\cos(\Omega_0 t + \theta)$

The power

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} [A\cos(\Omega_0 t + \theta)]^2 dt$$

$$= \lim_{T\to\infty} \frac{A^2}{2T} \left[\int_{-T}^{T} \frac{1+\cos(2\Omega_0 t + 2\theta)}{2} \right] dt$$

$$= \lim_{T\to\infty} \frac{A^2}{4T} \int_{-T}^{T} dt + \lim_{T\to\infty} \frac{A^2}{4T} \int_{-T}^{T} \cos(2\Omega_0 t + 2\theta) dt$$

$$= \lim_{T\to\infty} \frac{A^2}{4T} \int_{-T}^{T} dt + \underbrace{\lim_{T\to\infty} \frac{A^2}{4T} \int_{-T}^{T} \cos(2\Omega_0 t + 2\theta) dt}_{\text{equals to zero}}$$

$$= \frac{A^2}{2}$$

$$\text{The r.m.s value} = \sqrt{\frac{A^2}{2}} = \frac{A}{\sqrt{2}}$$

From the example we find that the power of a sinusoidal signal with amplitude A is equal $\frac{A^2}{2}$.

Solved Problem 1.17 Determine the power and RMS value of the following signals.

(i) $x_1(t) = 5\cos\left(50t + \frac{\pi}{3}\right)$

(ii) $x_2(t) = 10\sin\left(50t + \frac{\pi}{4}\right) + 16\cos\left(100t + \frac{\pi}{3}\right)$

(iii) $x_3(t) = 10\cos 5t \cos 10t$

(iv) $x_4(t) = e^{j\alpha t}\cos\Omega_0 t$

(v) $x_5(t) = Ae^{j\Omega_0 t}$

Solution:

(i) $x_1 = 5\cos\left(50t + \frac{\pi}{3}\right)$

The power of the signal is equal to $\frac{5^2}{2} = 12.500\,\text{W}$. The RMS value of the signal is $\sqrt{12.5}$.

(ii) $x_2(t) = 10\sin\left(50t + \frac{\pi}{4}\right) + 16\cos\left(100t + \frac{\pi}{3}\right)$

Power $= \frac{10^2}{2} + \frac{16^2}{2} = 178\,\text{W}$

The RMS value is equal to $\sqrt{178} = 13.341$

(iii)

$$x_3(t) = 10\cos 5t \cos 10t$$

$$= \frac{10}{2}[\cos 15t + \cos 5t]$$

$$= 5\cos 15t + 5\cos 5t$$

The power of the signal is

$$\frac{5^2}{2} + \frac{5^2}{2} = 25\,\text{W}$$

The RMS value is $\sqrt{25} = 5$

(iv)

$$x_4(t) = e^{j\alpha t}\cos\Omega_0 t$$

$$P = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|e^{j\alpha t}\cos\Omega_0 t|^2$$

$\because\ |e^{j\alpha t}| = 1$

$\cos^2\Omega_0 t = \frac{1+\cos 2\Omega_0 t}{2}$

$$= \lim_{T\to\infty}\frac{1}{4T}\int_{-T}^{T}(1+\cos 2\Omega_0 t)\,dt$$

$$= \lim_{T\to\infty}\frac{1}{4T}\int_{-T}^{T}dt + \lim_{T\to\infty}\frac{1}{4T}\underbrace{\int_{-T}^{T}\cos 2\Omega_0 t\,dt}_{\text{is equals to zero}}$$

$$= \lim_{T\to\infty}\frac{1}{4T}[2T]$$

$$= \frac{1}{2}$$

(v) Given $x_5(t) = Ae^{j\Omega_0 t}$

The power of the signal

$$P = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2\,dt$$

$$= \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|A^2 e^{j2\Omega_0 t}|\,dt$$

$$= \lim_{T\to\infty}\frac{A^2}{2T}\int_{-T}^{T}|e^{j2\Omega_0 t}|\,dt$$

$$= \frac{A^2}{2T}(2T) = A^2$$

RMS value of the signal $= A$

Solved Problem 1.18 Show that the power of signal

$$x(t) = \sum_{k=-n}^{n} c_k e^{j\Omega_k t} \text{ is } P = \sum_{k=-n}^{n}|c_k|^2$$

assuming all frequencies to be distinct, that is $\Omega_i \neq \Omega_k$ for $i \neq k$.

Solution:

We know

$$P = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2\,dt$$

$$P = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}\left|\sum_{k=-n}^{n}c_k e^{j\Omega_k t}\right|^2 dt$$

Interchanging the order of integration and summation we get

$$P = \sum_{k=-n}^{n}|c_k|^2 \lim_{T\to\infty}\frac{1}{2T}\underbrace{\int_{-T}^{T}|e^{j\Omega_k t}|^2\,dt}_{2T}$$

$$= \sum_{k=-n}^{n}|c_k|^2$$

$|e^{j\Omega_k t}| = 1$

Solved Problem 1.19 Sketch the following signals and calculate their energies

(i) $e^{-10t}u(t)$
(ii) $u(t)-u(t-15)$
(iii) $\cos(10\pi t)\,u(t)u(2-t)$

Solution:

(i) Given $x(t)=e^{-10t}u(t)$

The signal $x(t)$ is sketched in Fig. 1.88.

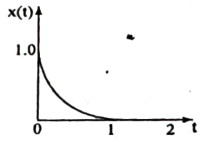


Fig. 1.88

The energy

$$E=\lim_{T\to\infty}\int_{-T}^{T}|x(t)|^2dt$$

$$=\lim_{T\to\infty}\int_{0}^{T}(e^{-10t})^2\,dt$$

$$=\lim_{T\to\infty}\int_{0}^{T}(e^{-20t})dt=-\frac{1}{20}[-1]=\frac{1}{20}\text{ Joules}$$

(ii) Given $x(t)=u(t)-u(t-15)$

The signal is sketched in Fig.1.89

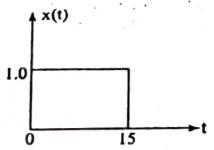


Fig. 1.89

$$E=\int_{0}^{15}(1)^2\,dt=15\text{J}$$

(iii) Given $x(t)=\cos 2\pi t\,u(t)u(2-t)$

The given signal is sketched in Fig.1.90

The period of the signal $\cos 2\pi t$ is $\frac{2\pi}{2\pi}=1$

$$E=\int_{0}^{2}\cos^2 2\pi t\,dt$$

$$=\frac{1}{2}\left[\int_{0}^{2}(1+\cos 4\pi t)\,dt\right]$$

$=1$ Joule

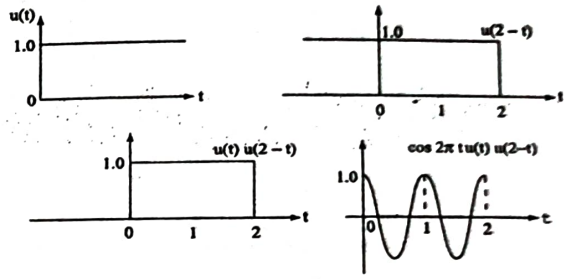


Fig. 1.90

Solved Problem 1.20 Which of the following signals are energy signals?

(i) $u(t)-u(t-1)$
(ii) $r(t)-r(t-2)$
(iii) $(1+e^{-5t})u(t)$

Solution:

(i) Given

$$x(t)=u(t)-u(t-1)$$

The signal $x(t)$ is shown in Fig. 1.91.

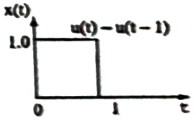


Fig. 1.91

$$E=\lim_{T\to\infty}\int_{-T}^{T}|x(t)|^2\,dt$$

We have

$$x(t)=\begin{cases}1 & \text{for } 0\le t\le 1\\ 0 & \text{otherwise}\end{cases}$$

SO, $E=\int_{0}^{1}(1)^2dt=1$ Joule

$$P=\lim_{T\to\infty}\frac{1}{2T}\left[\int_{0}^{1}(1)^2\,dt\right]=0$$

The energy of the signal is finite and the power is zero. Therefore the signal is energy signal.

(ii) Given

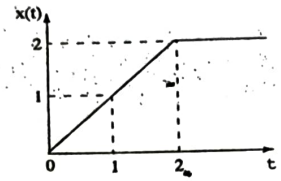


Fig. 1.92

$$x(t) = r(t) - r(t-2)$$

$$E = \lim_{T\to\infty}\left[\int_{-T}^{2}|x(t)|^2\,dt + \int_{2}^{T}|x(t)|^2 dt\right]$$

$$= \lim_{T\to\infty}\left[\int_{0}^{2} t^2 dt + \int_{2}^{T} 4\,dt\right]$$

$$= \lim_{T\to\infty}\left[\frac{8}{3} + 4(T-2)\right]$$

$$= \infty$$

Note: Limits of the integrals from 0 to T instead of $-T$ to T since the magnitude of signal is zero from $-T$ to zero.

Power

$$P = \lim_{T\to\infty}\frac{1}{2T}\left[\int_{0}^{2} t^2 dt + \int_{2}^{T} 4dt\right]$$

$$= \lim_{T\to\infty}\frac{1}{2T}\left[\frac{8}{3} + 4(T-2)\right]$$

$$= \lim_{T\to\infty}\frac{8}{6T} + \lim_{T\to\infty}\left(\frac{4T}{2T} - \frac{8}{2T}\right)$$

$$= 2\text{W}$$

The energy of the signal is infinite and power is finite. Therefore the signal is a power signal.

(iii) $$x(t) = (1+e^{-5t})\,u(t)$$

$$E = \lim_{T\to\infty}\int_{0}^{T}(1+e^{-5t})^2\,dt$$

$$E = \lim_{T\to\infty}\int_{0}^{T}(1+e^{-10t}+2e^{-5t})\,dt$$

$$= \infty$$

The power of the signal is given by

$$P = \lim_{T\to\infty}\frac{1}{2T}\int_{0}^{T}(1+e^{-10t}+2e^{-5t})\,dt$$

$$= \lim_{T\to\infty}\frac{1}{2T}\left[T - \frac{1}{10}e^{-10t}\Big|_0^T - \frac{2}{5}e^{-5t}\Big|_0^T\right]dt$$

$$= \lim_{T\to\infty}\left[\frac{T}{2T} - \frac{1}{20T}(e^{-10T}-1) - \frac{2}{10T}(e^{-5T}-1)\right]$$

$$= \frac{1}{2}\text{W}$$

The energy of the signal $E = \infty$.

The power of the signal $P = \frac{1}{2}$W.

Therefore the signal is a power signal.

Solved Problem 1.21 Find which of the following signals are energy signals, power signals, neither energy nor power signals.

(i) $x_1(t) = e^{-3t}u(t)$

(ii) $x_2(t) = e^{j\left(2t+\frac{\pi}{4}\right)}$

(iii) $x_3(t) = \cos t$

(iv) $x_1(n) = \left(\frac{1}{3}\right)^n u(n)$

(v) $x_2(n) = e^{j\left(\frac{\pi}{2}n+\frac{\pi}{8}\right)}$ AU, April 2008

(vi) $x_3(n) = \cos\left(\frac{\pi}{4}n\right)$

Solution:

(i) Given $x_1(t) = e^{-3t}u(t)$

The energy of the signal

$$E = \lim_{T\to\infty}\int_{-T}^{T}|x_1(t)|^2\,dt$$

$$= \lim_{T\to\infty}\int_{0}^{T}e^{-6t}\,dt$$

$$= -\frac{1}{6}e^{-6t}\Big|_0^\infty = \frac{1}{6}$$

The power of the signal

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x_1(t)|^2\, dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{0}^{T} e^{-6t}\, dt$$

$$= \lim_{T\to\infty} \left(-\frac{1}{12T}\right) e^{-6t}\Big|_0^T = \lim_{T\to\infty} \frac{(e^{-6T}-1)}{12T} = 0$$

The energy of the signal is finite, and the power is 0. Therefore the signal $x_1(t)$ is an energy signal.

(ii) Given $x_2(t) = e^{j(2t+\frac{\pi}{4})}$

The energy of the signal

$$E = \lim_{T\to\infty} \int_{-T}^{T} |x_2(t)|^2\, dt$$

$$= \lim_{T\to\infty} \int_{-T}^{T} \left|e^{j(2t+\frac{\pi}{4})}\right|^2 dt \qquad \boxed{\because \left|e^{j(2t+\frac{\pi}{4})}\right| = 1}$$

$$= \lim_{T\to\infty} (2T) = \infty$$

The power of the signal

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|e^{j(2t+\frac{\pi}{4})}\right|^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{2T}[2T] = 1$$

The power of the signal is finite and the energy of the signal is infinite. Therefore $x_2(t)$ is a power signal.

(iii)

$$x_3(t) = \cos(t)$$

$$E = \lim_{T\to\infty} \int_{-T}^{T} \cos^2 dt = \frac{1}{2} \lim_{T\to\infty} \int_{-T}^{T} (1+\cos 2t)\, dt$$

$$= \frac{1}{2}\left[\lim_{T\to\infty} \int_{-T}^{T} dt + \underbrace{\lim_{T\to\infty} \int_{-T}^{T} \cos 2t\, dt}_{\text{equals to zero}}\right]$$

$$= \infty$$

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \cos^2 t\, dt$$

$$= \lim_{T\to\infty} \frac{1}{4T} \int_{-T}^{T} (1+\cos 2t)\, dt$$

$$= \frac{1}{2}$$

The energy of the signal is infinite and the power of the signal is finite. Therefore $x_3(t)$ is power signal.

(iv) $$x_1(n) = \left(\frac{1}{3}\right)^n u(n)$$

$$E = \lim_{N\to\infty} \sum_{n=-N}^{N} |x(n)|^2$$

$$= \lim_{N\to\infty} \sum_{n=-N}^{N} \left[\left(\frac{1}{3}\right)^n\right]^2 u(n)$$

$$= \lim_{N\to\infty} \sum_{n=0}^{N} \left(\frac{1}{9}\right)^n$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{9}\right) = \frac{1}{1-\frac{1}{9}} = \frac{9}{8}$$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} \left(\frac{1}{9}\right)^n$$

$$= \lim_{N\to\infty} \frac{1}{2N+1} \left[\frac{1-(\frac{1}{9})^{N+1}}{1-\frac{1}{9}}\right]$$

$$= 0$$

The energy is finite and power is zero. Therefore $x_1(n)$ is an energy signal.

(v) $x_2(n) = e^{j\left(\frac{5}{2}n+\frac{\pi}{6}\right)}$

$$E = \lim_{N\to\infty} \sum_{n=-N}^{N} \left|e^{j\left(\frac{5}{2}n+\frac{\pi}{6}\right)}\right|^2$$

$$= \lim_{N\to\infty} \sum_{n=-N}^{N} 1$$

$$= \lim_{N\to\infty} (2N+1) = \infty$$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} \left|e^{j\left(\frac{5}{2}n+\frac{\pi}{6}\right)}\right|^2$$

$$= \lim_{N\to\infty} \frac{1}{2N+1}(2N+1) = 1$$

The energy is infinite and the power is finite. Therefore $x_2(n)$ is a power signal.

(vi)

$$x_3(n) = \cos\left(\frac{\pi}{4}n\right)$$

$$E = \lim_{N\to\infty} \sum_{n=-N}^{N} \cos^2\left(\frac{\pi}{4}n\right)$$

$$E = \lim_{N\to\infty} \sum_{n=-N}^{N} \frac{1+\cos\left(\frac{\pi}{2}n\right)}{2}$$

$$= \frac{1}{2} \lim_{N\to\infty} \sum_{n=-N}^{N} \left(1+\cos\frac{\pi}{2}n\right)$$

$$= \infty$$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} \cos^2\frac{\pi}{4}n$$

$$= \frac{1}{2} \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} \left[1+\cos\left(\frac{\pi}{2}n\right)\right]$$

$$= \frac{1}{2} \lim_{N\to\infty} \frac{1}{2N+1}(2N+1) = \frac{1}{2}$$

The energy is infinite and the power is finite. Therefore $x_3(n)$ is a power signal

Solved Problem 1.22 Determine the energy and power of the following signals

(i) $x(t) = tu(t)$ (ii) $x(n) = 2e^{j3\pi n}$

Solution:

(i) Given $x(t) = tu(t)$

$$E = \lim_{T\to\infty} \int_{-T}^{T} \left|x(t)\right|^2 dt$$

$$= \lim_{T\to\infty} \int_{0}^{T} t^2 dt = \infty$$

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} \left|x(t)\right|^2 dt$$

$$= \lim_{T\to\infty} \frac{1}{2T} \int_{0}^{T} t^2 dt = \infty$$

The energy and power of the signal are infinity. Therefore the signal is neither energy nor power signal.

(ii) $x(n) = 2e^{j3\pi n}$

$$E = \lim_{N\to\infty} \sum_{n=-N}^{N} \left|2e^{j3\pi n}\right|^2$$

$$= 4 \lim_{N\to\infty} \sum_{n=-N}^{N} \left|e^{j3\pi n}\right|^2$$

$$= 4 \lim_{N\to\infty} (2N+1) = \infty$$

The energy is infinity

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} \left|2e^{j2\pi n}\right|^2$$

$$= 4 \lim_{N\to\infty} \frac{1}{2N+1}(2N+1) = 4$$

The power is finite.

The signal is power signal.

Practice Problem 1.11

Find which of the following are energy signals? Which are power signals? Which are neither energy nor power signals?

(i) $e^{-5t}u(t)$ (ii) $\Pi\left(\frac{t+1}{2}\right)$ (iii) $r(t)-2r(t-1)+r(t-2)$

(iv) $u(t)u(2-t)$ Ans: (i) Energy signal (ii) Energy signal

1.8.6 Causal and non-causal signals

A continuous-time signal $x(t)$ is said to be causal if $x(t)=0$ for $t<0$, otherwise the signal is non-causal. For an anticausal signal $x(t)=0$ for $t>0$.

Similarly a discrete-time signal $x(n)$ is said to be causal if $x(n)=0$ for $n<0$; otherwise the signal is non-causal. For an anticausal discrete-time signal $x(n)=0$ for $n>0$.

Solved Problem 1.23 Find which of the following signals are causal or non-causal?

(i) $x_1(t)=e^{at}u(t)$;

(ii) $x_2(t)=e^{-2t}u(-t)$

(iii) $x_3(t)=\text{sinc}\,t$

(iv) $x_1(n)=u(n+3)-u(n-2)$

(v) $x_2(n)=\left(\frac{1}{2}\right)^n u(n+2)$

Solution:

(i) Given $x_1(t)=e^{at}u(t)$

The signal $x_1(t)$ is causal, because $x_1(t)=0$ for $t<0$.

(ii) $x_2(t)=e^{-2t}u(-t)$

The signal $x_2(t)$ is a non-causal signal because $x_2(t)\neq 0$ for $t<0$.

(iii) $x_3(t)=\text{sinc}\,t$

The sinc t signal is shown in Fig. 1.93. It has values for $t<0$. Therefore, it is a non-causal signal.

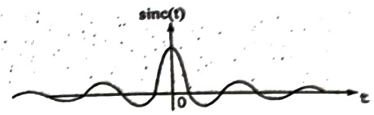


Fig. 1.93

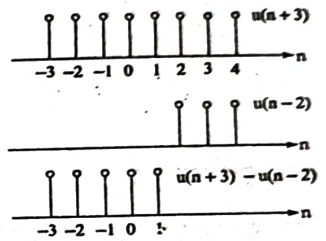


Fig. 1.94

(iv) The signal $x_1(n)=u(n+3)-u(n-2)$ is shown in Fig. 1.94.

The signal $x_1(n)$ is defined for values $n<0$. Therefore $x_1(n)$ is a non-causal signal.

(v) $x_2(n)=\left(\frac{1}{2}\right)^n u(n+2)$

This signal is non-causal as it is defined for values $n<0$.

Practice Problem 1.12

Find which of the following signals are causal

(i) $\left(\frac{1}{2}\right)^n u(n+3)$ (ii) $e^{-2t}u(t-2)$ (iii) $\sin tu(t)$

(iv) $u(n+2)-u(n-2)$

Ans: (i) Non-causal (ii) Causal (iii) Causal (iv) Non-Causal

Solved Problem 1.24 Consider the signal $x_1(t)$ shown in Fig.1.95. Plot $x_1(t-1)+x_1(-t+2)$. AU Nov 2008

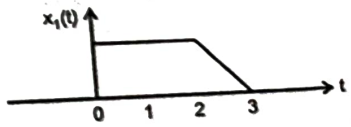


Fig.1.95

Solution:

The signal $x_1(t-1)$ can be obtained by shifting $x_1(t)$ right by one unit as shown in Fig.1.96.

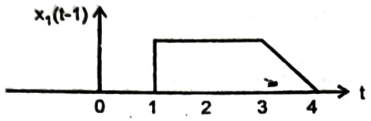


Fig.1.96

The signal $x_1(-t+2)$ can be obtained by folding the signal $x_1(t)$ and then shifting folded signal to right by 2 units.

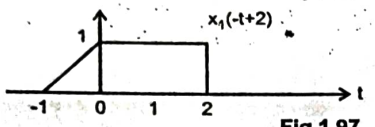


Fig.1.97

The addition of signals $x_1(t-1)$ and $x_1(-t+2)$ is shown in Fig.1.98.

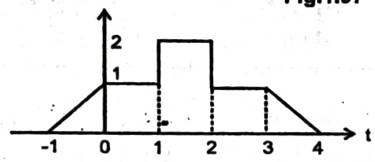


Fig.1.98

Solved Problem 1.25 Express the signal in Fig.1.99 in terms of step and ramp functions

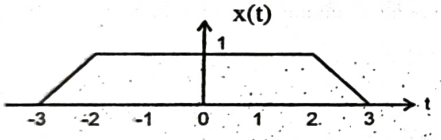


Fig. 1.99

Solution:

From Fig. 1.100 we can find that the signal $x(t)$ can be expressed as

$$x(t) = r(t+3) - r(t+2) - r(t-2) + r(t-3)$$

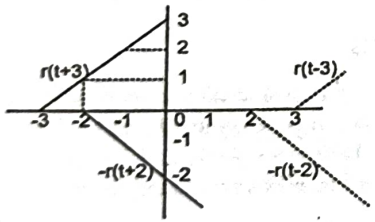


Fig. 1.100

Solved Problem 1.26 A continuous-time signal $x(t)$ is shown in Fig. 1.101 Sketch and label carefully each of the following signals.

(i) $x(t-1)$ (ii) $x(2-t)$ (iii) $x(t)[\delta(t+\frac{3}{2}) - \delta(t-\frac{3}{2})]$ (iv) $x(2t-1)$

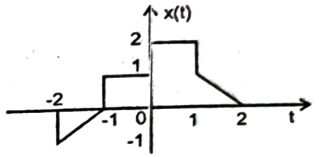


Fig. 1.101

Solution:

(i) The signal $x(t-1)$ can be obtained by shifting $x(t)$ by 1 unit.

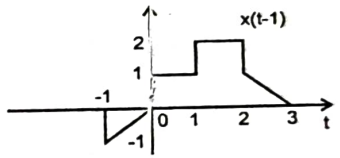


Fig. 1.102

(ii) The signal $x(2-t)$ can be obtained by folding $x(t)$ and then shifting right by two units.

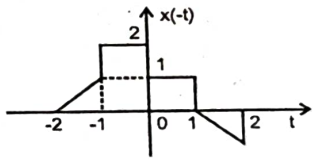


Fig.1.103

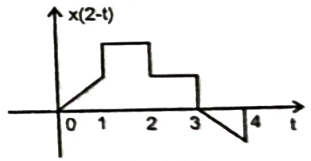


Fig.1.104

(iii) The signal $\delta(t+\frac{3}{2}) - \delta(t-\frac{3}{2})$ is shown below

$$x(t)\left[\delta\left(t+\frac{3}{2}\right) - \delta\left(t-\frac{3}{2}\right)\right]$$

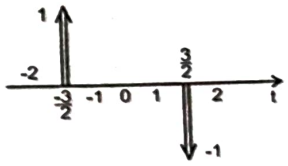


Fig.1.105

is equal to

$$x\left(-\frac{3}{2}\right)\delta\left(t+\frac{3}{2}\right)-x\left(\frac{3}{2}\right)\delta\left(t-\frac{3}{2}\right)$$

$$=-0.5\,\delta\left(t+\frac{3}{2}\right)-0.5\,\delta\left(t-\frac{3}{2}\right)$$

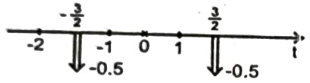


Fig. 1.106

(iv) The signal $x(2t-1)$ can be obtained by shifting $x(t)$ by one unit and then applying time scaling by factor 2

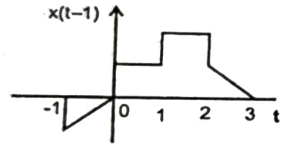


Fig.1.107

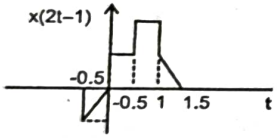


Fig.1.108

Solved Problem 1.27 Determine and sketch the even and odd part of the signals.

AU Nov 2006

(i)

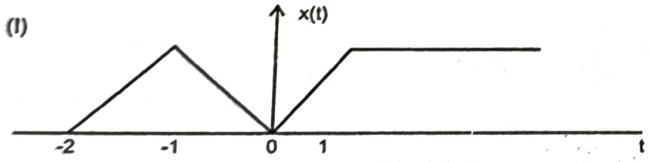


Fig.1.109

(ii)

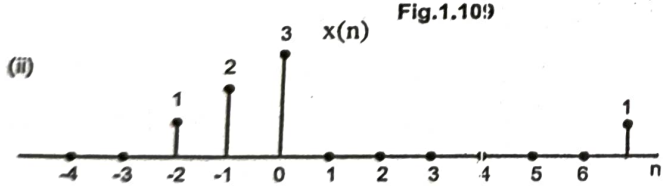


Fig. 1.110

Solution:

(i) The signal $x(-t)$ is shown in Fig.1.111

The signal $x(t)+x(-t)$ is shown in Fig.1.112

The even part of the signal is $x_e(t)=\frac{1}{2}[x(t)-x(-t)]$ shown in Fig.1.113

The signal $x(t)-x(-t)$ is shown in Fig.1.114

The signal $x_0(t)=\frac{1}{2}[x(t)-x(-t)]$ is shown in Fig.1.115

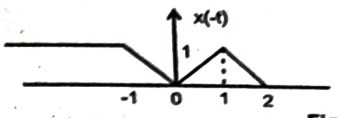


Fig.1.111

Fig.1.112

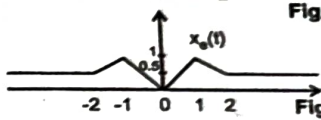


Fig.1.113

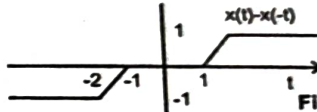


Fig.1.114

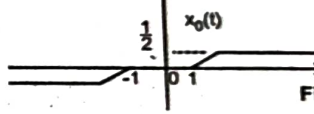


Fig.1.115

(ii) The sequence

$x(n)=\{1\,2\,3,0,0,0,0,0,0,1\}$

↑

The time reversal of $x(n)$ is

$x(-n)=\{1,0,0,0,0,0,0,3,2,1\}$

↑

$$x_e(n)=\frac{1}{2}[x(n)+x(-n)]$$

$$=\{0.5,0,0,0,0,0.5,1,3,1,0.5,0,0,0,0,0.5\}$$

↑

$$x_0(n)=0.5[x(n)-x(-n)]$$

$$=\{-0.5,0,0,0,0,0.5,1,0,-1,-0.5,0,0,0,0,0.5\}$$

↑

1.9 Signals and vectors

In physics we studied that a vector **u** is a quantity with magnitude and direction. It is represented by means of an arrow or directed line segment as shown in Fig.1.116. There is perfect analogy between signals and vectors. All signals are basically vectors. A vector shown in Fig.1.116 can be represented in terms of its co-ordinates (1,0) and (0,1). That is

$$(2,2)=2(1,0)+2(0,1)$$

The co-ordinates (1,0) and (0,1) are known as basis vectors. The same vector can be represented with some other co-ordinate system. That is a vector can be represented as a sum of its components in a variety of ways. Similarly a signal can be represented as a sum of its components in a variety of ways. For example any periodic signal (ex; square wave) can be represented as a linear combination of basis functions like sinusoids or exponentials. In this section we will study basic vector concepts and then apply these concepts to signals.

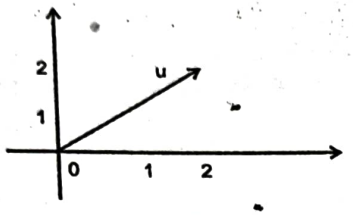


Fig. 1.116

1.9.1 Vector

A vector $\mathbf{u}$ in vector space $\mathbb{R}^n$ is an ordered set of n real numbers;
$\mathbf{u} = (u_1, u_2, u_3, \ldots, u_n)$. The real number u_k is called the components of $\mathbf{u}$.

1.9.2 Vector addition

Let $\mathbf{u} = (u_1, u_2 \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ be any vectors in $\mathbb{R}^n$.

Then the sum $\mathbf{u}+\mathbf{v}$, is a vector

$$\mathbf{u}+\mathbf{v} = (u_1+v_1, u_2+v_2, \ldots, u_n+v_n)$$

1.9.3 Scalar multiplication

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ be any vector in $\mathbb{R}^n$, and let k be any scalar in $\mathbb{R}^n$. Then the product vector $k\mathbf{u} = (ku_1, ku_2, \ldots, ku_n)$

1.9.4 Dot product (inner product)

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are vectors in $\mathbb{R}^n$. Then inner product (or dot product) of $\mathbf{u}$ and $\mathbf{v}$ denoted by $\mathbf{u}.\mathbf{v}$ is

$$\mathbf{u}.\mathbf{v} = \sum_{k=1}^{n} u_k v_k$$

The inner product is a scalar obtained by multiplying corresponding components.

1.9.5 Norm (length) in $\mathbb{R}^n$

Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ is a vector in $\mathbb{R}^n$, the norm or length of $\mathbf{u}$, denoted by $|\mathbf{u}|$ is given by

$$|\mathbf{u}| = \sqrt{\mathbf{u}.\mathbf{u}} = \sqrt{u_1^2+u_2^2+\ldots.+u_n^2}$$

The norm of a vector is equal of to length of a vector.

1.9.6 Distance

Let $\mathbf{u}$ and $\mathbf{v}$ be any vectors in $\mathbb{R}^n$. The distance between $\mathbf{u}$ and $\mathbf{v}$, is denoted as

$$d(\mathbf{u},\mathbf{v}) = |\mathbf{u}-\mathbf{v}|$$

If $\mathbf{u} = (a,b)$ and $\mathbf{v} = (c,d)$, then $d(\mathbf{u},\mathbf{v}) =$

$$|a-c, b-d| = \sqrt{(a-c)^2+(b-d)^2}$$

1.9.7 Angle

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$. The angle θ between $\mathbf{u}$ and $\mathbf{v}$ is defined as

$$\cos\theta = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}$$

1.9.8 Projection

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$ and $\mathbf{u}, \mathbf{v} \neq 0$. The vector projection of $\mathbf{u}$ on to $\mathbf{v}$ is the vector

$$\text{Proj}\,(\mathbf{u},\mathbf{v}) = \frac{\mathbf{u}.\mathbf{v}}{||\mathbf{v}||^2}\,\mathbf{v}$$

Fig.1.117 shows projection of $\mathbf{u}$ on to $\mathbf{v}$. The projection is perpendicular and let this vector be $\mathbf{x}$. From Fig.1.117

$$|\mathbf{x}| = |\mathbf{u}|\cos\theta$$

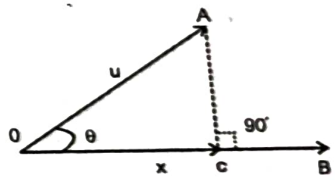


Fig.1.117

If θ is the angle between two vectors, then

$$\cos\theta = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|}$$

$$\text{Therefore } |\mathbf{x}| = |\mathbf{u}|\frac{\mathbf{u}.\mathbf{v}}{|\mathbf{u}|\,|\mathbf{v}|} = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{v}|}$$

To obtain $\mathbf{x}$, multiply its magnitude by the unit vector in the direction of $\mathbf{v}$

$$\mathbf{x} = |\mathbf{x}|\frac{\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{v}|^2}\,\mathbf{v}$$

1.10 Vector space

A vector space **V** is a set of vectors (**u**,**v**,**w**) with rules of addition and scalar multiplication. The addition and multiplication must produce vectors with in the space, and they must satisfy the following axioms.

1. For any vectors $\mathbf{u},\mathbf{v} \in \mathbf{V}, \mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$
2. For any vectors $\mathbf{u},\mathbf{v},\mathbf{w} \in \mathbf{V}, \mathbf{u}+(\mathbf{v}+\mathbf{w}) = (\mathbf{u}+\mathbf{v})+\mathbf{w}$.
3. There is a unique "zero vector" such that $\mathbf{u}+\mathbf{0} = \mathbf{u}$ for all **u**.
4. For each vector $\mathbf{u} \in \mathbf{V}$, there is a vector in **V**, denoted by $-\mathbf{u}$, for which $\mathbf{u}+(-\mathbf{u}) = \mathbf{0}$.
5. For the unit scalar '**1**', $1\mathbf{u} = \mathbf{u}$ for any vector $\mathbf{u} \in \mathbf{V}$.
6. For any scalar k and vector $\mathbf{u},\mathbf{v} \in \mathbf{V}$, $k(\mathbf{u}+\mathbf{v}) = k\mathbf{u}+k\mathbf{v}$.
7. For any scalars c_1 and c_2, and any vectors $\mathbf{u},\mathbf{v} \in \mathbf{V}$.
$$(c_1+c_2)\mathbf{u} = c_1\mathbf{u}+c_2\mathbf{u}.$$
8. For any scalars c_1 and c_2, and any vectors $\mathbf{u},\mathbf{v} \in \mathbf{V}$.
$$(c_1\ c_2)\mathbf{u} = c_1(c_2\mathbf{u}).$$

1.10.1 Subspace

A subspace of a vector space is a nonempty subset that satisfies the following requirements. Let **W** is a subspace of **V**. Then **W** has to satisfy the following requirements.

(i) **W** is a nonempty subset.
(ii) **W** is closed under vector addition. If we add any two vectors **u** and **v** in the subspace, then their sum $\mathbf{u}+\mathbf{v}$ is in the subspace.
(iii) If we multiply any vector **v** in the subspace with any scalar c, then the multiple $c\mathbf{v}$ is in the subspace.

1.10.2 Linear dependence and independence

Let **V** be a vector space. The set of vectors $\mathbf{v}_1,\mathbf{v}_2\ldots\mathbf{v}_n \in \mathbf{V}$ are said to be linearly dependent, if their exists scalars $c_1,c_2\ldots,c_n$, not all of them zero, such that

$$c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_n\mathbf{v}_n = 0$$

If the linear combination produces a zero vector, only when $c_1 = c_2, = \ldots = c_n = 0$ then the vectors $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n$ are linearly independent

In a set of vectors $\mathbf{v}_1,\mathbf{v}_2\ldots,\mathbf{v}_n$, if any two vectors are equal, then the vectors are linearly dependent. Similarly in a set of vectors $\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_n$ if one of them is a multiple of the other, then the vectors are linearly dependent.

1.10.3 Spanning a subspace

In a Vector space **V**, if every vector **v** can be expressed as some combination of **w**s, then these vectors span the space. That is every vector **v** can be expressed as

$$\mathbf{v} = c_1\mathbf{w}_1+c_2\mathbf{w}_2+\ldots+c_i\mathbf{w}_i$$

for some coefficients c_i.

For example the vectors $\mathbf{w}_1 = (1,0,0), \mathbf{w}_2 = (0,1,0), \mathbf{w}_3 = (-3,0,0)$ space a plane ($x-y$ plane) with in three dimensional space, where as the vectors $\mathbf{w}_1$ and $\mathbf{w}_3$ span only a line.

1.10.4 Basis

A sequence of vectors $(\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_n)$ is a basis of **V** if

(i) $\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_n$ are linearly independent and
(ii) $\mathbf{u}_1,\mathbf{u}_2,\ldots,\mathbf{u}_n$ span **V**.

Consider the following three vectors in $\mathbb{R}^3$.

$$\mathbf{x}_1 = (1,0,0)\ ;\ \mathbf{x}_2 = (0,1,0)\ ;\ \mathbf{x}_3 = (0,0,1)$$

These vectors are linearly independent and span $\mathbb{R}^3$. Thus the vectors form a basis of $\mathbb{R}^3$ called the usual basis (standard basis) of $\mathbb{R}^3$. Any vector in a three dimensional space can be expressed as a linear combination of basis vectors x_1,x_2 and x_3

Example : (3,2,1)= 3(1,0,0)+ 2(0,1,0)+ 1(0,0,1)

1.10.5 Orthogonality in vectors

Let **u** and **v** be vectors in $\mathbb{R}^n$. The **u** is said to be orthogonal (perpendicular) to **v** if **u**.**v** = 0. Consider projection of **u** on **v** as shown in Fig.1.118. Let **x** is the projection of **u** along **v**. Let $\mathbf{x} = c\mathbf{v}$. That is $c\mathbf{v}$ is the projection of **u** on **v**. Using vector addition we can write $\mathbf{u} = c\mathbf{v}+\mathbf{e}$, where **e** is an error vector. of **v**. Two other possibilities are shown in Fig.1.118 from which we can write $\mathbf{u} = c_1\mathbf{v}+\mathbf{e}_1 = c_2\mathbf{v}+\mathbf{e}_2$.

It can be easily observed from figures that the values $\mathbf{e}_1$ and $\mathbf{e}_2$ are greater than e. That is the error vector in minimum when it is perpendicular to $c\mathbf{v}$. We can write

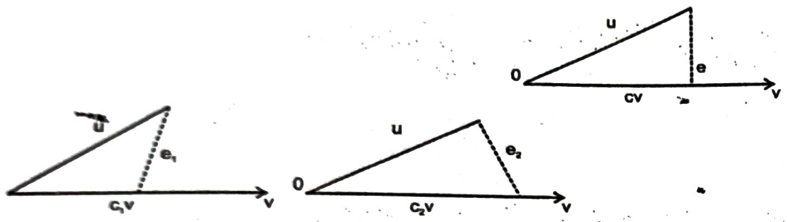


Fig. 1.118

$$\mathbf{x} = c\mathbf{v}$$

We know

$$\mathbf{x} = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}$$

$$c\mathbf{v} = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}$$

$$c = \frac{\mathbf{u}.\mathbf{v}}{|\mathbf{v}|^2}$$

From the above equation we find that the product $c\mathbf{v}$ is zero when the dot product between vector $\mathbf{u}$ and $\mathbf{v}$ is zero. Also we find that the dot product is zero when the angle $\theta = 90°$. That is when the angle between vectors $\mathbf{u}$ and $\mathbf{v}$ is 90°, $\mathbf{u}$ will not have component along $\mathbf{V}$.

Thus vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be orthogonal vectors if their dot product is zero.

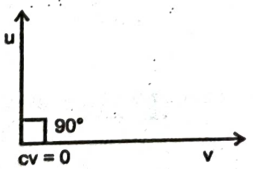


Fig.1.119

1.10.6 Orthonormal vectors

The vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ are orthonormal if

$$\mathbf{u}_i.\mathbf{v}_i = 0 \text{ when } i \neq j$$
$$= 1 \text{ when } i = j$$

That is for orthonormal vectors the length is equal to 1.

1.10.7 Orthogonal subspace

Two vectors $\mathbf{u}$ and $\mathbf{v}$ are said to be orthogonal if their dot product is zero. For example

$$\mathbf{u} = (2,2,-1) \text{ is orthogonal to } \mathbf{v} = (-1,2,2).$$

If every vector in one subspace is orthogonal to every vector in the other subspace then the two subspaces are orthogonal.

1.10.8 Orthogonal basis

In an orthogonal basis, every vector is perpendicular to every other vector. The co-ordinate axes are mutually orthogonal.

1.11 Orthogonal vectors space

In a three dimensional cartesian vectors space the orthogonal vectors $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$ represent a complete set of orthogonal vectors, where $\mathbf{x}_1 = (1,0,0)$; $\mathbf{x}_2 = (0,1,0)$and $\mathbf{x}_3 = (0,0,1)$. Any vector in this vector space can be represented using these three vectors. That is vectors $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$ are basis vectors for this space. Consider a vector $\mathbf{v}$ in this space. It can be expressed as a linear combination of $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_3$ as

$$\mathbf{v} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3$$

The coefficients c_i are calculated by projecting the vectors $\mathbf{v}$ on the basis vectors. The projection of $\mathbf{v}$ on $\mathbf{x}_1$ is given by $\frac{\mathbf{v}.\mathbf{x}_1}{\mathbf{x}_1.\mathbf{x}_1}$ which is equal to c_1. That is

$$c_1 = \frac{\mathbf{v}.\mathbf{x}_1}{\mathbf{x}_1.\mathbf{x}_1}$$

Similarly

$$c_2 = \frac{\mathbf{v}.\mathbf{x}_2}{\mathbf{x}_2.\mathbf{x}_2}$$

and

$$c_3 = \frac{\mathbf{v}.\mathbf{x}_3}{\mathbf{x}_3.\mathbf{x}_3}$$

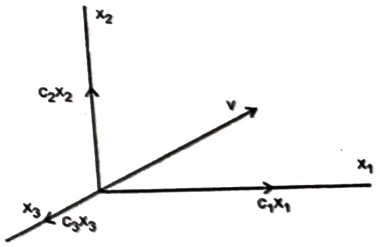


Fig.1.120

1.12 Orthogonality in real signals

Now let us extend the orthogonality concept to signals. Consider a real signal $y(t)$ to be approximated in terms of another real signal $x(t)$ over an interval t_1 and t_2.

Let the error in this approximation be $e(t)$. Then we can write

$$y(t) = cx(t) + e(t)$$
$$e(t) = y(t) - cx(t) \text{ for } t_1 \le t \le t_2$$
$$= 0 \text{ otherwise}$$

The minimum value of $e(t)$ will give best approximation. This value can be obtained by minimizing the mean square value E_r, which is the energy of $e(t)$ over the interval $[t_1, t_2]$. That is

$$E_r = \int_{t_1}^{t_2} e^2(t)dt$$
$$= \int_{t_1}^{t_2} [y(t) - cx(t)]^2 dt$$

The value of E_r depends on how we choose c. We have to select the value of c such that $cx(t)$ gives best approximation of $y(t)$. Therefore we differentiate E_r with respect to c and equate to zero. For minimum E_r, $\frac{dE_r}{dc} = 0$

$$\frac{d}{dc}\left[\int_{t_1}^{t_2} [y(t) - cx(t)]^2 dt\right] = 0$$

$$\frac{d}{dc}\left[\int_{t_1}^{t_2} y^2(t)dt - 2c\int_{t_1}^{t_2} y(t)x(t)dt + c^2\int_{t_1}^{t_2} x^2(t)dt\right] = 0$$

The first integral is independent of c. Therefore it is equal to zero. Differentiating remaining two integrals we get

$$-2\int_{t_1}^{t_2} y(t)x(t)dt + 2c\int_{t_1}^{t_2} x^2(t)dt = 0$$

$$\Rightarrow c = \frac{\int_{t_1}^{t_2} x(t)y(t)dt}{\int_{t_1}^{t_2} x^2(t)dt}$$

The denominator of above equation is the energy of the signal over the interval (t_1, t_2). Now we can obtain

$$y(t) = \frac{\int_{t_1}^{t_2} x(t)y(t)dt}{\int_{t_1}^{t_2} x^2(t)dt} \cdot x(t)$$

When c is zero there will be no component of $y(t)$ along $x(t)$, then $y(t)$ and $x(t)$ are orthogonal. The value 'c' is zero when

$$\int_{t_1}^{t_2} x(t)y(t)\,dt = 0$$

Therefore $x(t)$ and $y(t)$ are orthogonal when $\int_{t_1}^{t_2} x(t)\,y(t)\,dt = 0$

1.13 Orthogonality in complex signal

Let $x(t)$ and $y(t)$ be two complex signals. The approximation of $y(t)$ in $x(t)$ is

$$y(t) = cx(t) + e(t)$$

Since $x(t)$ and $y(t)$ are complex signals the error signal $e(t)$ also complex signal.

$$e(t) = y(t) - cx(t)$$

The mean square error E_r is given by

$$E_r = \int_{t_1}^{t_2} |e(t)|^2 dt = \int_{t_1}^{t_2} |y(t) - cx(t)|^2 dt$$
$$= \int_{t_1}^{t_2} [y(t) - cx(t)][y(t) - cx(t)]^* dt$$
$$= \int_{t_1}^{t_2} y(t)y^*(t)dt - c\int_{t_1}^{t_2} x(t)y^*(t)dt - c\int_{t_1}^{t_2} y(t)x^*(t)dt + c^2\int_{t_1}^{t_2} x(t)x^*(t)dt$$

Differentiating the above equation with respect to c and equating to zero we get

$$c = \frac{\int_{t_1}^{t_2} y(t)x^*(t)dt}{\int_{t_1}^{t_2} x(t)x^*(t)dt}$$

$$= \frac{\int_{t_1}^{t_2} y(t)x^*(t)dt}{\int_{t_1}^{t_2} |x(t)|^2 dt}$$

Another expression to find the value of c is

$$c = \frac{\int_{t_1}^{t_2} x(t)y^*(t)dt}{\int_{t_1}^{t_2} |x(t)|^2 dt}$$

For orthogonality the value of c is equal to zero, which implies that

$$\int_{t_1}^{t_2} y(t)x^*(t)dt = 0 \text{ (or) } \int_{t_1}^{t_2} y^*(t)x(t)dt = 0$$

The above equation is the condition for orthogonality of complex signals.

1.14 Orthogonal signal space

We studied that any vector **v** in vector space **V** can be represented as a linear combination of orthogonal basis vectors. Similarly let us define a orthogonal signal set $x_1(t), x_2(t) \ldots, x_n(t)$ over the interval $[t_1, t_2]$. If the orthogonal signal set is complete, then any signal $f(t)$ can be represented as linear combination of orthogonal signals given by

$$f(t) = c_1x_1(t) + c_2x_2(t) + \ldots + c_nx_n(t)$$

The coefficients c_i can be calculated by projecting the signal $f(t)$ on to the orthogonal signal $x_i(t)$. That is

$$c = \frac{\int_{t_1}^{t_2} f(t)x_i^*(t)}{\int_{t_1}^{t_2} x_i^2(t)dt}$$

The orthogonal signal set $[x_i(t)]$ is also known as basis function or basis signals.

Solved Problem 1.28 If $x(t)$ and $y(t)$ are orthogonal, then show that $E_{x+y} = E_x + E_y$, where E_{x+y} is energy of the signal $x(t) + y(t)$, E_x is the energy of the signal $x(t)$ and E_y is the energy of the signal $y(t)$.

The energy of the signals $x(t)$ and $y(t)$ are given by

$$E_x \int_{-\infty}^{\infty} x^2(t)dt \quad \text{and} \quad E_y = \int_{-\infty}^{\infty} y^2(t)dt$$

The energy of the signal $x(t) + y(t)$ is

$$E_{x+y} = \int_{-\infty}^{\infty} (x(t) + y(t))^2 dt$$

$$= \int_{-\infty}^{\infty} x^2(t)dt + \int_{-\infty}^{\infty} y^2(t)dt + \int_{-\infty}^{\infty} 2x(t)y(t)dt$$

$$= \int_{-\infty}^{\infty} x^2(t)dt + \int_{-\infty}^{\infty} y^2(t)dt + 2\int_{-\infty}^{\infty} x(t)y(t)dt$$

Since $x(t)$ and $y(t)$ are orthogonal

$$\int_{-\infty}^{\infty} x(t)y(t)dt = 0$$

Therefore

$$E_{x+y} = \int_{-\infty}^{\infty} x^2(t)dt + \int_{-\infty}^{\infty} y^2(t)dt$$

$$= E_x + E_y$$

Solved Problem 1.29 A rectangular function is defined as

$$f(t) = \begin{cases} A \text{ for } 0 \leq t \leq \frac{\pi}{2} \\ -A \text{ for } \frac{\pi}{2} \leq t \leq \frac{3\pi}{2} \\ A \text{ for } \frac{3\pi}{2} \leq t \leq 2\pi \end{cases}$$

Approximate the above function by $A \cos t$ between the intervals $(0, 2\pi)$ such that mean square error is minimum. JNTU, May 2005, Nov 2005

Solution:

Let us assume, the approximation $x(t) = A \cos t$.

The function $f(t)$ can be written as $f(t) = C\,x(t)$ where

$$C = \frac{\int_0^{2\pi} f(t)x(t)\,dt}{\int_0^{2\pi} x^2(t)\,dt}$$

$$= \frac{\int_0^{\pi/2} A^2\cos t\,dt + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (-A^2)\cos t\,dt + \int_{\frac{3\pi}{2}}^{2\pi} A^2\cos t\,dt}{\int_0^{2\pi} A^2\cos^2 t\,dt}$$

$$\int_0^{\pi/2} A^2\cos t\,dt = A^2 \sin t\,|_0^{\pi/2} = A^2$$

$$\int_{\pi/2}^{3\pi/2} A^2\cos t\,dt = A^2 \sin t\,|_{\pi/2}^{3\pi/2} = -2A^2$$

$$\int_{3\pi/2}^{2\pi} A^2\cos t\,dt = A^2 \sin t\,|_{3\pi/2}^{\pi/2} = A^2$$

$$\int_0^{2\pi} A^2\cos^2 t\,dt = \frac{A^2}{2}\int_0^{2\pi}(1+\cos 2t)dt = \pi A^2$$

$$\Rightarrow \qquad C = \frac{A^2-(-2A^2)+A^2}{\pi A^2} = \frac{4}{\pi}$$

Therefore $f(t) = \frac{4}{\pi} A\cos t$ is the required approximation.

Solved Problem 1.30 Show that the signal set

$$\{1, \cos\Omega_0 t,\ \cos 2\Omega_0 t \cdots, \cos n\Omega_0 t, \cdots \sin\Omega_0 t,\ \sin 2\Omega_0 t, \cdots \sin n\Omega_0 t \ldots\}$$

are orthogonal over an interval $T_0 = \frac{2\pi}{\Omega_0}$. JNTU, May 2004

Solution:

First we check orthogonality of cosine waves. Let the cosine waves are $\cos n\Omega_0 t$ and $\cos m\Omega_0 t$. The orthogonality of $\cos n\Omega_0 t$ and $\cos m\Omega_0 t$ over the interval $T_0 = \frac{2\pi}{\Omega_0}$ is

$$\int_t^{t+T_0} \cos(n\Omega_0 t)\cos(m\Omega_0 t)\,dt$$

$$= \frac{1}{2}\left[\int_t^{t+T_0} \cos(n-m)\Omega_0 t\,dt + \int_t^{t+T_0} \cos(n+m)\Omega_0 t\,dt\right]$$

For $n \neq m$, the terms $(n-m)$ and $(m+n)$ are integers. Therefore the first and second term integrates for full cycles of cosine wave over one period. Hence its value is zero.

That is

$$\int_t^{t+T_0} \cos(n-m)\Omega_0 t\,dt = \int_t^{t+T_0} \cos(n+m)\Omega_0 t\,dt = 0 \text{ for } n \neq m$$

For $n = m$

$$\int_t^{t+T_0} \cos^2 n\Omega_0 t\,dt = \frac{1}{2}\left[\int_t^{t+T_0} dt + \int_t^{t+T_0} \cos 2n\Omega_0 t\,dt\right]$$

$$= \frac{1}{2} t\,\Big|_t^{t+T_0} = \frac{T_0}{2}$$

From above result we find that $\cos n\Omega_0 t$ and $\cos m\Omega_0 t$ are orthogonal over one period.

Next we check orthogonality of sine waves. That is

$$\int_t^{t+T_0} \sin(n\Omega_0 t)\sin(m\Omega_0 t)dt = \frac{1}{2}\left[\int_t^{t+T_0} \cos(n-m)\Omega_0 t\,dt - \int_t^{t+T_0} \cos(n+m)\Omega_0 t\,dt\right]$$

The above integral is zero for $n \neq m$, since both integrations are of full cycles over one period.

If $n = m$, then

$$\int_t^{t+T_0} \sin^2 n\Omega_0 t\,dt = \frac{T_0}{2}$$

Therefore sine waves are orthogonal over one period. The orthogonality of $\sin n\Omega_0 t$ and $\cos m\Omega_0 t$ is

$$\int_t^{t+T_0} \sin(n\Omega_0 t)\cos(m\Omega_0 t)dt = \frac{1}{2}\left[\int_t^{t+T_0} \sin(n+m)\Omega_0 t\, dt + \int_t^{t+T_0} \sin(n-m)\Omega_0 t\, dt\right]$$

If $n \neq m$ the first and second integrations are zero.

If $n = m$

$$\int_t^{t+T_0} \sin(n\Omega_0 t)\cos(n\Omega_0 t)dt = \frac{1}{2}\int_t^{t+T_0} \sin(2n\Omega_0 t)\, dt = 0$$

So the integration

$$\int_t^{t+T_0} \sin(n\Omega_0 t)\cos(m\Omega_0 t)dt = 0 \text{ for all } n \text{ and } m$$

Therefore sine and cosine waves are orthogonal for all n and m.

Solved Problem 1.31 Prove that the set of exponentials $\{1, e^{\pm j\Omega_0 t}, e^{\pm j2\Omega_0 t}, \ldots\}$ is orthogonal over any interval T_0. JNTU May 2004

Solution:

The given set of exponentials are complex signals. For complex signals the orthogonality is evaluated using the expression

$$\int_{t_1}^{t_2} x_m(t)x_n^*(t)dt$$

Let $x_m(t) = e^{jm\Omega_0 t}$ and $x_n(t) = e^{jn\Omega_0 t}$

For any interval T_0

$$\int_t^{t+T_0} e^{jm\Omega_0 t}\left(e^{jn\Omega_0 t}\right)^* dt = \int_t^{t+T_0} e^{j(m-n)\Omega_0 t}dt$$

$$= \frac{1}{j(m-n)\Omega_0} e^{j(m-n)\Omega_0 t}\Big|_t^{t+T_0}$$

$$= \frac{1}{j(m-n)\Omega_0}\left[e^{j(m-n)\Omega_0(t+T_0)} - e^{j(m-n)\Omega_0 t}\right]$$

$$= \frac{1}{j(m-n)\Omega_0} e^{j(m-n)\Omega_0 t}\left[e^{j(m-n)\Omega_0 T_0} - 1\right]$$

Since $\Omega_0 T_0 = 2\pi$, we have $e^{j(m-n)\Omega_0 T_0} = e^{j(m-n)2\pi} = 1$ for any m and n

$$\text{Hence } \int_t^{t+T_0} e^{jm\Omega_0 t}\left(e^{jn\Omega_0 t}\right)^* dt = 0$$

If $m = n$, then

$$\int_t^{t+T_0} e^{jm\Omega_0 t}\left(e^{jn\Omega_0 t}\right)^* dt = \int_t^{t+T_0} dt = T_0$$

$$\text{Hence } \int_t^{t+T_0} e^{jm\Omega_0 t}\left(e^{jn\Omega_0 t}\right)^* dt = 0 \text{ for } m \neq n$$

$$= T_0 \text{ for } m = n$$

Therefore the set of exponentials is orthogonal.

Solved Problem 1.32 Show that the sequence $e^{j2\pi kn/N}$ is periodic with period N. Also prove that the sequence is orthogonal over a period N.

Solution:

$$\text{Let } x(n) = e^{j2\pi kn/N}$$

A sequence is periodic if it satisfies the relation

$$x(n+N) = x(n)$$

$$x(n+N) = e^{j2\pi k(n+N)/N} = e^{j2\pi k}e^{j2\pi kn/N}$$

$$x(n+N) = e^{j2\pi kn/N}$$

$$= x(n)$$

Thus $x(n)$ is periodic with period N.

Two discrete-time sequence $x_m(n)$ and $x_p(n)$ are orthogonal

$$\text{if } \sum_{n=0}^{N-1} x_m(n)x_p^*(n) = 0 \text{ for } m \neq p$$

$$= \text{for } m = p$$

$$\text{Let } x_m(n) = e^{j2\pi mn/N} \text{ and } x_p(n) = e^{j2\pi pn/N}$$

$$\sum_{n=0}^{N-1} x_m(n)x_p^*(n) = \sum_{n=0}^{N-1} e^{j2\pi mn/N} e^{-j2\pi pn/N}$$

$$= \sum_{n=0}^{N-1} e^{j2\pi(m-p)n/N}$$

$$= \frac{1-e^{j2\pi(m-p)}}{1-e^{j2\pi(m-p)/N}} \qquad \boxed{\sum_{n=0}^{n-1} a^n = \frac{1-a^N}{1-a}}$$

Since m and p are integers $e^{j2\pi(m-p)} = 1$.

Hence $\sum_{n=0}^{N-1} x_m(n)x_p^*(n) = 0$ for $m \neq p$

When $m = p$

$$\sum_{n=0}^{N-1} x_m(n)x_m^*(n) = \sum_{n=0}^{N-1} 1 = N$$

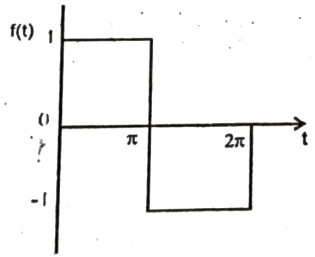
Thus $x_m(n)$ and $x_p(n)$ are orthogonal. Hence the sequence $e^{j2\pi kn/N}$ is orthogonal.

Solved Problem 1.33 The function $f(t)$ is defined as

$$f(t) = 1 \text{ for } 0 \leq t \leq \pi$$
$$= 1 \text{ for } \pi \leq t \leq 2\pi$$

Show that the above function is orthogonal to signals $\cos t, \cos 2t \ldots \cos nt$ for all non-zero integer values of n over the interval 0 to 2π.

Solution:



The signal $f(t)$ is shown in figure.
The set of cosine signals is given by
$x_n(t) = \cos nt$
The orthogonality of $f(t)$ and $\cos nt$ in the interval 0 to 2π is expressed as

$$\int_0^{2\pi} f(t)\cos nt\ dt = \int_0^{\pi} \cos nt\ dt + \int_{\pi}^{2\pi} (-1)\cos nt\ dt$$

$$= \frac{\sin nt}{n}\Big|_0^{\pi} - \frac{\sin nt}{n}\Big|_{\pi}^{2\pi}$$

$$= 0 \text{ for any } n \neq 0$$

Hence the set of functions $\cos nt$ for all non-zero integer value of n, will be orthogonal to $f(t)$.

Practice Problem 1.13

Show that the set of functions
$\frac{1}{\sqrt{T}}, \sqrt{\frac{2}{T}}\cos\omega_0 t, \sqrt{\frac{2}{T}}\cos 2\omega_0 t, \ldots \sqrt{\frac{2}{T}}\sin\omega_0 t, \sqrt{\frac{2}{T}}\sin 2\omega_0 t \ldots$ forms an orthogonal set of functions over the interval $-\frac{T}{2}$ to $\frac{T}{2}$.

Questions and Answers

1. **Define a signal.**

   A signal is defined as any physical quantity that varies with time, space or any other independent variable.

2. **Define deterministic and random signals.**

   A deterministic signal is a signal exhibiting no uncertainity of its value at all instants. Its instantaneous value can be accurately predicted by specifying a mathematical relation.

   A random signal is a signal characterized by uncertainity before its actual occurance. (or) A random signal can not be predicted before its actual occurance.

3. **Define step function and Delta function.** AU Nov. 2002

   A step function is defined as

$$x(t) = A \quad \text{for} \quad t \geq 0$$
$$= 0 \quad \text{for} \quad t < 0$$

A delta function is defined as

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

and

$$\delta(t) = 0 \quad \text{for} \quad t \neq 0$$

4. Define periodic and aperiodic signals.

A signal $x(t)$ is periodic with period T if and only if

$$x(t+T) = x(t) \quad \text{for all} \quad t$$

If there is no value of T that satisfies the above equation the signal is called aperiodic.

5. Define symmetric and antisymmetric signals.

A real valued signal $x(t)$ is called symmetric if

$$x(-t) = x(t)$$

on the otherhand, a signal $x(t)$ is called antisymmetric if

$$x(-t) = -x(t)$$

6. Define energy and power signals.

The energy of a signal $x(t)$ is defined as

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

A signal $x(t)$ is called an energy signals if the energy is finite and power is zero.

The average power of a signal $x(t)$ is defined as

$$P = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

A signal $x(t)$ is called a power signal if the average power P is finite and Energy is infinite.

7. What is the period T of the signal $x(t) = 2\cos\left(\frac{t}{4}\right)$? AU April 2003

$$\Omega_0 = \frac{1}{4}$$

$$T = \frac{2\pi}{\Omega_0} = \frac{2\pi}{1/4} = 8\pi \text{ sec}$$

8. Define continuous-time, discrete-time and digital signals.

Continuous-time signals: The signals that are defined at all instants of time are known as continuous-time signals.

Discrete-time signals: The signals that are defined at discrete instants of time are known as discrete-time signals.

Digital signals: The signals that are discrete in time and quantized in amplitude are digital signals.

9. What is the value of the following integral?

$$\int_{-\infty}^{\infty} x(t)\delta(at-b)dt$$

$$\text{Ans}: \frac{1}{|a|} x\left(\frac{b}{a}\right)$$

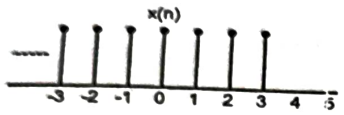
10. What are the different types of representation of discrete-time signals?

1. Graphical representation
2. Functional representation
3. Tabular representation
4. Sequence representation

11. Consider the discrete-time signal $x(n) = 1 - \sum_{k=3}^{\infty} \delta(n-1-k)$. Determine the value of the signals M and k so that $x(n)$ may be expressed as

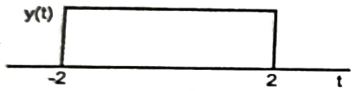
$x(n) = u(Mn-k)$ AU Nov 2006

The signal $x(n)$ is sketched in figure.



The signal $x(n)$ is equivalent to folding $x(n)$ and then shifting the folded signal by 3 to the right. Therefore $x(n) = u(-n+3)$. Comparing $u(-n-3)$ with $u(Mn - n_k)$ we get $M = -1$ and $k = -3$

12. Consider a continuous-time signal $x(t) = \delta(t-2) - \delta(t-2)$. Calculate the value E_∞ for the signal $y(t) = \int_{-\infty}^{t} x(t)dt$



$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau = \int_{-\infty}^{t} [\delta(\tau+2) - \delta(\tau-2)]dt$$

$$y(t) = u(t+2) - u(t-2)$$

$$E_\infty = \int_{-2}^{2} |y(t)|^2 dt = t\Big|_{-2}^{2}$$

$$= 4$$

13. Find the even and odd components of the signal $x(t) = e^{jt}$ AU Nov 2008

$$x_e(t) = \frac{x(t)+x(-t)}{2} = \frac{e^{jt}+e^{-jt}}{2} = \cos t$$

$$x_0(t) = \frac{x(t)+x(-t)}{2} = \frac{e^{jt}-e^{-jt}}{2} = j\sin t$$

Multiple Choice Questions

1. Which of the following signals is/are periodic?

(a) $s(t) = \cos 2t + \cos 3t + \cos 5t$ (b) $s(t) = \exp(j8\pi t)$

(c) $s(t) = e^{-7t}\sin 10\pi t$ (d) $s(t) = \cos 2t \cos t\, u(t)$

2. Let $\delta(t)$ denote the delta function. The value of the integral $\int_{-\infty}^{\infty} \delta(t)\cos\left(\frac{3t}{2}\right)dt$ is

(a) 1 (b) -1 (c) 0 (d) $\frac{\pi}{2}$

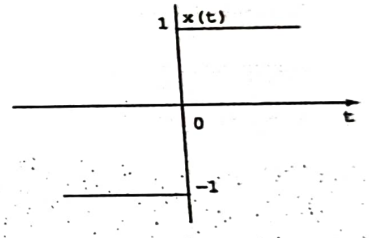
3. If a signal $f(t)$ has energy E, the energy of the signal $f(2t)$ is equal to

(a) E (b) $\frac{E}{2}$ (c) $2E$ (d) $4E$

4. The function $x(t)$ is shown in figure. Even and odd parts of a unit-step function $u(t)$ are respectively

(a) $\frac{1}{2}, \frac{1}{2}x(t)$ (b) $-\frac{1}{2}, \frac{1}{2}x(t)$

(c) $\frac{1}{2}, -\frac{1}{2}x(t)$ (d) $-\frac{1}{2}, \frac{1}{2}x(t)$



5. The period of the function $\cos\frac{\pi}{4}(t-1)$ is

(a) $\frac{1}{8}$ sec (b) 8 sec (c) 4 sec (d) $\frac{1}{4}$ sec

6. A signal $x(t) = A\cos(\Omega_0 t + \phi)$ is

(a) An energy signal

(b) A power signal

(c) An energy as well as a power signal

(d) Neither an energy nor a power signal

7. $x_1(t) = 2\sin\pi t + \cos 4\pi t$ and $x_2(t) = \sin 5\pi t + 3\sin 3\pi t$ then

(a) x_1 and x_2 both are periodic.

(b) x_1 and x_2 both are not periodic.

(c) x_1 is periodic, but x_2 is not periodic.
(d) x_1 is not periodic, but x_2 is periodic.

8. The signal $x(t) = 2(\cos \pi t) + 3\sin(2t)$
(a) is periodic (b) is not periodic (c) may or may not be periodic

9. The odd and even components of signal $u(t)$ are
(a) $\cos t, \sin t$ (b) $\sin t, -\cos t$
(c) $\cos t, j\sin t$ (d) $\cos t, -j\sin t$

10. The odd and even components of sequence $x(n) = \{1, \underset{\uparrow}{1}, 0.5\}$
(a) $\{0.25, \underset{\uparrow}{0}, -0.25\}, \{0.75, \underset{\uparrow}{1}, 0.75\}$
(b) $\{1.5, \underset{\uparrow}{0}, -1.5\}, \{1.5, \underset{\uparrow}{1}, 1.5\}$
(c) $\{0.25, \underset{\uparrow}{0}, -0.25\}, \{0.5, \underset{\uparrow}{1}, 0.5\}$
(d) $\{0.25, \underset{\uparrow}{0}, -0.25\}\ \{1.5, \underset{\uparrow}{1}, 1.5\}$

11. The signal $x(t) = e^{-2t}u(t)$ is
(a) power signal (b) energy signal
(c) neither power nor energy signal (d) either power or energy signal

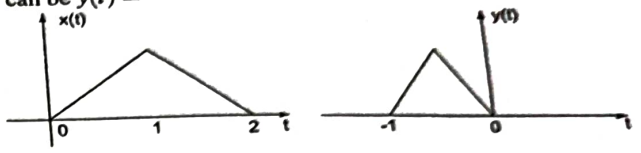
12. The signal $x(n) = \text{ccs}\frac{2\pi}{3}nu(n)$ is
(a) causal, aperiodic (b) causal, periodic
(c) non causal, periodic (d) non-causal, aperiodic

13. The value of the integral $\int_{-\infty}^{\infty} \delta(at - b) =$
(a) $\frac{1}{a^2}$ (b) $\frac{1}{|a|}$ (c) $\frac{1}{a}$ (d) $\frac{-1}{a}$

14. The time period of the signal $x(t) = \cos 2\pi t + \sin 5\pi t$ is
(a) 2 sec (b) 5 sec (c) 10 sec (d) 2.5 sec

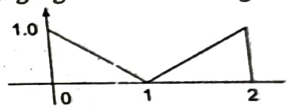
15. The signal $x(n) = \cos 2n$ is
(a) periodic with period π (b) periodic with period 2
(c) periodic with period 4π (d) aperiodic

16. Consider the following signals $x(t)$ and $y(t)$. The relation between $x(t)$ and $y(t)$ can be $y(t) =$



(a) $x(t+1)$ (b) $x(2t+2)$ (c) $x(t-1)$ (d) $x(2t-1)$

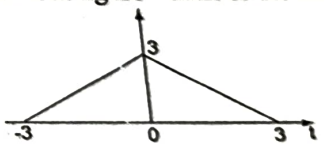
17. Represent the following signal as sum of singularity functions



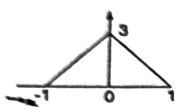
(a) $-r(t) + 2r(t) - u(t-1)$
(b) $r(-t+1) + r(t-1) + 2u(t-2) - r(t-2)$
(c) $r(-t+1)u(t) + 2r(t-1) - r(t-2) - u(t-2)$
(d) $r(-t+1)u(t) + r(t-1) - r(t-2) - u(t-2)$

18. The rms value of the signal $x(t) = 10\cos\left(100t + \frac{\pi}{3}\right) + 16\sin\left(150t + \frac{\pi}{3}\right)$ is
(a) $\sqrt{160}$ (b) $\sqrt{356}$ (c) $\sqrt{89}$ (d) $\sqrt{178}$

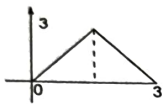
19. For the signal $x(t)$ shown in figure which of the following is $x(3t)$?



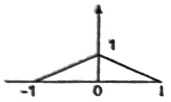
(a)



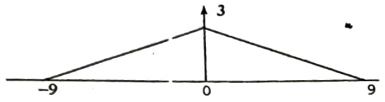
(b)



(c)



(d)



Multiple Choice Answers

1. (a) and (b)

   Explanation

   (a) $s(t) = \cos 2t + \cos 3t + cos5t$
   $T_1 = \pi$; $T_2 = \frac{2\pi}{3}$; $T_3 = \frac{2\pi}{5}$
   $T = 2\pi$, periodic

   (b) $s(t) = exp(j8\pi t)$
   $T = \frac{2\pi}{8\pi} = \frac{1}{4}$ periodic

   (c) $s(t) = e^{-7t}\sin 10\pi t$, Nonperiodic.

   (d) $s(T) = \cos 2t \cos u(t)$, nonperiodic, since $u(t) = 0$ for $t < 0$.

2. (a) $\int_{-\infty}^{\infty} \delta(t)\cos\frac{3t}{2}dt = \cos\left(\frac{3t}{2}\right)\Big|_{t=0}^{\cdot} = 1$

3. (a)

4. (a)

$$u_e(t) = \frac{u(t)+u(-t)}{2} = \frac{1}{2}$$

$$u_0(t) = \frac{u(t)-u(-t)}{2}$$

$$= 0.5 \text{ for } t \geq 0$$

$$= -0.5 \text{ for } t < 0$$

$$\text{hence } u_0(t) = \frac{1}{2}x(t)$$

5. (b) $T = \frac{2\pi}{\pi/4} = 8sec$

6. (b) Power signal

7. (a) Period of $x_1(t)$ is 4
   period of $x_2(t)$ is 2

8. (b)

$$x(t) = 2\cos\pi t + 3\sin 2t$$

$$T_1 = \frac{2\pi}{\pi} = 2$$

$$T_2 = \frac{2\pi}{2} = \pi$$

$$\frac{T_1}{T_2} = \frac{2}{\pi} \text{ not a rational number}$$

9. (c) $\cos t, j\sin t$

10. (a)

11. (b) $E = \lim_{T\to\infty}\int_{-T}^{T}|x(t)|^2 dt = \lim_{T\to\infty}\int_{0}^{T}e^{-2t}dt = \frac{1}{2}$

12. (a) Causal Because $x(n) = 0$ for $n < 0$. Aperiodic because $x(n) = 0$ for $n < 0$.

13. (b) $\frac{1}{|a|}$

14. (a)

$$T_1 = \frac{2\pi}{2\pi} = 1; T_2 = \frac{2\pi}{5\pi} = \frac{2}{5}$$

$$\frac{T_1}{T_2} = \frac{1}{2/5} = \frac{5}{2} \Rightarrow T = 2T_1 = 5T_2 = 2\text{sec}$$

15. (d)Aperiodic since ω is not multiple of π

16. (b) $x(2t+2)$

17. (c) $r(-t+1)u(t) + 2r(t-1) - r(t-2) - u(t-2)$

18. (d) rms value $\sqrt{\frac{10^2}{2} + \frac{16^2}{2}} = \sqrt{50+128} = \sqrt{178}$

19. (a)

2

Introduction to Systems

2.1 A System

A system is defined as a physical device that performs an operation on an input signal and produces another signal as output. The concept of system is applicable to the field of electrical, electronics, communication, chemical, mechanical, economics etc. Physical devices such as motor, amplifier, filter, boiler and turbine are examples of systems. All the above systems accepts a signal as input and produces another signal as output. For example, an amplifier increases the strength of the signal present at the input and a filter rejects unwanted frequencies that are present in the input signal. Similarly for a car, travelling along a highway, the pressure on the accelator pedal is input signal and the speed of the car is output signal.

Definition: A system is defined as a physical device that generates a response or output signal, for a given input signal.

Systems can be represented by block diagrams. The box shown in Fig. 2.1 represents a system. An arrow entering the box and arrow leaving the box are representing input signal and output signal respectively. An input is denoted by $x(t)$ and the corresponding output is denoted by $y(t)$.

The relation between the input $x(t)$ and corresponding output $y(t)$ of a system has the form

$$y(t) = \text{operation on } x(t).$$

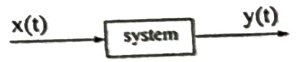


Fig. 2.1 A System

Mathematically

$$y(t) = T[x(t)] \tag{2.1}$$

which represents that the input $x(t)$ is transformed to $y(t)$. In otherwords $y(t)$ is the transformed form of $x(t)$.

2.2 Classification of Systems

The systems may be classified as follows.

1. Continuous-time and discrete-time systems
2. Lumped-parameter and distributed-parameter systems
3. Static and dynamic systems
4. Causal and Non-causal systems
5. Linear and Non-linear systems
6. Time-invariant and time-variant systems
7. Stable and unstable systems

2.2.1 Continuous-time and discrete-time systems

Continuous-time system

Definition: A continuous-time system is one which operates on a continuous-time input signal and produces a continuous-time output signal.

If the input and output of continuous-time system are $x(t)$ and $y(t)$, then we can say that $x(t)$ is transformed to $y(t)$. That is

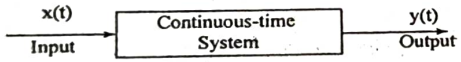


Fig. 2.2 A Continuous-time system

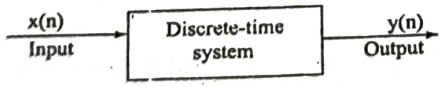
$$y(t) = T[x(t)] \tag{2.2a}$$

Examples for continuous-time system are amplifiers, filters, motors etc.

Discrete-time system

Definition: A discrete-time system is one which operates on a discrete-time input signal and produces a discrete-time output signal.

If the input and output of discrete-time system are $x(n)$ and $y(n)$ respectively, then we can write



$$y(n) = T[x(n)] \tag{2.2b}$$

Fig. 2.3 A Discrete-time system

2.2.2 Lumped-parameter and distributed-parameter systems

In lumped-parameter systems, each and every component is lumped at one-point in space. These systems are described by ordinary differential equations. In distributed-parameter systems, the signals are functions of space as well as time. These systems are described by partial differential equations.

2.2.3 Static and dynamic systems

A system is called static or memoryless if its output at any instant depends on the input at that instant but not on past or future values of input. Otherwise, the system is said to be dynamic or with memory.

Examples	$y(t) = x^2(t)$	in continuous-time	Static systems (or) memory less system
	$y(n) = nx(n)$	in discrete-time	
	$y(t) = \frac{dx(t)}{dt}$	in continuous-time	Dynamic systems
	$y(n) = x(n-1)$	in discrete-time	

Solved Problem 2.1 Find whether the following systems are dynamic or not.

(i) $y(t) = x(t-2)$ (ii) $y(n) = x(n+2)$ (iii) $y(t) = x^2(t)$

Solution:

(i) Given $y(t) = x(t-2)$.

The output depends on past value of input. Therefore the system is dynamic.

(ii) Given $y(n) = x(n+2)$.

The output depends on future value of input. Therefore, the system is dynamic.

(iii) Given $y(t) = x^2(t)$.

The output at any time depends on the input at that time. Therefore the system is static.

2.2.4 Causal and non-causal systems

A causal system is one for which the output at any time t depends on the present and past inputs but not future inputs. These systems are also known as non-anticipative systems.

A non-causal system is one whose output depends on future values.

Examples for causal systems are

$$y(t) = x(t) + x(t-1)$$
$$y(n) = nx(n) + x(n-3) \qquad (2.3)$$

Example for non-causal system are

$$y(t) = x(t+3) + x^2(t)$$
$$y(n) = x(2n) \qquad (2.4)$$

Solved Problem 2.2 Check whether the following systems are causal or not.

(i) $y(n) = x(n) + \dfrac{1}{x(n-1)}$
(ii) $y(t) = x^2(t) + x(t-2)$
(iii) $y(t) = x(t-2) + x(2-t)$
(iv) $y(t) = \int_{-\infty}^{2t} x(\tau)d\tau$
(v) $y(n) = x(-n)$
(vi) $y(n) = x(n^2)$

Solution:

(i) $y(n) = x(n) + \dfrac{1}{x(n-1)}$

For $n = 0$, $\quad y(0) = x(0) + \dfrac{1}{x(-1)}$

For $n = 1$ $\quad y(1) = x(1) + \dfrac{1}{x(0)}$

For $n = -1$ $\quad y(-1) = x(-1) + \dfrac{1}{x(-2)}$

For all values on n the output depends on present and past values of input. Therefore the system is causal.

(ii) $y(t) = x^2(t) + x(t-2)$

For $t = 0$ $\quad y(0) = x^2(0) + x(-2)$

For $t = 1$ $\quad y(1) = x^2(1) + x(-1)$

For $t = -1$ $\quad y(-1) = x^2(-1) + x(-3)$

For all values of t, the output depends an present and past values of input. Therefore the system is causal.

(iii) $y(t) = x(t-2) + x(2-t)$

Let $t = 0$, then $\quad y(0) = x(-2) + x(2)$

Here the output $y(0)$ depends on the future input $x(2)$. Therefore the system is non-causal.

(iv) $y(t) = \int_{-\infty}^{2t} x(\tau)d\tau$; Let $\int x(\tau)d\tau = z(t)$

Let $t = 0$, then $\quad y(0) = \int_{-\infty}^{0} x(\tau)d\tau = z(0) - z(-\infty).$

Let $t = 1$, then $\quad y(1) = \int_{-\infty}^{2} x(\tau)d\tau = z(2) - z(-\infty)$

The output $y(1)$ depends on the future value $z(2)$. Therefore the system is non-causal.

(v) $y(n) = x(-n)$

Let $n = 0$, then, $y(0) = x(0)$; Let $n = 1$, then $y(1) = x(-1)$; Let $n = -1$ then, $y(-1) = x(1)$.

Let $n = -2$, then $\quad y(-2) = x(2)$

For negative value of n the output depends on the future input. Therefore the system is non-causal.

(vi) $y(n) = x(n^2)$

For $n = 0$ and $n = 1$, the output does not require future values of input.

For $n = -1; y(-1) = x(1)$. The output depends on future value of input.

For $n = 2$, then $y(2) = x(4)$.

The output depends on the future values of input. Therefore the system is non-causal.

Review Questions

1. Define a system. How the systems are classified?
2. When a system is said to be static system?
3. What are causal and non-causal systems?
4. How you can find whether the system is causal or not?

Practice Problem 2.1

Check whether the following systems are causal or not

(i) $y(t) = x(t)x(t-1)$ **Ans:** Causal.

(ii) $y(t) = x(t/2)$ **Ans:** Non-causal. Hint: take negative values of t

(iii) $y(t) = x(t^{1/2})$; 't' takes only positive values **Ans:** Non-causal. Hint: take values of $t < 1$ Ex : $y(0.36) = x(0.6)$

(iv) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$ **Ans:** Non-causal.

(v) $y(n) = \cos[x(n)]$ **Ans:** Causal.

2.2.5 Linear and non-linear systems

Superposition principle states that the response to a weighted sum of input signals be equal to the weighted sum of the outputs corresponding to each of the individual input signals. A system that satisfies the superposition principle is said to be a linear system.

If an arbitrary input $x_1(t)$ produces output $y_1(t)$ and an arbitrary input $x_2(t)$ produces output $y_2(t)$ then the system is linear if the input $ax_1(t) + bx_2(t)$ produces an output $ay_1(t) + by_2(t)$, where a and b are constants.

That is,

$$T[ax_1(t) + bx_2(t)] = aT[x_1(t)] + bT[x_2(t)]$$

Similarly for a discrete-time linear system

$$T[ax_1(n) + bx_2(n)] = aT[x_1(n)] + bT[x_2(n)]$$

From the superposition property we can find that a zero input results a zero output.

Solved Problem 2.3 Check whether the following system are linear or not.

(i) $\frac{dy}{dt} + 3t\, y(t) = t^2 x(t)$ **AU Nov 2009**

(ii) $\frac{dy(t)}{dt} + 2y(t) = x^2(t)$

(iii) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

(iv) $\frac{dy}{dt} + 2y(t) = x(t)\frac{dx(t)}{dt}$

(v) $y(n) = Ax(n) + B$

(vi) $y(n) = 2x(n) + \frac{1}{x(n-1)}$ **AU Nov 2009**

(vii) $y(n) = nx(n)$

Solution:

(i) Given $\frac{dy(t)}{dt} + 3ty(t) = t^2 x(t)$

For an input $x_1(t)$ and for the corresponding output $y_1(t)$ the differential equation can be written as

$$\frac{dy_1(t)}{dt} + 3ty_1(t) = t^2 x_1(t) \tag{2.5}$$

Similarly for an input $x_2(t)$ and for the corresponding output $y_2(t)$ the differential equation can be written as

$$\frac{dy_2(t)}{dt} + 3ty_2(t) = t^2 x_2(t) \tag{2.6}$$

Multiply Eq. (2.5) with 'a' and Eq. (2.6) with 'b' and add to obtain

$$a\frac{dy_1(t)}{dt} + 3aty_1(t) + b\frac{dy_2(t)}{dt} + 3bty_2(t) = at^2 x_1(t) + bt^2 x_2(t) \tag{2.7}$$

$$\Rightarrow \frac{d}{dt}\underbrace{[ay_1(t) + by_2(t)]}_{\substack{\text{weighted sum}\\ \text{of outputs}}} + 3t\underbrace{[ay_1(t) + by_2(t)]}_{\substack{\text{weighted sum}\\ \text{of outputs}}} = t^2\underbrace{[ax_1(t) + bx_2(t)]}_{\substack{\text{weighted sum}\\ \text{of inputs}}} \tag{2.8}$$

From Eq. (2.8) we can note that the weighted sum inputs to the system produces an output which is also equal to weighted sum of outputs corresponding to each of the individual inputs.

Therefore, the system is linear.

Note

If RHS is function of weighted sum of inputs and LHS is function of weighted sum of outputs, then the system is linear.

(ii) Given $\frac{dy(t)}{dt} + 2y(t) = x^2(t)$

For an input $x_1(t)$, the output is $y_1(t)$. Then

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1^2(t) \tag{2.9}$$

Similarly, for an input $x_2(t)$, the output is $y_2(t)$. Then

$$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2^2(t) \tag{2.10}$$

Multiply Eq. (2.9) with a and Eq. (2.10) with b and add. We obtain

$$a\frac{dy_1(t)}{dt} + 2ay_1(t) + b\frac{dy_2(t)}{dt} + 2by_2(t) = ax_1^2(t) + bx_2^2(t)$$

$$\Rightarrow \underbrace{\frac{d}{dt}(ay_1(t) + by_2(t))}_{\text{weighted sum of outputs}} + 2\underbrace{\{ay_1(t) + by_2(t)\}}_{\text{weighted sum of outputs}} = \underbrace{ax_1^2(t) + bx_2^2(t)}_{\text{not a linear function of weighted sum of inputs}} \tag{2.11}$$

Superposition principle is not satisfied, therefore the system is nonlinear.

(iii) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$

For an input $x_1(t)$ and output $y_1(t)$

$$y_1(t) = \int_{-\infty}^{t} x_1(\tau)d\tau \tag{2.12}$$

Similarly

$$y_2(t) = \int_{-\infty}^{t} x_2(\tau)d\tau \tag{2.13}$$

Linear combination of the above equations equal to

$$\underbrace{ay_1(t) + by_2(t)}_{\substack{\text{Linear} \\ \text{combination of} \\ \text{outputs}}} = a\int_{-\infty}^{t} x_1(\tau)d\tau + b\int_{-\infty}^{t} x_2(\tau)d\tau$$

$$= \int_{-\infty}^{t} \underbrace{(ax_1(\tau) + bx_2(\tau))}_{\substack{\text{Linear} \\ \text{combination of} \\ \text{inputs}}} d\tau \tag{2.14}$$

From Eq. (2.14) we can find that for input $ax_1(t) + bx_2(t)$ the output is $ay_1(t) + by_2(t)$. Therefore, the system is linear.

(iv) $\frac{dy(t)}{dt} + 2y(t) = x(t)\frac{dx(t)}{dt}$

For an input $x_1(t)$ and the corresponding output $y_1(t)$, the differential equation can be written as

$$\frac{dy_1(t)}{dt} + 2y_1(t) = x_1(t)\frac{dx_1(t)}{dt} \tag{2.15}$$

Similarly for an input $x_2(t)$ and the corresponding output $y_2(t)$

$$\frac{dy_2(t)}{dt} + 2y_2(t) = x_2(t)\frac{dx_2(t)}{dt} \tag{2.16}$$

The linear combination of the above equations can be written as

$$a\frac{dy_1(t)}{dt} + 2ay_1(t) + b\frac{dy_2(t)}{dt} + 2by_2(t) = ax_1(t)\frac{dx_1(t)}{dt} + bx_2(t)\frac{dx_2(t)}{dt}$$

$$\Rightarrow \frac{d}{dt}[ay_1(t) + by_2(t)] + 2[ay_1(t) + by_2(t)] = ax_1(t)\frac{dx_1(t)}{dt} + bx_2(t)\frac{dx_2(t)}{dt} \tag{2.17}$$

From Eq. (2.17) we find that RHS is not a function of weighted sum of inputs. Superposition principle is not satisfied. Therefore the system is nonlinear.

(v) $y(n) = Ax(n) + B$

For an input $x_1(n)$ the corresponding output $y_1(n)$ is

$$y_1(n) = T[x_1(n)] = Ax_1(n) + B \tag{2.18}$$

For an input $x_2(n)$ the corresponding output $y_2(n)$ is

$$y_2(n) = T[x_2(n)] = Ax_2(n) + B \tag{2.19}$$

The weighted sum of outputs is

$$ay_1(n) + by_2(n) = a[Ax_1(n) + B] + b[Ax_2(n) + B] \tag{2.20}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = A[ax_1(n) + bx_2(n)] + B \tag{2.21}$$

$$y_3(n) \neq ay_1(n) + by_2(n)$$

Therefore, the system is non-linear.

(vi) $y(n) = 2x(n) + \dfrac{1}{x(n-1)}$

Given

$$y(n) = T[x(n)] = 2x(n) + \frac{1}{x(n-1)} \tag{2.22}$$

For an input $x_1(n)$

$$y_1(n) = T[x_1(n)] = 2x_1(n) + \frac{1}{x_1(n-1)} \tag{2.23}$$

Similarly for an input $x_2(n)$

$$y_2(n) = T[x_2(n)] = 2x_2(n) + \frac{1}{x_2(n-1)} \tag{2.24}$$

Weighted sum of outputs is given by

$$ay_1(n) + by_2(n) = 2ax_1(n) + \frac{a}{x_1(n-1)} + 2bx_2(n) + \frac{b}{x_2(n-1)} \tag{2.25}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)]$$

$$= 2[ax_1(n) + bx_2(n)] + \frac{1}{ax_1(n-1) + bx_2(n-1)}$$

$$y_3(n) \neq ay_1(n) + by_2(n) \tag{2.26}$$

Therefore, the system is non-linear.

(vii) Given $y(n) = nx(n)$

$$y(n) = T[x(n)] = nx(n) \tag{2.27}$$

For an input $x_1(n)$, the output $y_1(n)$ is

$$y_1(n) = T[x_1(n)] = nx_1(n) \tag{2.28}$$

For an input $x_2(n)$, the output $y_2(n)$ is

$$y_2(n) = T[x_2(n)] = nx_2(n) \tag{2.29}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = n[ax_1(n) + bx_2(n)] = anx_1(n) + bnx_2(n) \tag{2.30}$$

The weighted sum of outputs is given by

$$ay_1(n) + by_2(n) = anx_1(n) + bnx_2(n)$$

$$\Rightarrow y_3(n) = ay_1(n) + by_2(n) \tag{2.31}$$

Therefore, the system is linear.

Solved Problem 2.4 Check whether the following systems are linear or not

(i) $y(t) = e^{x(t)}$ (ii) $y(t) = x^2(t)$ (iii) $y(t) = t^2x(t)$

(i) Given $y(t) = e^{x(t)}$,

For inputs $x_1(t)$ and $x_2(t)$, the corresponding output are

$$y_1(t) = e^{x_1(t)} \tag{2.32}$$

$$y_2(t) = e^{x_2(t)} \tag{2.33}$$

The output due to weighted sum of inputs is

$$y_3(t) = T[ax_1(t) + bx_2(t)] = e^{[ax_1(t) + bx_2(t)]} = e^{ax_1(t)}e^{bx_2(t)} \tag{2.34}$$

The weighted sum of outputs is

$$ay_1(t) + by_2(t) = ae^{x_1(t)} + be^{x_2(t)}$$

$$\Rightarrow y_3(t) \neq ay_1(t) + by_2(t) \tag{2.35}$$

Therefore, the system is non-linear.

(ii) $y(t) = x^2(t)$

The responses due to inputs $x_1(t)$ and $x_2(t)$ are given by

$$y_1(t) = x_1^2(t) \tag{2.36}$$

$$y_2(t) = x_2^2(t) \tag{2.37}$$

The output due to weighted sum of inputs is

$$y_3(t) = T[ax_1(t) + bx_2(t)] = [ax_1(t) + bx_2(t)]^2 \tag{2.38}$$

The weighted sum of outputs is

$$ay_1(t) + by_2(t) = ax_1^2(t) + bx_2^2(t)$$

$$y_3(t) \neq ay_1(t) + by_2(t) \tag{2.39}$$

Therefore, the system is non-linear.

(iii) $y(t) = t^2x(t)$

The output due to inputs $x_1(t)$ and $x_2(t)$ is given by

$$y_1(t) = t^2x_1(t) \tag{2.40}$$

$$y_2(t) = t^2x_2(t) \tag{2.41}$$

The output due to weighted sum of inputs is

$$y_3(t) = T[ax_1(t) + bx_2(t)] = t^2[ax_1(t) + bx_2(t)] \qquad (2.42)$$

The weighted sum of outputs is

$$ay_1(t) + by_2(t) = at^2x_1(t) + bt^2x_2(t)$$

$$y_3(t) = ay_1(t) + by_2(t) \qquad (2.43)$$

Therefore, the system is linear.

Practice Problem 2.2

Check whether the following systems are linear or not.

(i) $\frac{dy}{dt} + t^2y(t) = 2x(t)$ (ii) $y(n) = x(n)\cos \omega n$

Ans: (i) linear (ii) linear

2.2.6 Time-invariant and time-variant systems

A system is said to be time-invariant if its input-output characteristics do not change with time. Suppose that we applied a signal $x(t)$ to a system and obtained an output $y(t)$ as shown in Fig. 2.4. If we delay the input by T seconds, then for a time-invariant system the output also will be delayed by T seconds.

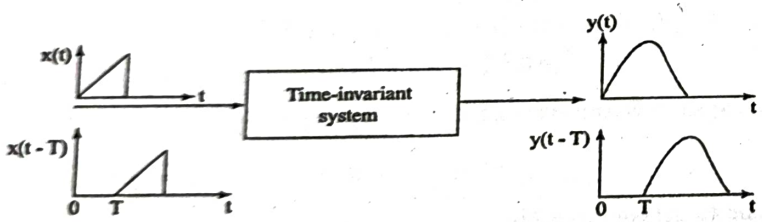


Fig. 2.4 A Time-invariant system.

The output due to the delayed input $x(t-T)$ for a time invariant system is given by

$$y(t-T) = T[x(t-T)]$$

If the output due to input $x(t-T)$, is not equal to $y(t-T)$ then the system is time-variant.

For a Linear Time-invariant system (LTI) the coefficiencts of differential equation describing the system are constants. If the coefficients are function of time then the system is a linear time-variant system.

Example 1:

$$2\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 5y(t) = 5x(t)$$

all coefficients are constants

Since the coefficients are constants, the given system is an LTI system.

Example 2:

$$\frac{d^2y(t)}{dt^2} + 4t\frac{dy(t)}{dt} + 5y(t) = x(t)$$

coefficient is a function of time

It is a linear time varying system, since all coefficients are not constant.

To test the time-invariant property of a system apply a signal $x(t)$ and find $y(t)$. Now delay the input signal by T and find the output signal and denote it as

$$y(t,T) = T[x(t-T)]$$

Delay the output signal by T and denote it as $y(t-T)$. If $y(t,T) = y(t-T)$ for any value of T, then the system is time-invariant. On the other hand if the output

$$y(t,T) \neq y(t-T)$$

then the system is time-variant.

For discrete-time systems the time-invariance can be found by observing the coefficients of a difference equation. If the coefficients of difference equation are constants then the system is time-invariant. If the coefficients are function of time then the system is time-variant.

To test the time-invariant property of the given discrete-time system, first apply an arbitrary input sequence $x(n)$ and find $y(n)$. Delay the input sequence by k samples and find the output sequence, denote it as

$$y(n,k) = T[x(n-k)]$$

Delay the output sequence by k samples denote it as $y(n-k)$

If $y(n,k) = y(n-k)$

for all possible values of k, the system is time-invariant, otherwise the system is time- variant. Example for linear time-invariant system (LTI) and Linear time-variant systems are shown below.

Example 1:

$$y(n) - 4y(n-1) + 3y(n-2) = x(n) - A \text{ LTI system}$$

Example 2:

$$y(n) - 3ny(n-1) + n^2y(n-2) = x(n) + x(n-1) -$$

$$A \quad \text{Linear time-variant system}$$

Solved Problem 2.5 For each of the following systems, determine whether or not the system is time-invariant.

(i) $y(t) = tx(t)$ AU Nov 2009

(ii) $y(t) = x(t)x(t-1)$

(iii) $y(t) = x(t^2)$

(iv) $y(t) = x(-t)$

(v) $y(t) = e^{x(t)}$

(vi) $y(n) = x(2n)$

(vii) $y(n) = x(n) + nx(n-1)$

(viii) $y(n) = x^2(n-1)$

Solution:

(i) Given $y(t) = tx(t)$

$$y(t) = T[x(t)] = tx(t)$$

The output due to delayed input is

$$y(t,T) = T[x(t-T)] = tx(t-T)$$

If the output is delayed by T, we get

$$y(t-T) = (t-T)x(t-T)$$

$$y(t,T) \neq y(t-T)$$

Therefore, the system is time-variant.

(ii) $y(t) = x(t)x(t-1)$

$$y(t) = T[x(t)] = x(t)x(t-1)$$

If the input is delayed by T sec, the output

$$y(t,T) = T[x(t-T)] = x(t-T)x(t-T-1)$$

If the output is delayed by T sec, we get

$$y(t-T) = x(t-T)x(t-T-1)$$

We can observe that $y(t,T) = y(t-T)$

Therefore, the system is time-invariant.

(iii) $y(t) = x(t^2)$

$$y(t) = T[x(t)] = x(t^2)$$

The output due to delayed input is

$$y(t,T) = T[x(t-T)] = x(t^2-T)$$

The delayed output

$$y(t-T) = x[(t-T)^2]$$

$$y(t,T) \neq y(t-T)$$

Therefore, the system is time-variant.

(iv) $y(t) = x(-t)$

$$y(t) = T[x(t)] = x(-t)$$

The output due to delayed input is

$$y(t,T) = T[x(t-T)] = x(-t-T)$$

The delayed output is

$$y(t-T) = x[-(t-T)] = x(-t+T)$$

$$y(t,T) \neq y(t-T)$$

Therefore, the system is time-variant.

(v) $y(t) = e^{x(t)}$

$$y(t) = T[x(t)] = e^{x(t)}$$

The output due to delayed input is

$$y(t,T) = T[x(t-T)] = e^{x(t-T)}$$

The delayed output is

$$y(t-T) = e^{x(t-T)}$$

$$y(t,T) = y(t-T)$$

Therefore, the system is time-invariant.

(vi)

$$y(n) = x(2n)$$

$$y(n) = T[x(n)] = x(2n)$$

If the input is delayed by k units of time then the output is

$$y(n,k) = T[x(n-k)] = x(2n-k)$$

The output, delayed by k units of time is

$$y(n-k) = x[2(n-k)] = x[2n-2k]$$

$$y(n,k) \neq y(n-k)$$

Therefore, system is time-variant.

(vii)

$$y(n) = x(n) + nx(n-1)$$

$$y(n) = T[x(n)] = x(n) + nx(n-1)$$

The output due to delayed input is

$$y(n,k) = T[x(n-k)] = x(n-k) + nx(n-1-k)$$

The delayed output is

$$y(n-k) = x(n-k) + (n-k)x(n-1-k)$$

$$y(n,k) \neq y(n-k)$$

Therefore, the system is time-variant.

(viii) $y(n) = x^2(n-1)$

The output due to delayed input

$$y(n,k) = x^2(n-1-k)$$

The delayed output

$$y(n-k) = x^2(n-1-k)$$

$$y(n,k) = y(n-k)$$

Therefore, the system is time-invariant.

Practice Problem 2.3

Check whether the following systems are time-invariant.

(i) $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$ **Ans:** Time-invariant

(ii) $\frac{dy(t)}{dt} + 5ty(t) = x(t)$ **Ans:** Time-variant

(iii) $y(t) = x(t+10) + x^2(t)$ **Ans:** Time-invariant

(iv) $y(n) = x(-n)$ **Ans:** Time-variant

(v) $y(n) = x(n+1) + x(n) + x(n-1)$ **Ans:** Time-invariant

(vi) $y(n) = \sin[x(n)]$ **Ans:** Time-invariant

2.2.7 Stable and unstable systems

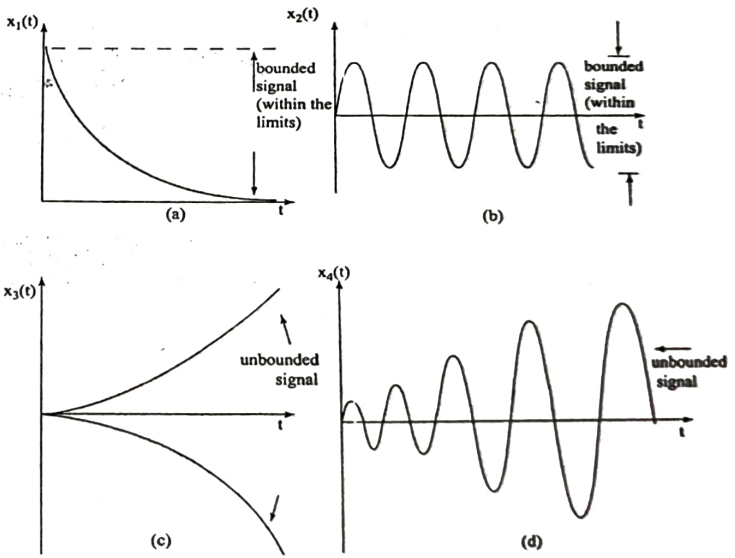


Fig. 2.5 Examples for bounded and unbounded signals.

An input signal $x(t)$ is said to be bounded if it satisfies the condition $|x(t)| \leq M_x < \infty$ for all t. Similarly the output signal is bounded if it satisfies the condition $|y(t)| \leq M_y < \infty$. The examples for bounded and unbounded signals are shown in Fig. 2.5.

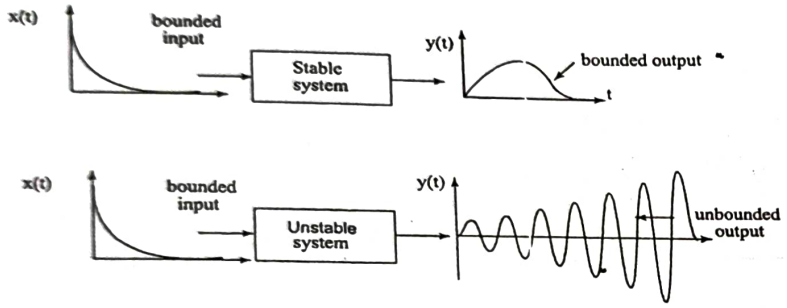


Fig. 2.6 Typical stable and unstable systems.

A system is said to be bounded input and bounded output stable if and only if every bounded input produces a bounded output. Fig.2.6 shows typical stable and unstable systems.

It is very important to know whether a system is stable or not. This can be found from the basic knowledge of impulse response of the system. Therefore, a detailed discussion on this topic will be covered in the forth coming chapters.

Solved Problem 2.6

Check whether the following systems are

(a) Static or dynamic.
(b) linear or non-linear.
(c) causal or noncausal.
(d) time invariant or time-variant.

(i) $y(t)\frac{d^2y(t)}{dt^2} + 3t\frac{dy(t)}{dt} + y(t) = x(t)$

(ii) $\frac{d^3y(t)}{dt^3} + 4\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 2y^2(t) = x(t)$

(iii) $y(t) = od\{x(t)\}$

(iv) $y(n) = x(n)x(n-1)$

(v) $y(n) = \cos[x(n)]$

(vi) $y(n) = \log_{10}|x(n)|$

(vii) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$

(viii) $y(n) = sgn[x(n)]$

(ix) $y(n) = x(n)u(n)$

i) Given $y(t)\frac{d^2y(t)}{dt^2} + 3t\frac{dy(t)}{dt} + y(t) = x(t)$

(a) The system is described by differential equation. Hence it is a dynamic system.

(b) The input $x_1(t)$ and output $y_1(t)$ can be described by the differential equation as

$$y_1(t)\frac{d^2y_1(t)}{dt^2} + 3t\frac{dy_1(t)}{dt} + y_1(t) = x_1(t) \qquad (2.44)$$

Similarly for input $x_2(t)$ and output $y_2(t)$ the differential equation can be written as

$$y_2(t)\frac{d^2y_2(t)}{dt^2} + 3t\frac{dy_2(t)}{dt} + y_2(t) = x_2(t) \qquad (2.45)$$

Multiply Eq. (2.44) with a and Eq. (2.45) with b and add. Then we get

$$ay_1(t)\frac{d^2y_1(t)}{dt^2} + 3at\frac{dy_1(t)}{dt} + ay_1(t) + by_2(t)\frac{d^2y_2(t)}{dt^2} + 3bt\frac{dy_2(t)}{dt} + by_2(t) = ax_1(t) + bx_2(t)$$

$$ay_1(t)\frac{d^2y_1(t)}{dt^2} + by_2(t)\frac{d^2y_2(t)}{dt^2} + 3t\frac{d}{dt}[ay_1(t) + by_2(t)] + ay_1(t) + by_2(t) = ax_1(t) + bx_2(t) \qquad (2.46)$$

Shaded part of the Eq. (2.46) is not a function of weight sum of outputs. Hence superposition principle is not satisfied, and the system is non-linear.

(c) The output depends on the present input only. Hence the system is causal.

(d) The coefficients of the differential equation are function of time. Hence the system is time-variant.

The system is dynamic, non-linear, causal and time variant.

ii) $\frac{d^3y(t)}{dt^3}+\frac{4d^2y(t)}{dt^2}+\frac{5dy(t)}{dt}+2y^2(t)=x(t)$

(a) The system is described by a differential equation. Hence it is a dynamic system.

(b) For input $x_1(t)$, let the output be $y_1(t)$. Then the differential equation can be written as

$$\frac{d^3y_1(t)}{dt^3}+\frac{4d^2y_1(t)}{dt^2}+\frac{5dy_1(t)}{dt}+2y_1^2(t)=x_2(t) \tag{2.47}$$

Similarly for an input $x_2(t)$, let the output be $y_2(t)$. Then the differential equation can be written as

$$\frac{d^3y_2(t)}{dt^3}+\frac{4d^2y_2(t)}{dt^2}+\frac{5dy_2(t)}{dt}+2y_2^2(t)=x_2(t) \tag{2.48}$$

The linear combination of Eq. (2.47) and Eq. (2.48) can be written as

$$a\frac{d^3y_1(t)}{dt^3}+4a\frac{d^2y_1(t)}{dt^2}+5a\frac{dy_1(t)}{dt}+2ay_1^2(t)+b\frac{d^3y_2(t)}{dt^3}$$
$$+4b\frac{d^2y_2(t)}{dt^2}+5b\frac{dy_2(t)}{dt}+2by_2^2(t)=ax_1(t)+bx_2(t)$$
$$\Rightarrow \frac{d^3[ay_1(t)+by_2(t)]}{dt^3}+4\frac{d^2[ay_1(t)+by_2(t)]}{dt^2}+5\frac{d[ay_1(t)+by_2(t)]}{dt}$$
$$+2ay_1^2(t)+2by_2^2(t)=ax_1(t)+bx_2(t) \tag{2.49}$$

Shaded portion in the LHS is function of weighted sum of squares of outputs. Superposition principle is not satisfied. Therefore the system is non- linear.

(c) The output depends on present input only. Therefore, the system is causal.

(d) The coefficient of the differential equation are constants. Hence the system is time-invariant.

The system is dynamic, non-linear, causal and time-invariant

iii) Given

$$y(t)=od\{x(t)\}$$
$$=\frac{1}{2}[x(t)-x(-t)] \tag{2.50}$$

(a) For any value of t the output depends on the stored value of input. Hence the system is dynamic.

(b) For an input $x_1(t)$

$$y_1(t)=T[x_1(t)]=\frac{1}{2}[x_1(t)-x_1(-t)] \tag{2.51}$$

For an input $x_2(t)$

$$y_2(t)=T[x_2(t)]=\frac{1}{2}[x_2(t)-x_2(-t)] \tag{2.52}$$

The output due to weighted sum of inputs is

$$y_3(t)=T[ax_1(t)+bx_2(t)]=\frac{1}{2}[ax_1(t)+bx_2(t)-(ax_1(-t)+bx_2(-t))] \tag{2.53}$$

The weighted sum of outputs is

$$ay_1(t)+by_2(t)=\frac{1}{2}a[x_1(t)-x_1(-t)]+\frac{1}{2}b[x_2(t)-x_2(-t)] \tag{2.54}$$
$$y_3(t)=ay_1(t)+by_2(t)$$

Eq (2.53) and Eq (2.54) are equal. Therefore, the system is linear.

(c) For negative values of t the output depends on future values of input. Hence the system is non-causal.

(d) Given

$$y(t)=T[x(t)]=\frac{1}{2}[x(t)-x(-t)]$$

The output due to delayed input is

$$y(t,T)=T[x(t-T)]=\frac{1}{2}[x(t-T)-x(-t-T)] \tag{2.55}$$

The delayed output is

$$y(t-T)=\frac{1}{2}[x(t-T)-x[-(t-T)]$$
$$=\frac{1}{2}[x(t-T)-x(-t+T)] \tag{2.56}$$
$$y(t,T)\neq y(t-T)$$

Hence the system is time-variant.

The system is dynamic, linear, non-causal and time-variant.

iv) $y(n) = x(n)x(n-1)$

(a) The output depends on past values of input. It requires memory. Therefore, the system is dynamic.

(b) The output $y_1(n)$ due to input $x_1(n)$ is

$$y_1(n) = x_1(n)x_1(n-1) \tag{2.57}$$

For an input $x_2(n)$

$$y_2(n) = x_2(n)x_2(n-1) \tag{2.58}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = [ax_1(n) + bx_2(n)][ax_1(n-1) + bx_2(n-1)] \tag{2.59}$$

The weighted sum of outputs is given by

$$ay_1(n) + by_2(n) = ax_1(n)x_1(n-1) + bx_2(n)x_2(n-1)$$
$$y_3(n) \neq ay_1(n) + by_2(n) \tag{2.60}$$

Eq(2.59) and Eq (2.60) are not equal. Hence the system is non-linear.

(c) The output depends on present and past values of the input. Hence the system is causal.

(d) Given $y(n) = T[x(n)] = x(n)x(n-1)$

The output due to delayed input is

$$y(n,k) = T[x(n-k)] = x(n-k)x(n-1-k) \tag{2.61}$$

The delayed output

$$y(n-k) = x(n-k)x(n-1-k) \tag{2.62}$$
$$y(n,k) = y(n-k)$$

Therefore, the system is time-invariant.

The system is dynamic, non-linear, causal and time-invariant.

v) $y(n) = \cos[x(n)]$

(a) The output at any instant depends on the input at that instant. Therefore, the system is static.

(b) For an input $x_1(n)$

$$y_1(n) = \cos[x_1(n)] \tag{2.63}$$

For an input $x_2(n)$

$$y_2(n) = \cos[x_2(n)] \tag{2.64}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[a_1x_1(n) + bx_2(n)]$$
$$= \cos[a_1x_1(n) + bx_2(n)] \tag{2.65}$$

The weighted sum of outputs is given by

$$ay_1(n) + by_2(n) = a\cos[x_1(n)] + b\cos[x_2(n)] \tag{2.66}$$
$$y_3(n) \neq ay_1(n) + by_2(n)$$

Eq(2.65) and Eq (2.66) are not equal. Hence the system is non-linear.

(c) The output depends on present inputs only. Hence the system is causal.

(d) Given $y(n) = T[x(n)] = \cos[x(n)]$

The output due to delayed input

$$y(n,k) = T[x(n-k)] = \cos[x(n-k)] \tag{2.67}$$

The delayed output is

$$y(n-k) = \cos[x(n-k)] \tag{2.68}$$
$$y(n,k) = y(n-k)$$

Therefore, the system is time-invariant.

The system is static, non-linear, causal and time-invariant.

vi) Given $y(n) = \log_{10}|x(n)|$

(a) The output at any instant depends on the input at that instant. Hence the system is static.

(b) Given $y(n) = \log_{10}|x(n)|$

For input $x_1(n)$

$$y_1(n) = \log_{10}|x_1(n)| \tag{2.69}$$

For input $x_2(n)$

$$y_2(n) = \log_{10}|x_2(n)| \tag{2.70}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = \log_{10}|ax_1(n) + bx_2(n)| \quad (2.71)$$

The weighted sum of outputs

$$ay_1(n) + by_2(n) = a\log_{10}|x_1(n)| + b\log_{10}|x_2(n)| \quad (2.72)$$
$$y_3(n) \neq ay_1(n) + by_2(n)$$

Eq (2.71) and Eq (2.72) are not equal. Hence the system is non-linear.

(c) The output depends on present input only. Hence the system is causal.

(d) $y(n) = T[x(n)] = \log_{10}|x(n)|$

The output due to delayed input

$$y(n,k) = T[x(n-k)] = \log_{10}|x(n-k)| \quad (2.73)$$

The delayed output

$$y(n-k) = \log_{10}|x(n-k)| \quad (2.74)$$
$$\Rightarrow \quad y(n,k) = y(n-k)$$

Hence, the system is time-invariant.

The system is static, non-linear, causal and time-invariant.

vii) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$

(a) The output

$$y(n) = x(-\infty) + \ldots\ldots + x(0) + x(1) + \ldots x(n) + x(n+1) \text{ for } n > 0$$

The output depends on stored values of input. Hence the system is dynamic.

(b) For input $x_1(n)$

$$y_1(n) = \sum_{k=-\infty}^{n+1} x_1(k) \quad (2.75)$$

For input $x_2(n)$

$$y_2(n) = \sum_{k=-\infty}^{n+1} x_2(k) \quad (2.76)$$

The input due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = \sum_{k=-\infty}^{n+1} [ax_1(k) + bx_2(k)] \quad (2.77)$$

The weighted sum of outputs is

$$ay_1(n) + by_2(n) = a\sum_{k=-\infty}^{n+1} x_1(k) + b\sum_{k=-\infty}^{n+1} x_2(k)$$
$$= \sum_{k=-\infty}^{n+1} ax_1(k) + \sum_{k=-\infty}^{n+1} bx_2(k) \quad (2.78)$$
$$y_3(n) = ay_1(n) + by_2(n)$$

Eq (2.77) and Eq (2.78) are equal. Hence the system is linear.

(c) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$

For $n = 0$

$$y(0) = \sum_{k=-\infty}^{1} x(k) = \underbrace{x(-\infty) + \ldots +}_{\text{past inputs}} \underset{\text{present input}}{x(0)} + \underset{\text{future input}}{x(1)}$$

The output depends on future value of input. Hence the system is noncausal.

viii) $y(n) = T[x(n)] = \sum_{k=-\infty}^{n+1} x(k)$

The output due to shifted input

$$y(n,k_1) = T[x(n-k_1)] = \sum_{k=-\infty}^{n+1-k_1} x(k) \quad (2.79)$$

The delayed output is

$$y(n-k_1) = \sum_{k=-\infty}^{n+1} x(k-k_1) \quad (2.80)$$
$$= \sum_{k=-\infty}^{n+1-k_1} x(k)$$
$$y(n,k_1) = y(n-k_1)$$

Hence the system is time-invariant.

ix) $y(n) = sgn[x(n)]$

$$sgn[x(n)] = 1 \text{ for } n > 0$$
$$= 1 \text{ for } n < 0$$

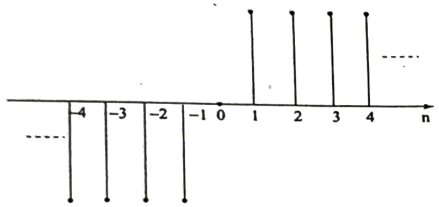


Fig. 2.7 Signum function.

(a) The output depends on the input at that instant only. Therefore the system is static.

(b) For an input $x_1(n)$

$$y_1(n) = T[x_1(n)] = sgn[x_1(n)] \tag{2.81}$$

For an input $x_2(n)$

$$y_2(n) = T[x_2(n)] = sgn[x_2(n)] \tag{2.82}$$

The output due to weighted sum of inputs is

$$y_3(n) = T[ax_1(n) + bx_2(n)] = sgn[ax_1(n) + bx_2(n)] \tag{2.83}$$

The weighted sum of outputs is

$$ay_1(n) + by_2(n) = asgn[x_1(n)] + bsgn[x_2(n)] \tag{2.84}$$
$$y_3(n) \neq ay_1(n) + by_2(n).$$

Eq (2.83) and Eq (2.84) are not equal. Therefore, the system is non-linear.

(c) The output $y(n)$ depends on present input. Hence the system is causal.

(d) Given $y(n) = sgn[x(n)]$

The output due to delayed input

$$y(n,k) = T[x(n-k)] = sgn[x(n-k)] \tag{2.85}$$

The delayed output

$$y(n-k) = sgn[x(n-k)] \tag{2.86}$$
$$y(n,k) = y(n-k).$$

Hence the system is time-invariant.

x) Given $y(n) = x(n)u(n)$

(a) The output depends on present input only. The system won't require any memory. Hence the system is static.

(b) For input $x_1(n)$

$$y_1(n) = T[x_1(n)] = x_1(n)u(n) \tag{2.87}$$

For input $x_2(n)$

$$y_2(n) = T[x_2(n)] = x_2(n)u(n) \tag{2.88}$$

The output due to weighted sum of inputs

$$y_3(n) = T[ax_1(n) + bx_2(n)] = [ax_1(n) + bx_2(n)]u(n) \tag{2.89}$$

The weighted sum of outputs is

$$ay_1(n) + by_2(n) = ax_1(n)u(n) + bx_2(n)u(n)$$
$$= [ax_1(n) + bx_2(n)]u(n)$$
$$y_3(n) = ay_1(n) + by_2(n) \tag{2.90}$$

Eq (2.89) and Eq (2.90) are equal. Therefore, the system is linear.

(c) The output depends on present input. Hence the system is causal.

(d) Given $y(n) = x(n)u(n)$

The output due to delayed input

$$y(n,k) = T[x(n-k)] = x(n-k)u(n) \tag{2.91}$$

The delayed output by k samples is

$$y(n-k) = x(n-k)u(n-k) \tag{2.92}$$
$$y(n,k) \neq y(n-k)$$

Hence the system is time-variant.

Practice Problem 2.4

Check whether the following systems are

(a) static/dynamic.
(b) linear/non-linear.
(c) causal/non-causal.
(d) time variant/time-invariant.

(i) $y(n) = x(n+1) - x(n-1)$

(ii) $\frac{d^3y(t)}{dt^3} + 4\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 2y^2(t) = x(t+5)$

(iii) $y(t) = Ev[x(t)]$

(iv) $y(t) = 10x(t) + 5$

(v) $y(n) = |x(n)|$

(vi) $y(n) = a^n x(n)$

Ans: (i) Dynamic, Linear, Non-causal, Time-invariant
(ii) Dynamic, Nonlinear, Non-causal, Time-invariant
(iii) Dynamic, Linear, Non-causal, Time-variant
(iv) Static, Non-linear, Causal, Time-invariant
(v) Static, Non-linear, Causal, Time-invariant
(vi) Static, Linear, Causal, Time-variant

Solved Problem 2.7 Check whether the following systems are

(a) Static/dynamic.
(b) linear/non-linear.
(c) Time-invariant or time-variant.

(i) A full wave rectifier whose output is the modulus of the input

$$y(t) = |x(t)|$$

(ii) A modulator whose carrier frequency is Ω_c, giving an output

$$y(t) = x(t)\cos\Omega_c t$$

Solution:

(i) (a) The output at any instant depends on the input at that instant. Therefore the system is static.

(b) Given $y(t) = |x(t)|$

For an input $x_1(t)$

$$y_1(t) = T[x_1(t)] = |x_1(t)| \tag{2.93}$$

For an input $x_2(t)$

$$y_2(t) = T[x_2(t)] = |x_2(t)| \tag{2.94}$$

The output due to weighted sum of inputs is

$$y_3(t) = T[ax_1(t) + bx_2(t)] = |ax_1(t) + bx_2(t)| \tag{2.95}$$

The weighted sum of outputs is

$$ay_1(t) + by_2(t) = a|x_1(t)| + b|x_2(t)| \tag{2.96}$$

$$y_3(t) \neq ay_1(t) + by_2(t)$$

Therefore, the system is non-linear.

(c) For an input $x(t)$

$$y(t) = |x(t)|$$

If the input is delayed by T seconds, the output is

$$y(t, T) = |x(t - T)| \tag{2.97}$$

If the output is delayed by T seconds we get

$$y(t - T) = |x(t - T)| \tag{2.98}$$

$$y(t, T) = y(t - T)$$

Hence the system is time-invariant.

(ii) (a) Given $y(t) = x(t)\cos\Omega_c t$

The output at any instant depends on the input at that instant. Hence the system is static.

(b) Given $y(t) = T[x(t)] = x(t)\cos\Omega_c t$

For an input $x_1(t)$

$$y_1(t) = T[x_1(t)] = x_1(t)\cos\Omega_c t \tag{2.99}$$

For an input $x_2(t)$

$$y_2(t) = T[x_2(t)] = x_2(t)\cos\Omega_c t \tag{2.100}$$

The output due to weighted sum of inputs is

$$\begin{aligned} y_3(t) &= T[ax_1(t) + bx_2(t)] \\ &= [ax_1(t) + bx_2(t)]\cos\Omega_c t \\ &= ax_1(t)\cos\Omega_c t + bx_2(t)\cos\Omega_c t \end{aligned} \tag{2.101}$$

The weighted sum of outputs is

$$ay_1(t) + by_2(t) = ax_1(t)\cos\Omega_c t + bx_2(t)\cos\Omega_c t \tag{2.102}$$

$$y_3(t) = ay_1(t) + by_2(t)$$

Eq (2.101) and Eq (2.102) are equal. Hence the system is linear.

(c) $y(t) = T[x(t)] = x(t)\cos\Omega_c t$

The output due to delayed input is

$$y(t,T) = T[x(t-T)] = x(t-T)\cos\Omega_c t \qquad (2.103)$$

The delayed output is

$$y(t-T) = x(t-T)\cos\Omega_c(t-T) \qquad (2.104)$$

$$y(t,T) \neq y(t-T)$$

Hence the system is time-variant.

Practice Problem 2.5

Find whether the following systems are

a) Static/dynamic.
b) Linear/non-linear.
c) Time-invariant or time-variant.

(i) $y(t) = at^2x(t) + btx(t-2)$ **Ans:** Dynamic, Linear, Time-invariant

(ii) $y(t) = Ax(t) + B$ **Ans:** Static, Non-linear, Time-invariant

(iii) $y(n) = 2x(n) + \dfrac{1}{x(n-1)}$ **Ans:** Dynamic, Non linear, Time-invariant

(iv) $y(n) = x^2(n) + \dfrac{1}{x^2(n-1)}$ **Ans:** Dynamic, Non linear, Time-invariant

Review Questions

1. State superposition principle.
2. How do you find whether the system is linear or nonlinear?
3. How do you prove that the system is time-invariant? **AU Nov, 2008**
4. Explain the following properties of the system.
   (a) Stability
   (b) Linearity
   (c) Causality
   (d) Time-invariance

2.3 System Modeling

The first step in analysis of any physical system is finding mathematical model. Mathematical model of a system describes the system by explicit mathematical expressions. These equations can be obtained based on the fundamental laws of physical systems. For example, to find mathematical model of mechanical systems Newton's laws of motion are applied; for electrical systems fundamental laws like Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) are applied.

Once a model is obtained, we can solve the equations to find outputs for different inputs. The inputs that are used to test the performance of the system are known as test signals. Some of the test signals are impulse, step, ramp and sinusoidal signals.

2.3.1 Modeling of mechanical and electrical elements

A system consists of many components. To model such a system, it is necessary to find a representation for each and every component present in the system. For example, to obtain model of an RLC series circuit, first we find the mathematical representation of variable across or through resistor, inductor and capacitor, and then we have to apply KVL or KCL. In this section we will study about modeling of mechanical and electrical elements.

Translation mechanical elements

Mass, spring and dashpot are the fundamental mechanical elements. Two of them, the spring and the mass are energy storage elements, and the dashpot is an energy absorbing element.

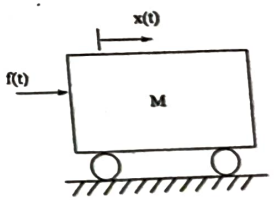


Fig. 2.8 Mass element

Mass: The dynamics of a mass element are described by Newton's second law of motion

$$f(t) = M\frac{d^2x(t)}{dt^2} = Ma, \qquad (2.105)$$

where M is mass in kg, and a is acceleration due to gravity, $f(t)$ is applied force and $x(t)$ is the displacement.

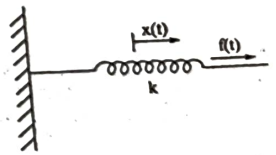


Fig. 2.9 Spring element

The spring: A spring element is one that stores energy during the variation of its shape due to elastic deformation resulting from the application of a force. Over its linear region, the spring satisfies Hooke's law which relates the linear force to the displacement by the expression

$$f(t) = kx(t), \qquad (2.106)$$

where k is the spring constant with units N/m.

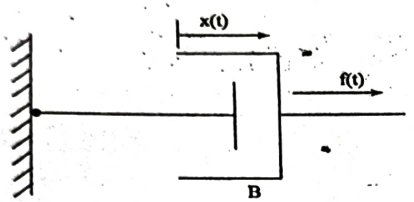


Fig. 2.10 Dashpot

The Dashpot: For a damper, the relationship between the applied force and velocity is given by

$$f(t) = Bv(t) = B\frac{dx(t)}{dt}, \qquad (2.107)$$

where B is the damping constant, and $x(t)$ is the displacement

Electrical elements

Resistor: Resistor is an energy dissipative element. For a linear resistor, the relationship between the current and voltage is given by

$$v(t) = Ri(t), \qquad (2.108)$$

where R = Resistance in ohm

$v(t)$ = The voltage across the resistor

$i(t)$ = The current through the resistor

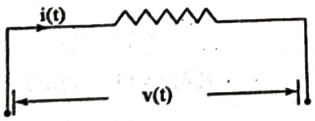


Fig.2.11 A Resistor

Capacitor: A capacitor is an electrical element which stores energy in electrical form. If $v(t)$ is the voltage across capacitance and $i(t)$ is the current through the capacitance then the relation between $v_c(t)$ and $i_c(t)$ is given by

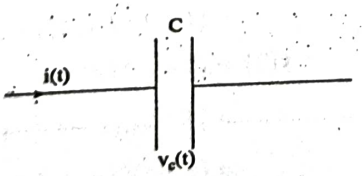


Fig. 2.11 A Capacitor

$$v_c(t) = \frac{1}{C}\int_{-\infty}^{t} i_c(t)dt \qquad (2.109)$$

(or)

$$i_c(t) = C\frac{dv_C(t)}{dt} \qquad (2.110)$$

The inductor: An inductor stores energy in the form of magnetic field. The relationship between voltage across inductor $v_L(t)$ and current through inductor $i_L(t)$ for an inductor is

$$i_L(t) = \frac{1}{L}\int_{-\infty}^{t} v_L(t)dt \qquad (2.111)$$

or

$$v_L(t) = L\frac{di_L(t)}{dt} \qquad (2.112)$$

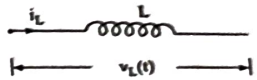


Fig. 2.13 An Inductor

Solved Problem 2.8 Obtain differential equations governing following systems

(i)

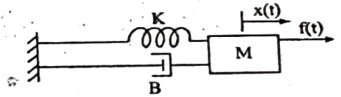


Fig. 2.14

(ii)

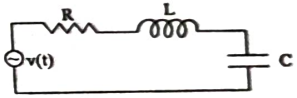


Fig. 2.15

(iii)

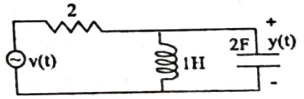


Fig. 2.16

Solution:

(i) The forces acting on the mass are shown in Fig. 2.17.

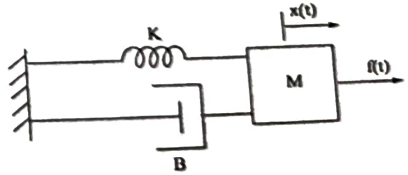


Fig. 2.17

Now we can write

$$f(t) = M\frac{d^2x(t)}{dt^2} + B\frac{dx(t)}{dt} + kx(t) \quad (2.113)$$

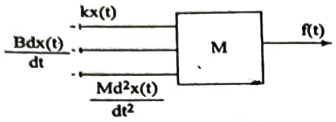


Fig. 2.18

(ii) Applying KVL to the circuit we obtain

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(t)dt \quad (2.114)$$

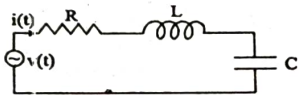


Fig.2.19

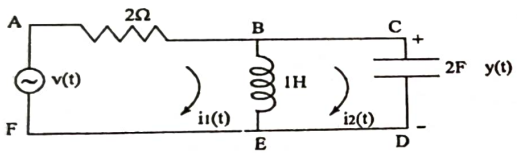


Fig. 2.20

(iii) Applying KVL for the loop ABEFA we get

$$v(t) = 2i_1(t) + 1\left[\frac{di_1(t)}{dt} - \frac{di_2(t)}{dt}\right]$$

$$\Rightarrow v(t) = 2i_1(t) + \frac{di_1(t)}{dt} - \frac{di_2(t)}{dt} \quad (2.115)$$

Similarly applying KVL for the loop BCDEB we get

$$\frac{1}{2}\int_{-\infty}^{t} i_2(t)dt + 1\left[\frac{di_2(t)}{dt} - \frac{di_1(t)}{dt}\right] = 0$$

$$\Rightarrow -\frac{di_1(t)}{dt} + \frac{di_2(t)}{dt} + \frac{1}{2}\int_{-\infty}^{t} i_2(t)dt = 0 \quad (2.116)$$

2.3.2 Examples of discrete-time system models

Population model

Let $x(n)$ be the population of India in a year n and let $b(n)$ and $d(n)$ are birth rate and death rate respectively. If $u(n)$ is the net number of immigrants entering India then the population in a year $(n+1)$ is given by

$$y(n+1) = y(n) + b(n)y(n) - d(n)y(n) + u(n)$$

$$= [1 + b(n) - d(n)]y(n) + u(n) \quad (2.117)$$

The above system is time-variant.

If the values of $b(n)$ and $d(n)$ are constants, then

$$y(n+1) = [1 + b - d]y(n) + u(n) \quad (2.118)$$

In this case the system is time-invariant.

Savings account

Another example of a discrete-time system is a savings account. Let $y(n)$ be the amount in the account at n-th month, $x(n)$ be the amount deposited at a month n and r be the interest rate per month. Then at $(n+1)$ th month the amount in the account is

$$y(n+1) = y(n) + ry(n) + x(n+1)$$

$$= (1+r)y(n) + x(n+1) \quad (2.119)$$

If the amount deposited in every month is same and equal to A,then

$$y(n+1) = (1+r)y(n) + A$$

If $y(0) = I$, the initial amount, then

$$y(1) = (1+r)y(0) + A$$

$$= (1+r)I + A$$

$$y(2) = (1+r)y(1) + A$$

$$= (1+r)[(1+r)I + A] + A$$

$$= (1+r)^2 I + A(1+r) + A$$

$$y(3) = (1+r)y(2) + A$$

$$= (1+r)[(1+r)^2 I + A(1+r) + A] + A$$

$$= (1+r)^3 I + A(1+r)^2 + A(1+r) + A$$

In general for any n

$$y(n) = (1+r)^n I + A(1+r)^{n-1} + A(1+r)^{n-2} + \ldots + A$$
$$= (1+r)^n I + A\left[\sum_{k=0}^{n-1}(1+r)^k\right]$$
$$= (1+r)^n I + A\left[\frac{(1+r)^n - 1}{r}\right] \quad (2.120)$$

Amortization

Let an housing loan of M rupees be paid in N monthly instalments of u rupees per month. This is known as amortization, where M is the initial amount of money sanctioned, and the amount we pay in month n is $u(n)$. The unpaid debt will carry a monthly interest of r%; where r is the interest rate per rupee per month. If $y(n)$ is the amount of debt at n^{th} month, the debt at $(n+1)^{th}$ month is

$$y(n+1) = y(n) + ry(n) - u(n+1)$$
$$= (1+r)y(n) - u(n+1) \quad (2.121)$$

This is a linear time invariant difference equation. If we want to pay off the loan in N instalments and in equal monthly payments, then the monthly instalment can be computed as follows.

$$\text{Let } u(n) = P \quad \text{for} n = 1,2\ldots N$$

Substituting $n = 0$ in Eq. (2.121) yields

$$y(1) = (1+r)y(0) - u(1)$$
$$= (1+r)y(0) - P$$

and

$$y(2) = (1+r)y(1) - P$$
$$= (1+r)[(1+r)y(0) - P] - P$$
$$= (1+r)^2 y(0) - (1+r)P - P$$
$$y(3) = (1+r)y(2) - P$$
$$= (1+r)[(1+r)^2 y(0) - (1+r)P - P] - P$$

$$= (1+r)^3 y(0) - (1+r)^2 P - (1+r)P - P$$
$$y(n) = (1+r)^n y(0) - P[(1+r)^{n-1} + (1+r)^{n-2} + \ldots + (1+r) + 1]$$
$$= (1+r)^n y(0) - P\left[\frac{1-(1+r)^n}{1-(1+r)}\right]$$

$\because 1 + a + a^2 \ldots a^n = \frac{1-a^{n+1}}{1-a}$

If we pay off debt in N instalments, then $n = N$ and $y(N) = 0$

$$\Rightarrow y(N) = (1+r)^N y(0) - P\left[\frac{1-(1+r)^N}{1-(1+r)}\right] = 0$$
$$\Rightarrow (1+r)^N y(0) = P\left[\frac{1-(1+r)^N}{1-(1+r)}\right]$$
$$P = \frac{r(1+r)^N y(0)}{(1+r)^N - 1} \quad (2.122)$$

This is the monthly instalment to be paid.

Practice Problem 2.6

Write the differential equations for the following systems.

(i)

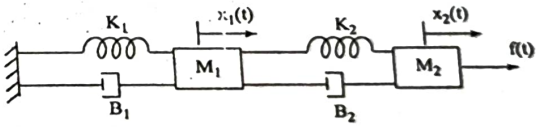


Fig.2.21

Ans: For mass M_1; $M_1\frac{d^2x_1}{dt^2} + k_1x_1 + B_1\frac{dx_1}{dt} + k_2(x_1 - x_2) + B_2\frac{d}{dt}(x_1 - x_2) = 0$

For mass M_2; $f(t) = M_2\frac{d^2x_2}{dt^2} + k_2(x_2 - x_1) + B_2\frac{d}{dt}(x_2 - x_1)$

(ii)

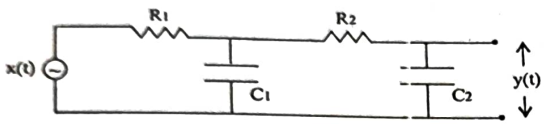


Fig.2.22

Ans: $R_1 i_1 + \frac{1}{C_1}\int (i_1 - i_2)dt = x(t)$

$$R_2 i_2 + \frac{1}{C_2} i_2 dt + \frac{1}{C_1}\int (i_2 - i_1)dt = 0$$

(iii)

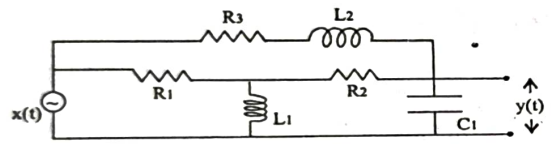


Fig.2.23

Ans: $R_1(i_1 - i_3) + L_1\frac{d}{dt}(i_1 - i_2) = x(t)$

$$R_2(i_2 - i_3) + \frac{1}{C_1}\int i_2 dt + L_1\frac{d}{dt}(i_2 - i_1) = 0$$

$$R_3 i_3 + L_2\frac{di_3}{dt} + R_2(i_3 - i_2) + R_1(i_3 - i_1) = 0$$

Solved Problem 2.9 Consider a system with input-output relationship given by

$$y(n) = x(n)x(n-1)$$

1. Is the system memory less?
2. Determine the output of the system when the input is $\delta(n)$.

Solution:

1. The output depends on past input. Hence the system is not memoryless.
2. The output of the system for input $\delta(n)$ is

$$y(n) = \delta(n)\,\delta(n-1)$$

We know

$\delta(n) = 1$ for $n = 0$ and $\delta(n-1) = 1$ for $n = 1$

$= 0$ for $n \neq 0$ $= 0$ for $n \neq 1$

Therefore, $y(n) = 0$ for all n.

Solved Problem 2.10 Check whether or not the given systems are linear, time invariant, causal, memory less and stable. AU April 2008

(i) $y(t) = x(t-2) + x(2-t)$ (ii) $y(t) = \frac{dx(t)}{dt}$

Solution:

(i) Given $y(t) = x(t-2) + x(2-t)$

The system is not memory less since output depends on past and future values of the input.

The system is noncausal since the output depends on future values of input.

$$y(t,\tau) = x(t-2-\tau) + x(2-t-\tau)$$
$$y(t-\tau) = x(t-2-\tau) + x(2-t+\tau)$$
$$y(t,\tau) \neq y(t-\tau)$$

The system is time variant.

To check Linearity

$$y_1(t) = x_1(t-2) + x_1(2-t)$$
$$y_2(t) = x_2(t-2) + x_2(2-t)$$
$$a_1 y_1(t) + a_2 y_2(t) = a_1 x_1(t-2) + a_1 x_1(2-t) + a_2 x_2(t-2) + a_2 x_2(2-t)$$
$$= a_1 x_1(t-2) + a_2 x_2(t-2) + a_1 x_1(2-t) + a_2 x_2(2-t)$$

Superposition principle is satisfied. Hence the system is linear.

For bounded input, the output also bounded. Hence the system is stable.

(ii) $y(t) = \frac{d}{dt}x(t)$

$y(t, \tau) = \frac{d}{dt}x(t-\tau)$ and $y(t-\tau) = \frac{d}{dt}x(t-\tau)$. Hence the system is time invariant.

$$a_1 y_1(t) + a_2 y_2(t) = a_1\frac{d x_1(t)}{dt} + a_2\frac{d x_2(t)}{dt} = \frac{d}{dt}[a_1 x_1(t) + a_2 x_2(t)]$$

Superposition Principle is satisfied
Therefore the system is linear.
The system is not memoryless.
For bounded input the output also bounded hence system is stable.
The system is causal.

Practice Problem 2.7

The input and output relation of a discrete-time system is

$$y(n) = \sum_{k=n-n_0}^{n+n_0} x(k)$$

where n_0 is a finite positive integer.
(i) Is the system is linear? (ii) Is the system is time-invariant.

2.4 Invertibility and Inverse Systems

A system is said to be invertible if the input applied to the system can be recovered from the output of the system. For such systems distinct inputs leads to distinct outputs. Consider a system 1 shown in Fig.2.24 with input $x(t)$ and output $y(t)$. Now if we connect a system 2 in cascade with system 1 such that the output of system 2 is equal to $x(t)$, then the system 2 is called as inverse system of system1.

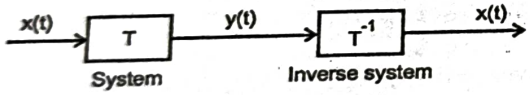


Fig. 2.24

From Fig.2.24 the output of system 1 is

$$y(t) = T[x(t)].$$

The output of system 2 is

$$y'(t) = T^{-1}T[x(t)].$$

The output of system 2 is equal to the original input $x(t)$ if $T^{-1}T = I$, where I denotes the identity operator. An example for invertible continuous system is

$$y(t) = k\,x(t)$$

for which the inverse system is

$$w(t) = \frac{1}{k}y(t)$$

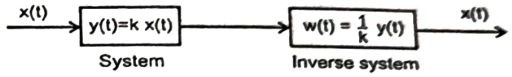


Fig. 2.25

To test whether a system is invertible or not, do the following.

1. If a system produces zero output for any input then that system is non-invertible.
2. If a system gives same output for different inputs then that system is non-invertible.

Solved Problem 2.11 Determine if each of the following systems is invertible. If the system is invertible, find the inverse system.

(i) $y(t) = \cos[x(t)]$ (ii) $y(t) = \dfrac{dx(t)}{dt}$

(iii) $y(t) = x(t-3)$ (iv) $y(n) = x(n)x(n-2)$

Solution:

(i) For input signals separated by 2π, the system gives same output. Therefore the system is non-invertible.

(ii) If $x(t)$ is a constant, then the output is zero. Therefore the system is non-invertible.

(iii) Invertible. The inverse system is $y(t) = x(t+3)$.

(iv) For inputs $\delta(n)$ and $\delta(n-2)$ the output is zero. Therefore the system is non-invertible.

Short Questions and Answers

1. **Define system.**

   A system is defined as a physical device that generates a response, or output signal, for a given input signal.

2. **Define (i) Continuous-time system (ii) Discrete-time system.**

   i) A continuous-time system is one which operates on a continuous time input signal and produces a continuous-time output signal.
   ii) A discrete-time system is one which operates on discrete-time input signal and produces a discrete-time input signal.

3. **What are the classification of system?**

   1) Static and dynamic systems.
   2) Time-variant and time-invariant systems.
   3) Linear and non-linear systems.
   4) Stable and unstable systems.
   5) Causal and non-causal system.

4. **Define linear-time invariant system.** AU April 2003

   An LTI system is one which posses two basic properties, linearity and time-invariance.

   Linearity: An LTI system obeys superposition principle which states that the output of the system to a weighted sum of inputs is equal to the weighted sum of the outputs corresponding to each of the individual inputs.

   Time invariance: If the input output relation of a system does not vary with time, then the system is said to be time-invariant

5. **Is the system $y(t) = y(t-1) + 2t\, y(t-2)$ time-invariant?** AU OCT-2002

   No. The system is time-variant. The coefficient $2t$ of the difference equation is function of time.

6. **Define: static and dynamic system.**

   Static system: A static or memoryless system is a system whose output at any instant depends only on the input value at that instant.

   Dynamic system: A dynamic is one in which the output at any instant depends on past or future values of inputs

7. **Define a causal system.**

   A system is said to be causal if the output of the system at any time depends only on present and past inputs, but does not depend on future inputs.

8. **What is a stable system?**

   Any relaxed system is said to be bounded input-bounded output (BIBO) stable if and only if every bounded input yields a bounded output.

Multiple Choice Questions

1. Consider the following statements regarding a discrete system with the output input relationship $y(n) = x(n) + 3$.

   The system

   1) is linear 2) is causal 3) have bounded input for bounded output 4) is non-reliable

   (a) 1, 2, and 3 are correct (b) 2 and 3 are correct
   (c) 1 and 2 are correct (d) 1, 3 and 4 are correct

2. Which are of the following system is nonlinear?

   (a) $y(t) = 2x(t-1) - 3x(t-2) + x(t-3)$ (b) $y(t) = 5x(t)$
   (c) $y(t) = 2x(t-1) - x(t-2) - x(t-4)$ (d) $y(t) = 2x(t) + 3.6$

3. Which are of the following systems are invertible?

   (a) $y(n) = n\, x(n)$
   (b) $y(n) = x(n)\, x(n-1)$
   (c) $y(t) = \dfrac{dx(t)}{dt}$
   (d) $y(n) = x(1-n)$

4. Which are of the following pairs is NOT correctly matched?

   (a) unstable system $\dfrac{dy}{dt} - 0.1y(t) = x(t)$
   (b) Non-linear system $\dfrac{dy}{dt} + 2t^2 y(t) = x(t)$
   (c) Non-causal system $y(t) = x(t+2)$
   (d) Non-dynamic system $y(t) = 3x^2(t)$

5. The discrete-time equation $y(n+1) + 0.5ny(n) = 0.5x(n+1)$ is not at tributable to a

   (a) memory less system (b) time-varying system (c) linear system (d) caus system

6. A continuous-time system is governed by the equation $3y^3(t)+2y^2(t)+y(t)=x^2(t)+x(t)$. The system is

   (a) linear and dynamic (b) linear and non-dynamic

   (c) non-linear and dynamic (d) non-linear and non-dynamic

7. Which one of the following systems is a causal system?

   (a) $y(t)=\sin[u(t+3)]$ (b) $y(t)=5u(t)+3u(t-1)$

   (c) $y(t)=5u(t)+3u(t+1)$ (d) $y(t)=\sin[u(t-3)]+\sin[u(t+3)]$

8. Which one of the following is the response $y(t)$ of a causal LTI system described by $H(s)=\frac{s+1}{s^2+2s+2}$ for a given input $e^{-t}u(t)$?

   (a) $y(t)=e^{-t}\sin tu(t)$ (b) $y(t)=e^{-(t-1)}\sin(t-1)u(t-1)$

   (c) $y(t)=\sin(t-1)u(t-1)$ (d) $y(t)=e^{-t}\cos tu(t)$

9. The discrete time system described by $y(n)=x(n^2)$ is

   (a) causal, linear and time variant

   (b) causal, linear and time variant

   (c) non-causal, linear and time invariant

   (d) non-causal, linear and time-variant

10. The system $y(n)=x(Mn), -\infty<n<\infty$ where M is a positive constant is

    (a) shift variant (b) shift invariant (c) non-causal (d) causal

11. The system $y(n+2)+y(n+1)=x(n+2)$ is

    (a) causal and memory less

    (b) causal and has memory

    (c) is causal

    (d) is non-causal

12. For the systems shown below which of the systems are linear?

    (a) $\frac{dy}{dt}+2y(t)=f(t)\frac{df}{dt}$ (b) $\frac{dy}{dt}+y^2(t)=f(t)$

    (c) $\frac{dy}{dt}+2y(t)=f(t)$ (d) $\frac{dy}{dt}+2ty(t)=f(t)$

13. Which of the following systems are time-invariant?

    (a) $y(t)=x(2t)$ (b) $y(t)=x(t)+x(t-1)$

    (c) $y(t)=x(t/2)$ (d) $y(t)=x(-t)$

14. The system $y(t)=x(3t-6)$ is

    (a) linear, time variant (b) linear, time invariant

    (c) nonlinear, time variant (d) nonlinear, time invariant

15. The system $y(n)=x(n)x(n-1)$ is

    (a) dynamic and linear (b) dynamic, nonlinear

    (c) causal and time invariant (d) noncausal, time variant

16. Which of the following systems are invertible?

    (a) $y(t)=\int_{-\infty}^{t}x(t)d\tau$ (b) $y(t)=x(2t-3)$

    (c) $y(t)=\cos[x(t)]$ (d) $y(t)=x^n(t)$ n, integer

17. The system $y(t)=e^{x(t)}$ is

    (a) Stable, causal (b) noncausal, stable

    (c) unstable, causal (d) unstable, noncausal

18. The system $y(n)=\cos[x(n)]$ is

    (a) stable, linear, time variant (b) stable, nonlinear, time invariant

    (c) unstable, nonlinear, time variant (c) unstable, linear, time invariant

19. Which of the following system are BIBO stable?

    (a) $y(n)=\sum_{k=-\infty}^{\infty}x(k)$ (b) $y(n)=a^{x(n)}$ (c) $y(n)=|x(n)|$ (d) $y(n)=\log_{10}[x(n)]$

20. If $x(t)$ is the input and $y(t)$ is zero-state response to $y(t)$ then the system $y(t)=\int_{-\infty}^{t}tx(\lambda)d\lambda$ is

    (a) causal, static (b) causal, dynamic (c) noncausal, dynamic d) non-causal, static

Answers to Multiple Choice Questions

1. (b) The system $y(n) = x(n) + 3$ is causal, stable and realizable.
2. (d) Hint: $y(t) = ax(t) + b$ is a nonlinear system
3. d
4. (b) Nonlinear system $\frac{dy}{dt} + 2t^2 y(t) = x(t)$
5. (a) memory less system
6. (d) non-linear and non-dynamic
7. (b) $y(t) = 5u(t) + 3u(t-1)$
8. (a) $y(t) = e^{-t} \sin t\, u(t)$
9. (d) non-causal, linear and time-variant
10. (a) shift variant (c) non-causal
11. (b) causal and has memory
12. (c) $\frac{dy}{dt} + 2y(t) = f(t)$
    (d) $\frac{dy}{dt} + 2ty(t) = f(t)$
13. (b) $y(t) = x(t) + x(t-1)$
14. (a) linear, time variant
15. (b) dynamic, nonlinear
    (c) causal and time invariant
16. a,b
17. (a) Stable, causal
18. (b) $y(n) = a^{x(n)}$
19. b, c, d
20. b

3

Time Domain Analysis of Discrete-Time Systems

3.1 Introduction

In chapter 2 we studied that a discrete-time system is one that accepts a discrete-time signal as input and produces a discrete-time signal as output. Further we extended our discussion on classification of systems. In this chapter we will study about the time domain analysis of linear time-invariant discrete-time systems for different forcing functions. There are two basic methods for analysing the response of a linear system to a given input signal. First method is based on the direct solution of the difference equation representing the system. The difference equation for an N^{th} order system is

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{k=0}^{M} b_k x(n-k) \tag{3.1}$$

where $\{a_k\}$ and $\{b_k\}$ are constant parameters. The solution of the difference equation provides us the system response.

In second method the input signal is first resolved into sum of elementary signals (ex: impulse). By using the linear property of the system, the responses of the system to the elementary signals are added to obtain the total response of the system.

3.2 Solution of Difference Equations

In this section we consider the classical technique for solving the difference equations. The response of any discrete-time system can be decomposed as

$$\text{Total response} = \text{zero-state response} + \text{zero-input response}$$

The zero-state response of the system is the response of the system due to input alone when the initial state of the system is zero. That is the system is initially

relaxed at time $n = 0$. On the other hand the zero input response depends only on the initial state of the system, that is the input is zero.

For an example, let us consider a discrete-time system with difference equation

$$y(n) = ay(n-1) + x(n), \tag{3.2}$$

where $x(n)$ and $y(n)$ are input and output respectively. Let the input sequence $x(n)$ is zero for $n < 0$ and the initial condition of $y(n)$ for $n = -1$ exists. That is $y(-1) \neq 0$. Now let us compute successive values of $y(n)$ for $n \geq 0$ as follows.

For $n = 0$

$$\begin{aligned}
y(0) &= ay(-1) + x(0) \\
y(1) &= ay(0) + x(1) \\
&= a[ay(-1) + x(0)] + x(1) \\
&= a^2y(-1) + ax(0)] + x(1) \\
y(2) &= ay(1) + x(2) \\
&= a[a^2y(-1) + ax(0) + x(1)] + x(2) \\
&= a^3y(-1) + a^2x(0) + ax(1) + x(2)
\end{aligned}$$

For any n

$$\begin{aligned}
y(n) &= a^{n+1}y(-1) + a^nx(0) + a^{n-1}x(1) + \ldots + x(n) \\
&= a^{n+1}y(-1) + \sum_{k=0}^{n} a^kx(n-k) \qquad n \geq 0
\end{aligned} \tag{3.3}$$

The response $y(n)$ includes two parts. The first part depends on the initial condition of the system and the second term depends on the input. When $y(-1) = 0$, the output $y(n)$ depends only on the input applied, and hence $y(n)$ is known as the zero state response or forced response of the system given by

$$y_f(n) = \sum_{k=0}^{n} a^kx(n-k) \qquad n \geq 0 \tag{3.4}$$

If the system is initially not relaxed that is $y(-1) \neq 0$, and the input $x(n) = 0$ for all n, the output of the system $y(n)$, depends only on the initial state of the system. Then the response of the system is called the zero input response or natural response and is denoted by

$$y_n(n) = a^{n+1}y(-1) \qquad n \geq 0 \tag{3.5}$$

In brief, the zero-input response (natural response) is obtained by letting the input signal to zero. It depends on the nature of the system and initial conditions. On the other hand the zero-state response (forced response) depends on the nature of the input signal.

Any discrete-time system can be represented by an N^{th} order difference equation of the form

$$\sum_{k=0}^{N} a_ky(n-k) = \sum_{k=0}^{M} b_kx(n-k), \tag{3.6}$$

where N is called the order of the difference equation.

If the coefficient a_0 is not equal to one, we can divide Eq. (3.6) throughout by a_0 to normalize the equation.

The solution of the difference equation can be expressed as sum of two parts

$$y(n) = y_h(n) + y_p(n) \tag{3.7}$$

where $y_h(n)$ is known as homogeneous or complimentary solution and $y_p(n)$ is called particular solution.

3.3 Natural Response (zero-input response)

The natural response of the system is the solution of Eq. (3.6) when $x(n) = 0$. Hence for a discrete-time system the natural response is the solution of homogeneous equation

$$\sum_{k=0}^{N} a_ky(n-k) = 0 \tag{3.8}$$

The solution of Eq. (3.8) is of the form

$$y_h(n) = \lambda^n \tag{3.9}$$

Substituting Eq. (3.9) in Eq. (3.8) yields

$$\sum_{k=0}^{N} a_k\lambda^{n-k} = 0$$

or

$$\begin{aligned}
&\lambda^n + a_1\lambda^{n-1} + \ldots + a_{N-1}\lambda^{n-N-1} + a_N\lambda^{n-N} = 0 \qquad (a_0 = 1) \\
&\lambda^{n-N}\left[\lambda^N + a_1\lambda^{N-1} + \ldots + a_{N-1}\lambda + a_N\right] = 0
\end{aligned}$$

which gives

$$\lambda^N + a_1\lambda^{N-1} + \ldots + a_{N-1}\lambda + a_N = 0 \qquad (3.10)$$

The Eq. (3.10) is known as characteristic equation and has N roots which we denote as $\lambda_1, \lambda_2 \ldots \lambda_N$.

The factorized form of Eq. (3.10) is

$$(\lambda - \lambda_1)(\lambda - \lambda_2)\ldots(\lambda - \lambda_N) = 0 \qquad (3.11)$$

The nature of the natural response depends on the type of roots: real, imaginary and complex. The real roots leads to real exponential response, imaginary roots lead to sinusoidal oscillatory response and complex roots lead to exponentially damped or undamped sinusoidal response.

Distinct roots

If the roots $\lambda_1, \lambda_2 \ldots \lambda_N$ of Eq. (3.11) are distinct, then it has N solutions $c_1\lambda_1^k, c_2\lambda_2^k \ldots, c_N\lambda_N^k$. In that case the general solution is of the form

$$y_h(n) = c_1\lambda_1^n + c_2\lambda_2^n + \ldots + c_N\lambda_N^n \qquad (3.12)$$

where $c_1, c_2 \ldots c_N$ are arbitrary constants.

These constants are determined by applying initial conditions.

For example if the roots are $\lambda_1 = 3$ and $\lambda_2 = 4$, then

$$y_h(n) = c_1(3)^n + c_2(4)^n$$

Repeated roots

If a root λ_1 is repeated m times (roots of multiplicity m) and remaining $(N-m)$ roots are distinct, then the general solution is

$$y_h(n) = \left(c_1 + c_2 n + c_3 n^2 + \ldots + c_m n^{m-1}\right)(\lambda_1)^n + c_{m+1}(\lambda_{m+1})^n + c_{m+2}(\lambda_{m+2})^n + \ldots + c_N\lambda_N^n \qquad (3.13)$$

For example if the roots of the characteristic equation are $\lambda_1 = 2, \lambda_2 = 2$ and $\lambda_3 = 3$, then the solution

$$y_h(n) = [c_1 + c_2 n](2)^n + c_3(3)^n$$

Complex roots

If the roots are complex say λ and λ^*, then we can write

$$\lambda_1 = \lambda = a + jb$$
$$\lambda_2 = \lambda^* = a - jb.$$

Then the homogeneous solution is of the form

$$y_h(n) = r^n[A_1\cos n\theta + A_2\sin n\theta] \qquad (3.14)$$

where $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}\frac{b}{a}$,

A_1 and A_2 are constants.

Solved Problem 3.1 Find the natural response of the system described by difference equation

$$y(n) - 1.5y(n-1) + 0.5y(n-2) = x(n)$$
$$y(-1) = 1; y(-2) = 0$$

Solution:

Given

$$y(n) - 1.5y(n-1) + 0.5y(n-2) = x(n) \qquad (3.15)$$

The homogeneous equation can be obtained by equating the input terms to zero. That is

$$y(n) - 1.5y(n-1) + 0.5y(n-2) = 0 \qquad (3.16)$$

The homogeneous solution

$$y_h(n) = \lambda^n. \qquad (3.17)$$

Substituting this solution in Eq. (3.16) we get

$$\lambda^n - 1.5\lambda^{n-1} + 0.5\lambda^{n-2} = 0$$
$$\Rightarrow \lambda^{n-2}[\lambda^2 - 1.5\lambda + 0.5] = 0$$
$$\lambda^2 - 1.5\lambda + 0.5 = 0$$
$$(\lambda - 1)(\lambda - 0.5) = 0.$$
$$\lambda_1 = 1; \lambda_2 = 0.5$$

The general form of homogeneous solution is

$$y_h(n) = c_1(1)^n + c_2(0.5)^n \tag{3.18}$$

From Eq. (3.18)

$$y(0) = c_1 + c_2$$
$$y(1) = c_1 + 0.5c_2 \tag{3.19}$$

From the homogeneous equation

$$y(0) - 1.5y(-1) + 0.5y(-2) = 0.$$

Given $y(-1) = 1$ and $y(-2) = 0$.
Therefore

$$y(0) - 1.5(1) = 0$$
$$\Rightarrow \quad y(0) = 1.5$$

Similarly

$$y(1) - 1.5y(0) + 0.5y(-1) = 0$$
$$y(1) - 1.5(1.5) + 0.5(1) = 0$$
$$\Rightarrow \quad y(1) = 1.75 \tag{3.20 a}$$
$$y(0) = 1.5 \tag{3.20 b}$$

Comparing Eq. (3.19) and Eq. (3.20) we get

$$c_1 + c_2 = 1.5$$
$$c_1 + 0.5c_2 = 1.75$$
$$\Rightarrow \quad c_2 = -0.5$$
$$c_1 = 2$$

The natural response

$$y_n(n) = 2(1)^n - 0.5(0.5)^n \quad \text{for} \quad n \geq 0$$
$$= 2u(n) - 0.5(0.5)^n u(n) \tag{3.21}$$

3.4 Forced Response (zero state response)

The forced response is the solution of the difference equation for the given input when the initial conditions are zero. That is forced response is the response of a relaxed system for any input $x(n)$. It consists of two parts, homogeneous solution and particular solution. The homogeneous solution can be obtained from the roots of characteristic equation. The particular solution $y_p(n)$ is to satisfy the difference equation for the specific input signal $x(n), n \geq 0$. In otherwords, $y_p(n)$ is a solution satisfying

$$1 + \sum_{k=1}^{N} a_k y_p(n-k) = \sum_{k=0}^{M} b_k x(n-k) \tag{3.22}$$

The general form of the particular solution for several inputs are given in Table (3.1). From the Table (3.1) we can find that, if the input $x(n) = A\cos \omega n$, then $y_p(n) = c_1 \cos \omega n + c_2 \sin \omega n$, where c_1 and c_2 are obtained by substituting $y_p(n)$ and $x(n)$ in the difference equation.

Table 3.1 General form of particular solution for several types of inputs

$x(n)$ input signal	$y_p(n)$ particular solution
A (Step input)	k
AM^n	kM^n
An^M	$k_0 n^M + k_1 n^{M-1} + \ldots k_M$
$A^n N^M$	$A^n[k_0 n^M + k_1 n^{M-1} + \ldots k_M]$
$\left.\begin{matrix} A\cos \omega n \\ A\sin \omega n \end{matrix}\right\}$	$c_1 \cos \omega n + c_2 \sin \omega n$

Note: Here A, k, M, k_i, c_1 and c_2 are constants.

If the input applied to the system and one of the components of the homogeneous solution are equal, then multiply the particular solution by the lower power of n that will give a response components not included in the homogeneous solution. For example, if the homogeneous solution contain the term $c_1(\lambda_1)^n$ and the input is $x(n) = (\lambda_1)^n$ then we assume a particular solution of the form $y_p(n) = c_2 n(\lambda_1)^n$.

The forced response of the system is obtained by summing the particular solution and homogeneous solution and finding the coefficients in the homogeneous solution so that the combined response $y_h(n) + y_p(n)$, satisfies the zero initial conditions.

Solved Problem 3.2 Find the forced response of the system described by the difference equation

$$y(n) - 1.5y(n-1) + 0.5y(n-2) = x(n)$$

for an input $x(n) = 2^n u(n)$

Solution:

From solved problem (3.1) we can find the homogeneous solution of the system is of the form

$$y_h(n) = c_1(1)^n + c_2(0.5)^n. \tag{3.23}$$

For input $x(n) = 2^n u(n)$ the particular solution is of the form (see table 3.1) $y_p(n) = k2^n u(n)$

Substituting $y_p(n)$ and $x(n)$ in the difference equation we get

$$k2^n u(n) - 1.5k2^{n-1}u(n-1) + 0.5k2^{n-2}u(n-2) = 2^n u(n)$$

For $n = 2$ where none of the terms vanish

$$\begin{aligned} k(2)^2 - 1.5k(2) + 0.5(k) &= 2^2 \\ \Rightarrow \quad 4k - 3k + 0.5k &= 4 \\ 1.5k &= 4 \\ \Rightarrow \quad k &= \frac{8}{3} \end{aligned}$$

Select the value of n such that no term vanishes

Therefore the particular solution

$$y_p(n) = \frac{8}{3}2^n u(n)$$

The forced response

$$\begin{aligned} y_f(n) &= y_h(n) + y_p(n) \\ &= c_1(1)^n + c_2(0.5)^n + \frac{8}{3}2^n u(n) \end{aligned} \tag{3.24}$$

From Eq. (3.24)

$$\begin{aligned} y(0) &= c_1 + c_2 + \frac{8}{3} \\ y(1) &= c_1 + 0.5c_2 + \frac{16}{3} \end{aligned} \tag{3.25}$$

From the difference equation

$$y(0) - 1.5y(-1) + 0.5(y)(-2) = x(0)$$

$$\Rightarrow \quad y(0) = 1 \qquad \boxed{\therefore \quad y(-1) = y(-2) = 0}$$

Similarly

$$\begin{aligned} y(1) - 1.5y(0) + 0.5y(-1) &= x(1) \\ y(1) - 1.5 &= 2 \\ y(1) &= 3.5 \end{aligned} \tag{3.26}$$

Comparing Eq. (3.25) and Eq. (3.26) we get

$$c_1 + c_2 + \frac{8}{3} = 1 \quad \Rightarrow \quad c_1 + c_2 = \frac{-5}{3}$$

and

$$c_1 + 0.5c_2 + \frac{16}{3} = \frac{7}{2} \quad \Rightarrow \quad c_1 + 0.5c_2 = \frac{-11}{6}$$

That is

$$\begin{aligned} c_1 + c_2 &= \frac{-5}{3} \\ c_1 + 0.5c_2 &= \frac{-11}{6} \end{aligned}$$

Solving for c_1 and c_2 we get

$$\begin{aligned} c_1 &= -2 \\ c_2 &= \frac{1}{3} \end{aligned}$$

The forced response

$$\begin{aligned} y_f(n) &= -2(1)^n + \frac{1}{3}(0.5)^n + \frac{8}{3}(2)^n \quad \text{for} \quad n \geq 0 \\ &= -2u(n) + \frac{1}{3}(0.5)^n u(n) + \frac{8}{3}(2)^n u(n) \end{aligned} \tag{3.27}$$

3.5 Total Response

The total response is obtained by adding the natural response and forced response

$$y(n) = y_n(n) + y_f(n)$$

If there is no need to find the forced response and natural response separately, the total response can be found in the same way as forced response by using the actual initial conditions instead of zero initial conditions.

Solved Problem 3.3 For the given system

$$y(n) - 1.5y(n-1) + 0.5y(n-2) = x(n)$$

with initial conditions $y(-1) = 1; y(-2) = 0$ find the response due to input $x(n) = 2^n u(n)$

Solution:

The total response of the system

$$y(n) = y_n(n) + y_f(n)$$

The natural response of the given system is already found in solved problem (3.1), that is,

$$y_n(n) = 2u(n) - 0.5(0.5)^n u(n)$$

and the forced response for input $x(n) = 2^n u(n)$ is found in solved problem (3.2) given by

$$y_f(n) = -2u(n) + \frac{1}{3}(0.5)^n u(n) + \frac{8}{3}(2)^n u(n)$$

The total response

$$\begin{aligned} y(n) &= y_f(n) + y_n(n) \\ &= \frac{-1}{6}(0.5)^n u(n) + \frac{8}{3}(2)^n u(n) \end{aligned}$$

If the natural and forced response are not in need to find separately then we can find the total response by proceeding in the same way as forced response upto Eq. (3.24). That is

$$y(n) = c_1(1)^n + c_2(0.5)^n + \frac{8}{3}2^n u(n) \tag{3.28}$$

Now we find c_1 and c_2 by substituting actual initial conditions rather than zero initial conditions.

From Eq. (3.28)

$$\begin{aligned} y(0) &= c_1 + c_2 + \frac{8}{3} \\ y(1) &= c_1 + 0.5c_2 + \frac{16}{3} \end{aligned} \tag{3.29}$$

From the difference equation, with initial conditions $y(-1) = 1$ and $y(-2) = 0$ we get

$$\begin{aligned} y(0) &= 1.5y(-1) - 0.5y(-2) + x(0) \\ &= 1.5 + 1 = 2.5 \\ y(1) &= 1.5y(0) - 0.5y(-1) + x(1) \\ &= 1.5(2.5) - 0.5 + 2 \\ &= 5.25 \end{aligned} \tag{3.30}$$

Comparing Eq. (3.29) and Eq. (3.30)

$$\begin{aligned} c_1 + c_2 + \frac{8}{3} &= 2.5 \\ c_1 + 0.5c_2 + \frac{16}{3} &= 5.25 \\ \Rightarrow \quad c_1 + c_2 &= \frac{-1}{6} \end{aligned}$$

and

$$c_1 + 0.5c_2 = \frac{-1}{12}$$

from which

$$c_1 = 0; c_2 = \frac{-1}{6}$$

Therefore

$$\begin{aligned} y(n) &= -\frac{1}{6}(0.5)^n + \frac{8}{3}(2)^n \quad \text{for } n \geq 0 \\ &= -\frac{1}{6}(0.5)^n u(n) + \frac{8}{3}(2)^n u(n) \end{aligned} \tag{3.31}$$

Practice Problem 3.1

i) Find the natural response of the system described by difference equation

$$y(n) + 2y(n-1) + y(n-2) = x(n) + x(n-1)$$

with initial conditions $y(-1) = y(-2) = 1$

Ans: $-3(-1)^n - 2n(-1)^n$

ii) Find the forced response of the system described by difference equation

$$y(n)+2y(n-1)+y(n-2)=x(n)+x(n-1)$$

for the input $x(n)=\left(\frac{1}{2}\right)^n u(n)$ Ans: $\frac{2}{3}(-1)^n+\frac{1}{3}\left(\frac{1}{2}\right)^n$

iii) Find the response of the system with difference equation

$$y(n)+2y(n-1)+y(n-2)=x(n)+x(n-1)$$

for the input $x(n)=\left(\frac{1}{2}\right)^n u(n)$ with initial conditions $y(-1)=y(-2)=1$

Ans: $-\frac{7}{3}(-1)^n-2n(-1)^n+\frac{1}{3}\left(\frac{1}{2}\right)^n$

3.6 Impulse Response

The general form of difference equation of a N^{th} order system is given by

$$1+\sum_{k=1}^{N} a_k y(n-k)=\sum_{k=0}^{M} b_k x(n-k) \qquad N>M \tag{3.32}$$

For input $x(n)=\delta(n)$, the system equation can be written as

$$1+\sum_{k=1}^{N} a_k y(n-k)=\sum_{k=0}^{M} b_k \delta(n-k) \tag{3.33}$$

For $n>M$, Eq. (3.33) reduces to homogeneous equation

$$\sum_{k=1}^{N} a_k y(n-k)=0 \qquad a_0=1. \tag{3.34}$$

We can obtain $h(n)$ by solving Eq. (3.34) and imposing the initial conditions to determine the arbitrary constants.

Note: If $N=M$, we have to add an impulse function to the homogeneous solution.

Solved Problem 3.4 Determine the impulse response $h(n)$ for the system described by the second-order difference equation

$$y(n)=0.6y(n-1)-0.08y(n-2)+x(n)$$

Solution:

Given

$$y(n)=0.6y(n-1)-0.08y(n-2)+x(n)$$

We know the total response

$$y(n)=y_h(n)+y_p(n). \tag{3.35}$$

For impulse $x(n)=\delta(n)$, the particular solution

$$y_p(n)=0$$
$$\Rightarrow \quad y(n)=y_h(n).$$

The homogeneous solution can be found by substituting $x(n)=0$

$$\Rightarrow \quad y(n)-0.6y(n-1)+0.08y(n-2)=0 \tag{3.36}$$

Let the solution be

$$y_h(n)=\lambda^n \tag{3.37}$$

Substituting Eq. (3.37) in Eq. (3.36) we obtain

$$\lambda^n-0.6\lambda^{n-1}+0.08\lambda^{n-2}=0$$
$$\lambda^{n-2}[\lambda^2-0.6\lambda+0.08]=0$$
$$\Rightarrow \quad \lambda^2-0.6\lambda+0.08=0$$

The roots of the characteristic equation are

$$\lambda_1=0.4;\lambda_2=0.2$$

The general form of the solution of the homogeneous equation is

$$y_h(n)=c_1\lambda_1^n+c_2\lambda_2^n$$
$$=c_1(0.4)^n+c_2(0.2)^n \tag{3.38}$$

From Eq. (3.38)

$$y(0)=c_1+c_2$$
$$y(1)=0.4c_1+0.2c_2 \tag{3.39}$$

From the difference equation we have

$$y(0) = 0.6y(-1) - 0.08y(-2) + x(0)$$
$$= 1 \qquad \boxed{\begin{matrix} y(-1) = y(-2) = 0 \\ x(0) = \delta(0) = 1 \end{matrix}}$$
$$y(1) = 0.6y(0) - 0.08y(-1) + x(1)$$
$$= 0.6(1) - 0.08(0) + 0$$
$$= 0.6$$

$$\Rightarrow \quad y(0) = 1$$
$$y(1) = 0.6 \qquad (3.40)$$

Comparing Eq. (3.39) and Eq. (3.40)

$$c_1 + c_2 = 1$$
$$0.4c_1 + 0.2c_2 = 0.6$$

and solving for c_1 and c_2 we get

$$c_1 = 2$$
$$c_2 = -1$$

Substituting the values in Eq. (3.38) yields

$$y(n) = 2(0.4)^n u(n) - (0.2)^n u(n) \qquad (3.41)$$

Solved Problem 3.5 Determine the impulse response $h(n)$ for the system described by difference equation

$$y(n) + y(n-1) - 2y(n-2) = x(n-1) + 2x(n-2)$$

Solution:

Given

$$y(n) + y(n-1) - 2y(n-2) = x(n-1) + 2x(n-2).$$

Since $M = N = 2$, the homogeneous solution includes an impulse term.

The total response is given by

$$y(n) = y_h(n) + y_p(n)$$

For input $x(n) = \delta(n)$, the particular solution $y_p(n) = 0$

$$\Rightarrow \quad y(n) = y_h(n)$$

The homogeneous solution can be found by equating the input terms to zero, that is

$$y(n) + y(n-1) - 2y(n-2) = 0 \qquad (3.42)$$

Let the homogeneous solution be $y_h(n) = \lambda^n$. Substituting this solution in Eq. (3.42), we obtain the characteristic equation

$$\lambda^n + \lambda^{n-1} - 2\lambda^{n-2} = 0$$
$$\lambda^{n-2}[\lambda^2 + \lambda - 2] = 0$$
$$\Rightarrow \quad \lambda^2 + \lambda - 2 = 0$$

Therefore, the roots are 1, -2 and the general form of the solution to the homogeneous equation is

$$y_h(n) = c_1(1)^n + c_2(-2)^n + A\delta(n) \qquad (3.43)$$

From the difference equation

$$y(0) + y(-1) - 2y(-2) = x(-1) + 2x(-2)$$
$$y(0) = 0$$
$$y(1) + y(0) - 2y(-1) = x(0) + 2x(-1)$$
$$y(1) = 1$$
$$\Rightarrow \quad y(0) = 0$$
$$y(1) = 1 \qquad (3.44)$$
$$y(2) = 1$$

Substituting $n = 0$ and $n = 1$ in Eq. (3.43) we get

$$y(0) = c_1 + c_2 + A$$
$$y(1) = c_1 - 2c_2 \qquad (3.45)$$
$$y(2) = c_1 - 4c_2$$

Comparing Eq. (3.44) and Eq. (3.45) we get

$$c_1 + c_2 + A = 0$$
$$c_1 - 2c_2 = 1$$
$$c_1 - 4c_2 = 1$$

from which $c_1 = 1; c_2 = 0; A = -1$.
Substituting these values in Eq. (3.43) yields

$$\begin{aligned} y(n) &= u(n) - \delta(n) \\ &= u(n-1) \end{aligned} \tag{3.46}$$

Practice Problem 3.2

Determine the impulse response $h(n)$ for the systems described by second-order difference equation as below

(i) $y(n) - 4y(n-1) + 4y(n-2) = x(n) - x(n-1)$ UPTU
Ans: $[(2)^n + \frac{n}{2}(2)^n]\,u(n)$

(ii) $y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$
Ans: $[\frac{6}{5}(4)^n - \frac{1}{5}(-1)^n]\,u(n)$

(iii) $y(n) - 0.1y(n-1) - 0.3y(n-2) = x(n) + x(n-1)$
Ans: $\frac{16}{11}(0.6)^n - \frac{5}{11}(0.5)^n$

Solved Problem 3.6 Determine the step response of the system described by difference equation

$$y(n) + 4y(n-1) + 4y(n-2) = x(n)$$

Solution:

Given $$y(n) + 4y(n-1) + 4y(n-2) = x(n)$$

The total solution is

$$y(n) = y_h(n) + y_p(n)$$

The homogeneous solution, $y_h(n)$ can be obtained by equating $x(n) = 0$.
That is

$$y(n) + 4y(n-1) + 4y(n-2) = 0 \tag{3.47}$$

Let the homogeneous solution

$$y_h(n) = \lambda^n. \tag{3.48}$$

Substituting Eq. (3.48) in Eq. (3.47) yields

$$\begin{aligned} \lambda^n + 4\lambda^{n-1} + 4\lambda^{n-2} &= 0 \\ \lambda^{n-2}[\lambda^2 + 4\lambda + 4] &= 0 \\ \Rightarrow \quad \lambda^2 + 4\lambda + 4 &= 0 \\ (\lambda+2)^2 &= 0 \\ \Rightarrow \quad \lambda_1 = \lambda_2 &= -2 \end{aligned}$$

The homogeneous solution for the repeated roots is given by

$$y_h(n) = (c_1 + c_2 n)(-2)^n \tag{3.49}$$

For input $x(n) = u(n)$, the particular solution is of the form (see table 3.1)

$$y_p(n) = ku(n) \tag{3.50}$$

Substituting the particular solution in the difference equation we get

$$\begin{aligned} & y_p(n) + 4y_p(n-1) + 4y_p(n-2) = x(n) \\ \Rightarrow\ & ku(n) + 4ku(n-1) + 4ku(n-2) = u(n) \end{aligned}$$

For $n = 2$, where none of the terms vanish, we get

$$\begin{aligned} k + 4k + 4k &= 1 \\ \Rightarrow \quad k &= \frac{1}{9} \\ \Rightarrow y_p(n) &= \frac{1}{9}u(n) \end{aligned} \tag{3.51}$$

The total solution

$$\begin{aligned} y(n) &= y_h(n) + y_p(n) \\ &= (c_1 + nc_2)(-2)^n + \frac{1}{9}u(n) \quad \text{For } n \geq 0 \end{aligned} \tag{3.52}$$

From Eq. (3.52)

$$\begin{aligned} y(0) &= c_1 + \frac{1}{9} \\ y(1) &= (c_1 + c_2)(-2) + \frac{1}{9} \end{aligned} \tag{3.53}$$

From the difference equation

$$y(0)+4y(-1)+4y(-2)=x(n) \qquad \boxed{x(n)=u(n)}$$

$$\Rightarrow \quad y(0)+4(0)+4(0)=1$$

$$y(0)=1$$

$$y(1)+4y(0)+4y(-1)=x(1)$$

$$y(1)+4(1)+4(0)=1$$

$$y(1)=1-4=-3 \tag{3.54}$$

Comparing Eq. (3.53) and Eq. (3.54) we get

$$c_1+\frac{1}{9}=1$$

$$\Rightarrow \quad c_1=\frac{8}{9}$$

$$(c_1+c_2)(-2)+\frac{1}{9}=-3$$

$$\left(\frac{8}{9}+c_2\right)(-2)+\frac{1}{9}=-3$$

$$-\frac{16}{9}-2c_2+\frac{1}{9}=-3$$

$$\Rightarrow \quad c_2=\frac{2}{3}$$

Substitute the values of c_1 and c_2 in Eq. (3.52) yields

$$y(n)=\left(\frac{8}{9}+\frac{2}{3}n\right)(-2)^n u(n)+\frac{1}{9}u(n) \tag{3.55}$$

Solved Problem 3.7 Determine the natural response (zero-input response) of the system described by the following difference equations

i) $y(n)+\frac{3}{4}y(n-1)+\frac{1}{8}y(n-2)=x(n)+x(n-1);\ y(-1)=0;y(-2)=1$

ii) $y(n)-\frac{9}{16}y(n-2)=x(n-1);y(-1)=1;y(-2)=-1$

Solution:

i) The natural response (zero-input response) can be found by equating the input terms to zero. That is

$$y(n)+\frac{3}{4}y(n-1)+\frac{1}{8}y(n-2)=0 \tag{3.56}$$

Let the homogeneous solution $y_h(n)=\lambda^n$. Then substituting this solution in Eq. (3.56) yields

$$\lambda^n+\frac{3}{4}\lambda^{n-1}+\frac{1}{8}\lambda^{n-2}=0$$

$$\lambda^{n-2}\left[\lambda^2+\frac{3}{4}\lambda+\frac{1}{8}\right]=0$$

$$\Rightarrow \quad \lambda^2+\frac{3}{4}\lambda+\frac{1}{8}=0$$

$$\left(\lambda+\frac{1}{2}\right)\left(\lambda+\frac{1}{4}\right)=0$$

$$\lambda_1=-\frac{1}{2} \quad \text{and} \quad \lambda_2=-\frac{1}{4}$$

The general form of homogeneous solution is

$$y_h(n)=c_1\left(-\frac{1}{2}\right)^n+c_2\left(-\frac{1}{4}\right)^n \tag{3.57}$$

From Eq. (3.57)

$$y(0)=c_1+c_2$$

$$y(1)=\frac{-1}{2}c_1-\frac{1}{4}c_2$$

From Eq. (3.56)

$$y(0)+\frac{3}{4}y(-1)+\frac{1}{8}y(-2)=0$$

Given $y(-1)=0;y(-2)=1$

$$\Rightarrow y(0)=-\frac{1}{8} \tag{3.58}$$

Similarly

$$y(1)+\frac{3}{4}y(0)+\frac{1}{8}y(-1)=0$$

$$y(1)+\frac{3}{4}\left(-\frac{1}{8}\right)+\frac{1}{8}(0)=0$$

$$y(1)=\frac{3}{32} \tag{3.59}$$

Comparing Eq. (3.58) and Eq. (3.59)

$$c_1 + c_2 = -\frac{1}{8}$$
$$-\frac{1}{2}c_1 - \frac{1}{4}c_2 = \frac{3}{32}$$
$$\Rightarrow \quad c_1 = \frac{-1}{4} \quad \text{and} \quad c_2 = \frac{1}{8}$$

Therefore the natural response

$$\Rightarrow \quad y_n(n) = \frac{-1}{4}\left(-\frac{1}{2}\right)^n - \frac{1}{8}\left(-\frac{1}{4}\right)^n \tag{3.60}$$

ii) Given

$$y(n) - \frac{9}{16}y(n-2) = x(n-1).$$

The homogeneous equation can be obtained by equating the input term to zero

$$y(n) - \frac{9}{16}y(n-2) = 0 \tag{3.61}$$

The homogeneous solution

$$y_h(n) = \lambda^n \tag{3.62}$$

Substituting this solution in Eq. (3.61) we get

$$\lambda^n - \frac{9}{16}\lambda^{n-2} = 0$$
$$\lambda^{n-2}\left[\lambda^n - \frac{9}{16}\right] = 0$$
$$\lambda = \pm\frac{3}{4}$$
$$\lambda_1 = \frac{3}{4}; \lambda_2 = \frac{-3}{4}$$

The general form of homogeneous solution is

$$y_h(n) = c_1\left(\frac{3}{4}\right)^n u(n) + c_2\left(\frac{-3}{4}\right)^n u(n) \tag{3.63}$$

For $n = 0$

$$y(0) = c_1 + c_2$$

and for $n = 1$

$$y(1) = \frac{3}{4}c_1 - \frac{3}{4}c_2 \tag{3.64}$$

From the difference equation

$$y(0) = \frac{9}{16}y(-2) = \frac{-9}{16}$$
$$y(1) = \frac{9}{16}y(-1) = \frac{9}{16} \tag{3.65}$$

comparing Eq. (3.64) and Eq. (3.65)

$$c_1 + c_2 = -\frac{9}{16}$$
$$\frac{3}{4}c_1 - \frac{3}{4}c_2 = \frac{9}{16}$$
$$c_1 = \frac{3}{32}; c_2 = \frac{-21}{32}$$
$$\Rightarrow \quad y(n) = \frac{3}{32}\left(\frac{3}{4}\right)^n u(n) - \frac{21}{32}\left(\frac{-3}{4}\right)^n u(n)$$

Solved Problem 3.8 Find the forced response of the following difference equations

i) $y(n) - y(n-1) = x(n)$ where $x(n) = \cos 2n$

ii) $y(n) - 7y(n-1) + 12y(n-2) = x(n)$ for $x(n) = n$

Solution:

Given

$$y(n) - y(n-1) = \cos 2n \tag{3.66}$$

The total response

$$y(n) = y_h(n) + y_p(n). \tag{3.67}$$

For the input $x(n) = \cos 2n$ the particular solution is of the form (see table 3.1)

$$y_p(n) = A\cos 2n + B\sin 2n. \tag{3.68}$$

Substituting Eq. (3.68) in the difference equation we have

$$(A\cos 2n + B\sin 2n) - [A\cos 2(n-1) + B\sin 2(n-1)] = \cos 2n$$
$$[A\cos 2n + B\sin 2n] - [A\cos(2n)\cos 2 + A\sin(2n)\sin 2 + B\sin(2n)\cos 2 - B\cos(2n)\sin 2] = \cos 2n$$

Comparing L.H.S and R.H.S terms we have

$$A(1-\cos 2) + B\sin 2 = 1$$
$$-A\sin 2 + B(1-\cos 2) = 0$$

from which

$$B = \frac{\sin 2}{\sin^2 2 + (1-\cos 2)^2} = \frac{\sin 2}{\sin^2 2 + 1 + \cos^2 2 - 2\cos 2}$$
$$= \frac{\sin 2}{2(1-\cos 2)} = \frac{2\sin 1\cos 1}{2\cdot 2\sin^2 1} = \frac{1}{2}\cot 1$$
$$A = \frac{1-\cos 2}{(1-\cos 2)^2 + \sin^2 2} = \frac{1-\cos 2}{1+\cos^2 2 + \sin^2 2 - 2\cos 2}$$
$$= \frac{1-\cos 2}{2(1-\cos 2)} = \frac{1}{2}$$
$$\Rightarrow \quad y_p(n) = \frac{1}{2}\cos 2n + \frac{1}{2}(\cot 1)\sin 2n$$
$$= \frac{1}{2}\cos 2n + \frac{1}{2}\frac{\cos 1}{\sin 1}\sin 2n$$
$$= \frac{1}{2}\left[\frac{\cos 2n\sin 1 + \cos 1\sin 2n}{\sin 1}\right]$$
$$= \frac{1}{2}\frac{\sin(2n+1)}{\sin 1} \qquad (3.69)$$

The homogeneous solution is $y_h(n) = \lambda^n$.

Substituting λ^n in difference equation yields

$$y(n) - y(n-1) = 0 \text{ we get}$$
$$\Rightarrow \quad \lambda^n - \lambda^{n-1} = 0$$
$$\lambda - 1 = 0$$
$$\lambda = 1$$
$$y_h(n) = Cu(n)$$

The total solution is

$$y(n) = Cu(n) + \frac{1}{2}\frac{\sin(2n+1)}{\sin 1} \qquad (3.70)$$

Let $n = 0$ in the Eq. (3.70), we have

$$y(0) = C + \frac{1}{2}$$

Similarly from the difference equation we have

$$y(0) = 1$$
$$\Rightarrow C + \frac{1}{2} = 1 \quad C = 1/2$$
$$\Rightarrow y(n) = \frac{1}{2}u(n) + \frac{1}{2}\frac{\sin(2n+1)}{\sin 1} \qquad (3.71)$$

ii) Given

$$y(n) - 7y(n-1) + 12y(n-2) = n$$

The homogeneous solution can be obtained from the characteristic equation

$$y(n) - 7y(n-1) + 12y(n-2) = 0$$

Let $y(n) = \lambda^n$, then the characteristic equation is

$$\lambda^2 - 7\lambda + 12 = 0$$
$$\Rightarrow \quad \lambda_1 = 4; \lambda_2 = 3$$
$$y_h(n) = C_1(4)^n + C_2(3)^n \qquad (3.72)$$

For input $x(n) = n$, the particular solution is of the form (see table 3.1)

$$y_p(n) = k_0 n + k_1 \qquad (3.73)$$

Substituting $y_p(n)$ in the difference equation we get

$$(k_0 n + k_1) - 7[k_0(n-1) + k_1] + 12[k_0(n-2) + k_1] = n$$

Comparing the coefficients of n on both sides, we get

$$6k_0 = 1; \quad \Rightarrow \quad k_0 = \frac{1}{6}.$$

Similarly comparing the constant terms on both sides, we get

$$k_1 + 7k_0 - 7k_1 - 24k_0 + 12k_1 = 0$$

$$\Rightarrow \quad 6k_1 - 17k_0 = 0$$

$$6k_1 = \frac{17}{6}$$

$$k_1 = \frac{17}{36}$$

$$\Rightarrow \quad y_p(n) = \frac{n}{6} + \frac{17}{36} \tag{3.74}$$

The total response

$$y(n) = c_1(4)^n + c_2(3)^n + \frac{n}{6} + \frac{17}{36} \tag{3.75}$$

From Eq. (3.75)

$$y(0) = C_1 + C_2 + \frac{17}{36}$$

$$y(1) = 4C_1 + 3C_2 + \frac{23}{36}$$

From the difference equation

$$y(0) = 0$$

$$y(1) = 1$$

$$\Rightarrow \quad C_1 + C_2 + \frac{17}{36} = 0$$

$$4C_1 + 3C_2 + \frac{23}{36} = 1$$

$$\Rightarrow \quad C_1 + C_2 = \frac{-17}{36}$$

$$4C_1 + 3C_2 = \frac{13}{36}$$

$$C_2 = \frac{-81}{36} = \frac{-9}{4}$$

$$C_1 = \frac{-17}{36} + \frac{9}{4}$$

$$= \frac{64}{36} = \frac{16}{9}$$

$$\Rightarrow \quad y(n) = \frac{16}{9}(4)^n u(n) - \frac{9}{4}(3)^n u(n) + \frac{n}{6} + \frac{17}{36}$$

Practice Problem 3.3

Determine the step response of the following systems

(i) $y(n) + 3y(n-1) + 2y(n-2) = x(n)$

Ans: $\frac{4}{3}(-2)^n - \frac{1}{2}(-1)^n + \frac{1}{2}$

(ii) $y(n) + 0.4y(n-1) - 0.21y(n-2) = x(n) + 2x(n-1)$

Ans: $-0.5353(-0.7)^n - 0.9857(0.3)^n + 2.521u(n)$

(iii) $y(n) - 2\cos\theta y(n-1) + y(n-2) = x(n) - \cos\theta x(n-1)$

Ans: $\frac{1}{2}\left[1 + \frac{\sin(n+\frac{1}{2})\theta}{\sin\frac{\theta}{2}}\right]$

Solved Problem 3.9 Determine the response $y(n), n \geq 0$, of the system described by the second-order difference equation

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1) \tag{3.76}$$

for the input $x(n) = 4^n u(n)$.

Solution:

Given

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

The total solution

$$y(n) = y_h(n) + y_p(n).$$

The homogeneous solution, $y_h(n)$ can be obtained by equating the input terms to zero. That is

$$y(n) - 3y(n-1) - 4y(n-2) = 0.$$

Let $y_h(n) = \lambda^n$

The difference equation becomes

$$\lambda^n - 3\lambda^{n-1} - 4\lambda^{n-2} = 0$$

$$\lambda^{n-2}[\lambda^2 - 3\lambda - 4] = 0$$

$$\lambda^2 - 3\lambda - 4 = 0$$

$$(\lambda - 4)(\lambda + 1) = 0$$

$$\lambda_1 = 4;\ \lambda_2 = -1$$

The general form of homogeneous solution is

$$y_h(n) = C_1(\lambda_1)^n + C_2(\lambda_2)^n$$
$$= C_1(4)^n + C_2(-1)^n$$

For input $x(n) = 4^n u(n)$, the particular solution is of the form (see table 3.1)

$$y_p(n) = k\,4^n u(n)$$

But the particular solution is having the same term as natural response.
Therefore multiply the particular solution by n

$$y_p(n) = kn\,4^n u(n). \tag{3.77}$$

Substituting the particular solution $y_p(n)$ in Eq. (3.76) yields

$$y_p(n) - 3y_p(n-1) - 4y_p(n-2) = x(n) + 2x(n-1)$$

$$\Rightarrow kn4^n u(n) - 3k(n-1)4^{n-1}u(n-1) - 4(n-2)k(4)^{n-2}u(n-2)$$
$$= 4^n u(n) + 2(4)^{n-1}u(n-1)$$

For $n = 2$ where none of the terms varnish

$$k(2)(16) - 3k(4) = 16 + 2(4)$$
$$32k - 12k = 24$$
$$k = 6/5$$
$$y_p(n) = \frac{6n}{5}(4)^n u(n)$$
$$y(n) = C_1(4)^n + C_2(-1)^n + \frac{6n}{5}(4)^n u(n) \tag{3.78}$$

From Eq. (3.78), we have

$$y(0) = C_1 + C_2$$
$$y(1) = 4C_1 - C_2 + \frac{24}{5} \tag{3.79}$$

From the difference Eq. (3.76), we have

$$y(0) - 3y(-1) - 4y(-2) = x(0) + 2x(-1) \quad \boxed{x(n) = 4^n u(n)}$$
$$y(0) = 1$$
$$y(1) - 3y(0) - 4y(-1) = x(1) + 2x(0)$$
$$y(1) - 3 = 4 + 2$$
$$y(1) = 9 \tag{3.80}$$

Comparing Eq. (3.79) and Eq. (3.80) we get

$$C_1 + C_2 = 1$$

and

$$4C_1 - C_2 + \frac{24}{5} = 9$$
$$\Rightarrow \quad 4C_1 - C_2 = \frac{21}{5}$$

Solving for C_1 and C_2 we get

$$C_1 = \frac{26}{25}\ ;\ C_2 = \frac{-1}{25}$$

The solution is given by

$$y(n) = \frac{26}{25}(4)^n u(n) - \frac{1}{25}(-1)^n u(n) + \frac{6n}{5}(4)^n u(n)$$

Solved Problem 3.10 Find the solution to the following linear constant coefficient difference equation

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = \left(\frac{1}{4}\right)^n \quad \text{for} \quad n \geq 0 \tag{3.81}$$

with initial conditions $y(-1) = 4$ and $y(-2) = 10$.

Solution:

Given

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = \left(\frac{1}{4}\right)^n$$

The homogeneous solution can be obtained by equating input to zero

$$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = 0$$

Let the solution be

$$y_h(n) = \lambda^n,$$

then we get

$$\lambda^n - \frac{3}{2}\lambda^{n-1} + \frac{1}{2}\lambda^{n-2} = 0$$
$$\lambda^{n-2}\left[\lambda^2 - \frac{3}{2}\lambda + \frac{1}{2}\right] = 0$$
$$\Rightarrow \quad \lambda^2 - \frac{3}{2}\lambda + \frac{1}{2} = 0$$

which is characteristic equation.
The roots are

$$\lambda_1 = 1; \lambda_2 = \frac{1}{2}$$

The homogeneous solution is of the form

$$y_h(n) = C_1(1)^n + C_2\left(\frac{1}{2}\right)^n \tag{3.82}$$

For the input $x(n) = \left(\frac{1}{4}\right)^n$, the particular solution is of the form (see table 3.1)

$$y_p(n) = k\left(\frac{1}{4}\right)^n. \tag{3.83}$$

Substituting Eq. (3.83) in Eq. (3.81), we have

$$k\left(\frac{1}{4}\right)^n - \frac{3}{2}k\left(\frac{1}{4}\right)^{n-1} + \frac{1}{2}k\left(\frac{1}{4}\right)^{n-2} = \left(\frac{1}{4}\right)^n$$

For $n \geq 2$ where none one of the terms varnish

$$k\left(\frac{1}{4}\right)^2 - \frac{3}{2}k\left(\frac{1}{4}\right) + \frac{k}{2} = \frac{1}{16}$$

$$\frac{1}{16}k - \frac{3k}{8} + \frac{k}{2} = \frac{1}{16}$$

$$\frac{3}{16}k = \frac{1}{16}$$

$$\Rightarrow \quad k = \frac{1}{3}$$

The total solution

$$y(n) = y_h(n) + y_p(n)$$

$$y(n) = C_1(1)^n + C_2\left(\frac{1}{2}\right)^n + \frac{1}{3}\left(\frac{1}{4}\right)^n \tag{3.84}$$

From Eq. (3.84)

$$y(0) = C_1 + C_2 + \frac{1}{3}$$

$$y(1) = C_1 + \frac{1}{2}C_2 + \frac{1}{12}$$

From the difference equation, for $n = 0$

$$y(0) - \frac{3}{2}y(-1) + \frac{1}{2}y(-2) = 1$$

$$\Rightarrow \quad y(0) = 1 + \frac{3}{2}y(-1) - \frac{1}{2}y(-2)$$

$$= 1 + \frac{3}{2}(4) - \frac{1}{2}(10) \tag{3.85}$$

$$= 2$$

Similarly when $n = 1$

$$y(1) - \frac{3}{2}y(0) + \frac{1}{2}y(-1) = \frac{1}{4}$$

$$\Rightarrow y(1) = \frac{1}{4} + \frac{3}{2}(2) - \frac{1}{2}(4) = \frac{5}{4} \tag{3.86}$$

Comparing Eq. (3.85) and Eq. (3.86) we get

$$C_1 + C_2 + \frac{1}{3} = 2$$

$$\Rightarrow \quad C_1 + C_2 = \frac{5}{3}.$$

$$\boxed{\begin{aligned} C_1 + C_2 &= \frac{5}{3} \\ C_1 + \frac{1}{2}C_2 &= \frac{14}{12} \end{aligned}}$$

Similarly

$$C_1 + \frac{1}{2}C_2 + \frac{1}{12} = \frac{5}{4}$$

$$\Rightarrow C_1 + \frac{1}{2}C_2 = \frac{14}{12}$$

$$\boxed{\begin{aligned} \frac{1}{2}C_2 &= \frac{5}{3} - \frac{14}{12} \\ \frac{1}{2}C_2 &= \frac{2-14}{12} \\ C_2 &= 1 \\ C_1 &= \frac{2}{3} \end{aligned}}$$

Solving for C_1 and C_2 we get

$$C_1 = \frac{2}{3}$$

$$C_2 = 1.$$

Substituting C_1 and C_2 in Eq. (3.84) yields. The complete solution

$$y(n) = \frac{2}{3}u(n) + \left(\frac{1}{2}\right)^n u(n) + \frac{1}{3}\left(\frac{1}{4}\right)^n u(n) \tag{3.87}$$

Solved Problem 3.11 Find the response of the system described by difference equation

$$y(n)-\frac{1}{2}y(n-1)-\frac{1}{2}y(n-2)=\left(\frac{1}{2}\right)^n \quad \text{for} \quad n\geq 0$$
$$y(-1)=1;y(-2)=0$$

Solution:

Let the solution be $y_h(n)=\lambda^n$

The characteristic equation is

$$\lambda^2-\frac{1}{2}\lambda-\frac{1}{2}=0$$
$$(\lambda-1)\left(\lambda+\frac{1}{2}\right)=0$$
$$\lambda_1=1;\lambda_2=\frac{-1}{2}$$

The general form of homogeneous solution is

$$y_h(n)=C_1(1)^n+C_2\left(\frac{-1}{2}\right)^n \tag{3.88}$$

If the input $x(n)=\left(\frac{1}{2}\right)^n u(n)$, then the particular solution is (see table 3.1)

$$y_p(n)=k\left(\frac{1}{2}\right)^n u(n) \tag{3.89}$$

Substituting this solution in difference equation, we get

$$k\left(\frac{1}{2}\right)^n-\frac{1}{2}k\left(\frac{1}{2}\right)^{n-1}-\frac{1}{2}k\left(\frac{1}{2}\right)^{n-2}=\left(\frac{1}{2}\right)^n$$

When $n=2$, we have

$$\frac{1}{4}k-\frac{1}{4}k-\frac{1}{2}k=\frac{1}{4}$$
$$-\frac{1}{2}k=\frac{1}{4}$$
$$k=-\frac{1}{2}$$

The particular solution is

$$y_p(n)=\frac{-1}{2}\left(\frac{1}{2}\right)^n \tag{3.90}$$

The total response is

$$y(n)=y_h(n)+y_p(n)$$
$$=C_1(1)^n+C_2\left(\frac{-1}{2}\right)^n-\frac{1}{2}\left(\frac{1}{2}\right)^n \tag{3.91}$$

From Eq. (3.91), we have

$$y(0)=C_1+C_2-\frac{1}{2}$$
$$y(1)=C_1-\frac{1}{2}C_2-\frac{1}{4} \tag{3.92}$$

Similarly from the difference equation, we have

$$y(0)-\frac{1}{2}y(-1)-\frac{1}{2}y(-2)=1$$
$$y(0)-\frac{1}{2}(1)-\frac{1}{2}(0)=1$$
$$y(0)=\frac{3}{2}$$
$$y(1)=\frac{1}{2}\left(\frac{3}{2}\right)+\frac{1}{2}(1)+\frac{1}{2}$$
$$=\frac{7}{4}$$
$$C_1+C_2-\frac{1}{2}=\frac{3}{2}$$
$$C_1-\frac{1}{2}C_2-\frac{1}{4}=\frac{7}{4}$$

$$C_1+C_2=2$$
$$C_1-\frac{1}{2}C_2=2$$
$$\Rightarrow \quad C_1=2;C_2=0$$

Therefore

$$y(n)=2u(n)-\frac{1}{2}\left(\frac{1}{2}\right)^n u(n) \tag{3.93}$$

Solved Problem 3.12 Find the impulse response of the system

$$y(n)+2y(n-1)+2y(n-2)=x(n) \tag{3.94}$$

Solution:

Equate input term to zero to get

$$y(n)+2y(n-1)+2y(n-2)=0 \tag{3.95}$$

The characteristic equation is

$$\lambda^2+2\lambda+2=0$$

and roots are $-1\pm j$.

The homogeneous solution is of the form

$$y_h(n)=r^n[A_1\cos n\theta+A_2\sin n\theta] \tag{3.96}$$

where $r=\sqrt{(-1)^2+(1)^2}=\sqrt{2}$.

$$\theta=\tan^{-1}(-1)=135^\circ=\frac{3\pi}{4}$$

$$\Rightarrow\quad y_h(n)=2^{n/2}\left[A_1\cos\frac{3n\pi}{4}+A_2\sin\frac{3n\pi}{4}\right] \tag{3.97}$$

From Eq. (3.94) we have

$$\begin{aligned}y(0)&=1\\ y(1)&=-2\end{aligned} \tag{3.98}$$

From Eq. (3.97)

$$\begin{aligned}y(0)&=A_1\\ y(1)&=2^{1/2}\left[A_1\cos\frac{3\pi}{4}+A_2\sin\frac{3\pi}{4}\right]\end{aligned} \tag{3.99}$$

Comparing Eq. (3.98) and Eq. (3.99) we get

$$A_1=1$$

$$2^{1/2}\left[A_1\cos\frac{3\pi}{4}+A_2\sin\frac{3\pi}{4}\right]=-2$$

$$\Rightarrow\quad 2^{1/2}\left[\frac{-1}{\sqrt{2}}+\frac{A_2}{\sqrt{2}}\right]=-2$$

$$-1+A_2=-2$$

$$A_2=-1$$

$$\Rightarrow\quad y_h(n)=2^{n/2}\left[\cos\frac{3n\pi}{4}-\sin\frac{3n\pi}{4}\right]$$

$$=\sqrt{2}(2^{n/2})\left[\frac{1}{\sqrt{2}}\cos\frac{3n\pi}{4}-\frac{1}{\sqrt{2}}\sin\frac{3n\pi}{4}\right]$$

$$\sqrt{2}(2^{n/2})\left[\cos\left(\frac{3n\pi}{4}+\theta\right)\right]\quad\text{where }\theta=\tan^{-1}(1)=45^\circ=\frac{\pi}{4}$$

$$=2^{(n+1)/2}\left[\cos(3n+1)\frac{\pi}{4}\right] \tag{3.100}$$

Practice Problem 3.4

Find the response of the following difference equations

(i) $y(n)+2y(n-1)+y(n-2)=x(n)$ where $x(n)=\cos 2n$

(ii) $y(n)-4y(n-1)+3y(n-2)=x(n)$ for $x(n)=n$

(iii) $y(n)-7y(n-1)+12y(n-2)=x(n)$ for $x(n)=2^n u(n)$

In the second method of analyzing the behaviour of a linear system, first we decompose the input signal into sum of elementary signals. Then using the linearity property, the response of the system to the elementary signals are added to obtain the total response of the system.

Review Questions

1. Define natural response and forced response.
2. Explain how you can find the total response of a system using classical method.

3.7 Representation of Discrete-Time Signals in Terms of Impulses

The unit-impulse can be used as a basic signal to construct any discrete-time signal. That is any discrete time signal can be represented as a sum of shifted, scaled unit impulse sequences. Consider a discrete-time signal shown in Fig. 3.1a.

The sample at $n = 0$ can be obtained by multiplying $x(0)$, the magnitude with impulse $\delta(n)$ as shown in Fig. 3.1c.

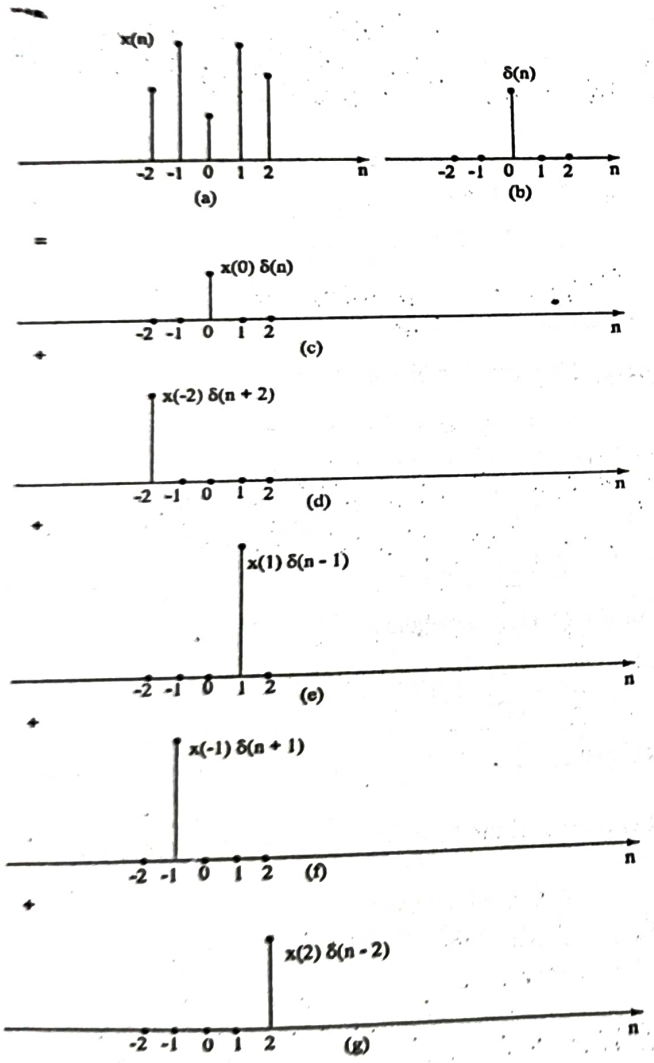


Fig. 3.1 Representation of a sequence as a sum of delayed impulses.

That is

$$\begin{aligned} x(0)\delta(n) &= x(0) \quad \text{for} \quad n = 0 \\ &= 0 \quad \text{for} \quad n \neq 0 \end{aligned} \tag{3.101}$$

Similarly the sample $x(-2)$ can be obtained by multiplying $x(-2)$, the magnitude, with two samples advanced unit impulse $\delta(n+2)$ as shown in Fig. 3.1d. That is

$$\begin{aligned} x(-2)\delta(n+2) &= x(-2) \quad \text{for} \quad n = -2 \\ &= 0 \quad \text{for} \quad n \neq -2 \end{aligned} \tag{3.102}$$

In the same way we can obtain the following

$$\begin{aligned} x(-1)\delta(n+1) &= x(-1) \quad \text{for} \quad n = -1 \\ &= 0 \quad \text{for} \quad n \neq -1 \end{aligned} \tag{3.103}$$

$$\begin{aligned} x(1)\delta(n-1) &= x(1) \quad \text{for} \quad n = 1 \\ &= 0 \quad \text{for} \quad n \neq 1 \end{aligned} \tag{3.104}$$

$$\begin{aligned} x(2)\delta(n-2) &= x(2) \quad \text{for} \quad n = 2 \\ &= 0 \quad \text{for} \quad n \neq 2 \end{aligned} \tag{3.105}$$

The sum of the above five sequences is

$$x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) + x(2)\delta(n-2)$$

equals to the given sequence $x(n)$.

In general, we can write $x(n)$ for $-\infty \leq n \leq \infty$ as

$$\begin{aligned} x(n) &= \ldots\ldots x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) \\ &\quad + x(2)\delta(n-2) + \ldots\ldots \\ &= \sum_{k=-\infty}^{\infty} x(n)\delta(n-k) \end{aligned} \tag{3.106}$$

where $\delta(n-k)$ is unity for $n = k$ and zero for all other n.

Review Question

1. Prove that any discrete-time signal can be represented as a sum of shifted, scaled unit impulse sequences.

Solved Problem 3.13 Represent the sequence $x(n) = \{3, \underset{\uparrow}{2}, -1, 2, 4, 1\}$ as sum of shifted unit impulses

Solution:

Given

$$x(n) = \{3, \underset{\uparrow}{2}, -1, 2, 4, 1\}$$

index n $\quad -1 \quad 0 \quad 1 \quad 2 \quad 3 \quad 4$

From $x(n)$ we can find

$$x(-1) = 3; x(0) = 2; x(1) = -1; x(2) = 2; x(3) = 4; x(4) = 1.$$

We know

$$x(n) = \sum_{n=-\infty}^{\infty} x(k)\delta(n-k)$$

But $x(n)$ has values from $n = -1$ to $n = 4$. Therefore

$$x(n) = \sum_{n=-1}^{4} x(k)\delta(n-k)$$

$$= x(-1)\delta(n+1) + x(0)\delta(n) + x(1)\delta(n-1) + x(2)\delta(n-2) + x(3)\delta(n-3) + x(4)\delta(n-4)$$

Substituting the sequence values yields

$$= 3\delta(n+1) + 2\delta(n) - \delta(n-1) + 2\delta(n-2) + 4\delta(n-3) + \delta(n-4)$$

3.8 Impulse Response and Convolution Sum

A discrete time system performs an operation on an input signal based on a predefined criteria to produce a modified output signal. The input signal to the system is $x(n)$ and the output of the system is $y(n)$. This transform operation is shown in Fig. 3.2.

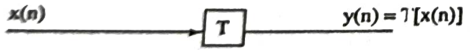


Fig. 3.2 A Discrete-time system

If the input to the system is a unit impulse then the output of the system is known as impulse response of the system denoted by $h(n)$. That is

$$h(n) = T[\delta(n)]. \tag{3.107}$$

We know that any arbitrary sequence $x(n)$ can be represented as a weighted sum of discrete impulses given by

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k). \tag{3.108}$$

Then the output

$$y(n) = T\left[\sum_{k=-\infty}^{\infty} x(k)\delta(n-k)\right] \tag{3.109}$$

Using the linearity property of the system, interchange the system operator T with the summation of $x(k)$ to obtain

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)T[\delta(n-k)] \tag{3.110}$$

The response to the shifted impulse sequence can be denoted by $h(n,k)$

$$h(n,k) = T[\delta(n-k)] \tag{3.111}$$

Now

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n,k) \tag{3.112}$$

For a time-invariant system

$$h(n,k) = h(n-k) \tag{3.113}$$

Substituting Eq. (3.113) in Eq. (3.112) we obtain

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \tag{3.114}$$

Thus the output of a LTI system is given by weighted sum of time-shifted responses.

The sum in Eq. (3.114) is termed the convolution sum and can be represented as

$$y(n) = x(n) * h(n), \tag{3.115}$$

where $*$ denotes the convolution operation.

For a linear time invariant system of the input sequence $x(n)$ and impulse response $h(n)$ are given, we can find the output $y(n)$ by using convolution sum. This is an extremely powerful result that allows us to compute the system output for any input signal excitation.

Review Questions

1. Define impulse response and convolution sum.
2. Prove that the output of a discrete-time system can be represented as a weighted sum of shifted impulses.

3.9 Properties of Convolution

3.9.1 The distributive property

$$x(n) * [h_1(n) + h_2(n)] = x(n) * h_1(n) + x(n) * h_2(n) \qquad (3.116)$$

Consider two LTI systems with impulse responses $h_1(n)$ and $h_2(n)$ connected in parallel as shown in Fig. 3.3.

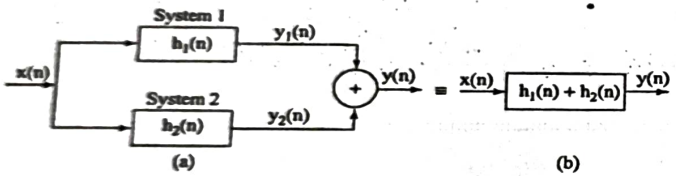


Fig. 3.3 (a) Parallel connection of two systems. (b) Equivalent system.

From Fig. 3.3, the output of system1 $y_1(n) = x(n) * h_1(n)$ and the output of system2 $y_2(n) = x(n) * h_2(n)$

$$\begin{aligned}
\text{The output } y(n) &= y_1(n) + y_2(n) \\
&= x(n) * h_1(n) + x(n) * h_2(n) \\
&= \sum_{k=-\infty}^{\infty} x(k)h_1(n-k) + \sum_{k=-\infty}^{\infty} x(k)h_2(n-k) \\
&= \sum_{k=-\infty}^{\infty} x(k)[h_1(n-k) + h_2(n-k)] \\
&= \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\
&= x(n) * h(n),
\end{aligned}$$

where $h(n) = h_1(n) + h_2(n)$.

Thus if the two systems are connected in parallel the overall impulse response is equal to sum of two impulse responses of the systems acting independently.

3.9.2 The associative property

$$[x(n) * h_1(n)] * h_2(n) = x(n) * [h_1(n) * h_2(n)] \qquad (3.117)$$

Consider two LTI systems with impulse responses $h_1(n)$ and $h_2(n)$ connected in cascade, as shown in Fig.3.4.

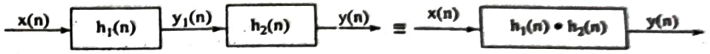


Fig. 3.4 (a) Cascade connection of two systems. (b) Equivalent system.

Let $y_1(n)$ is the output of the first system. Then

$$\begin{aligned}
y_1(n) &= x(n) * h_1(n) \\
&= \sum_{k=-\infty}^{\infty} x(k)h_1(n-k)
\end{aligned}$$

and

$$\begin{aligned}
y(n) &= y_1(n) * h_2(n) \\
&= \left[\sum_{k=-\infty}^{\infty} x(k)h_1(n-k)\right] * h_2(n) \\
&= \sum_{p=-\infty}^{\infty}\sum_{k=-\infty}^{\infty} x(p)h_1(k-p)h_2(n-k)
\end{aligned}$$

Let $l = k - p$ then

$$\begin{aligned}
y(n) &= \sum_{p=-\infty}^{\infty} x_1(p) \sum_{l=-\infty}^{\infty} h_1(l)h_2(n-p-l) \\
&= \sum_{p=-\infty}^{\infty} x_1(p)h(n-p) \\
y(n) &= x(n) * h(n)
\end{aligned}$$

where $h(n) = \sum_{l=-\infty}^{\infty} h_1(l)h_2(n-l)$

$$= h_1(n) * h_2(n)$$

Hence the impulse response of two LTI systems connected in cascade is the convolution of individual impulse responses.

Solution:

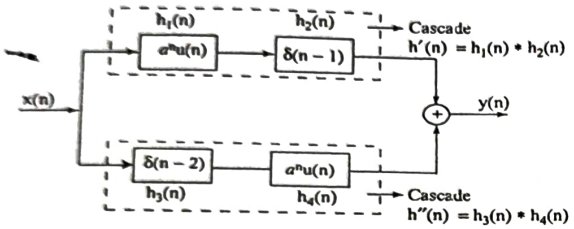


Fig. 3.6

The systems with impulse responses $h_1(n)$ and $h_2(n)$ are cascaded. Therefore

$$\begin{aligned} h'(n) &= h_1(n) * h_2(n) \\ &= a^n u(n) * \delta(n-1) \\ &= a^{n-1}\, u(n-1) \end{aligned}$$

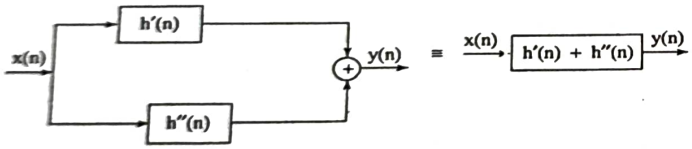


Fig. 3.7

Similarly

$$\begin{aligned} h''(n) &= h_3(n) * h_4(n) \\ &= \delta(n-2) * a^n u(n) \\ &= a^{n-2} u(n-2) \end{aligned}$$

$$\begin{aligned} \text{Now} \quad y(n) &= h'(n) + h''(n) \\ &= a^{n-1} u(n-1) + a^{n-2} u(n-2) \end{aligned}$$

Practice Problem 3.1

An interconnection of LTI systems is shown in Fig. 3.8.
The impulse responses are $h_1(n) = \left(\frac{1}{2}\right)^n [u(n) - u(n-3)]$; $h_2(n) = \delta(n)$ and $h_3(n) = u(n-1)$. Let the impulse response of the overall system from $x(n)$ to $y(n)$ be denoted as $h(n)$.

(a) Express $h(n)$ in terms of $h_1(n)$, $h_2(n)$ and $h_3(n)$.

(b) Evaluate $h(n)$

Ans: (a) $h(n) = h_1(n) * [h_2(n) + h_3(n)]$

(b) $2[1 - \left(\frac{1}{2}\right)^n]u(n-1) + \left[\frac{1}{4} - 2\left(\frac{1}{2}\right)^n\right] u(n-4) + \left(\frac{1}{2}\right)^n [u(n) - u(n-3)]$

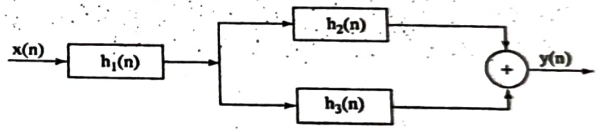


Fig. 3.8

3.10 Convolution of Two Sequences

The convolution sum of two sequences can be found by using the following steps.

Step 1: Choose an initial value of n; the starting time for evaluating the output sequence $y(n)$. If $x(n)$ start at $n = n_1$ and $h(n)$ starts at $n = n_2$, then $n = n_1 + n_2$ is a good choice.

Step 2: Express both sequences in terms of the index k.

Step 3: Fold $h(k)$ about $k = 0$ to obtain $h(-k)$ and shift right by n if n is positive and shift left if n is negative to obtain $h(n-k)$.

Step 4: Multiply the two sequences $x(k)$ and $h(n-k)$ element by element and sum the products to get $y(n)$.

Step 5: Increment the index n, shift the sequence $h(n-k)$ to right by one sample and do step 4.

Step 6: Repeat step 5 until the sum of products is zero for all remaining values of n.

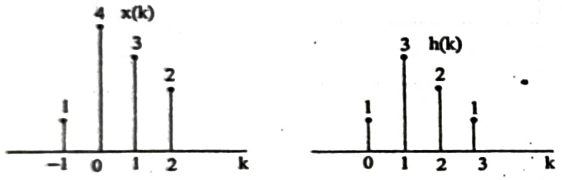
Solved Problem 3.16 Determine the convolution sum of two sequences

$$x(n) = \{1, 4, 3, 2\}; h(n) = \{1, 3, 2, 1\}$$
$$\uparrow$$

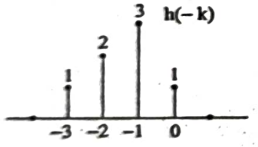
Solution:

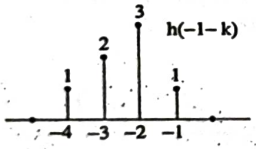
Step 1: The sequence $x(n)$ starts at $n_1 = -1$ and $h(n)$ starts at $n_2 = 0$. Therefore the starting time for evaluating the output sequence $y(n)$ is $n = n_1 + n_2 = -1 + 0 = -1$.

Step 2: Express both sequences in terms the index k.



Step 3: Fold $h(k)$ about $k = 0$, to obtain $h(-k)$





As starting time to evaluate $y(n)$ is -1, shift $h(k)$ by one unit to left to obtain $h(-1-k)$.

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$$

Multiply the two sequences $x(k)$ and $h(-1-k)$ element by element and sum the products

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k)$$
$$= (0)(1) + (0)(2) + 0(3) + (1)(1) + 4(0) + 3(0) + 2(0) = 1$$

Increment the index by 1, i.e., n to zero, shift the sequence to right to obtain $h(-k)$ and then multiply the two sequences $x(k)$ and $h(-k)$ element by element and sum the products we get

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$$
$$= (0)(1) + (0)(2) + 1(3) + 4(1) + 3(0) + 2(0) = 7$$

Similarly,

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$$
$$= 0(1) + 1(2) + 4(3) + 3(1) + 2(0) = 17$$

$$y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$$
$$= 1(1) + 4(2) + 3(3) + 2(1) = 20$$

$$y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$$
$$= 1(0) + 4(1) + 3(2) + 2(3) + 0(1) = 16$$

$$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$$
$$= 3.1 + 2.2 = 7$$

$$y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k) = 2.1 = 2$$

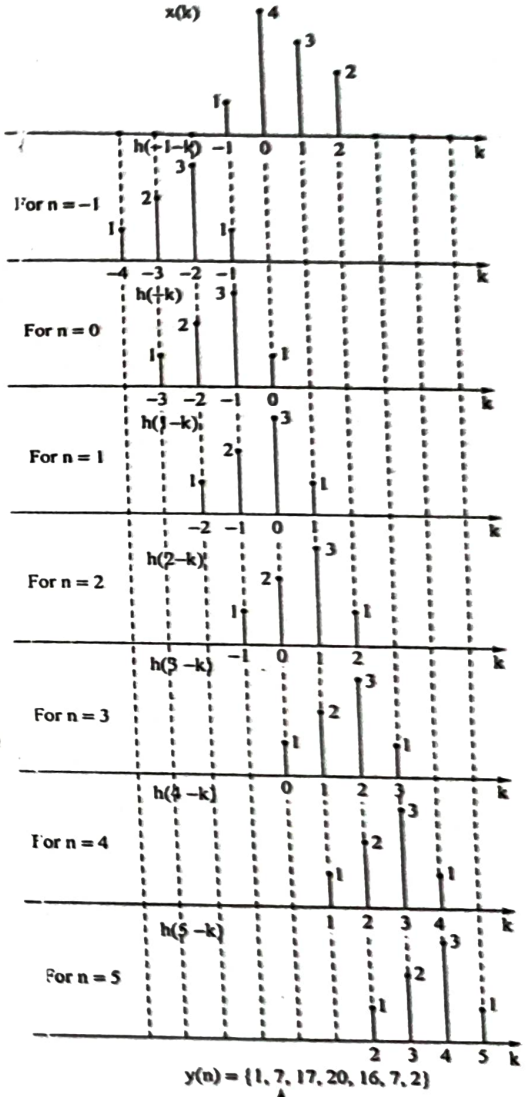


Fig. 3.9

For remaining values of n the value $y(n) = 0$. Therefore we stop at $n = 5$.

Solved Problem 3.17 Determine the output response $y(n)$ if

$$x(n) = \{1,2,3,2\}; \quad h(n) = \{1,2,2\}$$

Solution:

Given

$$x(n) = \{1,2,3,2\}; \quad h(n) = \{1,2,2\}$$

Both sequences starts at $n = 0$.

Therefore the output $y(n)$ also starts at $n = 0$.

We know

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k)$$

$$= 1.1 = 1$$

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k)$$

$$= 1.2 + 2.1 = 4$$

$$y(2) = \sum_{k=-\infty}^{\infty} x(k)h(2-k)$$

$$= 1.2 + 2.2 + 3.1 = 9$$

$$y(3) = \sum_{k=-\infty}^{\infty} x(k)h(3-k)$$

$$= 2.2 + 3.2 + 2.1 = 12$$

$$y(4) = \sum_{k=-\infty}^{\infty} x(k)h(4-k)$$

$$= 3.2 + 2.2 + 0.1 = 10$$

$$y(5) = \sum_{k=-\infty}^{\infty} x(k)h(5-k)$$

$$= 2.2 = 4$$

$$y(n) = \{1,4,9,12,10,4\}$$

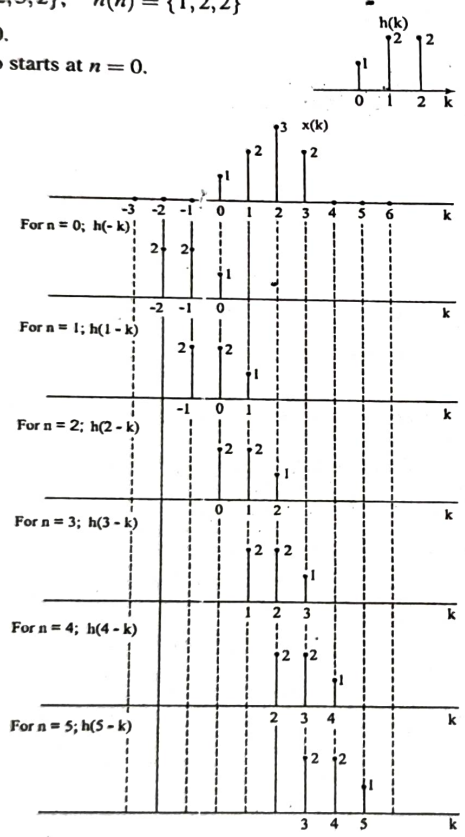


Fig. 3.10

The convolution of sequences $x_1(n)$ and $x_2(n)$ can also be found using following methods.

3.10.1 Matrix convolution

Method 1

The convolution of two sequences $x_1(n)$ and $x_2(n)$ can be obtained by representing the sequences in matrix form by using the steps given below.

Step 1: Write down the sequences $x_1(n)$ and $x_2(n)$ as shown. (Here it is assumed that the initial value of $n = 0$)

$x_2(n)$ \ $x_1(n)$	$x_1(0)$	$x_1(1)$	$x_1(2)$	$x_1(3)$	.....
$x_2(0)$					
$x_2(1)$					
$x_2(2)$					
$x_2(3)$					
⋮					

(3.121)

Step 2: Multiply each and every sample in $x_1(n)$ with the samples of $x_2(n)$ and formulate a matrix.

$x_2(n)$ \ $x_1(n)$	$x_1(0)$	$x_1(1)$	$x_1(2)$	$x_1(3)$	- -
$x_2(0)$	$x_1(0)\,x_2(0)$	$x_1(1)\,x_2(0)$	$x_1(2)\,x_2(0)$	$x_1(3)\,x_2(0)$	
$x_2(1)$	$x_1(0)\,x_2(1)$	$x_1(1)\,x_2(1)$	$x_1(2)\,x_2(1)$	$x_1(3)\,x_2(1)$	
$x_2(2)$	$x_1(0)\,x_2(2)$	$x_1(1)\,x_2(2)$	$x_1(2)\,x_2(2)$	$x_1(3)\,x_2(2)$	
$x_2(3)$	$x_1(0)\,x_2(3)$	$x_1(1)\,x_2(3)$	$x_1(2)\,x_2(3)$	$x_1(3)\,x_2(3)$	

(3.122)

Step 3: Divide the elements in matrix by drawing diagonal lines.
Step 4: Starting from left sum all the elements in each strip and write down in the same order.

If the sequences $x_1(n)$ and/or $x_2(n)$ are not starting at $n = 0$, then find the value of n for which $y(n)$ has initial value and mark the symbol ↑ at time origin ($n = 0$).

Solved Problem 3.18 Find the convolution of the sequences

$$x_1(n) = \{1, -1, 2, 3\}; x_2(n) = \{1, -2, 3, -1\}$$
$$\uparrow$$

Solution:

Step 1: Write down the sequence $x_1(n)$ and $x_2(n)$ as shown

$x_2(n)$ \ $x_1(n)$	1	−1	2	3
1				
−2				
3				
−1				

Step 2: Multiply each and every sample in $x_2(n)$ with the samples of $x_1(n)$ and form a matrix

$x_2(n)$ \ $x_1(n)$	1	−1	2	3
1	1×1	−1×1	2×1	3×1
−2	1×(−2)	−1×(−2)	2×(−2)	3×(−2)
3	1×3	(−1)×3	2×3	3×3
−1	1×(−1)	(−1)×(−1)	2×(−1)	3×(−1)

Step 3: Divide the elements in the matrix by drawing diagonal lines as shown below.

$x_2(n)$ \ $x_1(n)$	1	−1	2	3
1	1	−1	2	3
−2	−2	2	−4	−6
3	3	−3	6	9
−1	−1	1	−2	−3

Step 4: Starting from left sum all the elements in each strip and write down in the same order

$$1, -2+(-1), 3+2+2, (-1)+(-3)+(-4)+3, 1+6+(-6), (-2)+(9), -3$$
$$= 1, -3, 7, -5, 1, 7, -3$$

The starting value of n is -1, therefore mark the symbol ↑ at the time origin

$$y(n) = \{1, -3, 7, -5, 1, 7, -3\}$$
$$\uparrow$$

Method 2

If the number of elements in $x_1(n)$ are N_1 and that of $x_2(n)$ are N_2, then to find the convolution of $x_1(n)$ and $x_2(n)$ form the following matrices.

1. Matrix X_2 of order $(N_1 + N_2 - 1) \times N_1$ with the elements of $x_2(n)$.
2. A column matrix, X_1 of order $(N_1 \times 1)$ with the elements of $x_1(n)$.

Multiply the matrices X_2 and X_1 to get a column matrix Y of order $(N_1 + N_2 - 1) \times 1$, that has elements of $y(n)$, the convolution $x_1(n)$ and $x_2(n)$.

$$\underbrace{\begin{bmatrix} x_2(0) & 0 & 0 & \dots & 0 \\ x_2(1) & x_2(0) & 0 & \dots & 0 \\ \vdots & & & & \\ x_2(N_2-1) & x_2(N_2-2) & 0 & \dots & x_2(0) \\ 0 & x_2(N_2-1) & & \dots & x_2(1) \\ \vdots & \vdots & & & \\ 0 & 0 & & \dots & x_2(N_2-1) \end{bmatrix}}_{X_2} \underbrace{\begin{bmatrix} x_1(0) \\ x_1(1) \\ \vdots \\ x_1(N_1-1) \end{bmatrix}}_{X_1} = \underbrace{\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N_2-1) \\ y(N_2) \\ \vdots \\ y(N_1+N_2-1) \end{bmatrix}}_{Y} \quad (3.123)$$

Solved Problem 3.19 Find the convolution of

$$x_1(n) = \{1, -2, 3, 1\}$$

$$x_2(n) = \{2, -3, -2\}$$

Solution:

Given the sequences

$$x_1(n) = \{1, -2, 3, 1\}; x_2(n) = \{2, -3, -2\}$$

Both sequences are starting at $n = 0$ and

$$x_1(0) = 1; x_1(1) = -2; x_1(2) = 3; x_1(3) = 1$$
$$x_2(0) = 2; x_2(1) = -3; x_2(2) = -2$$

Substitute the sequence values in the matrix of Eq. (3.123) and multiply to get $y(n)$.

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ -3 & 2 & 0 & 0 \\ -2 & -3 & 2 & 0 \\ 0 & -2 & -3 & 2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \\ 10 \\ -3 \\ -9 \\ -2 \end{bmatrix}$$

$$y(n) = \{2, -7, 10, -7, -9, -2\}$$

Solved Problem 3.20 Find the convolution of the following sequence

$$x(n) = 2\delta(n+1) - \delta(n) + \delta(n-1) + 3\delta(n-2)$$
$$h(n) = 3\delta(n-1) + 4\delta(n-2) + 2\delta(n-3)$$

Solution:

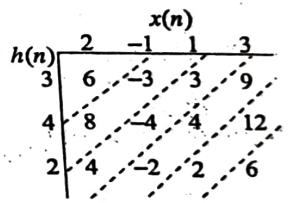
Given

$$x(n) = \{2, -1, 1, 3\}; \quad n_1 = -1$$
$$\uparrow$$
$$h(n) = \{3, 4, 2\}; \quad n_2 = 1$$

Method 1

$$n_1 + n_2 = 0.$$

Therefore, the sequence $y(n)$ starts at $n = 0$. By following the steps given for matrix convolution, we obtain



$$= 6, 8-3, 4-4+3, -2+4+9, 12+2, 6$$
$$y(n) = \{6, 5, 3, 11, 14, 6\}$$
$$\uparrow$$

Method 2

Given the sequences

$$x(n) = \{2, -1, 1, 3\}; h(n) = \{3, 4, 2\}$$

Writing the sequences in matrix form and multiplying we get

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 2 & 4 & 3 & 0 \\ 0 & 2 & 4 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 3 \\ 11 \\ 14 \\ 6 \end{bmatrix}$$

$$y(n) = \{6, 5, 3, 11, 14, 6\}$$

3.11 Causality

In section (2.2.4) we already studied that a causal system is one whose output depends on past or/and present values of the input. Using convolution sum we have

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \tag{3.124}$$

$$= \sum_{k=-\infty}^{-1} h(k)x(n-k) + \sum_{k=0}^{\infty} h(k)x(n-k)$$

$$= \ldots\ldots \underbrace{h(-2)x(n+2)+h(-1)x(n+1)}_{\text{depends on future inputs}} + \underbrace{h(0)x(n)}_{\text{present input}} + \underbrace{h(1)x(n-1)+\ldots}_{\text{depends on past inputs}}$$

From Eq. (3.124) we find that the output depends on the past and present values of the input if the index $k \geq 0$. If $k < 0$ then the output depends on the future values of the input. Therefore for a causal system whose output doesnot depends on future values of the input, the limits on the summation changes as

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) \tag{3.125}$$

From Eq. (3.125) we find that for a causal system $h(k)$ should be zero for $k < 0$. That is

$$h(k) = 0 \qquad \text{for} \quad k < 0 \tag{3.126}$$

> An LTI system is causal if and only if its impulse response is zero for negative values of n.

The limits in the convolution sum can be modified according to the type of sequence and system. For a causal system, the impulse response $h(n) = 0$ for $n < 0$. Therefore the limits in the convolution sum is modified as

$$y(n) = \sum_{k=-\infty}^{n} x(k)h(n-k) \tag{3.127}$$

$$= \sum_{k=0}^{\infty} h(k)x(n-k) \tag{3.128}$$

If the input to the causal system is a causal sequence (i.e., $x(n) = 0$ for $n < 0$) the limits in the convolution sum is modified as

$$y(n) = \sum_{k=0}^{n} x(k)h(n-k) \tag{3.129}$$

$$= \sum_{k=0}^{n} h(k)x(n-k) \tag{3.130}$$

Solved Problem 3.21 Find the convolution of two infinite duration sequences

$h(n) = a^n u(n)$ for all n

$x(n) = b^n u(n)$ for all n

(i) when $a \neq b$

(ii) when $a = b$

Solution:

The impulse response $h(n) = 0$ for $n < 0$, so the given system is causal and $x(n) = 0$ for $n < 0$, hence the sequence is a causal sequence. Therefore $y(n) = 0$ for $n < 0$ and

Using Eq. (3.129) we have

$$y(n) = \sum_{k=0}^{n} x(k)h(n-k)$$

$$= \sum_{k=0}^{n} b^k a^{n-k} = a^n \sum_{k=0}^{n} \left(\frac{b}{a}\right)^k \tag{3.131}$$

$$= a^n \left[1 + \frac{b}{a} + \frac{b^2}{a^2} \cdots (n+1)\text{terms}\right] \qquad \because \sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$$

$$= a^n \left[\frac{1-(b/a)^{n+1}}{1-b/a}\right] \quad \text{for } n \geq 0 \tag{3.132}$$

when $a = b$, the Eq. (3.132) reduces to indeterminate form.

Therefore, applying L Hospital's rule we get

$$y(n) = a^n \lim_{b \to a} \left[\frac{-(-1/a)^{n+1}(n+1)b^n}{(-1/a)}\right]$$

$$= a^n \lim_{b \to a} (n+1)\left(\frac{b}{a}\right)^n = a^n(n+1) \tag{3.133}$$

when $a = b$ the Eq. (3.130) reduces to

$$y(n) = a^n \sum_{k=0}^{n} (1)^k = a^n(n+1) \quad \text{for } n \geq 0 \tag{3.134}$$

$$\Rightarrow \quad y(n) = a^n(n+1)u(n)$$

$$\because \quad 1+1+1\cdots(n+1) \text{ terms} = (n+1)$$

Solved Problem 3.22 Determine the response of the relaxed system characterized by the impulse response $h(n) = \left(\frac{1}{3}\right)^n u(n)$ to the input signal $x(n) = 2^n u(n)$.

Solution:

Given

$$x(n) = 2^n u(n) \quad \rightarrow \text{causal signal}$$

$$h(n) = \left(\frac{1}{3}\right)^n u(n) \quad \rightarrow \text{causal system}$$

A causal signal is applied to a causal system.

Therefore,

$$y(n) = \sum_{k=0}^{n} x(k)h(n-k)$$

$$= \sum_{k=0}^{n} 2^k \left(\frac{1}{3}\right)^{n-k}$$

$$= \left(\frac{1}{3}\right)^n \sum_{k=0}^{n} 2^k (3)^k$$

$$= \left(\frac{1}{3}\right)^n \sum_{k=0}^{n} (6)^k$$

$$= \left(\frac{1}{3}\right)^n \left[\frac{(6)^{n+1}-1}{6-1}\right]$$

$$\Rightarrow \quad y(n) = \left(\frac{1}{3}\right)^n \left[\frac{(6)^{n+1}-1}{5}\right] u(n) \tag{3.135}$$

Solved Problem 3.23 Given the causal system

$$y(n) - y(n-1) = x(n) + x(n-1)$$

Find the response to $x(n) = u(n)$. Use convolution sum.

Solution:

The response $y(n) = x(n) * h(n)$. The impulse response $h(n)$ can be obtained from the difference equation. The homogeneous equation can be obtained by equating input terms to zero. That is

$$y(n) - y(n-1) = 0 \tag{3.136}$$

Let homogeneous solution $y_h(n) = \lambda^n$. Substituting this solution in Eq. (3.136) we obtain

$$\lambda^n - \lambda^{n-1} = 0$$

$$\lambda^{n-1}[\lambda - 1] = 0$$

$$\lambda = 1$$

$$y_h(n) = c_1(1)^n$$

$$y(0) = c_1$$

From the difference equation for $n = 0$

$$y(0) - y(-1) = x(0) + x(-1)$$

$$y(0) = 1$$

$$\Rightarrow \quad c_1 = 1$$

$x(-1) = y(-1) = 0$
$x(0) = 1$ for unit step

Therefore the impulse response $h(n) = u(n)$.

The response due to $x(n) = u(n)$ is

$$y(n) = x(n) * h(n)$$

$$= u(n) * u(n) = \sum_{k=0}^{n} 1 = n+1$$

$$\Rightarrow \quad y(n) = (n+1)u(n)$$

Solved Problem 3.24 Obtain the convolution of the following sequences

(i) $x(n) = u(n)$; $\quad h(n) = 2^n u(n)$

(ii) $x(n) = u(n)$; $\quad h(n) = u(n-3)$

(iii) $x(n) = u(n) - u(n-7)$; $\quad h(n) = u(n-1) - u(n-4)$

Solution:

i) Given $\quad x(n) = u(n)$; $\quad h(n) = 2^n u(n)$

$x(n) = 0$ for $n < 0$; therefore the input sequence is causal. $h(n) = 0$ for $n < 0$; therefore the system is causal. A causal signal is applied to a causal system.

Therefore

$$y(n) = \sum_{k=0}^{n} x(k)h(n-k)$$

$$= \sum_{k=0}^{n} 2^{n-k} = 2^n \sum_{k=0}^{n} 2^{-k}$$

$$= 2^n \left[1 + 2^{-1} + 2^{-2} \cdots (n+1) \text{ terms}\right]$$

$$= 2^n \left[1 - \left(\frac{1}{2}\right)^{n+1} \Big/ \left(1 - \frac{1}{2}\right)\right]$$

$$= 2^{n+1}\left[1 - \left(\frac{1}{2}\right)^{n+1}\right]$$

$$= 2.2^n - 1 \qquad \text{for} \quad n \geq 0$$

$$\Rightarrow y(n) = 2(2)^n u(n) - u(n)$$

ii) Given $x(n) = u(n)$; $h(n) = u(n-3)$

From Fig. 3.11.

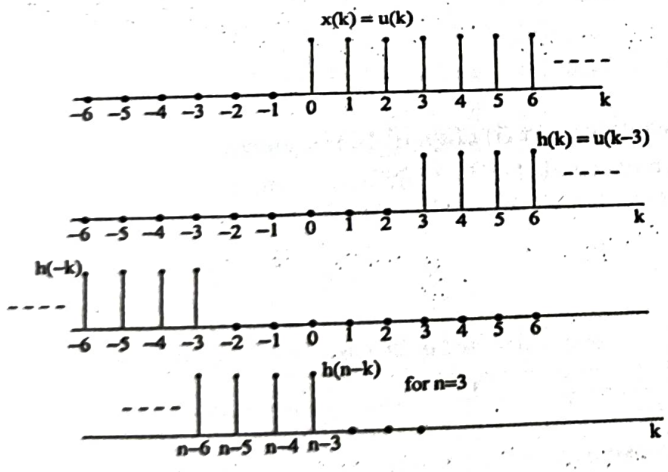


Fig. 3.11

we find that

$$x(k) = 0 \text{ for } k < 0$$

and

$$h(n-k) = 0 \text{ for } k > n-3$$

Therefore the limits in convolution sum is

$$k = 0 \text{ to } n-3$$

$$y(n) = \sum_{k=0}^{n-3} x(k)h(n-k)$$

$$= \sum_{k=0}^{n-3} 1$$

$$= n-2$$

iii) $x(n) = u(n) - u(n-7)$

$h(n) = u(n-1) - u(n-4)$

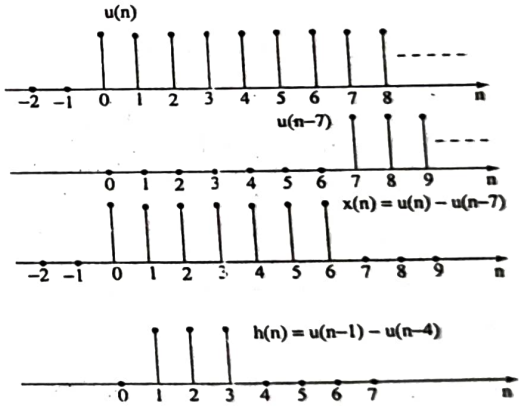
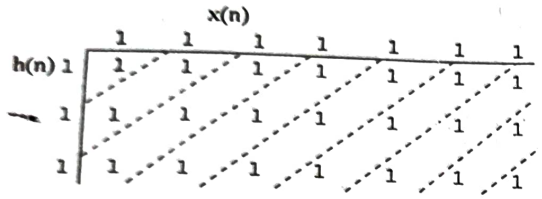


Fig. 3.12

From Fig. 3.12, we find that

$$x(n) = \{1,1,1,1,1,1,1,1\}$$

$$h(n) = \{1,1,1\}$$



Using matrix convolution

$$y(n) = \{1,2,3,3,3,3,3,2,1\}$$

Practice Problem 3.6

Find the convolution of the following signals

(i) $x(n) = \{1,0,-1,2,1\}; h(n) = \{\underset{\uparrow}{1},2,-1,2\}$ Ans: $\{1,2,-2,2,6,-2,3,2\}$

(ii) $x(n) = n/2 \quad 0 \le n \le 5; \quad h(n) = 1 \quad -3 \le n \le 3$
$= 0 \quad$ otherwise $\quad = 0 \quad$ otherwise

Ans: $\{0,0,5,1.5,3,5,7.5,7.5,7.5,7,6,4.5,2.5\}$

Practice Problem 3.7

Find the convolution of the following signals.

(i) $x(n) = \left(\frac{1}{3}\right)^n u(n); h(n) = u(n) - u(n-3)$ Ans: $\frac{3}{2} - \frac{1}{2}\left(\frac{1}{3}\right)^n \quad 0 \le n \le 2$

(ii) $x(n) = 3^n u(-n-1); h(n) = u(n-1)$ Ans: $\begin{cases} 13\left(\frac{1}{3}\right)^n & \text{for } n > 2 \\ 3^n/2 & \text{for } n < 0 \\ \frac{1}{2} & \text{for } n \ge 0 \end{cases}$

3.12 FIR and IIR Systems

Linear time-invariant systems can be classified according to the type of impulse response. They are 1. FIR system 2. IIR system.

FIR system

Definition:
If the impulse response of the system is of finite duration then the system is called a Finite Impulse Response (FIR) system.

An example of a FIR system is

$$h(n) = \begin{cases} 1 & \text{for } n = -1,2 \\ 2 & \text{for } n = 1 \\ 3 & \text{for } n = 0,3 \\ 0 & \text{otherwise.} \end{cases} \quad (3.137)$$

IIR system

Definition:
If the impulse response of the system is of infinite duration then the system is called an Infinite Impulse Response (IIR) system.

An example of an IIR system is

$$h(n) = a^n u(n) \quad (3.138)$$

3.13 Stability

Bounded sequence and unbounded sequence

A sequence is said to be bounded if it satisfy the property

$$|x(n)| \le M \quad (3.139)$$

where M is a constant.

Examples of bounded sequences are (i) $u(n)$ (ii) $\left(\frac{1}{2}\right)^n u(n)$ etc.

A sequence is said to be unbounded of its magnitude continuously increases to very large values without any bound. Example: $2^n u(n)$

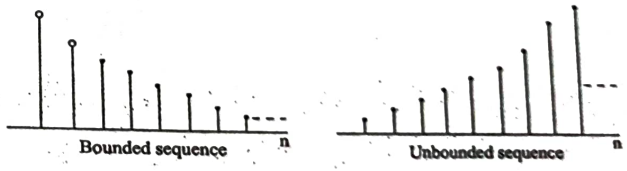


Fig. 3.13 (a) Bounded sequence. (b) Unbounded sequence.

3.14 BIBO Stability Criterion

A system is said to be stable if a bounded input sequence always produce a bounded output sequence. For a bounded input sequence $x(n)$, if the output is unbounded, the system is said to be unstable system.

Now we derive the condition for stability of a system on impulse response.

Let $x(n)$ be a bounded input sequence satisfying $|x(n)| \leq M_x < \infty$, $h(n)$ be the impulse response of the system, then the output of the system $y(n)$ can be found by using convolution sum. That is

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

$$= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \tag{3.140}$$

The magnitude of the output is given by

$$|y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right| \tag{3.141}$$

We know the magnitude of the sum of terms is less than or equal to the sum of the magnitudes, hence

$$|y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right| \leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)| \tag{3.142}$$

Let the bounded value of the input is equal to M, then Eq. (3.142) can be written us

$$|y(n)| \leq M \sum_{k=-\infty}^{\infty} |h(k)|$$

The above condition will be satisfied when

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty \tag{3.143}$$

Therefore the necessary and sufficient condition for stability is

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \tag{3.144}$$

That is, a LTI system is stable if its impulse response is absolutely summable.

Review Questions

1. What is the condition on impulse response for causality?
2. What is the condition on impulse response for stability?
3. Define FIR and IIR systems.

Solved Problem 3.25 From the given impulse responses $h(n)$ of the systems, find whether the systems are causal and stable.

(i) $h(n) = 2^n u(-n)$ (ii) $e^{2n}u(n-1)$ (iii) $5^n u(3-n)$ (iv) $e^{-6|n|}$

Solution:

(i) Given $h(n) = 2^n u(-n)$

The given system is noncausal since $h(n) \neq 0$ for $n < 0$. For stability the impulse response must be absolutely summable

$$\text{i.e.,} \quad \sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

$$\Rightarrow \quad \sum_{n=-\infty}^{\infty} 2^n u(-n) = \sum_{n=-\infty}^{0} 2^n$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = 1 + \frac{1}{2} + \cdots$$

$$= \frac{1}{1-\frac{1}{2}} = 2$$

Therefore, the system is stable.

(ii) Given $h(n) = e^{2n}u(n-1)$

The system is causal since $h(n) = 0$ for $n < 0$. The stability of the system is obtained by finding the absolute summation.

$$\sum_{n=0}^{\infty} e^{2n} = (1 + e^2 + e^4 + \cdots \infty) = \infty$$

Therefore the system is unstable.

(iii) $h(n) = 5^n u(3-n)$

The system is noncausal since $h(n) \neq 0$ for $n < 0$.

For stability the impulse response must be absolutely summable. That is

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty \qquad \boxed{u(3-n) = 0 \text{ for } n > 3}$$

$$= \sum_{n=-\infty}^{3} 5^n$$

$$= \sum_{n=-\infty}^{0} 5^n + \sum_{n=1}^{3} 5^n$$

$$= \sum_{n=0}^{\infty} (5)^{-n} + \sum_{n=1}^{3} 5^n = \frac{1}{1-\frac{1}{5}} + 155$$

$$= \left(\frac{5}{4} + 155\right) < \infty$$

Therefore the system is stable.

(iv) $h(n) = e^{-6|n|}$

The system is noncausal, since $h(n) \neq 0$ for $n < 0$

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=-\infty}^{\infty} e^{-6|n|} = \sum_{n=-\infty}^{-1} e^{6n} + \sum_{n=0}^{\infty} e^{-6n}$$

$$= \sum_{n=1}^{\infty} e^{-6n} + \sum_{n=0}^{\infty} e^{-6n}$$

$$= \frac{e^{-6}}{1-e^{-6}} + \frac{1}{1-e^{-6}}$$

$$= \frac{1+e^{-6}}{1-e^{-6}} < \infty$$

The system is stable.

3.15 Step Response

The step response can be easily expressed in terms of the impulse response using convolution sum. Let the impulse response of a discrete time system is $h(n)$. Then

the step response, $s(n)$ can be found using convolution sum

$$s(n) = h(n) * u(n) \qquad (\because x(n) = u(n))$$

$$= \sum_{n=-\infty}^{\infty} h(k)u(n-k)$$

Since $u(n-k) = 0$ for $k > n$ and $u(n-k) = 1$ for $k \leq n$ we have

$$s(n) = \sum_{k=-\infty}^{n} h(k)$$

That is, the step response is the running sum of the impulse response.
For a causal system

$$s(n) = \sum_{k=0}^{n} h(k)$$

Solved Problem 3.26 Find the step responses if the impulse responses are

(i) $\delta(n-2) + \delta(n-3)$ (ii) $(-a)^n u(n)$ (iii) $u(n)$

(iv) $\left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n)$

(i) Given $h(n) = \delta(n-2) + \delta(n-3)$

The step response

$$s(n) = h(n) * u(n)$$

$$= [\delta(n-2) + \delta(n-3) * u(n) \qquad \boxed{\delta(n) * x(n) = x(n)}$$

$$= \delta(n-2) * u(n) + \delta(n-3) * u(n)$$

$$= u(n-2) + u(n-3)$$

(ii) Given $h(n) = (-a)^n u(n)$

The step response

$$s(n) = h(n) * u(n)$$

$$= (-a)^n u(n) * u(n)$$

$$= \sum_{k=0}^{n} (-a)^k$$

$$= \frac{1-(-a)^{n+1}}{1-(-a)}$$

$$= \frac{1-(-a)^{n+1}}{1+a}$$

(iii) Given $h(n) = u(n)$

The step response

$$s(n) = h(n) * u(n)$$
$$= u(n) * u(n)$$
$$= \sum_{k=0}^{n} u(k)u(n-k)$$
$$\sum_{k=0}^{n} 1 = n+1$$
$$\Rightarrow s(n) = (n+1)u(n)$$

(iv) Given $h(n) = \left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n)$

$$\boxed{\sum_{k=0}^{n} a^k = \frac{1-(a)^{n+1}}{1-a}}$$

The step response $s(n) = h(n) * u(n)$

$$= \left(\frac{1}{2}\right)^n u(n) * u(n) + \left(-\frac{1}{3}\right)^n u(n) * u(n)$$
$$= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k + \sum_{k=0}^{n} \left(-\frac{1}{3}\right)^k$$
$$= \left[\frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}} + \frac{1-\left(-\frac{1}{3}\right)^{n+1}}{1-\left(-\frac{1}{3}\right)}\right]$$
$$= 2 - \left(\frac{1}{2}\right)^n + \frac{3}{4} + \frac{1}{4}\left(\frac{-1}{3}\right)^n$$
$$s(n) = \frac{11}{4}u(n) - \left(\frac{1}{2}\right)^n u(n) + \frac{1}{4}\left(\frac{-1}{3}\right)^n u(n)$$

Practice Problem 3.8

Find the stability of the following system with impulse response

(i) $h(n) = \sin\left(\frac{\pi n}{2}\right)$ (ii) $3^n u(-n)$ (iii) $[u(n) - u(n-10)]2^n$

Ans: (i) BIBO unsable (ii) Stable (iii) Stable

Practice Problem 3.9

Find the step response for LTI system represented by the following impulse responses

(i) $h(n) = e^{-2|n|}$ (ii) $h(n) = 2^n u(n)$ Ans: (i) $\frac{e^{-2}[1-e^{-2n}]+1}{1-e^{-2}}$

(iii) $h(n) = \left(\frac{1}{3}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(-n-1)$ Ans: $(2^{n+1}-1)u(n)$

Solved Problem 3.27 Find the convolution of the following signals

(i) $x(n) = \cos \pi n u(n)$; $h(n) = \left(\frac{1}{2}\right)^n u(n)$

(ii) $x(n) = \left(\frac{1}{3}\right)^{-n} u(-n-1)$; $h(n) = u(n-1)$

(iii) $x(n) = \left(\frac{1}{2}\right)^n u(n)$; $h(n) = u(n) - u(n-10)$

Solution:

i) Given

$$x(n) = \cos \pi n u(n) = (-1)^n u(n)$$
$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$
$$x(n) * h(n) = \sum_{k=-\infty}^{\infty} \cos \pi k u(k) \left(\frac{1}{2}\right)^{n-k} u(k)$$
$$x(n) = 0 \text{ for } n < 0;$$
$$h(n) = 0 \text{ for } n < 0$$

Therefore

$$x(n) * h(n) = \sum_{k=0}^{n} (-1)^k \left(\frac{1}{2}\right)^{n-k}$$

$$= \left(\frac{1}{2}\right)^n \sum_{k=0}^{n} (-1)^k (2)^k$$

$$= \left(\frac{1}{2}\right)^n \sum_{k=0}^{n} (-2)^k$$

$$= \left(\frac{1}{2}\right)^n \left[\frac{1-(-2)^{n+1}}{1-(-2)}\right] = \left(\frac{1}{2}\right)^n \left[\frac{1+2(-2)^n}{3}\right]$$

$$= \frac{1}{3}\left(\frac{1}{2}\right)^n + \frac{2}{3}(-1)^n \quad \text{for } n \geq 0$$

$$= \frac{1}{3}\left(\frac{1}{2}\right)^n u(n) + \frac{2}{3}(-1)^n u(n)$$

ii) Given

$$x(n) = \left(\frac{1}{3}\right)^{-n} u(-n-1) = 3^n u(-n-1)$$

$$h(n) = u(n-1)$$

Let

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

From Fig. 3.14a and Fig. 3.14d we can find that for $n-1 \leq -1$

$$x(k) = 0 \text{ for } k > -1$$

and

$$h(n-k) = 0 \text{ for } k > n-1$$

For $n-1 < -1$, the interval on summation is from $k = -\infty$ to $n-1$.

$$y(n) = \sum_{k=-\infty}^{n-1} 3^k \quad \text{for } n-1 \leq -1$$

or

$$\text{for} \quad n \leq 0$$

For $n-1 > -1$ the interval on summation is from $k = -\infty$ to -1.

Therefore

$$y(n) = \sum_{k=-\infty}^{-1} 3^k \quad \text{for } n > 0$$

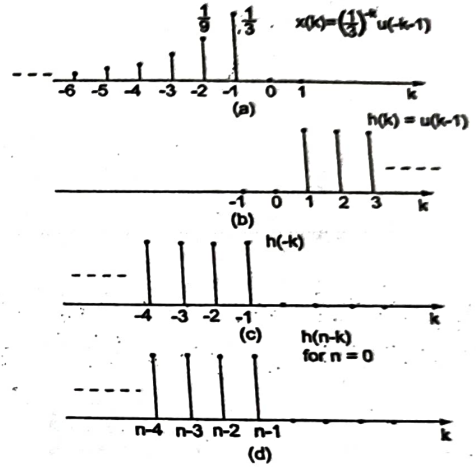


Fig. 3.14

For $n \leq 0$

$$y(n) = \sum_{k=-\infty}^{n-1} 3^k$$

$$y(n) = 3^{n-1} + 3^{n-2} + \cdots$$

$$= 3^{n-1}\left[1 + \frac{1}{3} + \cdots\right]$$

$$= 3^{n-1}\left[\frac{1}{1-\frac{1}{3}}\right]$$

$$= 0.5(3)^n$$

For $n > 0$

$$y(n) = \sum_{k=-\infty}^{-1} 3^k = \sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k = \frac{\frac{1}{3}}{1-\frac{1}{3}} = 0.5$$

iii) $x(n) = \left(\frac{1}{2}\right)^n u(n)$ and $h(n) = u(n) - u(n-10)$.

Let

$$y(n) = x(n) * h(n)$$

$$= \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

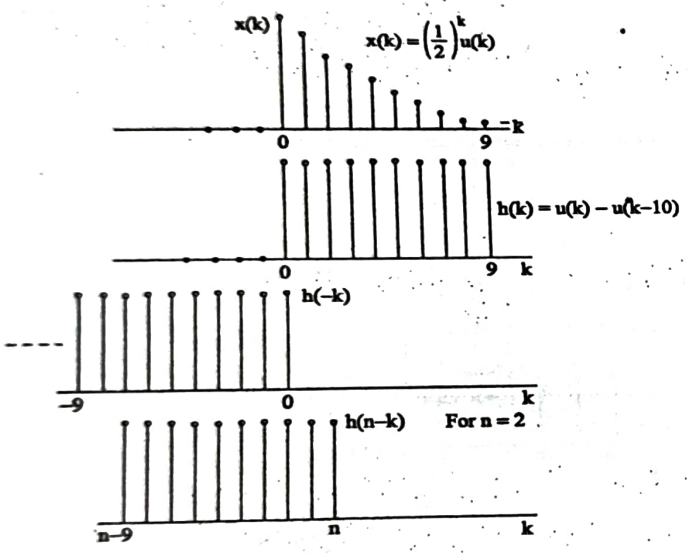


Fig. 3.15

From Fig. 3.16

$$y(n) = \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \quad \text{for} \quad 0 \le n \le 9$$

$$= \frac{1-\left(\frac{1}{2}\right)^{n+1}}{1-\frac{1}{2}}$$

$$= 2 - 2\left(\frac{1}{2}\right)^{n+1}$$

$$= \left(\frac{1}{2}\right)^n \left[2^{n+1} - 1\right]$$

$$y(n) = \left(\frac{1}{2}\right)^n \left[2^{n+1} - 1\right] u(n) \quad \text{for } 0 \le n \le 9$$

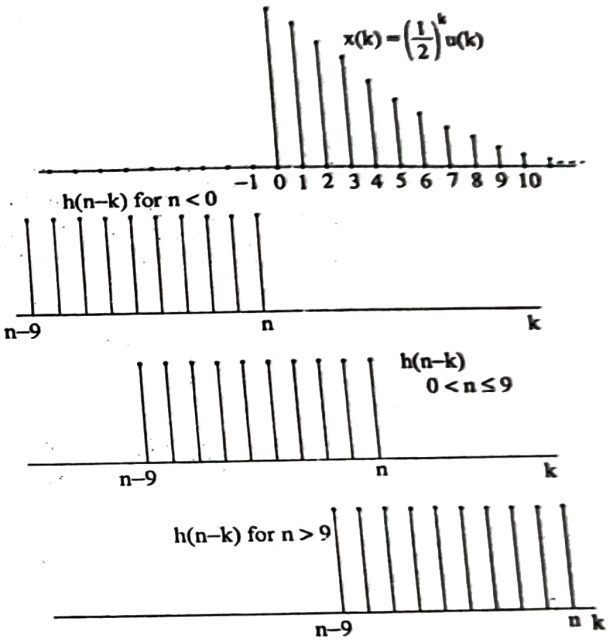


Fig. 3.16

For $n > 9$

$$y(n) = \sum_{k=n-9}^{n} \left(\frac{1}{2}\right)^k$$

$$= \left(\frac{1}{2}\right)^{n-9} \left[\frac{1-\left(\frac{1}{2}\right)^{10}}{1-\frac{1}{2}}\right]$$

$$= 2\left(\frac{1}{2}\right)^{n-9} - 2\left(\frac{1}{2}\right)^{n+1}$$

$$= \left(\frac{1}{2}\right)^n \left[2^{10} - 1\right]$$

Solved Problem 3.28 Find the convolution of

$$x(n) = u(n) - 2u(n-3) + u(n-6)$$
$$h(n) = u(n+2) - u(n-9)$$

Solution:

The signal $x(n)$ and its folded version are shown in Fig.3.17. The signal $h(k)$ is shown in Fig.3.18. From Fig.3.18 we can find the following.

For $n < -2$

$$h(k) = 0 \quad \text{for} \quad n < -2$$
$$x(n-k) = 0 \quad \text{for} \quad n > 0$$
$$\Rightarrow \quad y(n) = 0$$

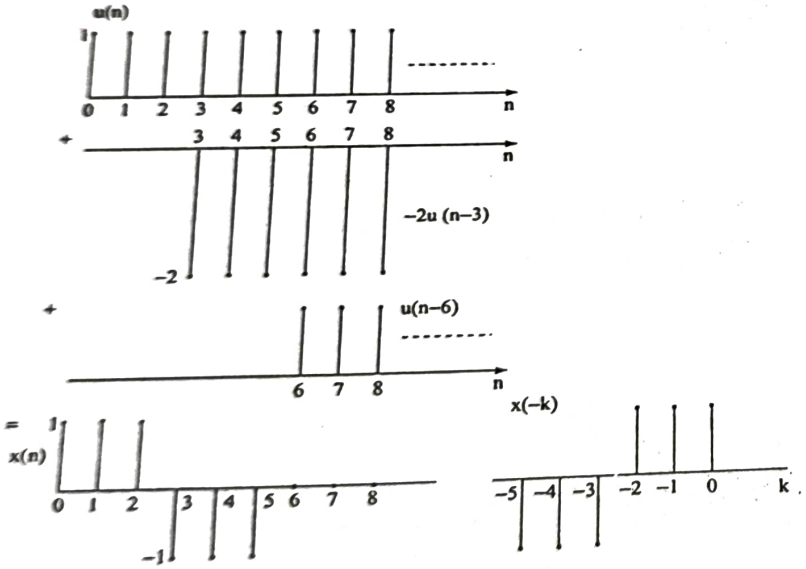


Fig. 3.17

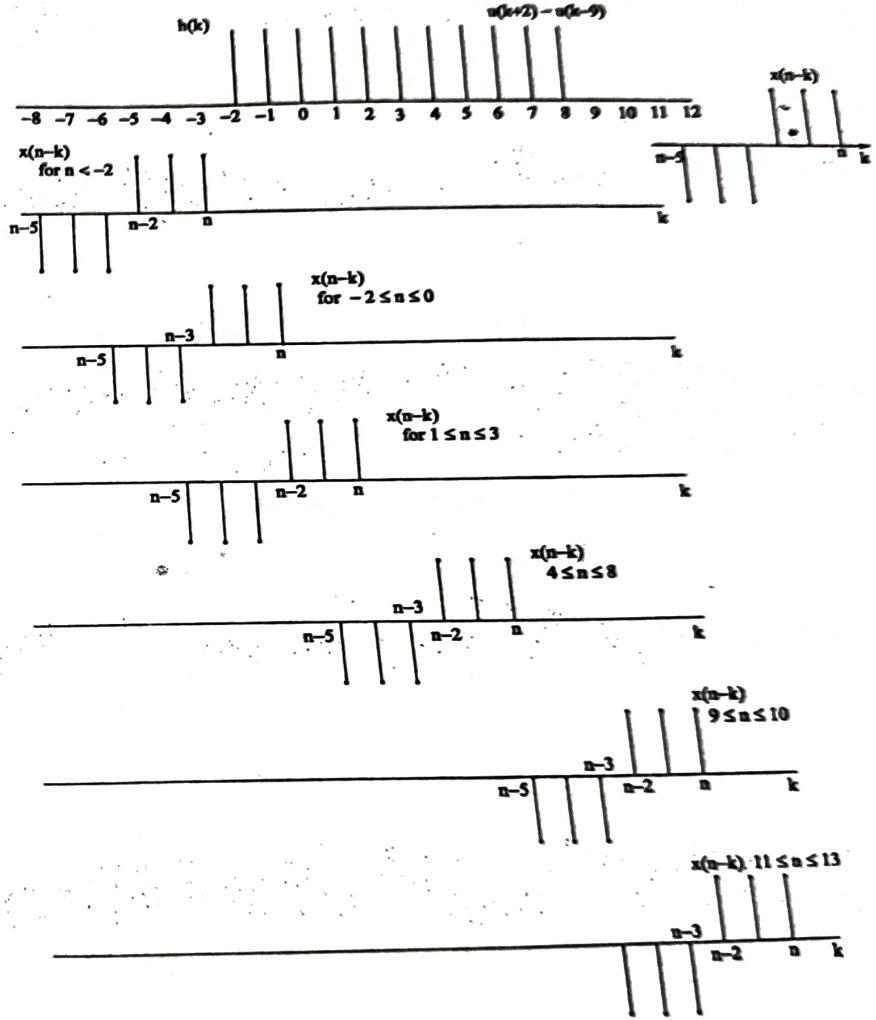


Fig. 3.18

For $-2 \le n < 0$

$$y(n) = \sum_{k=-2}^{n} 1 = n+3$$

$$\boxed{\sum_{n=n_1}^{n_2} 1 = n_2 - n_1 + 1}$$

For $1 \le n \le 3$

$$y(n) = \sum_{k=-2}^{n-3}(-1) + \sum_{k=n-2}^{n} 1 = -[(n-3)+2+1] + [n-(n-2)+1]$$
$$= -n+3$$

For $4 \le n \le 8$

$$y(n) = \sum_{k=n-5}^{n-3}(-1) + \sum_{k=n-2}^{n}(1) = (-3)+3$$
$$y(n) = 0$$

For $9 \le n \le 10$

$$y(n) = \sum_{k=n-5}^{n-3}(-1) + \sum_{k=n-2}^{8} 1$$
$$= 8-n$$

For $11 \le n \le 13$

$$y(n) \sum_{k=n-5}^{8}(-1) = n-14$$

For $n \ge 14$

$$y(n) = 0$$

$$\Rightarrow \quad y(n) = \begin{cases} 0 & n < -2 \\ n+3 & -2 \le n \le 0 \\ -n+3 & 1 \le n \le 3 \\ 0 & 4 \le n \le 8 \\ 8-n & 9 \le n \le 10 \\ n-14 & 11 \le n \le 13 \\ 0 & n > 14 \end{cases}$$

Solved Problem 3.29 Determine the stability of the system described by difference equations

i) $y(n) = 0.6y(n-1) - 0.08\,y(n-2) + x(n)$

ii) $y(n) - \frac{5}{2}y(n-1) + y(n-2) = x(n) - x(n-1)$

Solution:

i) Given

$$y(n) = 0.6y(n-1) - 0.08y(n-2) + x(n)$$

The condition for system to be stable, is impulse response of the system must be absolutely summable. That is

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

From Solved problem (3.4), the impulse response of the system is

$$h(n) = 2(0.4)^n u(n) - (0.2)^n u(n)$$

$$\sum_{n=-\infty}^{\infty} |h(n)| = 2\sum_{n=0}^{\infty}(0.4)^n - \sum_{n=0}^{\infty}(0.2)^n$$

$$= 2\left[\frac{1}{1-0.4}\right] - \frac{1}{1-0.2}$$

$$= 2.083$$

$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$. Therefore the system is stable

ii) Given

$$y(n) - \frac{5}{2}y(n-1) + y(n-2) = x(n) - x(n-1)$$

The homogeneous equation is

$$y(n) - \frac{5}{2}y(n-1) + y(n-2) = 0$$

The characteristic equation is given by

$$\lambda^2 - \frac{5}{2}\lambda + 1 = 0$$

$$(\lambda - 2)\left(\lambda - \frac{1}{2}\right) = 0$$

$$\lambda_1 = 2; \quad \lambda_1 = \frac{1}{2}$$

$$\Rightarrow \quad y_h(n) = c_1(2)^n + c_2\left(\frac{1}{2}\right)^n \tag{3.145}$$

$$y(0) = c_1 + c_2$$

$$y(1) = 2c_1 + \frac{1}{2}c_2$$

From the difference equation

$$y(0) = 1$$

$$y(1) = \frac{3}{2}$$

$$\Rightarrow \quad c_1 + c_2 = 1$$

$$2c_1 + \frac{1}{2}c_2 = 3/2$$

$$y(-1) = y(-2) = 0 \text{ and } x(-1) = 0$$
$$x(0) = \delta(0) = 1$$

Solving for c_1 and c_2, and substituting the values in Eq. (3.145) we get the impulse response.

$$h(n) = \frac{2}{3}(2)^n u(n) + \frac{1}{3}\left(\frac{1}{2}\right)^n u(n)$$

For the system to be stable

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

$$= \frac{2}{3}\sum_{n=0}^{\infty}(2)^n + \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^n = \infty$$

$$\downarrow \qquad \downarrow$$

diverges converges

Therefore, the system is unstable.

3.16 Correlation of Two Sequences

So far we discussed about the convolution of two signals which is used to find the output $y(n)$ of a system for the given impulse response $h(n)$ and input signal $x(n)$. In this section, we will study a mathematical operation known as correlation that closely resembles convolution. Correlation is basically used to compare two signals. It occupies a significant place in signal processing.

Definition: Correlation is a measure of the degree to which two signals are similar.

The correlation of two signals is divided into

(i) Cross-correlation,
(ii) Auto-correlation.

3.16.1 Cross-correlation

The cross-correlation between a pair of signals $x(n)$ and $y(n)$ is given by

$$\gamma_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y(n-l) \quad l = 0, \pm1, \pm2, \pm3, \ldots \tag{3.146}$$

The index l is the shift (lag) parameter. The order of subscripts xy indicates that $x(n)$ is the reference sequence that remains unshifted in time whereas the sequence $y(n)$ is shifted l units in time with respect to $x(n)$.

If we wish to fix $y(n)$ and to shift $x(n)$, then correlation of two sequences can be written as

$$\gamma_{yx}(l) = \sum_{n=-\infty}^{\infty} y(n)x(n-l)$$

$$= \sum_{n=-\infty}^{\infty} y(n+l)x(n) \tag{3.147}$$

If the time shift $l = 0$, then we get

$$\gamma_{xy}(0) = \gamma_{yx}(0) = \sum_{n=-\infty}^{\infty} x(n)y(n) \tag{3.148}$$

Comparing Eq. (3.146) with Eq. (3.147) we find that

$$\gamma_{xy}(l) = \gamma_{yx}(-l)$$

where $\gamma_{yx}(-l)$ is the folded version of $\gamma_{xy}(l)$ about $l = 0$.

We can rewrite Eq. (3.146) as

$$\gamma_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)y[-(l-n)]$$

$$= x(l) * y(-l) \tag{3.149}$$

From Eq. (3.149) we find that the correlation process is essentially the convolution of two data sequences in which one of the sequences has been reversed. Therefore, the same algorithm (procedure) can be used to compute convolution and correlation of two sequences.

3.16.2 Auto-correlation

The auto-correlation of a sequence is correlation of a sequence with itself. The auto-correlation of a sequence $x(n)$ is defined by

$$\gamma_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n)x(n-l) \tag{3.150}$$

or equivalently

$$\gamma_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n+l)x(n) \tag{3.151}$$

If the time shift $l = 0$, then we have

$$\gamma_{xx}(0) = \sum_{n=-\infty}^{\infty} x^2(n) \tag{3.152}$$

3.16.3 Properties of cross-correlaion and auto-correlation sequences

In this section, we discuss some of the important properties of cross-correlation and auto-correlation of sequences. Consider two finite energy sequences $x(n)$ and $y(n)$. Let us consider the linear combination of sequences

$$a_1x(n) + a_2y(n-l) \tag{3.153}$$

where a_1 and a_2 are constants and l is a time shift.

The energy of the signal is

$$E = \sum_{n=-\infty}^{\infty} [a_1x(n) + a_2y(n-l)]^2 \tag{3.154}$$

$$= a_1^2 \sum_{n=-\infty}^{\infty} x^2(n) + 2a_1a_2 \sum_{n=-\infty}^{\infty} x(n)y(n-l) + a_2^2 \sum_{n=-\infty}^{\infty} y^2(n-l) \tag{3.155}$$

$$= a_1^2\gamma_{xx}(0) + 2a_1a_2\gamma_{xy}(l) + a_2^2\gamma_{yy}(0) \tag{3.156}$$

The energy of the signals $x(n)$ and $y(n)$ are finite. Therefore, the energy E of the signal $a_1x(n) + a_2y(n-1)$ is also finite. Therefore,

$$a_1^2\gamma_{xx}(0) + 2a_1a_2\gamma_{xy}(l) + a_2^2\gamma_{yy}(0) \geq 0 \tag{3.157}$$

From Eq. (3.155) we find that $\gamma_{xx}(0)$ is the energy of the sequences $x(n)$. Therefore,

$$E_x = \gamma_{xx}(0) \tag{3.158}$$

Similarly,

$$E_y = \gamma_{yy}(0) \tag{3.159}$$

Dividing Eq. (3.157) by $a_2^2 (a_2 \neq 0)$, we obtain

$$\left(\frac{a_1}{a_2}\right)^2 \gamma_{xx}(0) + 2\frac{a_1}{a_2}\gamma_{xy}(l) + \gamma_{yy}(0) \geq 0$$

Let $\frac{a_1}{a_2} = k$

$$k^2\gamma_{xx}(0) + 2k\gamma_{xy}(l) + \gamma_{yy}(0) \geq 0 \tag{3.160}$$

We can rewrite Eq. (3.160) as

$$\begin{bmatrix} k & 1 \end{bmatrix} \begin{bmatrix} \gamma_{xx}(0) & \gamma_{xy}(l) \\ \gamma_{xy}(l) & \gamma_{yy}(0) \end{bmatrix} \begin{bmatrix} k \\ l \end{bmatrix} \geq 0 \tag{3.161}$$

for any finite value of k, from which we find that

$$\gamma_{xx}(0)\gamma_{yy}(0) - \gamma_{xy}^2(l) \geq 0 \tag{3.162}$$

$$\Rightarrow |\gamma_{xy}(l)| \leq \sqrt{\gamma_{xx}(0)\gamma_{yy}(0)} = \sqrt{E_x E_y} \tag{3.163}$$

If $y(n) = x(n)$, Eq. (3.163) reduces to

$$|\gamma_{xy}(l)| \leq \gamma_{xx}(0) = E_x \tag{3.164}$$

The Eq. (3.164) states that at zero lag the auto-correlation of sequence attains its maximum value.

The normalized expression for auto-correlation $\gamma_{xx}(l)$ is

$$\rho_{xx}(l) = \frac{\gamma_{xx}(l)}{\gamma_{xx}(0)} \tag{3.165}$$

Similarly the normalized cross-correlation sequence is

$$\rho_{xy}(l) = \frac{\gamma_{xy}(l)}{\sqrt{\gamma_{xx}(0)\gamma_{yy}(0)}} \tag{3.166}$$

$\rho_{xy}(l)$ is also known as the cross-correlation coefficient. Its value always lies between -1 and $+1$. A value 0 for cross correlation means no correlation.

We have

$$\gamma_{xy}(l) = \gamma_{yx}(-l) \qquad \text{for any } l$$

Properties of autocorrelation sequence

Property 1 $\gamma_{xx}(k) \leq \gamma_{xx}(0)$

Property 2 $\gamma_{xx}(-k) = \gamma_{xx}(k)$

3.16.4 Computation of correlation

To compute the cross correlation of sequences $x(n)$ and $y(n)$, the following steps are followed.

1. Obtain the sequence $y(n-l)$ by shifting the sequence right by the time lag l.
2. Multiply the shifted sequence $y(n-l)$ with $x(n)$ and sum all the values to obtain $\gamma_{xy}(l)$.
3. Repeat steps 1 and 2 for all values of the lag l.

The correlation of two sequences $x(n)$ and $y(n)$ can be obtained by using the convolution procedure discussed in section 3.10.

From Eq. (3.149) we have

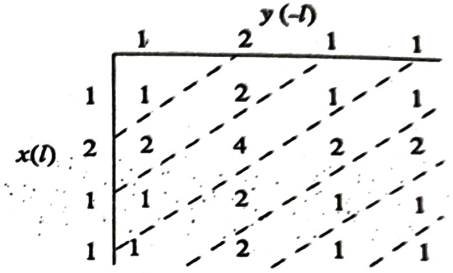
$$\gamma_{xy}(l) = x(l) * \gamma(-l)$$

To get $\gamma_{xy}(l)$ first fold the sequence $\gamma(l)$ to obtain $\gamma(-l)$. Then the convolution of $x(l)$ and $y(-l)$ gives the value $\gamma_{xy}(l)$.

Solved Problem 3.30 Find the cross-correlation of two finite length sequences $x(n) = \{1,2,1,1\}; y(n) = \{1,1,2,1\}$

Solution:

$$x(l) = \{1,2,1,1\}; y(l) = \{1,1,2,1\}; y(-l) = \{1,2,1,1\}$$
$$\gamma_{xy} = x(l) * y(-l)$$



$$\gamma_{xy}(l) = \{1, 4, 6, 6, 5, 2, 1\}$$

3.16.5 Correlation of power and periodic signals

Let $x(n)$ and $y(n)$ are power signals. Their cross-correlation sequence is defined as

$$\gamma_{xy}(l) = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} x(n)y(n-l) \qquad (3.167)$$

and the auto-correlation of a power signal $x(n)$ is defined as

$$\gamma_{xx}(l) = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} x(n)x(n-l) \qquad (3.168)$$

If $x(n)$ and $y(n)$ are periodic sequences, each with period N, then

$$\gamma_{xy}(l) = \frac{1}{N} \sum_{n=0}^{N-1} x(n)y(n-l) \qquad (3.169)$$

and

$$\gamma_{xx}(l) = \frac{1}{N} \sum_{n=0}^{N-1} x(n)x(n-l) \qquad (3.170)$$

Review Questions

1. Define (i) Crosscorrelation (ii) Autocorrelation
2. What are the properties of
   (i) autocorrelation function (ii) crosscorrelation function

3.17 Inverse System and Deconvolution

3.17.1 Inverse system

A system is said to be invertible if the input to the system can be recovered from the output. That is when the original system is cascaded with its inverse system, the output $\omega(n)$ is equal to the input $x(n)$ of the original system. This concept is illustrated in Fig. (3.19)

Fig. 3.19 Cascading of the system and its inverse

For example if the system $y(n) = ax(n)$ then the inverse system is $\omega(n) = \frac{1}{a}y(n)$

If the impulse response of the original system is $h(n)$, then the impulse response of inverse system is $h^{-1}(n)$.

From Fig. (3.19) we find that

$$y(n) = x(n) * h(n)$$

and

$$\begin{aligned}\omega(n) &= y(n) * h^{-1}(n)\\ &= [x(n) * h(n)] * h^{-1}(n)\\ &= x(n) * [h(n) * h^{-1}(n)]\\ &= x(n)\end{aligned}$$

where

$$h(n) * h^{-1}(n) = \delta(n) \tag{3.171}$$

3.17.2 Deconvolution

In certain applications like measurements, the knowledge of the impulse response $h(n)$ and output $y(n)$ may be known ahead and we may have to find the input applied to the system. For example, in a measurement system like temperature transducer, the impulse response $h(n)$ and the output $y(n)$ of transducer are known we have to find the input applied to the measurement system by deconvolving $y(n)$ and $h(n)$.

The process of recovering $x(n)$ from $x(n) * h(n)$ is known as deconvolution.

We have

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \tag{3.172}$$

Assuming $y(n)$ and $h(n)$ are one sided finite sequences. We have

$$y(n) = \sum_{k=0}^{N} x(k)h(n-k) \tag{3.173}$$

from which we get

$$\begin{aligned}
y(0) &= h(0)x(0)\\
y(1) &= h(1)x(0) + h(0)x(1)\\
y(2) &= h(2)x(0) + h(1)x(1) + h(0)x(2)\\
&\cdots \quad \cdots \quad \cdots \quad \cdots\\
&\cdots \quad \cdots \quad \cdots \quad \cdots\\
&\cdots \quad \cdots \quad \cdots \quad \cdots
\end{aligned} \tag{3.174}$$

In matrix form

$$\begin{bmatrix} y(0)\\ y(1)\\ y(2)\\ \cdots\\ \cdots\\ \cdots \end{bmatrix} = \begin{bmatrix} h(0) & 0 & 0 & \cdots & 0\\ h(1) & h(0) & 0 & \cdots & 0\\ h(2) & h(1) & h(0) & \cdots & 0\\ \cdots & \cdots & \cdots & \cdots & \cdots\\ \cdots & \cdots & \cdots & \cdots & \cdots\\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} x(0)\\ x(1)\\ x(2)\\ \cdots\\ \cdots\\ \cdots \end{bmatrix} \tag{3.175}$$

Y H X

$$Y = HX \tag{3.176}$$

$$\Rightarrow X = H^{-1}Y \tag{3.177}$$

From matrix we can find

$$y(0) = h(0)x(0)$$

$$\Rightarrow \quad x(0) = \frac{y(0)}{h(0)} \tag{3.178}$$

$$y(1) = h(1)x(0) + h(0)x(1) \tag{3.179}$$

$$\Rightarrow \quad x(1) = \frac{y(1) - h(1)x(0)}{h(0)} \tag{3.180}$$

Similarly,

$$x(2) = \frac{y(2) - x(0)h(2) - x(1)h(1)}{h(0)} \quad (3.181)$$

In general,

$$x(n) = \frac{y(n) - \sum_{k=0}^{n-1} x(k)h(n-k)}{h(0)} \quad (3.182)$$

Review Question

1. Define (i) Inverse system (ii) Deconvolution.

Solved Problem 3.31 What is the input signal $x(n)$ that will generate the output sequence.

$y(n) = \{\underset{\uparrow}{1},5,10,11,8,4,1\}$ for a system with impulse response $h(n) = \{\underset{\uparrow}{1},2,1\}$ **AU April 2003**

Solution:

Given

$$y(n) = \{\underset{\uparrow}{1},5,10,11,8,4,1\}$$

and

$$h(n) = \{\underset{\uparrow}{1},2,1\}$$

The number of samples in output is $N_1 + N_2 - 1 = 7$.

The number of samples in impulse response is $N_2 = 3$.

Therefore $N_1 = 5$.

That is $x(n)$ has five samples.

From Eq. (3.182) we have

$$x(n) = \frac{y(n) - \sum_{k=0}^{n-1} x(k)h(n-k)}{h(0)}$$

For $n = 0$

$$x(0) = \frac{y(0)}{h(0)} = 1$$

For $n = 1$

$$x(1) = \frac{y(1) - x(0)h(1)}{h(0)}$$

$$= \frac{5 - 1(2)}{1} = 3$$

$$x(2) = \left[y(2) - \sum_{k=0}^{1} x(k)h(2-k)\right] \Big/ h(0)$$

$$= \frac{y(2) - x(0)h(2) - x(1)h(1)}{h(0)}$$

$$= \frac{10 - 1(1) - 3(2)}{1} = 3$$

$$x(3) = \frac{y(3) - \sum_{k=0}^{2} x(k)h(3-k)}{h(0)}$$

$$= \frac{y(3) - x(0)h(3) - x(1)h(2) - x(2)h(1)}{h(0)}$$

$$= 11 - 1(0) - 3(1) - 3(2) = 2$$

$$x(4) = \left[y(4) - \sum_{k=0}^{3} x(k)h(4-k)\right] \Big/ h(0)$$

$$= \frac{y(4) - x(0)h(4) - x(1)h(3) - x(2)h(2) - x(3)h(1)}{h(0)}$$

$$= 8 - 1(0) - 3(0) - 3(1) - 2(2) = 1$$

$$\Rightarrow \quad x(n) = \{1,3,3,2,1\}$$

Solved Problem 3.32 Find the overall response of the system shown in Fig.3.20 below with $h_1(n) = \delta(n)$, $h_2(n) = (n-1)u(n)$ and $h_3(n) = \delta(n) + n\,u(n-1) + \delta(n-2)$.

(AU Dec 2008)

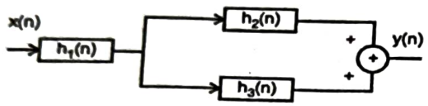


Fig. 3.20

Solution:

$$\begin{aligned} h(n) &= h_1(n) * [h_2(n) + h_3(n)] \\ &= \delta(n) * [(n-1)u(n) + \delta(n) + nu(n-1) + \delta(n-2)] \\ &= (n-1)u(n) + \delta(n) + n\,u(n-1) + \delta(n-2) \end{aligned}$$

Solved Problem 3.33 A periodic signal $x(n)$ with period N is applied as input to a discrete-time LTI system. Prove that the output $y(n)$ is also periodic with period N.

Solution:

The output of a discrete-time LTI system is given by

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

Where $h(n)$ is impulse response of the system.

Let $n = m+N$, then

$$y(m+N) = \sum_{k=-\infty}^{\infty} h(k)\, x(m+N-k) \qquad = \sum_{k=-\infty}^{\infty} h(k)\, x(m-k-N)$$

The input $x(n)$ is periodic with period N. Therefore $x(m-k+N) = x(m-k)$

$$\Rightarrow \quad y(m+N) = \sum_{k=-\infty}^{\infty} h(k)\, x(m-k)$$

$$= y(m)$$

$$\Rightarrow \quad y(m+N) = y(m)$$

Short Questions and Answers

1. Define the terms (i) Natural response (ii) Forced response.

(i) Natural response: It is the response of the system with zero input. It depends on the initial state of the system.

(ii) Forced response: It is the response of the system due to input alone when the initial state of the system is zero.

2. Define the impulse response and step response of a system.

The impulse response of a system is the output of the system when the input signal is an impulse function.

The step response of a system is the output of the system when the input signal is a unit step function.

3. Represent the sequence $x(n) = \{1, 2, \underset{\uparrow}{-3}, -1, 2\}$ as a sum of weighted impulse sequences.

Given

$$x(n) = \{1, 2, \underset{\uparrow}{-3}, -1, 2\}$$

$$n = -2, -1, 0, 1, 2$$

i.e., $x(-2) = 1; x(-1) = 2; x(0) = -3; x(1) = -1; x(2) = 2$

The sequence $x(n)$ is having values from $n = -2$ to $n = 2$. Therefore we can write

$$\begin{aligned} x(n) &= \sum_{k=-2}^{2} x(k)\delta(n-k) \\ &= x(-2)\delta(n+2) + x(-1)\delta(n+1) + x(0)\delta(n) \\ &\quad + x(1)\delta(n-1) + x(2)\delta(n-2) \\ &= \delta(n+2) + 2\delta(n+1) - 3\delta(n) - \delta(n-1) + 2\delta(n-2) \end{aligned}$$

4. What is meant by discrete convolution?

The convolution of discrete-time signals is known as discrete convolution. Let $x(n)$ be the input to an LTI system and $y(n)$ be the output of the system. If $h(n)$ is the impulse response $h(n)$ of the system, then the output $y(n)$ can be obtained by convoluting the impulse response $h(n)$ and the input signal $x(n)$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

or

$$= \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

5. What are the properties of convolution? **(AU Nov 2008)**

1. Commutative Property: $x(n) * h(n) = h(n) * x(n)$
2. Associative Property: $[x(n) * h_1(n)] * h_2(n) = x(n) * [h_1(n) * h_2(n)]$
3. Distribute Property: $x(n) * [h_1(n) + h_2(n)] = x(n) * h_1 + x(n) * h_2(n)$

6. What is the overall impulse response $h(n)$ when two systems with impulse responses $h_1(n)$ and $h_2(n)$ are in parallel and in series?

If the systems with impulse responses $h_1(n)$ and $h_2(n)$ are connected in parallel like in Fig. 1

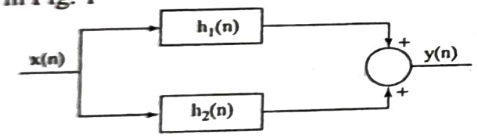


Fig. 1

then the output

$$\begin{aligned} y(n) &= x(n) * h_1(n) + x(n) * h_2(n) \\ &= x(n) * [h_1 + h_2(n)] \\ &= x(n) * h(n) \end{aligned}$$

Therefore the overall impulse response

$$h(n) = h_1(n) + h_2(n)$$

If the system with impusle responses $h_1(n)$ and $h_2(n)$ are connected in series like in Fig. 2

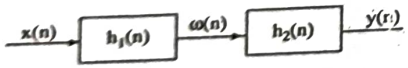


Fig. 2

then

$$\omega(n) = x(n) * h_1(n)$$

$$\begin{aligned} y(n) &= \omega(n) * h_2(n) \\ &= [x(n) * h_1(n)] * h_2(n) \\ &= x(n) * [h_1(n) * h_2(n)] \\ &= x(n) * h(n) \end{aligned}$$

Therefore the overall impulse response

$$h(n) = h_1(n) * h_2(n)$$

7. How you can find the step response of a system if the impulse response $h(n)$ is known?

We know that the convolution sum is given by

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} x(k)h(n-k) \\ &= \sum_{k=-\infty}^{\infty} x(n-k)h(k) \end{aligned}$$

For input $x(n) = u(n)$, we have step response

$$y(n) = s(n) = \sum_{k=-\infty}^{\infty} u(n-k)h(k)$$

$$\begin{aligned} u(n-k) &= 0 \quad \text{for} \quad k > n \\ &= 1 \quad \text{for} \quad k < n \end{aligned}$$

Therefore

$$s(n) = \sum_{k=-\infty}^{n} h(k)$$

8. What are FIR and IIR systems?

FIR system: The FIR system has an impulse response which is zero outside a finite time interval.

Example: $h(n) = 0 \quad \text{for } n < 0 \quad \text{and } n \geq N$

IIR system: An IIR system exhibits an impulse response of infinite duration.

9. Define a stable and causal system.

Stable system: Any relaxed system is said to be bounded input bounded output (BIBO) stable if and only if every bounded input yields a bounded output.

Causal system: A system is said to be causal if the output of the system at any time n depends only on present and past input, but does not depend on future inputs.

10. What is causality condition for an LTI discrete-time system?

(April 2008, Dec 2008)

The necessary and sufficient condition for causality of an LTI system is its, unit sample response $h(n) = 0$ for negative values of n. i,e.,

$$h(n) = 0 \quad \text{for} \quad n < 0$$

11. What is the necessary and sufficient condition on the impulse response for stability? **(April 2008, Dec 2008)**

The necessary and sufficient condition guranteeing the stability of a linear time-invariant system is that its impulse response is absolutely summable.

$$\text{i.e.,} \quad \sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

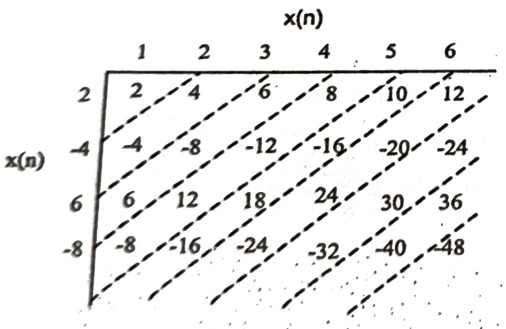
12. What is the overall impulse response $h(n)$ when two systems with impulse response $h_1(n)$ and $h_2(n)$ are connected in parallel and in series?

(Nov 2008)

For series connection of systems $h(n) = h_1(n) * h_2(n)$

For parallel connection of systems $h(n) = h_1(n) + h_2(n)$

13. Find the linear convolution of $x(n) = \{1,2,3,4,5,6\}$ with $y(n) = \{2,-4,6,-8\}$ **(AU May 2004)**



$x(n) * y(n)$

$= \{2,0,4,0 \quad -4,-8,-26,-4,-48\}$

Multiple Choice Questions

1. The impulse response $h(n)$ of a linear time invariant system is given by

$$h(n) = u(n+3) + u(n-2) - 2u(n-7)$$

where $u(n)$ is the unit step sequence. The above system is

(a) stable but not causal (b) stable and causal
(c) causal but unstable (d) unstable and not causal

2. Consider the sequence $x(n) = [-4 - j5 \quad \underset{\uparrow}{1 + j2} \quad 4]$. The conjugate anti-symmetric part of the sequence is

(a) $[-4 - j2.5 \quad j2 \quad 4 - j2.5]$ (b) $[-j2.5 \quad 1 \quad j2.5]$
(c) $[-j5 \quad j2 \quad 0]$ (d) $[-4 \quad 1 \quad 4]$

3. A causal LTI system is described by the difference equation $2y(n) = \alpha y(n-1) - 2x(n) + \beta x(n-1)$

The system is stable only if

(a) $|\alpha| = 2 \quad |\beta| < 2$
(b) $|\alpha| > 2 \quad |\beta| > 2$
(c) $|\alpha| < 2$ any value of β
(d) $|\beta| < 2$ any value of α

4. The impulse response $h(n)$ of a linear time invariant system is given as

$$h(n) = \begin{cases} -2\sqrt{2} & n = 1, -1 \\ 4\sqrt{2} & n = 2, -2 \\ 0 & \text{otherwise} \end{cases}$$

If the input to the above system is the sequence $e^{j\pi n/4}$, then the output is

(a) $4\sqrt{2}e^{j\pi n/4}$ (b) $4\sqrt{2}e^{-j\pi n/4}$
(c) $4e^{j\pi n/4}$ (d) $-4e^{j\pi n/4}$

5. The system described by the difference equation $y(n) - 2y(n-1) + y(n-2) = x(n) - x(n-1)$ has $y(n) = 0$ for $n < 0$. If $x(n) = \delta(n)$ then $y(2)$ will be

(a) 2 (b) 1 (c) zero (d) -1

6. If $a(n)$ is the response of a linear, time-invariant discrete-time system to a unit step input, then the response of the same system to a unit impulse input is

(a) $\frac{d}{dn}[a(n)]$ (b) $na(n)$ (c) $a(n)-a(n-1)$ (d) $a(n+1)-2a(n)-a(n-1)$

7. The system represented by $h(n)=(0.99)^n u(n+2)$ is

(a) unstable because it is an FIR system

(b) stable because it is an IIR system

(c) unstable because it does not obey BIBO stability criterion

(d) stable because it obeys BIBO stability criterion

8. If $x(n)=0$ for $n<N_1$ and $h(n)=0$ for $n<N_2$ then

(a) $x(n)*h(n)\neq 0$ for $n<N_1+N_2$

(b) $x(n)*h(n)=0$ for $n<N_1+N_2-1$

(c) $x(n)*h(n)=0$ for $n<N_1+N_2$

(d) $x(n)*h(n)\neq 0$ for $n<N_1+N_2-1$

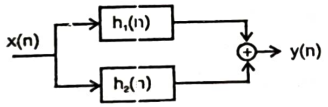
9. Which of the following equations represent the following system

(a) $y(n)=x(n)[h_1(n)+h_2(n)]$

(b) $y(n)=x(n)*[h_1(n)+h_2(n)]$

(c) $y(n)=x(n)*[h_1(n)*h_2(n)]$

(d) $y(n)=x(n)*[h_1(n)h_2(n)]$



10. Which of the following systems is are causal?

(a) $h(n)=\left(\frac{1}{2}\right)^n u(n)$ (b) $h(n)=a^n u(-n)$ (c) $h(n)=\cos\pi n$ (d) $h(n)=\sin\frac{n\pi}{4}$

11. If an input signal is applied to two LTI system with impulse responses $h(t)$ and $2h(t-2)$, then the response of the second system is the response of the first with

(a) amplitude scaled by 2 and delayed by 2

(b) amplitude scaled by 2 and advanced by 2

(c) delayed by 2

(d) amplitude scaled by 2

Answers to Multiple Choice Questions

1. (a) stable but not causal

2. (a) Hint: The conjugate antisymmetry part is

$$x_0(n)=\frac{1}{2}[x(n)-x^*(-n)]$$

3. (c) $|\alpha|<2$ any value of β

4. (d)

$$Y(e^{j\omega})=e^{j\omega n}H(e^{j\omega})$$

$$\omega=\frac{\pi}{4}$$

For the given $h(n)$, $H(e^{j\omega})=8\sqrt{2}\cos 2\omega-4\sqrt{2}\cos\omega$

$$H(e^{\frac{j\pi}{4}})=-4$$

$$Y(e^{j\omega})=e^{\frac{j\pi n}{4}}H(e^{\frac{j\pi}{4}})$$

$$=-4e^{\frac{j\pi n}{4}}$$

5. (b) 1

6. (c) $a(n)-a(n-1)$

7. (c) unstable because the system does not obey BIBO stability criterion

8. (c) $x(n)*h(n)=0$ for $n<N_1+N_2$

9. (b) $y(n)=x(n)*[h_1(n)+h_2(n)]$

10. (a)

11. (a) amplitude scaled by 2 and delayed by 2